Material Characterization Report

NanoVNA UTN Toolkit

2026-10-23 19:08:03

Measurement Details:

- **Sample:** Muestra14
- **Technique:** Open-ended coaxial probe (liquids)
- **Temperature:** 23.0 °C
- **Reference Liquids:** Distilled Water / Isopropyl Alcohol (IPA)
- **Probe:** 3 mm probe
- **Frequency Range:** 0.005–6.000 GHz, 1200 pts
- **Data sources:**
  - **Open:** kit:cal_sonda3mm_UBA_1200p
  - **Short:** kit:cal_sonda3mm_UBA_1200p (1 spike(s) replaced in the computation: 4885 MHz)
  - **Ref 1:** kit:cal_sonda3mm_UBA_1200p
  - **Ref 2:** kit:cal_sonda3mm_UBA_1200p
  - **DUT:** imported

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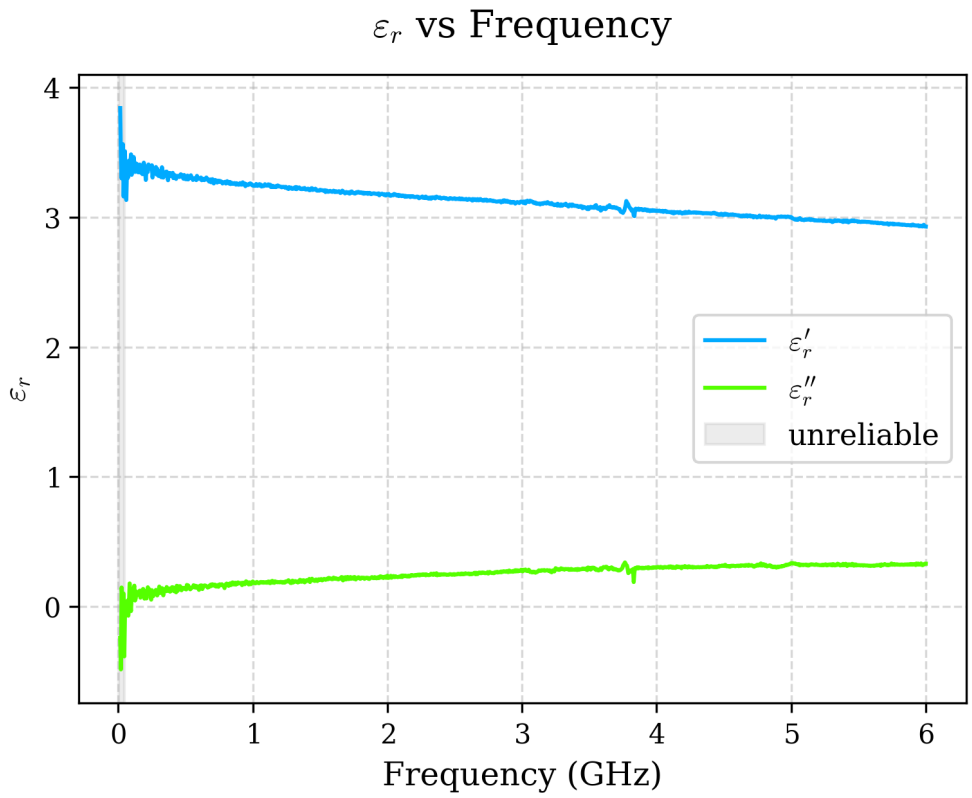
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1 Measurement Results

1.1 Complex Permittivity $\varepsilon_r(f)$



Points without a solution: 2 of 1200 (left blank in the plot and shown as — in the table). Non-physical points ($\varepsilon_r'' < 0$ or $\varepsilon_r' < 1$): 8 of 1200, kept and marked † in the table, not discarded.

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.005	—	—	—
0.010	—	—	—
0.015	3.8411	-0.2437†	-0.0634
0.020	3.4688	-0.4854†	-0.1399
0.025	3.2990	0.1458	0.0442
0.030	3.4649	0.0784	0.0226
0.035	3.5628	-0.3590†	-0.1008
0.040	3.1595	0.0988	0.0313
0.045	3.4261	-0.3851†	-0.1124
0.050	3.5084	-0.0576†	-0.0164
0.055	3.3056	0.0344	0.0104
0.060	3.1328	0.0076	0.0024
0.065	3.3774	-0.0227†	-0.0067
0.070	3.3006	0.0457	0.0139
0.075	3.3114	-0.0716†	-0.0216

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.080	3.4362	0.0062	0.0018
0.085	3.3506	0.1771	0.0529
0.090	3.4111	0.0195	0.0057
0.095	3.4845	-0.0360†	-0.0103
0.100	3.3267	0.0906	0.0272
0.105	3.4571	0.0512	0.0148
0.110	3.4225	0.0547	0.0160
0.115	3.4635	0.1418	0.0409
0.120	3.4164	0.1599	0.0468
0.125	3.3491	0.0456	0.0136
0.130	3.3801	0.0626	0.0185
0.135	3.4132	0.1147	0.0336
0.140	3.3503	0.1001	0.0299
0.145	3.3662	0.1007	0.0299
0.150	3.3459	0.1043	0.0312
0.155	3.4038	0.0884	0.0260
0.160	3.4058	0.1216	0.0357
0.165	3.3628	0.0675	0.0201
0.170	3.3414	0.0798	0.0239
0.175	3.3608	0.1063	0.0316
0.180	3.3318	0.1260	0.0378
0.185	3.4001	0.1345	0.0396
0.190	3.4206	0.0972	0.0284
0.195	3.4006	0.0600	0.0176
0.200	3.3532	0.1196	0.0357
0.205	3.2874	0.0978	0.0297
0.210	3.3781	0.0615	0.0182
0.215	3.3554	0.1072	0.0320
0.220	3.3484	0.1477	0.0441
0.225	3.4063	0.1033	0.0303
0.230	3.3840	0.1332	0.0394
0.235	3.3837	0.1177	0.0348
0.240	3.3745	0.0961	0.0285
0.245	3.3747	0.0523	0.0155
0.250	3.4004	0.1254	0.0369
0.255	3.3058	0.1023	0.0309
0.260	3.3666	0.0785	0.0233
0.265	3.3500	0.1017	0.0303
0.270	3.3732	0.1132	0.0336
0.275	3.3648	0.1112	0.0330
0.280	3.3870	0.1273	0.0376
0.285	3.3289	0.0972	0.0292
0.290	3.3735	0.0847	0.0251
0.295	3.3289	0.1186	0.0356
0.300	3.3175	0.1023	0.0308
0.305	3.3565	0.1528	0.0455

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.310	3.3358	0.1030	0.0309
0.315	3.3319	0.1057	0.0317
0.320	3.3082	0.1153	0.0348
0.325	3.3852	0.1407	0.0416
0.330	3.3530	0.1372	0.0409
0.335	3.3056	0.0855	0.0259
0.340	3.3416	0.1293	0.0387
0.345	3.3195	0.1225	0.0369
0.350	3.3436	0.1331	0.0398
0.355	3.3506	0.1169	0.0349
0.360	3.3515	0.1493	0.0445
0.365	3.3187	0.1196	0.0360
0.370	3.2851	0.1371	0.0417
0.375	3.3397	0.1253	0.0375
0.380	3.3272	0.1378	0.0414
0.385	3.3410	0.1135	0.0340
0.390	3.3172	0.1176	0.0355
0.395	3.3351	0.1313	0.0394
0.400	3.3261	0.1483	0.0446
0.405	3.3419	0.1386	0.0415
0.410	3.3014	0.1460	0.0442
0.415	3.3278	0.1172	0.0352
0.420	3.3022	0.1381	0.0418
0.425	3.3126	0.1309	0.0395
0.430	3.2952	0.1088	0.0330
0.435	3.3192	0.1592	0.0479
0.440	3.3116	0.1192	0.0360
0.445	3.3164	0.1276	0.0385
0.450	3.3186	0.1553	0.0468
0.455	3.3101	0.1417	0.0428
0.460	3.3128	0.1558	0.0470
0.465	3.3312	0.1294	0.0389
0.470	3.3272	0.1244	0.0374
0.475	3.3213	0.1306	0.0393
0.480	3.3111	0.1429	0.0432
0.485	3.2962	0.1422	0.0431
0.490	3.3255	0.1419	0.0427
0.495	3.2981	0.1360	0.0412
0.500	3.2934	0.1387	0.0421
0.505	3.2926	0.1304	0.0396
0.510	3.3053	0.1655	0.0501
0.515	3.3207	0.1487	0.0448
0.520	3.2914	0.1616	0.0491
0.525	3.3044	0.1503	0.0455
0.530	3.3072	0.1386	0.0419
0.535	3.3027	0.1314	0.0398

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.540	3.3231	0.1433	0.0431
0.545	3.3042	0.1410	0.0427
0.550	3.2963	0.1486	0.0451
0.555	3.2874	0.1480	0.0450
0.560	3.2965	0.1635	0.0496
0.565	3.3007	0.1595	0.0483
0.570	3.2974	0.1454	0.0441
0.575	3.2947	0.1591	0.0483
0.580	3.2859	0.1477	0.0450
0.585	3.3186	0.1428	0.0430
0.590	3.2977	0.1550	0.0470
0.595	3.3033	0.1529	0.0463
0.600	3.2953	0.1274	0.0386
0.605	3.3015	0.1433	0.0434
0.610	3.3019	0.1609	0.0487
0.615	3.3056	0.1474	0.0446
0.620	3.2851	0.1618	0.0492
0.625	3.2913	0.1337	0.0406
0.630	3.2903	0.1467	0.0446
0.635	3.2957	0.1552	0.0471
0.640	3.2858	0.1619	0.0493
0.645	3.2969	0.1615	0.0490
0.650	3.2924	0.1600	0.0486
0.655	3.3060	0.1565	0.0473
0.660	3.2903	0.1536	0.0467
0.665	3.2944	0.1449	0.0440
0.670	3.2890	0.1541	0.0468
0.675	3.2749	0.1499	0.0458
0.680	3.2772	0.1599	0.0488
0.685	3.2908	0.1667	0.0507
0.690	3.2696	0.1563	0.0478
0.695	3.2794	0.1696	0.0517
0.700	3.2932	0.1703	0.0517
0.705	3.2805	0.1620	0.0494
0.710	3.2602	0.1668	0.0512
0.715	3.2761	0.1602	0.0489
0.720	3.2872	0.1675	0.0510
0.725	3.2809	0.1538	0.0469
0.730	3.2820	0.1534	0.0467
0.735	3.2672	0.1608	0.0492
0.740	3.2817	0.1642	0.0500
0.745	3.2754	0.1514	0.0462
0.750	3.2803	0.1671	0.0509
0.755	3.2717	0.1702	0.0520
0.760	3.2783	0.1653	0.0504
0.765	3.2755	0.1667	0.0509

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
0.770	3.2722	0.1707	0.0522
0.775	3.2598	0.1629	0.0500
0.780	3.2680	0.1594	0.0488
0.785	3.2674	0.1552	0.0475
0.790	3.2672	0.1411	0.0432
0.795	3.2733	0.1666	0.0509
0.800	3.2596	0.1606	0.0493
0.805	3.2663	0.1763	0.0540
0.810	3.2777	0.1670	0.0510
0.815	3.2709	0.1718	0.0525
0.820	3.2622	0.1586	0.0486
0.825	3.2590	0.1728	0.0530
0.830	3.2721	0.1776	0.0543
0.835	3.2557	0.1637	0.0503
0.840	3.2593	0.1667	0.0512
0.845	3.2644	0.1824	0.0559
0.850	3.2573	0.1850	0.0568
0.855	3.2606	0.1713	0.0525
0.860	3.2567	0.1693	0.0520
0.865	3.2633	0.1738	0.0533
0.870	3.2558	0.1828	0.0561
0.875	3.2512	0.1685	0.0518
0.880	3.2621	0.1781	0.0546
0.885	3.2584	0.1732	0.0532
0.890	3.2545	0.1850	0.0568
0.895	3.2502	0.1822	0.0561
0.900	3.2563	0.1790	0.0550
0.905	3.2629	0.1788	0.0548
0.910	3.2588	0.1756	0.0539
0.915	3.2509	0.1816	0.0559
0.920	3.2577	0.1641	0.0504
0.925	3.2688	0.1762	0.0539
0.930	3.2518	0.1829	0.0562
0.935	3.2571	0.1805	0.0554
0.940	3.2603	0.1709	0.0524
0.945	3.2393	0.1795	0.0554
0.950	3.2551	0.1937	0.0595
0.955	3.2504	0.1866	0.0574
0.960	3.2503	0.1775	0.0546
0.965	3.2484	0.1844	0.0568
0.970	3.2551	0.1692	0.0520
0.975	3.2531	0.1852	0.0569
0.980	3.2610	0.1830	0.0561
0.985	3.2593	0.1712	0.0525
0.990	3.2516	0.1844	0.0567
0.995	3.2509	0.1875	0.0577

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.000	3.2431	0.1924	0.0593
1.005	3.2470	0.1803	0.0555
1.010	3.2463	0.1836	0.0566
1.015	3.2419	0.1830	0.0565
1.020	3.2451	0.1777	0.0548
1.025	3.2559	0.1830	0.0562
1.030	3.2439	0.1887	0.0582
1.035	3.2418	0.1820	0.0561
1.040	3.2467	0.1815	0.0559
1.045	3.2519	0.1787	0.0550
1.050	3.2549	0.1815	0.0558
1.055	3.2423	0.1870	0.0577
1.060	3.2507	0.1883	0.0579
1.065	3.2366	0.1875	0.0579
1.070	3.2419	0.1835	0.0566
1.075	3.2425	0.1876	0.0578
1.080	3.2397	0.1828	0.0564
1.085	3.2437	0.1851	0.0571
1.090	3.2543	0.1833	0.0563
1.095	3.2450	0.1857	0.0572
1.100	3.2450	0.1870	0.0576
1.105	3.2432	0.1821	0.0561
1.110	3.2493	0.1875	0.0577
1.115	3.2532	0.1918	0.0590
1.120	3.2435	0.1910	0.0589
1.125	3.2423	0.1803	0.0556
1.130	3.2502	0.1795	0.0552
1.135	3.2324	0.1917	0.0593
1.140	3.2333	0.1943	0.0601
1.145	3.2421	0.1969	0.0607
1.150	3.2339	0.1831	0.0566
1.155	3.2394	0.1862	0.0575
1.160	3.2325	0.1817	0.0562
1.165	3.2334	0.1935	0.0598
1.170	3.2319	0.1892	0.0586
1.175	3.2412	0.1874	0.0578
1.180	3.2357	0.1890	0.0584
1.185	3.2450	0.1860	0.0573
1.190	3.2398	0.1871	0.0578
1.195	3.2393	0.1870	0.0577
1.200	3.2363	0.1854	0.0573
1.205	3.2357	0.1895	0.0586
1.210	3.2343	0.1872	0.0579
1.215	3.2302	0.1887	0.0584
1.220	3.2333	0.1835	0.0568
1.225	3.2342	0.1884	0.0582

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.230	3.2339	0.1839	0.0569
1.235	3.2331	0.1887	0.0584
1.240	3.2270	0.1845	0.0572
1.245	3.2341	0.1926	0.0596
1.250	3.2303	0.1934	0.0599
1.255	3.2225	0.1935	0.0600
1.260	3.2233	0.1919	0.0595
1.265	3.2174	0.2019	0.0627
1.270	3.2321	0.1956	0.0605
1.275	3.2259	0.1896	0.0588
1.280	3.2252	0.1810	0.0561
1.285	3.2284	0.1800	0.0558
1.290	3.2191	0.1886	0.0586
1.295	3.2212	0.1987	0.0617
1.300	3.2195	0.1838	0.0571
1.305	3.2241	0.1930	0.0599
1.310	3.2289	0.1912	0.0592
1.315	3.2191	0.1953	0.0607
1.320	3.2327	0.1924	0.0595
1.325	3.2161	0.1884	0.0586
1.330	3.2298	0.1933	0.0598
1.335	3.2246	0.1844	0.0572
1.340	3.2250	0.1925	0.0597
1.345	3.2295	0.1908	0.0591
1.350	3.2205	0.2048	0.0636
1.355	3.2172	0.1910	0.0594
1.360	3.2183	0.1998	0.0621
1.365	3.2161	0.1901	0.0591
1.370	3.2209	0.2004	0.0622
1.375	3.2228	0.1967	0.0610
1.380	3.2183	0.2014	0.0626
1.385	3.2254	0.1853	0.0574
1.390	3.2145	0.1904	0.0592
1.395	3.2133	0.1960	0.0610
1.400	3.2142	0.2023	0.0629
1.405	3.2159	0.2025	0.0630
1.410	3.2201	0.1960	0.0609
1.415	3.2166	0.2009	0.0624
1.420	3.2080	0.2121	0.0661
1.425	3.2134	0.1880	0.0585
1.430	3.2104	0.2048	0.0638
1.435	3.2233	0.2053	0.0637
1.440	3.2235	0.2137	0.0663
1.445	3.2099	0.2015	0.0628
1.450	3.2139	0.2006	0.0624
1.455	3.2140	0.2003	0.0623

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.460	3.2152	0.1962	0.0610
1.465	3.2198	0.2054	0.0638
1.470	3.2144	0.2072	0.0645
1.475	3.2053	0.1999	0.0624
1.480	3.2049	0.1957	0.0611
1.485	3.2055	0.2054	0.0641
1.490	3.2056	0.2089	0.0652
1.495	3.2048	0.2031	0.0634
1.500	3.2114	0.2089	0.0650
1.505	3.2097	0.2083	0.0649
1.510	3.2085	0.2109	0.0657
1.515	3.2081	0.2005	0.0625
1.520	3.2064	0.2066	0.0644
1.525	3.2041	0.2072	0.0647
1.530	3.2048	0.2123	0.0662
1.535	3.2079	0.2081	0.0649
1.540	3.1940	0.2053	0.0643
1.545	3.2042	0.2075	0.0647
1.550	3.2052	0.2133	0.0666
1.555	3.1969	0.2149	0.0672
1.560	3.1960	0.2096	0.0656
1.565	3.1971	0.2201	0.0688
1.570	3.2011	0.2129	0.0665
1.575	3.2034	0.2145	0.0669
1.580	3.1995	0.2156	0.0674
1.585	3.1974	0.2102	0.0657
1.590	3.1954	0.2127	0.0666
1.595	3.1972	0.2104	0.0658
1.600	3.2047	0.2218	0.0692
1.605	3.2005	0.2181	0.0681
1.610	3.1936	0.2136	0.0669
1.615	3.1990	0.2178	0.0681
1.620	3.2002	0.2117	0.0662
1.625	3.1942	0.2135	0.0668
1.630	3.1959	0.2133	0.0667
1.635	3.1982	0.2080	0.0651
1.640	3.2003	0.2120	0.0662
1.645	3.1956	0.2131	0.0667
1.650	3.1869	0.2186	0.0686
1.655	3.1887	0.2190	0.0687
1.660	3.1963	0.2266	0.0709
1.665	3.2048	0.2113	0.0659
1.670	3.2002	0.2004	0.0626
1.675	3.1931	0.2188	0.0685
1.680	3.1921	0.2234	0.0700
1.685	3.1834	0.2083	0.0654

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.690	3.1918	0.2252	0.0706
1.695	3.1942	0.2089	0.0654
1.700	3.1884	0.2169	0.0680
1.705	3.1900	0.2271	0.0712
1.710	3.1914	0.2109	0.0661
1.715	3.1832	0.2111	0.0663
1.720	3.1818	0.2199	0.0691
1.725	3.1954	0.2267	0.0709
1.730	3.1879	0.2125	0.0666
1.735	3.1987	0.2171	0.0679
1.740	3.1897	0.2224	0.0697
1.745	3.1887	0.2230	0.0699
1.750	3.1869	0.2123	0.0666
1.755	3.1827	0.2270	0.0713
1.760	3.1883	0.2129	0.0668
1.765	3.1845	0.2079	0.0653
1.770	3.1926	0.2195	0.0688
1.775	3.1936	0.2174	0.0681
1.780	3.1900	0.2183	0.0684
1.785	3.1955	0.2249	0.0704
1.790	3.1908	0.2167	0.0679
1.795	3.1878	0.2175	0.0682
1.800	3.1878	0.2126	0.0667
1.805	3.1920	0.2215	0.0694
1.810	3.1864	0.2179	0.0684
1.815	3.1845	0.2201	0.0691
1.820	3.1898	0.2186	0.0685
1.825	3.1811	0.2211	0.0695
1.830	3.1881	0.2121	0.0665
1.835	3.1877	0.2173	0.0682
1.840	3.1835	0.2199	0.0691
1.845	3.1861	0.2212	0.0694
1.850	3.1880	0.2208	0.0692
1.855	3.1799	0.2154	0.0677
1.860	3.1865	0.2192	0.0688
1.865	3.1879	0.2114	0.0663
1.870	3.1868	0.2149	0.0674
1.875	3.1786	0.2192	0.0690
1.880	3.1819	0.2217	0.0697
1.885	3.1795	0.2258	0.0710
1.890	3.1828	0.2228	0.0700
1.895	3.1829	0.2177	0.0684
1.900	3.1843	0.2203	0.0692
1.905	3.1819	0.2258	0.0710
1.910	3.1812	0.2179	0.0685
1.915	3.1813	0.2230	0.0701

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
1.920	3.1788	0.2165	0.0681
1.925	3.1789	0.2202	0.0693
1.930	3.1766	0.2249	0.0708
1.935	3.1837	0.2182	0.0685
1.940	3.1690	0.2195	0.0693
1.945	3.1775	0.2315	0.0728
1.950	3.1782	0.2234	0.0703
1.955	3.1781	0.2306	0.0726
1.960	3.1813	0.2227	0.0700
1.965	3.1765	0.2237	0.0704
1.970	3.1749	0.2189	0.0689
1.975	3.1781	0.2279	0.0717
1.980	3.1701	0.2280	0.0719
1.985	3.1712	0.2301	0.0726
1.990	3.1679	0.2202	0.0695
1.995	3.1784	0.2233	0.0702
2.000	3.1731	0.2364	0.0745
2.005	3.1768	0.2191	0.0690
2.010	3.1794	0.2265	0.0712
2.015	3.1685	0.2218	0.0700
2.020	3.1660	0.2295	0.0725
2.025	3.1651	0.2360	0.0746
2.030	3.1702	0.2212	0.0698
2.035	3.1719	0.2322	0.0732
2.040	3.1731	0.2259	0.0712
2.045	3.1693	0.2342	0.0739
2.050	3.1794	0.2292	0.0721
2.055	3.1766	0.2253	0.0709
2.060	3.1759	0.2326	0.0732
2.065	3.1711	0.2284	0.0720
2.070	3.1668	0.2282	0.0720
2.075	3.1669	0.2257	0.0713
2.080	3.1724	0.2279	0.0719
2.085	3.1629	0.2273	0.0719
2.090	3.1696	0.2308	0.0728
2.095	3.1693	0.2306	0.0728
2.100	3.1655	0.2268	0.0717
2.105	3.1623	0.2378	0.0752
2.110	3.1675	0.2264	0.0715
2.115	3.1642	0.2383	0.0753
2.120	3.1654	0.2340	0.0739
2.125	3.1646	0.2365	0.0747
2.130	3.1572	0.2329	0.0738
2.135	3.1646	0.2376	0.0751
2.140	3.1590	0.2329	0.0737
2.145	3.1578	0.2304	0.0730

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.150	3.1615	0.2309	0.0730
2.155	3.1625	0.2369	0.0749
2.160	3.1616	0.2305	0.0729
2.165	3.1615	0.2307	0.0730
2.170	3.1553	0.2342	0.0742
2.175	3.1614	0.2408	0.0762
2.180	3.1588	0.2365	0.0749
2.185	3.1611	0.2416	0.0764
2.190	3.1619	0.2379	0.0752
2.195	3.1573	0.2364	0.0749
2.200	3.1549	0.2393	0.0759
2.205	3.1555	0.2375	0.0753
2.210	3.1551	0.2426	0.0769
2.215	3.1554	0.2382	0.0755
2.220	3.1592	0.2433	0.0770
2.225	3.1576	0.2443	0.0774
2.230	3.1570	0.2408	0.0763
2.235	3.1461	0.2450	0.0779
2.240	3.1502	0.2392	0.0759
2.245	3.1517	0.2452	0.0778
2.250	3.1546	0.2440	0.0773
2.255	3.1532	0.2421	0.0768
2.260	3.1597	0.2458	0.0778
2.265	3.1496	0.2429	0.0771
2.270	3.1505	0.2466	0.0783
2.275	3.1536	0.2460	0.0780
2.280	3.1477	0.2466	0.0784
2.285	3.1591	0.2399	0.0759
2.290	3.1519	0.2467	0.0783
2.295	3.1482	0.2434	0.0773
2.300	3.1450	0.2420	0.0770
2.305	3.1500	0.2416	0.0767
2.310	3.1435	0.2411	0.0767
2.315	3.1501	0.2435	0.0773
2.320	3.1518	0.2436	0.0773
2.325	3.1422	0.2522	0.0803
2.330	3.1496	0.2490	0.0791
2.335	3.1534	0.2467	0.0782
2.340	3.1502	0.2430	0.0771
2.345	3.1525	0.2532	0.0803
2.350	3.1544	0.2504	0.0794
2.355	3.1448	0.2474	0.0787
2.360	3.1519	0.2425	0.0769
2.365	3.1588	0.2555	0.0809
2.370	3.1552	0.2399	0.0760
2.375	3.1533	0.2444	0.0775

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.380	3.1503	0.2451	0.0778
2.385	3.1473	0.2423	0.0770
2.390	3.1533	0.2465	0.0782
2.395	3.1469	0.2449	0.0778
2.400	3.1433	0.2481	0.0789
2.405	3.1467	0.2456	0.0780
2.410	3.1481	0.2466	0.0783
2.415	3.1494	0.2455	0.0780
2.420	3.1504	0.2487	0.0789
2.425	3.1465	0.2457	0.0781
2.430	3.1478	0.2465	0.0783
2.435	3.1437	0.2457	0.0782
2.440	3.1454	0.2433	0.0774
2.445	3.1420	0.2440	0.0777
2.450	3.1395	0.2442	0.0778
2.455	3.1391	0.2438	0.0776
2.460	3.1418	0.2467	0.0785
2.465	3.1472	0.2491	0.0792
2.470	3.1463	0.2469	0.0785
2.475	3.1448	0.2466	0.0784
2.480	3.1428	0.2439	0.0776
2.485	3.1408	0.2452	0.0781
2.490	3.1437	0.2456	0.0781
2.495	3.1432	0.2492	0.0793
2.500	3.1460	0.2441	0.0776
2.505	3.1397	0.2465	0.0785
2.510	3.1422	0.2479	0.0789
2.515	3.1411	0.2465	0.0785
2.520	3.1420	0.2485	0.0791
2.525	3.1423	0.2517	0.0801
2.530	3.1394	0.2511	0.0800
2.535	3.1456	0.2490	0.0792
2.540	3.1426	0.2471	0.0786
2.545	3.1375	0.2477	0.0789
2.550	3.1437	0.2535	0.0806
2.555	3.1432	0.2489	0.0792
2.560	3.1380	0.2529	0.0806
2.565	3.1377	0.2476	0.0789
2.570	3.1388	0.2490	0.0793
2.575	3.1374	0.2498	0.0796
2.580	3.1376	0.2554	0.0814
2.585	3.1413	0.2522	0.0803
2.590	3.1351	0.2517	0.0803
2.595	3.1335	0.2519	0.0804
2.600	3.1362	0.2531	0.0807
2.605	3.1405	0.2537	0.0808

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.610	3.1401	0.2472	0.0787
2.615	3.1366	0.2516	0.0802
2.620	3.1376	0.2495	0.0795
2.625	3.1375	0.2524	0.0805
2.630	3.1285	0.2493	0.0797
2.635	3.1404	0.2515	0.0801
2.640	3.1314	0.2528	0.0807
2.645	3.1370	0.2584	0.0824
2.650	3.1392	0.2588	0.0825
2.655	3.1359	0.2481	0.0791
2.660	3.1305	0.2592	0.0828
2.665	3.1317	0.2563	0.0818
2.670	3.1395	0.2490	0.0793
2.675	3.1308	0.2541	0.0812
2.680	3.1304	0.2611	0.0834
2.685	3.1356	0.2607	0.0832
2.690	3.1399	0.2594	0.0826
2.695	3.1321	0.2556	0.0816
2.700	3.1344	0.2603	0.0830
2.705	3.1297	0.2576	0.0823
2.710	3.1309	0.2610	0.0833
2.715	3.1273	0.2608	0.0834
2.720	3.1350	0.2593	0.0827
2.725	3.1309	0.2559	0.0817
2.730	3.1342	0.2577	0.0822
2.735	3.1353	0.2573	0.0821
2.740	3.1263	0.2566	0.0821
2.745	3.1285	0.2644	0.0845
2.750	3.1280	0.2532	0.0810
2.755	3.1292	0.2603	0.0832
2.760	3.1312	0.2654	0.0848
2.765	3.1278	0.2618	0.0837
2.770	3.1300	0.2594	0.0829
2.775	3.1269	0.2575	0.0823
2.780	3.1255	0.2568	0.0822
2.785	3.1276	0.2589	0.0828
2.790	3.1244	0.2560	0.0819
2.795	3.1237	0.2575	0.0824
2.800	3.1233	0.2638	0.0845
2.805	3.1213	0.2630	0.0842
2.810	3.1200	0.2648	0.0849
2.815	3.1203	0.2664	0.0854
2.820	3.1220	0.2614	0.0837
2.825	3.1213	0.2641	0.0846
2.830	3.1229	0.2637	0.0844
2.835	3.1238	0.2646	0.0847

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
2.840	3.1242	0.2637	0.0844
2.845	3.1142	0.2693	0.0865
2.850	3.1187	0.2643	0.0847
2.855	3.1257	0.2666	0.0853
2.860	3.1183	0.2708	0.0869
2.865	3.1302	0.2687	0.0858
2.870	3.1298	0.2703	0.0864
2.875	3.1208	0.2694	0.0863
2.880	3.1248	0.2672	0.0855
2.885	3.1197	0.2697	0.0864
2.890	3.1266	0.2728	0.0872
2.895	3.1218	0.2650	0.0849
2.900	3.1173	0.2652	0.0851
2.905	3.1192	0.2767	0.0887
2.910	3.1208	0.2670	0.0856
2.915	3.1112	0.2705	0.0869
2.920	3.1146	0.2700	0.0867
2.925	3.1113	0.2690	0.0865
2.930	3.1179	0.2672	0.0857
2.935	3.1143	0.2737	0.0879
2.940	3.1172	0.2686	0.0862
2.945	3.1137	0.2724	0.0875
2.950	3.1047	0.2792	0.0899
2.955	3.1114	0.2693	0.0865
2.960	3.1127	0.2784	0.0894
2.965	3.1170	0.2780	0.0892
2.970	3.1141	0.2701	0.0867
2.975	3.1133	0.2772	0.0891
2.980	3.1092	0.2736	0.0880
2.985	3.1154	0.2778	0.0892
2.990	3.1201	0.2747	0.0880
2.995	3.1178	0.2826	0.0906
3.000	3.1195	0.2810	0.0901
3.005	3.1193	0.2702	0.0866
3.010	3.1098	0.2825	0.0908
3.015	3.1211	0.2768	0.0887
3.020	3.1168	0.2812	0.0902
3.025	3.1063	0.2847	0.0917
3.030	3.1224	0.2753	0.0882
3.035	3.1221	0.2751	0.0881
3.040	3.1274	0.2721	0.0870
3.045	3.1183	0.2746	0.0881
3.050	3.1203	0.2832	0.0908
3.055	3.1172	0.2716	0.0871
3.060	3.1273	0.2656	0.0849
3.065	3.1243	0.2689	0.0861

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.070	3.1230	0.2661	0.0852
3.075	3.1205	0.2618	0.0839
3.080	3.1169	0.2640	0.0847
3.085	3.1114	0.2677	0.0860
3.090	3.1147	0.2636	0.0846
3.095	3.1085	0.2658	0.0855
3.100	3.1129	0.2598	0.0835
3.105	3.1039	0.2629	0.0847
3.110	3.1074	0.2644	0.0851
3.115	3.1062	0.2664	0.0858
3.120	3.1038	0.2716	0.0875
3.125	3.1029	0.2776	0.0895
3.130	3.1060	0.2760	0.0889
3.135	3.0985	0.2751	0.0888
3.140	3.1046	0.2737	0.0882
3.145	3.1031	0.2788	0.0899
3.150	3.1066	0.2792	0.0899
3.155	3.1024	0.2768	0.0892
3.160	3.1085	0.2769	0.0891
3.165	3.1104	0.2804	0.0901
3.170	3.1074	0.2749	0.0885
3.175	3.1105	0.2769	0.0890
3.180	3.1058	0.2780	0.0895
3.185	3.1064	0.2790	0.0898
3.190	3.1164	0.2793	0.0896
3.195	3.1086	0.2751	0.0885
3.200	3.1082	0.2732	0.0879
3.205	3.1028	0.2783	0.0897
3.210	3.1035	0.2685	0.0865
3.215	3.1012	0.2694	0.0869
3.220	3.1091	0.2630	0.0846
3.225	3.1007	0.2761	0.0890
3.230	3.0988	0.2777	0.0896
3.235	3.0980	0.2771	0.0894
3.240	3.0927	0.2691	0.0870
3.245	3.0969	0.2820	0.0911
3.250	3.0974	0.2947	0.0952
3.255	3.0951	0.2885	0.0932
3.260	3.0903	0.2853	0.0923
3.265	3.1038	0.2879	0.0927
3.270	3.0903	0.2833	0.0917
3.275	3.0927	0.2816	0.0911
3.280	3.0934	0.2864	0.0926
3.285	3.0875	0.2854	0.0924
3.290	3.0993	0.2937	0.0947
3.295	3.0897	0.2886	0.0934

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.300	3.0888	0.2863	0.0927
3.305	3.0941	0.2889	0.0934
3.310	3.0976	0.2875	0.0928
3.315	3.0972	0.2831	0.0914
3.320	3.0966	0.2884	0.0931
3.325	3.1043	0.2882	0.0928
3.330	3.0924	0.2874	0.0929
3.335	3.0912	0.2882	0.0932
3.340	3.0934	0.2864	0.0926
3.345	3.0821	0.2884	0.0936
3.350	3.0902	0.2817	0.0912
3.355	3.0971	0.2862	0.0924
3.360	3.0963	0.2820	0.0911
3.365	3.0951	0.2889	0.0933
3.370	3.0860	0.2836	0.0919
3.375	3.0975	0.2829	0.0913
3.380	3.0943	0.2840	0.0918
3.385	3.0908	0.2833	0.0916
3.390	3.0874	0.2842	0.0920
3.395	3.0885	0.2826	0.0915
3.400	3.0889	0.2865	0.0927
3.405	3.0853	0.2875	0.0932
3.410	3.0883	0.2805	0.0908
3.415	3.0841	0.2888	0.0937
3.420	3.0793	0.2825	0.0918
3.425	3.0841	0.2856	0.0926
3.430	3.0816	0.2848	0.0924
3.435	3.0856	0.2855	0.0925
3.440	3.0853	0.2861	0.0927
3.445	3.0743	0.2873	0.0934
3.450	3.0692	0.2918	0.0951
3.455	3.0806	0.2911	0.0945
3.460	3.0755	0.2880	0.0936
3.465	3.0777	0.2889	0.0939
3.470	3.0741	0.2900	0.0944
3.475	3.0713	0.2849	0.0928
3.480	3.0681	0.2929	0.0955
3.485	3.0624	0.2798	0.0914
3.490	3.0776	0.3001	0.0975
3.495	3.0621	0.2987	0.0976
3.500	3.0694	0.2971	0.0968
3.505	3.0737	0.2946	0.0958
3.510	3.0802	0.2836	0.0921
3.515	3.0742	0.2980	0.0969
3.520	3.0770	0.2910	0.0946
3.525	3.0751	0.2990	0.0972

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.530	3.0754	0.2984	0.0970
3.535	3.0722	0.2978	0.0969
3.540	3.0858	0.2901	0.0940
3.545	3.0850	0.2915	0.0945
3.550	3.0933	0.2806	0.0907
3.555	3.0716	0.2850	0.0928
3.560	3.0733	0.2851	0.0928
3.565	3.0652	0.2779	0.0907
3.570	3.0647	0.2773	0.0905
3.575	3.0640	0.2819	0.0920
3.580	3.0779	0.2846	0.0925
3.585	3.0508	0.2948	0.0966
3.590	3.0556	0.2806	0.0918
3.595	3.0609	0.2908	0.0950
3.600	3.0725	0.2949	0.0960
3.605	3.0682	0.3035	0.0989
3.610	3.0729	0.3032	0.0987
3.615	3.0650	0.3019	0.0985
3.620	3.0621	0.2999	0.0979
3.625	3.0668	0.3059	0.0998
3.630	3.0735	0.3043	0.0990
3.635	3.0827	0.2965	0.0962
3.640	3.0837	0.2881	0.0934
3.645	3.0787	0.2848	0.0925
3.650	3.0927	0.2966	0.0959
3.655	3.0776	0.2939	0.0955
3.660	3.0787	0.2922	0.0949
3.665	3.0774	0.3016	0.0980
3.670	3.0740	0.2901	0.0944
3.675	3.0761	0.2944	0.0957
3.680	3.0741	0.2845	0.0926
3.685	3.0782	0.2887	0.0938
3.690	3.0727	0.2855	0.0929
3.695	3.0713	0.2877	0.0937
3.700	3.0748	0.2845	0.0925
3.705	3.0692	0.2815	0.0917
3.710	3.0681	0.2820	0.0919
3.715	3.0663	0.2776	0.0905
3.720	3.0614	0.2771	0.0905
3.725	3.0622	0.2785	0.0910
3.730	3.0483	0.2799	0.0918
3.735	3.0413	0.2847	0.0936
3.740	3.0424	0.2898	0.0952
3.745	3.0379	0.2946	0.0970
3.750	3.0330	0.3117	0.1028
3.755	3.0440	0.3224	0.1059

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.760	3.0505	0.3295	0.1080
3.765	3.0700	0.3392	0.1105
3.770	3.0916	0.3281	0.1061
3.775	3.1249	0.3194	0.1022
3.780	3.1147	0.3141	0.1008
3.785	3.1060	0.2888	0.0930
3.790	3.0974	0.2566	0.0829
3.795	3.0827	0.2661	0.0863
3.800	3.0754	0.2733	0.0889
3.805	3.0661	0.2671	0.0871
3.810	3.0639	0.2762	0.0901
3.815	3.0493	0.2806	0.0920
3.820	3.0573	0.2782	0.0910
3.825	3.0479	0.2879	0.0945
3.830	3.0135	0.1878	0.0623
3.835	3.0084	0.2642	0.0878
3.840	3.0593	0.2959	0.0967
3.845	3.0662	0.2972	0.0969
3.850	3.0555	0.2928	0.0958
3.855	3.0622	0.3015	0.0985
3.860	3.0596	0.2991	0.0978
3.865	3.0662	0.2929	0.0955
3.870	3.0562	0.3007	0.0984
3.875	3.0598	0.2977	0.0973
3.880	3.0634	0.2958	0.0966
3.885	3.0590	0.2954	0.0966
3.890	3.0624	0.2983	0.0974
3.895	3.0549	0.2938	0.0962
3.900	3.0571	0.2981	0.0975
3.905	3.0546	0.2949	0.0965
3.910	3.0616	0.2943	0.0961
3.915	3.0573	0.2976	0.0973
3.920	3.0596	0.2995	0.0979
3.925	3.0532	0.2990	0.0979
3.930	3.0499	0.2988	0.0980
3.935	3.0570	0.2940	0.0962
3.940	3.0524	0.3020	0.0989
3.945	3.0505	0.2953	0.0968
3.950	3.0466	0.2967	0.0974
3.955	3.0530	0.2977	0.0975
3.960	3.0492	0.2974	0.0975
3.965	3.0518	0.2961	0.0970
3.970	3.0546	0.3007	0.0984
3.975	3.0497	0.3013	0.0988
3.980	3.0499	0.2981	0.0978
3.985	3.0524	0.3003	0.0984

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
3.990	3.0515	0.3001	0.0984
3.995	3.0484	0.3019	0.0990
4.000	3.0485	0.2983	0.0978
4.005	3.0507	0.3027	0.0992
4.010	3.0478	0.3042	0.0998
4.015	3.0486	0.3035	0.0995
4.020	3.0511	0.3001	0.0984
4.025	3.0466	0.2983	0.0979
4.030	3.0485	0.3051	0.1001
4.035	3.0457	0.3024	0.0993
4.040	3.0479	0.3004	0.0986
4.045	3.0495	0.2963	0.0972
4.050	3.0410	0.3023	0.0994
4.055	3.0460	0.2993	0.0982
4.060	3.0438	0.3041	0.0999
4.065	3.0492	0.3034	0.0995
4.070	3.0467	0.3003	0.0986
4.075	3.0400	0.3008	0.0989
4.080	3.0418	0.3036	0.0998
4.085	3.0406	0.3036	0.0998
4.090	3.0399	0.2991	0.0984
4.095	3.0403	0.3077	0.1012
4.100	3.0399	0.3018	0.0993
4.105	3.0371	0.3085	0.1016
4.110	3.0411	0.2997	0.0986
4.115	3.0347	0.3040	0.1002
4.120	3.0400	0.2974	0.0978
4.125	3.0295	0.3049	0.1006
4.130	3.0406	0.3005	0.0988
4.135	3.0327	0.3007	0.0992
4.140	3.0378	0.2965	0.0976
4.145	3.0359	0.3011	0.0992
4.150	3.0381	0.3090	0.1017
4.155	3.0394	0.3071	0.1010
4.160	3.0375	0.3020	0.0994
4.165	3.0324	0.3041	0.1003
4.170	3.0350	0.3022	0.0996
4.175	3.0405	0.3045	0.1002
4.180	3.0337	0.2987	0.0985
4.185	3.0318	0.2943	0.0971
4.190	3.0280	0.3065	0.1012
4.195	3.0312	0.3048	0.1006
4.200	3.0366	0.3039	0.1001
4.205	3.0328	0.3047	0.1005
4.210	3.0305	0.3109	0.1026
4.215	3.0407	0.3017	0.0992

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.220	3.0388	0.3052	0.1004
4.225	3.0321	0.3067	0.1011
4.230	3.0352	0.2999	0.0988
4.235	3.0363	0.3066	0.1010
4.240	3.0429	0.3091	0.1016
4.245	3.0350	0.3023	0.0996
4.250	3.0362	0.3047	0.1004
4.255	3.0352	0.3012	0.0992
4.260	3.0351	0.3021	0.0996
4.265	3.0363	0.3055	0.1006
4.270	3.0363	0.2991	0.0985
4.275	3.0352	0.3033	0.0999
4.280	3.0374	0.3030	0.0998
4.285	3.0369	0.3011	0.0991
4.290	3.0368	0.3012	0.0992
4.295	3.0333	0.3011	0.0993
4.300	3.0358	0.3022	0.0996
4.305	3.0360	0.3021	0.0995
4.310	3.0341	0.3039	0.1001
4.315	3.0349	0.3011	0.0992
4.320	3.0347	0.3014	0.0993
4.325	3.0333	0.3027	0.0998
4.330	3.0341	0.3026	0.0997
4.335	3.0318	0.3017	0.0995
4.340	3.0317	0.3029	0.0999
4.345	3.0290	0.3037	0.1003
4.350	3.0300	0.3036	0.1002
4.355	3.0315	0.3028	0.0999
4.360	3.0308	0.3048	0.1006
4.365	3.0311	0.3036	0.1001
4.370	3.0318	0.3028	0.0999
4.375	3.0298	0.3121	0.1030
4.380	3.0305	0.3081	0.1017
4.385	3.0312	0.3061	0.1010
4.390	3.0296	0.3024	0.0998
4.395	3.0298	0.3050	0.1007
4.400	3.0304	0.3059	0.1009
4.405	3.0267	0.3093	0.1022
4.410	3.0293	0.3063	0.1011
4.415	3.0314	0.3062	0.1010
4.420	3.0307	0.3068	0.1012
4.425	3.0345	0.3032	0.0999
4.430	3.0338	0.3044	0.1003
4.435	3.0283	0.3071	0.1014
4.440	3.0289	0.3021	0.0998
4.445	3.0257	0.3029	0.1001

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.450	3.0263	0.3078	0.1017
4.455	3.0210	0.3054	0.1011
4.460	3.0212	0.3071	0.1016
4.465	3.0188	0.3135	0.1039
4.470	3.0214	0.3106	0.1028
4.475	3.0197	0.3111	0.1030
4.480	3.0268	0.3126	0.1033
4.485	3.0223	0.3153	0.1043
4.490	3.0242	0.3151	0.1042
4.495	3.0159	0.3124	0.1036
4.500	3.0177	0.3136	0.1039
4.505	3.0266	0.3118	0.1030
4.510	3.0201	0.3209	0.1063
4.515	3.0256	0.3158	0.1044
4.520	3.0194	0.3093	0.1024
4.525	3.0189	0.3116	0.1032
4.530	3.0219	0.3115	0.1031
4.535	3.0160	0.3089	0.1024
4.540	3.0190	0.3106	0.1029
4.545	3.0196	0.3057	0.1013
4.550	3.0210	0.3118	0.1032
4.555	3.0186	0.3136	0.1039
4.560	3.0189	0.3108	0.1030
4.565	3.0220	0.3136	0.1038
4.570	3.0182	0.3140	0.1040
4.575	3.0189	0.3121	0.1034
4.580	3.0165	0.3119	0.1034
4.585	3.0179	0.3099	0.1027
4.590	3.0166	0.3114	0.1032
4.595	3.0157	0.3134	0.1039
4.600	3.0149	0.3136	0.1040
4.605	3.0128	0.3140	0.1042
4.610	3.0152	0.3106	0.1030
4.615	3.0121	0.3139	0.1042
4.620	3.0105	0.3144	0.1044
4.625	3.0071	0.3118	0.1037
4.630	3.0120	0.3160	0.1049
4.635	3.0067	0.3163	0.1052
4.640	3.0072	0.3157	0.1050
4.645	3.0136	0.3148	0.1045
4.650	3.0112	0.3158	0.1049
4.655	3.0073	0.3131	0.1041
4.660	3.0097	0.3184	0.1058
4.665	3.0144	0.3150	0.1045
4.670	3.0148	0.3189	0.1058
4.675	3.0049	0.3163	0.1053

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.680	3.0100	0.3128	0.1039
4.685	3.0104	0.3209	0.1066
4.690	3.0133	0.3154	0.1047
4.695	3.0068	0.3161	0.1051
4.700	3.0120	0.3148	0.1045
4.705	3.0073	0.3187	0.1060
4.710	3.0089	0.3162	0.1051
4.715	3.0093	0.3138	0.1043
4.720	2.9986	0.3077	0.1026
4.725	3.0043	0.3061	0.1019
4.730	3.0071	0.3130	0.1041
4.735	3.0049	0.3124	0.1040
4.740	3.0051	0.3124	0.1040
4.745	2.9983	0.3197	0.1066
4.750	3.0018	0.3217	0.1072
4.755	3.0042	0.3145	0.1047
4.760	3.0024	0.3169	0.1055
4.765	3.0124	0.3239	0.1075
4.770	3.0049	0.3156	0.1050
4.775	2.9995	0.3318	0.1106
4.780	3.0025	0.3164	0.1054
4.785	3.0057	0.3148	0.1047
4.790	3.0066	0.3223	0.1072
4.795	3.0051	0.3257	0.1084
4.800	3.0054	0.3170	0.1055
4.805	3.0012	0.3124	0.1041
4.810	2.9993	0.3209	0.1070
4.815	3.0071	0.3150	0.1048
4.820	2.9984	0.3081	0.1027
4.825	2.9983	0.3112	0.1038
4.830	3.0020	0.3134	0.1044
4.835	3.0062	0.3146	0.1047
4.840	3.0010	0.3173	0.1057
4.845	2.9996	0.3104	0.1035
4.850	2.9979	0.3135	0.1046
4.855	3.0012	0.3054	0.1018
4.860	2.9970	0.3035	0.1013
4.865	2.9968	0.3051	0.1018
4.870	2.9998	0.3027	0.1009
4.875	3.0006	0.3048	0.1016
4.880	2.9984	0.3067	0.1023
4.885	2.9887	0.3041	0.1017
4.890	2.9986	0.2990	0.0997
4.895	3.0018	0.3020	0.1006
4.900	3.0038	0.3069	0.1022
4.905	3.0042	0.3020	0.1005

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
4.910	3.0034	0.3069	0.1022
4.915	3.0034	0.3064	0.1020
4.920	3.0031	0.3065	0.1021
4.925	3.0062	0.3071	0.1022
4.930	3.0028	0.3117	0.1038
4.935	3.0042	0.3099	0.1031
4.940	3.0041	0.3114	0.1037
4.945	3.0034	0.3113	0.1037
4.950	3.0040	0.3151	0.1049
4.955	3.0049	0.3163	0.1052
4.960	3.0009	0.3206	0.1068
4.965	3.0011	0.3189	0.1063
4.970	3.0039	0.3219	0.1072
4.975	3.0053	0.3204	0.1066
4.980	3.0029	0.3238	0.1078
4.985	3.0027	0.3243	0.1080
4.990	3.0014	0.3268	0.1089
4.995	2.9979	0.3304	0.1102
5.000	3.0019	0.3339	0.1112
5.005	2.9962	0.3340	0.1115
5.010	2.9910	0.3305	0.1105
5.015	2.9892	0.3336	0.1116
5.020	2.9876	0.3305	0.1106
5.025	2.9827	0.3308	0.1109
5.030	2.9815	0.3311	0.1110
5.035	2.9850	0.3269	0.1095
5.040	2.9812	0.3256	0.1092
5.045	2.9827	0.3259	0.1093
5.050	2.9798	0.3240	0.1087
5.055	2.9892	0.3209	0.1073
5.060	2.9864	0.3255	0.1090
5.065	2.9928	0.3162	0.1057
5.070	2.9866	0.3216	0.1077
5.075	2.9805	0.3222	0.1081
5.080	2.9796	0.3204	0.1075
5.085	2.9796	0.3185	0.1069
5.090	2.9808	0.3216	0.1079
5.095	2.9770	0.3197	0.1074
5.100	2.9807	0.3201	0.1074
5.105	2.9841	0.3216	0.1078
5.110	2.9809	0.3224	0.1082
5.115	2.9800	0.3170	0.1064
5.120	2.9819	0.3201	0.1074
5.125	2.9821	0.3152	0.1057
5.130	2.9814	0.3164	0.1061
5.135	2.9860	0.3210	0.1075

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.140	2.9849	0.3203	0.1073
5.145	2.9781	0.3233	0.1085
5.150	2.9839	0.3246	0.1088
5.155	2.9808	0.3225	0.1082
5.160	2.9803	0.3230	0.1084
5.165	2.9793	0.3224	0.1082
5.170	2.9762	0.3205	0.1077
5.175	2.9749	0.3207	0.1078
5.180	2.9768	0.3203	0.1076
5.185	2.9760	0.3248	0.1091
5.190	2.9743	0.3228	0.1085
5.195	2.9748	0.3198	0.1075
5.200	2.9739	0.3202	0.1077
5.205	2.9760	0.3195	0.1074
5.210	2.9737	0.3192	0.1073
5.215	2.9745	0.3176	0.1068
5.220	2.9749	0.3186	0.1071
5.225	2.9749	0.3194	0.1074
5.230	2.9742	0.3171	0.1066
5.235	2.9715	0.3204	0.1078
5.240	2.9722	0.3171	0.1067
5.245	2.9766	0.3190	0.1072
5.250	2.9730	0.3197	0.1075
5.255	2.9706	0.3242	0.1091
5.260	2.9703	0.3219	0.1084
5.265	2.9711	0.3162	0.1064
5.270	2.9708	0.3211	0.1081
5.275	2.9724	0.3194	0.1075
5.280	2.9728	0.3229	0.1086
5.285	2.9745	0.3203	0.1077
5.290	2.9705	0.3235	0.1089
5.295	2.9717	0.3191	0.1074
5.300	2.9720	0.3237	0.1089
5.305	2.9627	0.3164	0.1068
5.310	2.9693	0.3253	0.1096
5.315	2.9669	0.3256	0.1098
5.320	2.9732	0.3184	0.1071
5.325	2.9766	0.3161	0.1062
5.330	2.9705	0.3214	0.1082
5.335	2.9715	0.3227	0.1086
5.340	2.9704	0.3190	0.1074
5.345	2.9756	0.3292	0.1106
5.350	2.9732	0.3247	0.1092
5.355	2.9755	0.3247	0.1091
5.360	2.9713	0.3183	0.1071
5.365	2.9700	0.3155	0.1062

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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.370	2.9697	0.3207	0.1080
5.375	2.9718	0.3177	0.1069
5.380	2.9657	0.3184	0.1073
5.385	2.9631	0.3281	0.1107
5.390	2.9745	0.3174	0.1067
5.395	2.9675	0.3176	0.1070
5.400	2.9731	0.3209	0.1079
5.405	2.9662	0.3284	0.1107
5.410	2.9730	0.3254	0.1094
5.415	2.9648	0.3120	0.1052
5.420	2.9731	0.3224	0.1085
5.425	2.9600	0.3167	0.1070
5.430	2.9681	0.3245	0.1093
5.435	2.9710	0.3219	0.1084
5.440	2.9677	0.3198	0.1078
5.445	2.9712	0.3171	0.1067
5.450	2.9755	0.3271	0.1099
5.455	2.9705	0.3279	0.1104
5.460	2.9715	0.3215	0.1082
5.465	2.9749	0.3143	0.1057
5.470	2.9685	0.3235	0.1090
5.475	2.9705	0.3160	0.1064
5.480	2.9706	0.3234	0.1089
5.485	2.9697	0.3160	0.1064
5.490	2.9689	0.3183	0.1072
5.495	2.9639	0.3168	0.1069
5.500	2.9652	0.3180	0.1073
5.505	2.9673	0.3178	0.1071
5.510	2.9641	0.3178	0.1072
5.515	2.9633	0.3157	0.1065
5.520	2.9635	0.3179	0.1073
5.525	2.9665	0.3171	0.1069
5.530	2.9642	0.3178	0.1072
5.535	2.9647	0.3144	0.1061
5.540	2.9663	0.3176	0.1071
5.545	2.9650	0.3172	0.1070
5.550	2.9651	0.3159	0.1065
5.555	2.9651	0.3155	0.1064
5.560	2.9630	0.3157	0.1066
5.565	2.9633	0.3157	0.1065
5.570	2.9642	0.3142	0.1060
5.575	2.9630	0.3131	0.1057
5.580	2.9633	0.3145	0.1061
5.585	2.9622	0.3135	0.1058
5.590	2.9612	0.3119	0.1053
5.595	2.9580	0.3137	0.1060

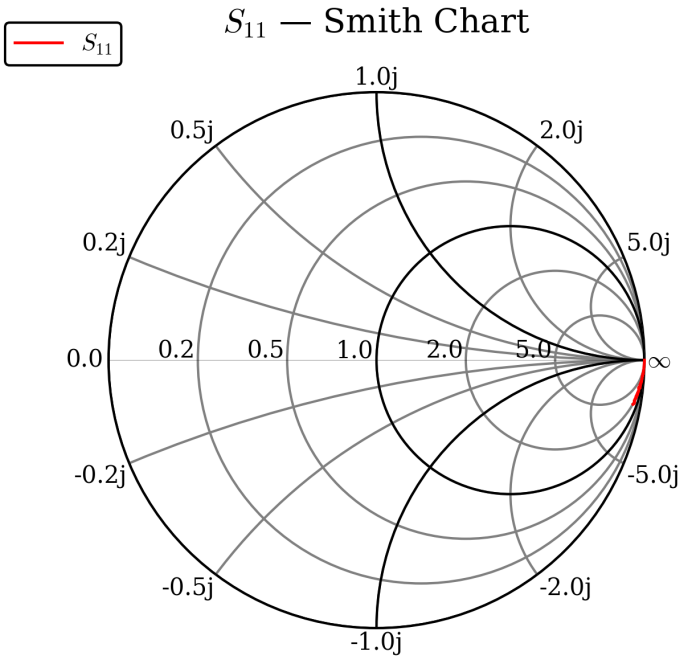
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Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.600	2.9609	0.3138	0.1060
5.605	2.9600	0.3135	0.1059
5.610	2.9572	0.3145	0.1063
5.615	2.9577	0.3123	0.1056
5.620	2.9577	0.3147	0.1064
5.625	2.9591	0.3148	0.1064
5.630	2.9572	0.3158	0.1068
5.635	2.9560	0.3189	0.1079
5.640	2.9538	0.3163	0.1071
5.645	2.9577	0.3181	0.1076
5.650	2.9541	0.3183	0.1077
5.655	2.9533	0.3166	0.1072
5.660	2.9546	0.3208	0.1086
5.665	2.9559	0.3118	0.1055
5.670	2.9548	0.3238	0.1096
5.675	2.9539	0.3175	0.1075
5.680	2.9511	0.3210	0.1088
5.685	2.9519	0.3199	0.1084
5.690	2.9487	0.3228	0.1095
5.695	2.9503	0.3264	0.1106
5.700	2.9508	0.3224	0.1093
5.705	2.9486	0.3198	0.1085
5.710	2.9502	0.3202	0.1085
5.715	2.9534	0.3176	0.1075
5.720	2.9461	0.3219	0.1093
5.725	2.9428	0.3181	0.1081
5.730	2.9475	0.3199	0.1085
5.735	2.9512	0.3213	0.1089
5.740	2.9484	0.3215	0.1090
5.745	2.9500	0.3310	0.1122
5.750	2.9476	0.3247	0.1102
5.755	2.9511	0.3211	0.1088
5.760	2.9444	0.3237	0.1099
5.765	2.9492	0.3267	0.1108
5.770	2.9461	0.3206	0.1088
5.775	2.9457	0.3228	0.1096
5.780	2.9461	0.3232	0.1097
5.785	2.9456	0.3249	0.1103
5.790	2.9456	0.3219	0.1093
5.795	2.9473	0.3219	0.1092
5.800	2.9442	0.3227	0.1096
5.805	2.9460	0.3212	0.1090
5.810	2.9428	0.3208	0.1090
5.815	2.9441	0.3224	0.1095
5.820	2.9432	0.3230	0.1098
5.825	2.9410	0.3213	0.1092

(continued on next page)

Frequency (GHz)	ε_r'	ε_r''	$\tan\delta$
5.830	2.9426	0.3224	0.1096
5.835	2.9421	0.3210	0.1091
5.840	2.9428	0.3210	0.1091
5.845	2.9412	0.3235	0.1100
5.850	2.9401	0.3228	0.1098
5.855	2.9392	0.3220	0.1095
5.860	2.9382	0.3237	0.1102
5.865	2.9396	0.3216	0.1094
5.870	2.9397	0.3235	0.1100
5.875	2.9387	0.3218	0.1095
5.880	2.9375	0.3242	0.1104
5.885	2.9374	0.3220	0.1096
5.890	2.9362	0.3271	0.1114
5.895	2.9409	0.3228	0.1098
5.900	2.9374	0.3219	0.1096
5.905	2.9433	0.3256	0.1106
5.910	2.9381	0.3243	0.1104
5.915	2.9365	0.3259	0.1110
5.920	2.9369	0.3258	0.1109
5.925	2.9374	0.3189	0.1085
5.930	2.9353	0.3310	0.1128
5.935	2.9355	0.3219	0.1097
5.940	2.9348	0.3236	0.1103
5.945	2.9377	0.3301	0.1124
5.950	2.9297	0.3241	0.1106
5.955	2.9341	0.3224	0.1099
5.960	2.9344	0.3229	0.1100
5.965	2.9314	0.3316	0.1131
5.970	2.9335	0.3222	0.1098
5.975	2.9287	0.3178	0.1085
5.980	2.9351	0.3198	0.1090
5.985	2.9415	0.3228	0.1097
5.990	2.9291	0.3275	0.1118
5.995	2.9286	0.3299	0.1126
6.000	2.9280	0.3296	0.1126

1.2 S_{11} — Smith Chart



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0000	0.00
0.010	1.0000	-0.0002	-0.00
0.015	1.0001	-0.0006	0.00
0.020	1.0001	-0.0007	0.00
0.025	0.9999	-0.0008	-0.00
0.030	1.0000	-0.0009	-0.00
0.035	1.0001	-0.0012	0.00
0.040	1.0000	-0.0012	0.00
0.045	1.0002	-0.0015	0.00
0.050	1.0001	-0.0017	0.00
0.055	1.0000	-0.0017	-0.00
0.060	1.0000	-0.0018	0.00
0.065	1.0001	-0.0021	0.00
0.070	0.9999	-0.0022	-0.00
0.075	1.0000	-0.0024	-0.00
0.080	1.0000	-0.0026	0.00
0.085	0.9999	-0.0027	-0.00
0.090	1.0001	-0.0029	0.00
0.095	1.0001	-0.0032	0.00
0.100	0.9999	-0.0032	-0.00
0.105	1.0000	-0.0035	0.00
0.110	0.9999	-0.0035	-0.00
0.115	0.9998	-0.0037	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.120	1.0000	-0.0038	-0.00
0.125	1.0000	-0.0040	0.00
0.130	0.9999	-0.0042	-0.00
0.135	0.9998	-0.0044	-0.00
0.140	0.9999	-0.0045	-0.00
0.145	1.0000	-0.0046	-0.00
0.150	1.0000	-0.0048	-0.00
0.155	0.9999	-0.0049	-0.00
0.160	0.9999	-0.0052	-0.00
0.165	1.0000	-0.0052	0.00
0.170	1.0000	-0.0053	0.00
0.175	0.9999	-0.0055	-0.00
0.180	0.9998	-0.0057	-0.00
0.185	0.9999	-0.0060	-0.00
0.190	0.9999	-0.0061	-0.00
0.195	1.0000	-0.0063	0.00
0.200	0.9999	-0.0064	-0.00
0.205	0.9999	-0.0064	-0.00
0.210	0.9999	-0.0067	-0.00
0.215	0.9999	-0.0069	-0.00
0.220	0.9998	-0.0070	-0.00
0.225	0.9998	-0.0073	-0.00
0.230	0.9998	-0.0074	-0.00
0.235	0.9999	-0.0076	-0.00
0.240	0.9998	-0.0077	-0.00
0.245	0.9998	-0.0079	-0.00
0.250	0.9998	-0.0080	-0.00
0.255	0.9998	-0.0081	-0.00
0.260	0.9998	-0.0083	-0.00
0.265	0.9998	-0.0085	-0.00
0.270	0.9998	-0.0087	-0.00
0.275	0.9997	-0.0088	-0.00
0.280	0.9997	-0.0090	-0.00
0.285	0.9998	-0.0091	-0.00
0.290	0.9998	-0.0094	-0.00
0.295	0.9997	-0.0094	-0.00
0.300	0.9998	-0.0095	-0.00
0.305	0.9996	-0.0098	-0.00
0.310	0.9998	-0.0099	-0.00
0.315	0.9997	-0.0101	-0.00
0.320	0.9996	-0.0102	-0.00
0.325	0.9996	-0.0106	-0.00
0.330	0.9996	-0.0106	-0.00
0.335	0.9997	-0.0107	-0.00
0.340	0.9995	-0.0109	-0.00
0.345	0.9997	-0.0111	-0.00

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.350	0.9996	-0.0112	-0.00
0.355	0.9997	-0.0114	-0.00
0.360	0.9995	-0.0116	-0.00
0.365	0.9996	-0.0117	-0.00
0.370	0.9996	-0.0117	-0.00
0.375	0.9996	-0.0120	-0.00
0.380	0.9995	-0.0122	-0.00
0.385	0.9995	-0.0123	-0.00
0.390	0.9996	-0.0125	-0.00
0.395	0.9995	-0.0127	-0.00
0.400	0.9995	-0.0128	-0.00
0.405	0.9994	-0.0129	-0.00
0.410	0.9994	-0.0130	-0.00
0.415	0.9995	-0.0132	-0.00
0.420	0.9994	-0.0133	-0.00
0.425	0.9994	-0.0136	-0.00
0.430	0.9995	-0.0136	-0.00
0.435	0.9994	-0.0139	-0.00
0.440	0.9995	-0.0140	-0.00
0.445	0.9994	-0.0142	-0.00
0.450	0.9993	-0.0143	-0.01
0.455	0.9994	-0.0145	-0.00
0.460	0.9993	-0.0146	-0.00
0.465	0.9994	-0.0149	-0.00
0.470	0.9994	-0.0149	-0.00
0.475	0.9993	-0.0151	-0.00
0.480	0.9993	-0.0153	-0.01
0.485	0.9993	-0.0154	-0.01
0.490	0.9993	-0.0156	-0.01
0.495	0.9993	-0.0156	-0.01
0.500	0.9993	-0.0158	-0.01
0.505	0.9993	-0.0159	-0.00
0.510	0.9992	-0.0161	-0.01
0.515	0.9993	-0.0164	-0.01
0.520	0.9991	-0.0165	-0.01
0.525	0.9992	-0.0166	-0.01
0.530	0.9992	-0.0167	-0.01
0.535	0.9992	-0.0168	-0.01
0.540	0.9992	-0.0172	-0.01
0.545	0.9993	-0.0172	-0.01
0.550	0.9992	-0.0174	-0.01
0.555	0.9991	-0.0175	-0.01
0.560	0.9991	-0.0177	-0.01
0.565	0.9991	-0.0179	-0.01
0.570	0.9992	-0.0180	-0.01
0.575	0.9990	-0.0182	-0.01

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.580	0.9991	-0.0182	-0.01
0.585	0.9991	-0.0185	-0.01
0.590	0.9991	-0.0186	-0.01
0.595	0.9990	-0.0188	-0.01
0.600	0.9992	-0.0189	-0.01
0.605	0.9991	-0.0191	-0.01
0.610	0.9990	-0.0192	-0.01
0.615	0.9990	-0.0194	-0.01
0.620	0.9990	-0.0194	-0.01
0.625	0.9991	-0.0196	-0.01
0.630	0.9990	-0.0198	-0.01
0.635	0.9990	-0.0199	-0.01
0.640	0.9989	-0.0200	-0.01
0.645	0.9990	-0.0203	-0.01
0.650	0.9989	-0.0204	-0.01
0.655	0.9990	-0.0206	-0.01
0.660	0.9989	-0.0207	-0.01
0.665	0.9989	-0.0208	-0.01
0.670	0.9989	-0.0209	-0.01
0.675	0.9990	-0.0210	-0.01
0.680	0.9989	-0.0212	-0.01
0.685	0.9988	-0.0214	-0.01
0.690	0.9989	-0.0215	-0.01
0.695	0.9988	-0.0216	-0.01
0.700	0.9988	-0.0218	-0.01
0.705	0.9988	-0.0219	-0.01
0.710	0.9989	-0.0220	-0.01
0.715	0.9989	-0.0222	-0.01
0.720	0.9988	-0.0224	-0.01
0.725	0.9988	-0.0225	-0.01
0.730	0.9989	-0.0227	-0.01
0.735	0.9988	-0.0227	-0.01
0.740	0.9987	-0.0230	-0.01
0.745	0.9988	-0.0231	-0.01
0.750	0.9988	-0.0233	-0.01
0.755	0.9987	-0.0235	-0.01
0.760	0.9986	-0.0236	-0.01
0.765	0.9987	-0.0237	-0.01
0.770	0.9987	-0.0238	-0.01
0.775	0.9987	-0.0239	-0.01
0.780	0.9987	-0.0241	-0.01
0.785	0.9987	-0.0243	-0.01
0.790	0.9988	-0.0244	-0.01
0.795	0.9986	-0.0247	-0.01
0.800	0.9987	-0.0247	-0.01
0.805	0.9986	-0.0249	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.810	0.9985	-0.0251	-0.01
0.815	0.9984	-0.0252	-0.01
0.820	0.9986	-0.0252	-0.01
0.825	0.9985	-0.0254	-0.01
0.830	0.9984	-0.0256	-0.01
0.835	0.9985	-0.0257	-0.01
0.840	0.9984	-0.0260	-0.01
0.845	0.9983	-0.0261	-0.01
0.850	0.9984	-0.0262	-0.01
0.855	0.9985	-0.0264	-0.01
0.860	0.9985	-0.0265	-0.01
0.865	0.9984	-0.0267	-0.01
0.870	0.9983	-0.0268	-0.01
0.875	0.9984	-0.0269	-0.01
0.880	0.9983	-0.0272	-0.01
0.885	0.9984	-0.0273	-0.01
0.890	0.9982	-0.0274	-0.01
0.895	0.9983	-0.0276	-0.01
0.900	0.9982	-0.0278	-0.01
0.905	0.9982	-0.0280	-0.01
0.910	0.9982	-0.0280	-0.01
0.915	0.9982	-0.0281	-0.01
0.920	0.9983	-0.0284	-0.01
0.925	0.9982	-0.0286	-0.01
0.930	0.9981	-0.0286	-0.01
0.935	0.9981	-0.0289	-0.01
0.940	0.9982	-0.0291	-0.01
0.945	0.9981	-0.0291	-0.01
0.950	0.9980	-0.0294	-0.01
0.955	0.9980	-0.0294	-0.01
0.960	0.9980	-0.0296	-0.01
0.965	0.9979	-0.0298	-0.01
0.970	0.9980	-0.0299	-0.01
0.975	0.9979	-0.0301	-0.01
0.980	0.9979	-0.0304	-0.01
0.985	0.9980	-0.0305	-0.01
0.990	0.9979	-0.0305	-0.01
0.995	0.9978	-0.0307	-0.01
1.000	0.9978	-0.0309	-0.02
1.005	0.9979	-0.0310	-0.01
1.010	0.9978	-0.0312	-0.02
1.015	0.9978	-0.0314	-0.02
1.020	0.9978	-0.0315	-0.01
1.025	0.9977	-0.0318	-0.02
1.030	0.9976	-0.0318	-0.02
1.035	0.9976	-0.0320	-0.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.040	0.9977	-0.0321	-0.02
1.045	0.9978	-0.0324	-0.01
1.050	0.9976	-0.0325	-0.02
1.055	0.9976	-0.0325	-0.02
1.060	0.9976	-0.0327	-0.02
1.065	0.9975	-0.0328	-0.02
1.070	0.9975	-0.0330	-0.02
1.075	0.9976	-0.0332	-0.02
1.080	0.9975	-0.0334	-0.02
1.085	0.9975	-0.0336	-0.02
1.090	0.9975	-0.0338	-0.02
1.095	0.9974	-0.0338	-0.02
1.100	0.9974	-0.0340	-0.02
1.105	0.9974	-0.0342	-0.02
1.110	0.9974	-0.0343	-0.02
1.115	0.9973	-0.0345	-0.02
1.120	0.9972	-0.0346	-0.02
1.125	0.9973	-0.0348	-0.02
1.130	0.9974	-0.0350	-0.02
1.135	0.9973	-0.0350	-0.02
1.140	0.9972	-0.0352	-0.02
1.145	0.9972	-0.0353	-0.02
1.150	0.9972	-0.0354	-0.02
1.155	0.9972	-0.0356	-0.02
1.160	0.9973	-0.0358	-0.02
1.165	0.9971	-0.0358	-0.02
1.170	0.9971	-0.0360	-0.02
1.175	0.9971	-0.0363	-0.02
1.180	0.9970	-0.0364	-0.02
1.185	0.9971	-0.0366	-0.02
1.190	0.9971	-0.0368	-0.02
1.195	0.9971	-0.0368	-0.02
1.200	0.9970	-0.0369	-0.02
1.205	0.9970	-0.0372	-0.02
1.210	0.9969	-0.0372	-0.02
1.215	0.9970	-0.0373	-0.02
1.220	0.9969	-0.0375	-0.02
1.225	0.9969	-0.0376	-0.02
1.230	0.9969	-0.0377	-0.02
1.235	0.9969	-0.0379	-0.02
1.240	0.9968	-0.0380	-0.02
1.245	0.9968	-0.0382	-0.02
1.250	0.9967	-0.0383	-0.02
1.255	0.9967	-0.0383	-0.02
1.260	0.9966	-0.0385	-0.02
1.265	0.9966	-0.0385	-0.02

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.270	0.9966	-0.0388	-0.02
1.275	0.9967	-0.0389	-0.02
1.280	0.9968	-0.0390	-0.02
1.285	0.9968	-0.0391	-0.02
1.290	0.9966	-0.0392	-0.02
1.295	0.9965	-0.0393	-0.02
1.300	0.9967	-0.0395	-0.02
1.305	0.9966	-0.0397	-0.02
1.310	0.9965	-0.0399	-0.02
1.315	0.9965	-0.0399	-0.02
1.320	0.9965	-0.0402	-0.02
1.325	0.9966	-0.0402	-0.02
1.330	0.9965	-0.0405	-0.02
1.335	0.9965	-0.0405	-0.02
1.340	0.9964	-0.0406	-0.02
1.345	0.9965	-0.0408	-0.02
1.350	0.9963	-0.0409	-0.02
1.355	0.9964	-0.0409	-0.02
1.360	0.9963	-0.0411	-0.02
1.365	0.9965	-0.0412	-0.02
1.370	0.9963	-0.0413	-0.02
1.375	0.9962	-0.0415	-0.03
1.380	0.9963	-0.0416	-0.02
1.385	0.9964	-0.0418	-0.02
1.390	0.9963	-0.0418	-0.02
1.395	0.9963	-0.0419	-0.02
1.400	0.9961	-0.0421	-0.03
1.405	0.9962	-0.0421	-0.03
1.410	0.9962	-0.0423	-0.02
1.415	0.9962	-0.0425	-0.03
1.420	0.9960	-0.0425	-0.03
1.425	0.9963	-0.0428	-0.02
1.430	0.9961	-0.0428	-0.03
1.435	0.9960	-0.0431	-0.03
1.440	0.9959	-0.0432	-0.03
1.445	0.9961	-0.0433	-0.03
1.450	0.9959	-0.0434	-0.03
1.455	0.9961	-0.0436	-0.03
1.460	0.9960	-0.0437	-0.03
1.465	0.9959	-0.0440	-0.03
1.470	0.9958	-0.0440	-0.03
1.475	0.9960	-0.0442	-0.03
1.480	0.9960	-0.0441	-0.03
1.485	0.9958	-0.0444	-0.03
1.490	0.9957	-0.0445	-0.03
1.495	0.9959	-0.0447	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.500	0.9957	-0.0449	-0.03
1.505	0.9957	-0.0451	-0.03
1.510	0.9957	-0.0452	-0.03
1.515	0.9958	-0.0453	-0.03
1.520	0.9957	-0.0455	-0.03
1.525	0.9957	-0.0456	-0.03
1.530	0.9956	-0.0458	-0.03
1.535	0.9956	-0.0460	-0.03
1.540	0.9957	-0.0460	-0.03
1.545	0.9956	-0.0462	-0.03
1.550	0.9955	-0.0464	-0.03
1.555	0.9955	-0.0465	-0.03
1.560	0.9954	-0.0466	-0.03
1.565	0.9953	-0.0468	-0.03
1.570	0.9954	-0.0471	-0.03
1.575	0.9954	-0.0473	-0.03
1.580	0.9952	-0.0473	-0.03
1.585	0.9953	-0.0476	-0.03
1.590	0.9952	-0.0477	-0.03
1.595	0.9952	-0.0479	-0.03
1.600	0.9951	-0.0482	-0.03
1.605	0.9951	-0.0483	-0.03
1.610	0.9951	-0.0483	-0.03
1.615	0.9950	-0.0486	-0.03
1.620	0.9951	-0.0488	-0.03
1.625	0.9952	-0.0489	-0.03
1.630	0.9950	-0.0492	-0.03
1.635	0.9951	-0.0493	-0.03
1.640	0.9950	-0.0495	-0.03
1.645	0.9949	-0.0496	-0.03
1.650	0.9948	-0.0498	-0.03
1.655	0.9948	-0.0499	-0.03
1.660	0.9946	-0.0501	-0.04
1.665	0.9948	-0.0505	-0.03
1.670	0.9949	-0.0507	-0.03
1.675	0.9947	-0.0507	-0.03
1.680	0.9945	-0.0508	-0.04
1.685	0.9947	-0.0510	-0.03
1.690	0.9945	-0.0512	-0.04
1.695	0.9946	-0.0515	-0.04
1.700	0.9945	-0.0516	-0.04
1.705	0.9943	-0.0517	-0.04
1.710	0.9945	-0.0520	-0.04
1.715	0.9944	-0.0520	-0.04
1.720	0.9943	-0.0522	-0.04
1.725	0.9940	-0.0526	-0.04

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.730	0.9942	-0.0526	-0.04
1.735	0.9943	-0.0530	-0.04
1.740	0.9941	-0.0530	-0.04
1.745	0.9940	-0.0531	-0.04
1.750	0.9941	-0.0534	-0.04
1.755	0.9938	-0.0534	-0.04
1.760	0.9940	-0.0537	-0.04
1.765	0.9941	-0.0537	-0.04
1.770	0.9938	-0.0540	-0.04
1.775	0.9938	-0.0542	-0.04
1.780	0.9937	-0.0544	-0.04
1.785	0.9936	-0.0546	-0.04
1.790	0.9938	-0.0547	-0.04
1.795	0.9937	-0.0548	-0.04
1.800	0.9937	-0.0551	-0.04
1.805	0.9936	-0.0553	-0.04
1.810	0.9935	-0.0553	-0.04
1.815	0.9934	-0.0554	-0.04
1.820	0.9933	-0.0556	-0.04
1.825	0.9933	-0.0557	-0.05
1.830	0.9935	-0.0560	-0.04
1.835	0.9934	-0.0562	-0.04
1.840	0.9934	-0.0562	-0.04
1.845	0.9933	-0.0565	-0.04
1.850	0.9932	-0.0565	-0.05
1.855	0.9932	-0.0566	-0.04
1.860	0.9931	-0.0567	-0.05
1.865	0.9932	-0.0569	-0.05
1.870	0.9931	-0.0570	-0.05
1.875	0.9930	-0.0571	-0.05
1.880	0.9929	-0.0573	-0.05
1.885	0.9929	-0.0574	-0.05
1.890	0.9928	-0.0577	-0.05
1.895	0.9929	-0.0578	-0.05
1.900	0.9929	-0.0579	-0.05
1.905	0.9927	-0.0580	-0.05
1.910	0.9927	-0.0581	-0.05
1.915	0.9928	-0.0583	-0.05
1.920	0.9928	-0.0584	-0.05
1.925	0.9928	-0.0585	-0.05
1.930	0.9925	-0.0586	-0.05
1.935	0.9926	-0.0589	-0.05
1.940	0.9926	-0.0588	-0.05
1.945	0.9924	-0.0590	-0.05
1.950	0.9926	-0.0592	-0.05
1.955	0.9924	-0.0593	-0.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.960	0.9925	-0.0595	-0.05
1.965	0.9924	-0.0596	-0.05
1.970	0.9925	-0.0597	-0.05
1.975	0.9924	-0.0598	-0.05
1.980	0.9923	-0.0598	-0.05
1.985	0.9922	-0.0600	-0.05
1.990	0.9924	-0.0601	-0.05
1.995	0.9923	-0.0603	-0.05
2.000	0.9921	-0.0603	-0.05
2.005	0.9924	-0.0605	-0.05
2.010	0.9922	-0.0606	-0.05
2.015	0.9922	-0.0606	-0.05
2.020	0.9922	-0.0606	-0.05
2.025	0.9919	-0.0607	-0.05
2.030	0.9922	-0.0610	-0.05
2.035	0.9921	-0.0612	-0.05
2.040	0.9922	-0.0613	-0.05
2.045	0.9920	-0.0613	-0.05
2.050	0.9920	-0.0615	-0.05
2.055	0.9922	-0.0616	-0.05
2.060	0.9920	-0.0618	-0.05
2.065	0.9920	-0.0617	-0.05
2.070	0.9921	-0.0619	-0.05
2.075	0.9921	-0.0620	-0.05
2.080	0.9921	-0.0622	-0.05
2.085	0.9920	-0.0621	-0.05
2.090	0.9919	-0.0624	-0.05
2.095	0.9919	-0.0626	-0.05
2.100	0.9920	-0.0626	-0.05
2.105	0.9918	-0.0626	-0.05
2.110	0.9919	-0.0628	-0.05
2.115	0.9917	-0.0629	-0.05
2.120	0.9918	-0.0631	-0.05
2.125	0.9919	-0.0632	-0.05
2.130	0.9919	-0.0632	-0.05
2.135	0.9917	-0.0634	-0.05
2.140	0.9917	-0.0636	-0.05
2.145	0.9918	-0.0637	-0.05
2.150	0.9918	-0.0640	-0.05
2.155	0.9917	-0.0642	-0.05
2.160	0.9919	-0.0643	-0.05
2.165	0.9917	-0.0644	-0.05
2.170	0.9917	-0.0644	-0.05
2.175	0.9915	-0.0646	-0.06
2.180	0.9916	-0.0648	-0.05
2.185	0.9915	-0.0650	-0.06

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.190	0.9915	-0.0652	-0.06
2.195	0.9916	-0.0653	-0.05
2.200	0.9915	-0.0654	-0.06
2.205	0.9915	-0.0657	-0.05
2.210	0.9914	-0.0657	-0.06
2.215	0.9915	-0.0659	-0.06
2.220	0.9913	-0.0662	-0.06
2.225	0.9913	-0.0663	-0.06
2.230	0.9914	-0.0665	-0.06
2.235	0.9912	-0.0665	-0.06
2.240	0.9913	-0.0667	-0.06
2.245	0.9912	-0.0669	-0.06
2.250	0.9910	-0.0673	-0.06
2.255	0.9912	-0.0673	-0.06
2.260	0.9911	-0.0677	-0.06
2.265	0.9911	-0.0676	-0.06
2.270	0.9909	-0.0680	-0.06
2.275	0.9910	-0.0682	-0.06
2.280	0.9909	-0.0683	-0.06
2.285	0.9910	-0.0687	-0.06
2.290	0.9908	-0.0687	-0.06
2.295	0.9909	-0.0689	-0.06
2.300	0.9908	-0.0689	-0.06
2.305	0.9910	-0.0693	-0.06
2.310	0.9907	-0.0693	-0.06
2.315	0.9907	-0.0697	-0.06
2.320	0.9906	-0.0699	-0.06
2.325	0.9906	-0.0700	-0.06
2.330	0.9906	-0.0702	-0.06
2.335	0.9906	-0.0705	-0.06
2.340	0.9905	-0.0707	-0.06
2.345	0.9904	-0.0709	-0.06
2.350	0.9904	-0.0712	-0.06
2.355	0.9904	-0.0713	-0.06
2.360	0.9905	-0.0715	-0.06
2.365	0.9901	-0.0718	-0.06
2.370	0.9904	-0.0720	-0.06
2.375	0.9902	-0.0722	-0.06
2.380	0.9902	-0.0723	-0.06
2.385	0.9902	-0.0723	-0.06
2.390	0.9900	-0.0727	-0.06
2.395	0.9901	-0.0727	-0.06
2.400	0.9901	-0.0729	-0.06
2.405	0.9900	-0.0732	-0.06
2.410	0.9898	-0.0735	-0.06
2.415	0.9900	-0.0737	-0.06

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.420	0.9899	-0.0738	-0.06
2.425	0.9899	-0.0740	-0.06
2.430	0.9898	-0.0741	-0.06
2.435	0.9898	-0.0742	-0.06
2.440	0.9898	-0.0744	-0.07
2.445	0.9897	-0.0746	-0.06
2.450	0.9897	-0.0747	-0.07
2.455	0.9896	-0.0749	-0.07
2.460	0.9896	-0.0751	-0.07
2.465	0.9894	-0.0753	-0.07
2.470	0.9894	-0.0755	-0.07
2.475	0.9894	-0.0756	-0.07
2.480	0.9894	-0.0757	-0.07
2.485	0.9893	-0.0758	-0.07
2.490	0.9892	-0.0761	-0.07
2.495	0.9892	-0.0763	-0.07
2.500	0.9891	-0.0764	-0.07
2.505	0.9891	-0.0765	-0.07
2.510	0.9890	-0.0767	-0.07
2.515	0.9890	-0.0768	-0.07
2.520	0.9889	-0.0770	-0.07
2.525	0.9888	-0.0771	-0.07
2.530	0.9888	-0.0773	-0.07
2.535	0.9887	-0.0775	-0.07
2.540	0.9887	-0.0776	-0.07
2.545	0.9887	-0.0777	-0.07
2.550	0.9885	-0.0780	-0.07
2.555	0.9885	-0.0781	-0.07
2.560	0.9884	-0.0781	-0.07
2.565	0.9885	-0.0782	-0.07
2.570	0.9884	-0.0784	-0.07
2.575	0.9883	-0.0784	-0.08
2.580	0.9880	-0.0785	-0.08
2.585	0.9882	-0.0786	-0.08
2.590	0.9882	-0.0786	-0.08
2.595	0.9883	-0.0787	-0.07
2.600	0.9882	-0.0789	-0.08
2.605	0.9883	-0.0791	-0.07
2.610	0.9882	-0.0792	-0.08
2.615	0.9882	-0.0794	-0.07
2.620	0.9884	-0.0795	-0.07
2.625	0.9882	-0.0796	-0.07
2.630	0.9883	-0.0794	-0.07
2.635	0.9882	-0.0797	-0.07
2.640	0.9882	-0.0797	-0.08
2.645	0.9879	-0.0800	-0.08

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.650	0.9881	-0.0801	-0.08
2.655	0.9882	-0.0801	-0.07
2.660	0.9880	-0.0801	-0.08
2.665	0.9881	-0.0802	-0.08
2.670	0.9882	-0.0805	-0.07
2.675	0.9881	-0.0807	-0.07
2.680	0.9879	-0.0806	-0.08
2.685	0.9879	-0.0809	-0.08
2.690	0.9879	-0.0811	-0.08
2.695	0.9880	-0.0811	-0.08
2.700	0.9879	-0.0813	-0.08
2.705	0.9878	-0.0812	-0.08
2.710	0.9880	-0.0813	-0.08
2.715	0.9878	-0.0815	-0.08
2.720	0.9879	-0.0817	-0.08
2.725	0.9881	-0.0817	-0.07
2.730	0.9880	-0.0818	-0.08
2.735	0.9879	-0.0819	-0.08
2.740	0.9878	-0.0819	-0.08
2.745	0.9877	-0.0820	-0.08
2.750	0.9880	-0.0820	-0.07
2.755	0.9878	-0.0822	-0.08
2.760	0.9876	-0.0822	-0.08
2.765	0.9877	-0.0824	-0.08
2.770	0.9877	-0.0824	-0.08
2.775	0.9877	-0.0826	-0.08
2.780	0.9877	-0.0826	-0.08
2.785	0.9876	-0.0828	-0.08
2.790	0.9877	-0.0828	-0.08
2.795	0.9875	-0.0829	-0.08
2.800	0.9874	-0.0830	-0.08
2.805	0.9874	-0.0831	-0.08
2.810	0.9872	-0.0832	-0.08
2.815	0.9872	-0.0834	-0.08
2.820	0.9873	-0.0836	-0.08
2.825	0.9872	-0.0836	-0.08
2.830	0.9871	-0.0837	-0.08
2.835	0.9871	-0.0839	-0.08
2.840	0.9871	-0.0840	-0.08
2.845	0.9869	-0.0840	-0.08
2.850	0.9870	-0.0842	-0.08
2.855	0.9869	-0.0845	-0.08
2.860	0.9868	-0.0846	-0.08
2.865	0.9868	-0.0850	-0.08
2.870	0.9867	-0.0852	-0.08
2.875	0.9867	-0.0850	-0.08

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.880	0.9868	-0.0852	-0.08
2.885	0.9867	-0.0853	-0.08
2.890	0.9867	-0.0854	-0.08
2.895	0.9867	-0.0855	-0.08
2.900	0.9867	-0.0855	-0.08
2.905	0.9863	-0.0856	-0.09
2.910	0.9865	-0.0858	-0.09
2.915	0.9864	-0.0857	-0.09
2.920	0.9862	-0.0859	-0.09
2.925	0.9861	-0.0862	-0.09
2.930	0.9861	-0.0863	-0.09
2.935	0.9860	-0.0865	-0.09
2.940	0.9859	-0.0866	-0.09
2.945	0.9858	-0.0868	-0.09
2.950	0.9856	-0.0868	-0.09
2.955	0.9858	-0.0870	-0.09
2.960	0.9857	-0.0874	-0.09
2.965	0.9853	-0.0876	-0.09
2.970	0.9857	-0.0878	-0.09
2.975	0.9855	-0.0879	-0.09
2.980	0.9854	-0.0879	-0.09
2.985	0.9853	-0.0881	-0.09
2.990	0.9856	-0.0885	-0.09
2.995	0.9852	-0.0886	-0.09
3.000	0.9853	-0.0888	-0.09
3.005	0.9854	-0.0888	-0.09
3.010	0.9851	-0.0891	-0.09
3.015	0.9852	-0.0893	-0.09
3.020	0.9850	-0.0891	-0.10
3.025	0.9851	-0.0891	-0.10
3.030	0.9852	-0.0896	-0.09
3.035	0.9854	-0.0897	-0.09
3.040	0.9855	-0.0900	-0.09
3.045	0.9857	-0.0899	-0.09
3.050	0.9850	-0.0901	-0.10
3.055	0.9857	-0.0898	-0.09
3.060	0.9856	-0.0902	-0.09
3.065	0.9854	-0.0903	-0.09
3.070	0.9856	-0.0900	-0.09
3.075	0.9856	-0.0900	-0.09
3.080	0.9854	-0.0899	-0.09
3.085	0.9851	-0.0900	-0.09
3.090	0.9852	-0.0900	-0.09
3.095	0.9850	-0.0898	-0.10
3.100	0.9850	-0.0902	-0.09
3.105	0.9848	-0.0901	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.110	0.9847	-0.0903	-0.10
3.115	0.9845	-0.0904	-0.10
3.120	0.9843	-0.0903	-0.10
3.125	0.9841	-0.0905	-0.10
3.130	0.9841	-0.0908	-0.10
3.135	0.9841	-0.0908	-0.10
3.140	0.9840	-0.0911	-0.10
3.145	0.9839	-0.0913	-0.10
3.150	0.9838	-0.0915	-0.10
3.155	0.9840	-0.0916	-0.10
3.160	0.9839	-0.0919	-0.10
3.165	0.9842	-0.0920	-0.10
3.170	0.9841	-0.0920	-0.10
3.175	0.9841	-0.0923	-0.10
3.180	0.9842	-0.0923	-0.10
3.185	0.9844	-0.0925	-0.10
3.190	0.9842	-0.0925	-0.10
3.195	0.9844	-0.0924	-0.10
3.200	0.9845	-0.0925	-0.10
3.205	0.9844	-0.0923	-0.10
3.210	0.9847	-0.0922	-0.10
3.215	0.9845	-0.0922	-0.10
3.220	0.9846	-0.0923	-0.10
3.225	0.9842	-0.0923	-0.10
3.230	0.9839	-0.0925	-0.10
3.235	0.9839	-0.0924	-0.10
3.240	0.9841	-0.0924	-0.10
3.245	0.9836	-0.0926	-0.11
3.250	0.9833	-0.0927	-0.11
3.255	0.9834	-0.0929	-0.11
3.260	0.9836	-0.0928	-0.11
3.265	0.9835	-0.0935	-0.11
3.270	0.9836	-0.0932	-0.11
3.275	0.9838	-0.0933	-0.10
3.280	0.9836	-0.0935	-0.10
3.285	0.9838	-0.0936	-0.10
3.290	0.9831	-0.0939	-0.11
3.295	0.9835	-0.0938	-0.10
3.300	0.9836	-0.0939	-0.10
3.305	0.9838	-0.0940	-0.10
3.310	0.9839	-0.0942	-0.10
3.315	0.9839	-0.0946	-0.10
3.320	0.9838	-0.0944	-0.10
3.325	0.9840	-0.0950	-0.10
3.330	0.9838	-0.0946	-0.10
3.335	0.9840	-0.0948	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.340	0.9841	-0.0949	-0.10
3.345	0.9839	-0.0946	-0.10
3.350	0.9842	-0.0950	-0.10
3.355	0.9840	-0.0954	-0.10
3.360	0.9842	-0.0953	-0.10
3.365	0.9841	-0.0952	-0.10
3.370	0.9844	-0.0952	-0.10
3.375	0.9842	-0.0955	-0.10
3.380	0.9841	-0.0954	-0.10
3.385	0.9842	-0.0955	-0.10
3.390	0.9842	-0.0955	-0.10
3.395	0.9842	-0.0956	-0.10
3.400	0.9841	-0.0957	-0.10
3.405	0.9840	-0.0956	-0.10
3.410	0.9841	-0.0957	-0.10
3.415	0.9839	-0.0957	-0.10
3.420	0.9840	-0.0956	-0.10
3.425	0.9838	-0.0960	-0.10
3.430	0.9838	-0.0960	-0.10
3.435	0.9837	-0.0961	-0.10
3.440	0.9835	-0.0963	-0.10
3.445	0.9835	-0.0960	-0.10
3.450	0.9833	-0.0962	-0.10
3.455	0.9832	-0.0966	-0.11
3.460	0.9834	-0.0966	-0.10
3.465	0.9829	-0.0970	-0.11
3.470	0.9829	-0.0970	-0.11
3.475	0.9830	-0.0972	-0.11
3.480	0.9827	-0.0972	-0.11
3.485	0.9832	-0.0974	-0.10
3.490	0.9826	-0.0981	-0.11
3.495	0.9826	-0.0978	-0.11
3.500	0.9827	-0.0983	-0.11
3.505	0.9827	-0.0983	-0.11
3.510	0.9831	-0.0991	-0.10
3.515	0.9829	-0.0988	-0.11
3.520	0.9830	-0.0991	-0.10
3.525	0.9832	-0.0993	-0.10
3.530	0.9833	-0.0994	-0.10
3.535	0.9833	-0.0994	-0.10
3.540	0.9838	-0.0997	-0.10
3.545	0.9836	-0.0996	-0.10
3.550	0.9838	-0.0998	-0.10
3.555	0.9838	-0.0992	-0.10
3.560	0.9835	-0.0988	-0.10
3.565	0.9837	-0.0987	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.570	0.9835	-0.0985	-0.10
3.575	0.9829	-0.0980	-0.11
3.580	0.9823	-0.0984	-0.11
3.585	0.9819	-0.0978	-0.12
3.590	0.9810	-0.0984	-0.12
3.595	0.9804	-0.0993	-0.13
3.600	0.9800	-0.1000	-0.13
3.605	0.9796	-0.1003	-0.13
3.610	0.9798	-0.1009	-0.13
3.615	0.9799	-0.1013	-0.13
3.620	0.9800	-0.1017	-0.13
3.625	0.9798	-0.1026	-0.13
3.630	0.9803	-0.1027	-0.13
3.635	0.9807	-0.1031	-0.12
3.640	0.9811	-0.1030	-0.12
3.645	0.9813	-0.1028	-0.12
3.650	0.9812	-0.1033	-0.12
3.655	0.9812	-0.1030	-0.12
3.660	0.9812	-0.1027	-0.12
3.665	0.9808	-0.1029	-0.12
3.670	0.9812	-0.1026	-0.12
3.675	0.9812	-0.1028	-0.12
3.680	0.9813	-0.1027	-0.12
3.685	0.9809	-0.1027	-0.12
3.690	0.9811	-0.1027	-0.12
3.695	0.9808	-0.1027	-0.12
3.700	0.9808	-0.1028	-0.12
3.705	0.9808	-0.1026	-0.12
3.710	0.9806	-0.1027	-0.12
3.715	0.9804	-0.1026	-0.12
3.720	0.9802	-0.1025	-0.13
3.725	0.9799	-0.1028	-0.13
3.730	0.9796	-0.1024	-0.13
3.735	0.9791	-0.1026	-0.14
3.740	0.9787	-0.1030	-0.14
3.745	0.9785	-0.1034	-0.14
3.750	0.9781	-0.1037	-0.14
3.755	0.9780	-0.1048	-0.14
3.760	0.9782	-0.1054	-0.14
3.765	0.9784	-0.1062	-0.14
3.770	0.9796	-0.1067	-0.13
3.775	0.9800	-0.1079	-0.12
3.780	0.9811	-0.1066	-0.11
3.785	0.9813	-0.1056	-0.11
3.790	0.9818	-0.1054	-0.11
3.795	0.9812	-0.1049	-0.12

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.800	0.9807	-0.1044	-0.12
3.805	0.9804	-0.1044	-0.12
3.810	0.9799	-0.1044	-0.13
3.815	0.9795	-0.1044	-0.13
3.820	0.9794	-0.1045	-0.13
3.825	0.9790	-0.1046	-0.14
3.830	0.9843	-0.1036	-0.09
3.835	0.9787	-0.1061	-0.14
3.840	0.9789	-0.1063	-0.13
3.845	0.9789	-0.1066	-0.13
3.850	0.9789	-0.1063	-0.13
3.855	0.9787	-0.1065	-0.14
3.860	0.9789	-0.1067	-0.13
3.865	0.9791	-0.1072	-0.13
3.870	0.9788	-0.1068	-0.13
3.875	0.9789	-0.1072	-0.13
3.880	0.9789	-0.1074	-0.13
3.885	0.9791	-0.1074	-0.13
3.890	0.9789	-0.1078	-0.13
3.895	0.9791	-0.1077	-0.13
3.900	0.9790	-0.1077	-0.13
3.905	0.9790	-0.1078	-0.13
3.910	0.9790	-0.1081	-0.13
3.915	0.9789	-0.1083	-0.13
3.920	0.9789	-0.1085	-0.13
3.925	0.9790	-0.1084	-0.13
3.930	0.9790	-0.1085	-0.13
3.935	0.9791	-0.1089	-0.13
3.940	0.9788	-0.1090	-0.13
3.945	0.9789	-0.1092	-0.13
3.950	0.9790	-0.1091	-0.13
3.955	0.9789	-0.1096	-0.13
3.960	0.9790	-0.1096	-0.13
3.965	0.9789	-0.1100	-0.13
3.970	0.9788	-0.1102	-0.13
3.975	0.9789	-0.1103	-0.13
3.980	0.9790	-0.1105	-0.13
3.985	0.9789	-0.1108	-0.13
3.990	0.9787	-0.1109	-0.13
3.995	0.9787	-0.1111	-0.13
4.000	0.9788	-0.1113	-0.13
4.005	0.9787	-0.1116	-0.13
4.010	0.9786	-0.1118	-0.13
4.015	0.9786	-0.1120	-0.13
4.020	0.9786	-0.1123	-0.13
4.025	0.9786	-0.1124	-0.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.030	0.9785	-0.1128	-0.13
4.035	0.9786	-0.1129	-0.13
4.040	0.9785	-0.1132	-0.13
4.045	0.9785	-0.1135	-0.13
4.050	0.9784	-0.1136	-0.13
4.055	0.9784	-0.1141	-0.13
4.060	0.9782	-0.1142	-0.13
4.065	0.9778	-0.1146	-0.14
4.070	0.9782	-0.1146	-0.13
4.075	0.9782	-0.1147	-0.13
4.080	0.9780	-0.1151	-0.13
4.085	0.9778	-0.1151	-0.14
4.090	0.9779	-0.1156	-0.13
4.095	0.9777	-0.1158	-0.14
4.100	0.9778	-0.1160	-0.13
4.105	0.9775	-0.1163	-0.14
4.110	0.9776	-0.1168	-0.14
4.115	0.9776	-0.1167	-0.14
4.120	0.9777	-0.1172	-0.13
4.125	0.9774	-0.1170	-0.14
4.130	0.9773	-0.1179	-0.14
4.135	0.9775	-0.1179	-0.14
4.140	0.9775	-0.1183	-0.13
4.145	0.9770	-0.1189	-0.14
4.150	0.9768	-0.1191	-0.14
4.155	0.9765	-0.1195	-0.14
4.160	0.9765	-0.1197	-0.14
4.165	0.9764	-0.1199	-0.14
4.170	0.9762	-0.1203	-0.14
4.175	0.9755	-0.1206	-0.15
4.180	0.9761	-0.1205	-0.14
4.185	0.9760	-0.1208	-0.14
4.190	0.9754	-0.1208	-0.15
4.195	0.9754	-0.1211	-0.15
4.200	0.9752	-0.1214	-0.15
4.205	0.9752	-0.1217	-0.15
4.210	0.9750	-0.1217	-0.15
4.215	0.9751	-0.1220	-0.15
4.220	0.9750	-0.1224	-0.15
4.225	0.9747	-0.1218	-0.16
4.230	0.9749	-0.1226	-0.15
4.235	0.9746	-0.1227	-0.16
4.240	0.9746	-0.1232	-0.16
4.245	0.9748	-0.1232	-0.15
4.250	0.9746	-0.1232	-0.15
4.255	0.9747	-0.1234	-0.15

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.260	0.9746	-0.1235	-0.15
4.265	0.9744	-0.1237	-0.16
4.270	0.9746	-0.1239	-0.15
4.275	0.9744	-0.1240	-0.16
4.280	0.9744	-0.1242	-0.16
4.285	0.9744	-0.1243	-0.16
4.290	0.9744	-0.1245	-0.16
4.295	0.9744	-0.1244	-0.16
4.300	0.9741	-0.1246	-0.16
4.305	0.9742	-0.1248	-0.16
4.310	0.9740	-0.1248	-0.16
4.315	0.9740	-0.1250	-0.16
4.320	0.9740	-0.1251	-0.16
4.325	0.9739	-0.1251	-0.16
4.330	0.9738	-0.1253	-0.16
4.335	0.9737	-0.1253	-0.16
4.340	0.9736	-0.1255	-0.16
4.345	0.9735	-0.1256	-0.16
4.350	0.9734	-0.1257	-0.16
4.355	0.9734	-0.1259	-0.16
4.360	0.9734	-0.1260	-0.16
4.365	0.9735	-0.1262	-0.16
4.370	0.9733	-0.1263	-0.16
4.375	0.9732	-0.1263	-0.16
4.380	0.9733	-0.1266	-0.16
4.385	0.9734	-0.1270	-0.16
4.390	0.9733	-0.1270	-0.16
4.395	0.9735	-0.1270	-0.16
4.400	0.9735	-0.1271	-0.16
4.405	0.9735	-0.1273	-0.16
4.410	0.9735	-0.1273	-0.16
4.415	0.9734	-0.1275	-0.16
4.420	0.9737	-0.1272	-0.16
4.425	0.9738	-0.1274	-0.16
4.430	0.9736	-0.1277	-0.16
4.435	0.9735	-0.1277	-0.16
4.440	0.9736	-0.1278	-0.16
4.445	0.9736	-0.1276	-0.16
4.450	0.9727	-0.1277	-0.17
4.455	0.9730	-0.1275	-0.16
4.460	0.9726	-0.1278	-0.17
4.465	0.9723	-0.1274	-0.17
4.470	0.9723	-0.1277	-0.17
4.475	0.9722	-0.1274	-0.17
4.480	0.9719	-0.1276	-0.17
4.485	0.9719	-0.1275	-0.17

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.490	0.9722	-0.1276	-0.17
4.495	0.9725	-0.1274	-0.17
4.500	0.9726	-0.1275	-0.17
4.505	0.9729	-0.1280	-0.16
4.510	0.9728	-0.1277	-0.17
4.515	0.9730	-0.1280	-0.16
4.520	0.9734	-0.1279	-0.16
4.525	0.9734	-0.1278	-0.16
4.530	0.9734	-0.1281	-0.16
4.535	0.9738	-0.1278	-0.16
4.540	0.9736	-0.1279	-0.16
4.545	0.9738	-0.1281	-0.16
4.550	0.9734	-0.1282	-0.16
4.555	0.9735	-0.1280	-0.16
4.560	0.9736	-0.1281	-0.16
4.565	0.9735	-0.1282	-0.16
4.570	0.9735	-0.1283	-0.16
4.575	0.9736	-0.1282	-0.16
4.580	0.9736	-0.1280	-0.16
4.585	0.9738	-0.1283	-0.16
4.590	0.9737	-0.1283	-0.16
4.595	0.9736	-0.1283	-0.16
4.600	0.9735	-0.1283	-0.16
4.605	0.9735	-0.1284	-0.16
4.610	0.9736	-0.1285	-0.16
4.615	0.9734	-0.1286	-0.16
4.620	0.9732	-0.1286	-0.16
4.625	0.9734	-0.1286	-0.16
4.630	0.9731	-0.1289	-0.16
4.635	0.9730	-0.1288	-0.16
4.640	0.9731	-0.1289	-0.16
4.645	0.9729	-0.1293	-0.16
4.650	0.9730	-0.1294	-0.16
4.655	0.9730	-0.1292	-0.16
4.660	0.9725	-0.1295	-0.17
4.665	0.9727	-0.1300	-0.16
4.670	0.9726	-0.1301	-0.16
4.675	0.9728	-0.1299	-0.16
4.680	0.9729	-0.1300	-0.16
4.685	0.9723	-0.1301	-0.17
4.690	0.9727	-0.1302	-0.16
4.695	0.9725	-0.1302	-0.17
4.700	0.9727	-0.1305	-0.16
4.705	0.9723	-0.1312	-0.17
4.710	0.9727	-0.1306	-0.16
4.715	0.9727	-0.1308	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.720	0.9730	-0.1306	-0.16
4.725	0.9728	-0.1304	-0.16
4.730	0.9723	-0.1309	-0.17
4.735	0.9726	-0.1306	-0.16
4.740	0.9724	-0.1309	-0.17
4.745	0.9718	-0.1304	-0.17
4.750	0.9716	-0.1308	-0.17
4.755	0.9719	-0.1313	-0.17
4.760	0.9717	-0.1315	-0.17
4.765	0.9712	-0.1319	-0.17
4.770	0.9713	-0.1318	-0.17
4.775	0.9705	-0.1319	-0.18
4.780	0.9711	-0.1324	-0.17
4.785	0.9712	-0.1323	-0.17
4.790	0.9705	-0.1328	-0.18
4.795	0.9704	-0.1329	-0.18
4.800	0.9709	-0.1332	-0.18
4.805	0.9706	-0.1331	-0.18
4.810	0.9706	-0.1329	-0.18
4.815	0.9706	-0.1335	-0.18
4.820	0.9708	-0.1328	-0.18
4.825	0.9706	-0.1327	-0.18
4.830	0.9702	-0.1333	-0.18
4.835	0.9703	-0.1334	-0.18
4.840	0.9700	-0.1331	-0.18
4.845	0.9702	-0.1331	-0.18
4.850	0.9700	-0.1329	-0.18
4.855	0.9703	-0.1328	-0.18
4.860	0.9704	-0.1329	-0.18
4.865	0.9703	-0.1324	-0.18
4.870	0.9702	-0.1330	-0.18
4.875	0.9701	-0.1330	-0.18
4.880	0.9700	-0.1330	-0.18
4.885	0.9701	-0.1324	-0.18
4.890	0.9701	-0.1333	-0.18
4.895	0.9701	-0.1332	-0.18
4.900	0.9698	-0.1335	-0.18
4.905	0.9701	-0.1336	-0.18
4.910	0.9699	-0.1336	-0.18
4.915	0.9699	-0.1337	-0.18
4.920	0.9699	-0.1338	-0.18
4.925	0.9698	-0.1341	-0.18
4.930	0.9697	-0.1340	-0.19
4.935	0.9697	-0.1342	-0.18
4.940	0.9697	-0.1343	-0.18
4.945	0.9696	-0.1343	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.950	0.9694	-0.1345	-0.19
4.955	0.9694	-0.1346	-0.19
4.960	0.9692	-0.1345	-0.19
4.965	0.9692	-0.1345	-0.19
4.970	0.9691	-0.1348	-0.19
4.975	0.9692	-0.1349	-0.19
4.980	0.9692	-0.1350	-0.19
4.985	0.9689	-0.1352	-0.19
4.990	0.9690	-0.1352	-0.19
4.995	0.9690	-0.1351	-0.19
5.000	0.9688	-0.1355	-0.19
5.005	0.9688	-0.1354	-0.19
5.010	0.9690	-0.1353	-0.19
5.015	0.9689	-0.1353	-0.19
5.020	0.9690	-0.1354	-0.19
5.025	0.9691	-0.1354	-0.19
5.030	0.9691	-0.1356	-0.19
5.035	0.9692	-0.1357	-0.19
5.040	0.9693	-0.1358	-0.19
5.045	0.9690	-0.1360	-0.19
5.050	0.9691	-0.1359	-0.19
5.055	0.9690	-0.1363	-0.19
5.060	0.9689	-0.1361	-0.19
5.065	0.9693	-0.1363	-0.19
5.070	0.9689	-0.1359	-0.19
5.075	0.9690	-0.1356	-0.19
5.080	0.9691	-0.1355	-0.19
5.085	0.9693	-0.1353	-0.19
5.090	0.9693	-0.1354	-0.19
5.095	0.9693	-0.1352	-0.19
5.100	0.9692	-0.1353	-0.19
5.105	0.9693	-0.1353	-0.19
5.110	0.9694	-0.1355	-0.19
5.115	0.9696	-0.1355	-0.18
5.120	0.9696	-0.1356	-0.18
5.125	0.9697	-0.1357	-0.18
5.130	0.9699	-0.1358	-0.18
5.135	0.9696	-0.1359	-0.18
5.140	0.9698	-0.1359	-0.18
5.145	0.9696	-0.1358	-0.18
5.150	0.9696	-0.1360	-0.18
5.155	0.9697	-0.1360	-0.18
5.160	0.9697	-0.1361	-0.18
5.165	0.9698	-0.1361	-0.18
5.170	0.9698	-0.1361	-0.18
5.175	0.9699	-0.1361	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.180	0.9698	-0.1362	-0.18
5.185	0.9697	-0.1362	-0.18
5.190	0.9698	-0.1362	-0.18
5.195	0.9698	-0.1364	-0.18
5.200	0.9699	-0.1364	-0.18
5.205	0.9699	-0.1366	-0.18
5.210	0.9699	-0.1366	-0.18
5.215	0.9700	-0.1368	-0.18
5.220	0.9700	-0.1370	-0.18
5.225	0.9699	-0.1371	-0.18
5.230	0.9700	-0.1371	-0.18
5.235	0.9699	-0.1372	-0.18
5.240	0.9700	-0.1374	-0.18
5.245	0.9698	-0.1377	-0.18
5.250	0.9698	-0.1377	-0.18
5.255	0.9695	-0.1376	-0.18
5.260	0.9693	-0.1378	-0.18
5.265	0.9696	-0.1378	-0.18
5.270	0.9695	-0.1381	-0.18
5.275	0.9694	-0.1385	-0.18
5.280	0.9690	-0.1386	-0.19
5.285	0.9689	-0.1388	-0.19
5.290	0.9688	-0.1390	-0.19
5.295	0.9688	-0.1392	-0.19
5.300	0.9685	-0.1393	-0.19
5.305	0.9687	-0.1390	-0.19
5.310	0.9684	-0.1397	-0.19
5.315	0.9681	-0.1397	-0.19
5.320	0.9683	-0.1406	-0.19
5.325	0.9685	-0.1407	-0.19
5.330	0.9680	-0.1405	-0.19
5.335	0.9681	-0.1409	-0.19
5.340	0.9683	-0.1409	-0.19
5.345	0.9679	-0.1414	-0.19
5.350	0.9677	-0.1415	-0.19
5.355	0.9675	-0.1417	-0.20
5.360	0.9678	-0.1421	-0.19
5.365	0.9675	-0.1418	-0.19
5.370	0.9676	-0.1422	-0.19
5.375	0.9674	-0.1421	-0.20
5.380	0.9674	-0.1421	-0.20
5.385	0.9668	-0.1421	-0.20
5.390	0.9671	-0.1429	-0.20
5.395	0.9668	-0.1428	-0.20
5.400	0.9665	-0.1432	-0.20
5.405	0.9661	-0.1433	-0.21

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.410	0.9660	-0.1437	-0.21
5.415	0.9663	-0.1441	-0.20
5.420	0.9659	-0.1443	-0.21
5.425	0.9661	-0.1440	-0.20
5.430	0.9654	-0.1445	-0.21
5.435	0.9653	-0.1450	-0.21
5.440	0.9657	-0.1448	-0.21
5.445	0.9654	-0.1453	-0.21
5.450	0.9649	-0.1458	-0.21
5.455	0.9648	-0.1458	-0.21
5.460	0.9650	-0.1459	-0.21
5.465	0.9652	-0.1462	-0.21
5.470	0.9649	-0.1462	-0.21
5.475	0.9650	-0.1465	-0.21
5.480	0.9646	-0.1466	-0.21
5.485	0.9649	-0.1469	-0.21
5.490	0.9646	-0.1468	-0.21
5.495	0.9645	-0.1469	-0.21
5.500	0.9643	-0.1469	-0.22
5.505	0.9642	-0.1472	-0.22
5.510	0.9640	-0.1473	-0.22
5.515	0.9639	-0.1475	-0.22
5.520	0.9637	-0.1477	-0.22
5.525	0.9636	-0.1479	-0.22
5.530	0.9636	-0.1480	-0.22
5.535	0.9636	-0.1483	-0.22
5.540	0.9634	-0.1484	-0.22
5.545	0.9633	-0.1487	-0.22
5.550	0.9632	-0.1488	-0.22
5.555	0.9632	-0.1489	-0.22
5.560	0.9631	-0.1490	-0.22
5.565	0.9631	-0.1491	-0.22
5.570	0.9631	-0.1492	-0.22
5.575	0.9631	-0.1493	-0.22
5.580	0.9630	-0.1494	-0.22
5.585	0.9630	-0.1496	-0.22
5.590	0.9629	-0.1496	-0.23
5.595	0.9627	-0.1496	-0.23
5.600	0.9625	-0.1499	-0.23
5.605	0.9624	-0.1501	-0.23
5.610	0.9624	-0.1502	-0.23
5.615	0.9623	-0.1505	-0.23
5.620	0.9621	-0.1508	-0.23
5.625	0.9622	-0.1511	-0.23
5.630	0.9620	-0.1513	-0.23
5.635	0.9618	-0.1515	-0.23

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.640	0.9619	-0.1516	-0.23
5.645	0.9619	-0.1521	-0.23
5.650	0.9621	-0.1520	-0.23
5.655	0.9620	-0.1522	-0.23
5.660	0.9619	-0.1523	-0.23
5.665	0.9624	-0.1527	-0.23
5.670	0.9618	-0.1528	-0.23
5.675	0.9620	-0.1530	-0.23
5.680	0.9620	-0.1530	-0.23
5.685	0.9617	-0.1534	-0.23
5.690	0.9618	-0.1536	-0.23
5.695	0.9617	-0.1536	-0.23
5.700	0.9618	-0.1541	-0.23
5.705	0.9619	-0.1542	-0.23
5.710	0.9620	-0.1544	-0.23
5.715	0.9622	-0.1548	-0.22
5.720	0.9620	-0.1546	-0.23
5.725	0.9623	-0.1548	-0.22
5.730	0.9622	-0.1553	-0.22
5.735	0.9622	-0.1556	-0.22
5.740	0.9625	-0.1557	-0.22
5.745	0.9619	-0.1556	-0.23
5.750	0.9625	-0.1561	-0.22
5.755	0.9625	-0.1563	-0.22
5.760	0.9626	-0.1561	-0.22
5.765	0.9623	-0.1565	-0.22
5.770	0.9626	-0.1566	-0.22
5.775	0.9627	-0.1565	-0.22
5.780	0.9627	-0.1569	-0.22
5.785	0.9628	-0.1571	-0.22
5.790	0.9630	-0.1572	-0.21
5.795	0.9629	-0.1574	-0.21
5.800	0.9629	-0.1576	-0.21
5.805	0.9631	-0.1577	-0.21
5.810	0.9631	-0.1577	-0.21
5.815	0.9629	-0.1579	-0.21
5.820	0.9628	-0.1580	-0.21
5.825	0.9630	-0.1582	-0.21
5.830	0.9629	-0.1583	-0.21
5.835	0.9629	-0.1584	-0.21
5.840	0.9628	-0.1587	-0.21
5.845	0.9626	-0.1588	-0.21
5.850	0.9626	-0.1588	-0.21
5.855	0.9626	-0.1590	-0.21
5.860	0.9623	-0.1591	-0.22
5.865	0.9624	-0.1593	-0.22

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.870	0.9620	-0.1595	-0.22
5.875	0.9621	-0.1597	-0.22
5.880	0.9619	-0.1598	-0.22
5.885	0.9617	-0.1601	-0.22
5.890	0.9615	-0.1602	-0.22
5.895	0.9615	-0.1609	-0.22
5.900	0.9616	-0.1608	-0.22
5.905	0.9609	-0.1612	-0.23
5.910	0.9612	-0.1613	-0.22
5.915	0.9608	-0.1613	-0.23
5.920	0.9608	-0.1618	-0.23
5.925	0.9610	-0.1620	-0.22
5.930	0.9603	-0.1619	-0.23
5.935	0.9607	-0.1621	-0.23
5.940	0.9605	-0.1623	-0.23
5.945	0.9597	-0.1624	-0.23
5.950	0.9600	-0.1624	-0.23
5.955	0.9599	-0.1625	-0.23
5.960	0.9597	-0.1629	-0.23
5.965	0.9592	-0.1629	-0.24
5.970	0.9593	-0.1632	-0.24
5.975	0.9586	-0.1632	-0.24
5.980	0.9587	-0.1637	-0.24
5.985	0.9581	-0.1646	-0.25
5.990	0.9579	-0.1641	-0.25
5.995	0.9573	-0.1645	-0.25
6.000	0.9571	-0.1646	-0.25

2 Measurement Quality

Reliable band: from 45 MHz (S11 error σ = 1.0e-03); the points below it are kept in the plot (shaded, outside the scale) and in the tables, but are not reliable.
Spike removal: on (isolated point, 2nd difference > 6 x local noise and > S11 error budget, not shared with another sweep); replaced 1 point(s): Short at 4885 MHz ($|\Delta S11|$ = 1.4e-03).
Outliers shared by several sweeps, left alone (instrument artefacts cancelled by the calibration): 3830 MHz, 3835 MHz.

- Unreliable below 45 MHz: 8 point(s) (3 mm probe, S11 error σ = 1.0e-03). There the reference liquids resolve ε_r only to $\pm$17.2 at 5 MHz … $\pm$2.1 at 40 MHz per σ (usable: $\leq \pm 2$); the sample differs from air by less than 1σ (|S11_dut - S11_open|) at 5 of them. Values kept, but left out of the chart scale and of the summary medians; start the sweep at $\geq$ 45 MHz.

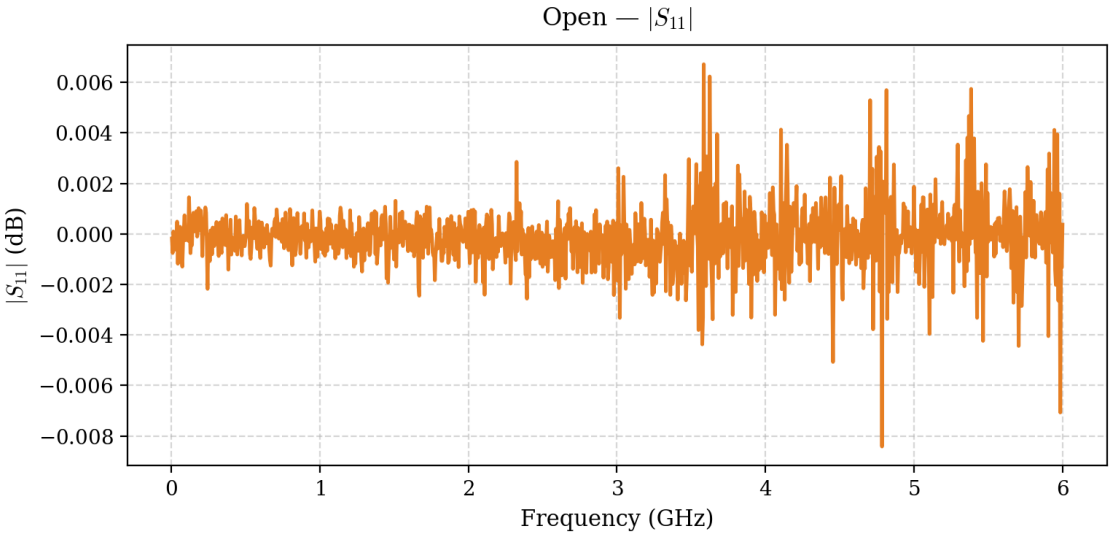
- Gn shows no radiation signature even at the top of the sweep (3 GHz – 6 GHz): mean arg(Gn) = -108° (radiation: -90°, median deviation 20°) and $|Gn| \propto f^{0.4}$ (radiation: f^3). Over the whole sweep it holds noise and the inconsistency between the reference liquids instead (its term is ≈7% of the Ref 1 (Distilled Water) admittance), so the full method's radiation correction is empirical here: compare with the simplified cross-check, especially for samples whose ε_r lies outside the range of the reference liquids.

- 2 frequency point(s) excluded: reference liquids nearly indistinguishable (|S11_ref1-S11_ref2| < 1e-02). Consider starting the sweep above the affected range.

- 8 frequency point(s) with $\varepsilon'' < 0$ (non-passive, worst ε'' = -0.485), attributable to measurement noise on a near-lossless liquid. Values kept and flagged, not replaced by another root.

3 Calibration Standard Measurements

3.1 Open



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	+0.0001	-0.00
0.010	0.9999	-0.0000	-0.00
0.015	1.0000	+0.0001	0.00
0.020	1.0000	+0.0000	-0.00
0.025	0.9999	+0.0000	-0.00
0.030	1.0000	+0.0001	-0.00
0.035	0.9999	+0.0001	-0.00
0.040	1.0001	+0.0000	0.00
0.045	0.9999	+0.0000	-0.00
0.050	1.0000	+0.0000	0.00
0.055	0.9999	+0.0000	-0.00
0.060	1.0000	-0.0001	0.00
0.065	1.0000	+0.0000	0.00
0.070	0.9999	-0.0000	-0.00
0.075	0.9999	-0.0000	-0.00
0.080	1.0000	+0.0001	-0.00
0.085	1.0000	-0.0000	0.00
0.090	1.0001	+0.0001	0.00
0.095	1.0000	+0.0000	-0.00
0.100	1.0000	-0.0000	0.00
0.105	1.0000	+0.0000	0.00
0.110	1.0000	+0.0001	-0.00
0.115	1.0000	+0.0001	-0.00
0.120	1.0002	+0.0001	0.00
0.125	1.0000	-0.0000	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.130	0.9999	-0.0001	-0.00
0.135	1.0000	-0.0001	-0.00
0.140	0.9999	-0.0000	-0.00
0.145	1.0001	+0.0000	0.00
0.150	1.0001	-0.0000	0.00
0.155	1.0000	+0.0002	0.00
0.160	1.0001	+0.0001	0.00
0.165	1.0000	+0.0000	0.00
0.170	1.0001	+0.0000	0.00
0.175	1.0000	+0.0001	-0.00
0.180	1.0000	-0.0001	0.00
0.185	1.0001	+0.0000	0.00
0.190	1.0000	+0.0001	0.00
0.195	1.0000	+0.0001	0.00
0.200	1.0001	+0.0000	0.00
0.205	1.0001	-0.0001	0.00
0.210	0.9999	+0.0001	-0.00
0.215	1.0000	+0.0000	0.00
0.220	1.0000	-0.0000	0.00
0.225	1.0000	+0.0001	0.00
0.230	1.0001	+0.0001	0.00
0.235	1.0001	-0.0000	0.00
0.240	1.0000	+0.0001	-0.00
0.245	0.9997	+0.0000	-0.00
0.250	1.0000	+0.0002	-0.00
0.255	1.0000	-0.0001	-0.00
0.260	0.9999	+0.0000	-0.00
0.265	1.0000	+0.0000	0.00
0.270	1.0000	+0.0001	0.00
0.275	0.9999	+0.0000	-0.00
0.280	1.0000	+0.0001	0.00
0.285	1.0000	-0.0001	-0.00
0.290	0.9999	+0.0000	-0.00
0.295	1.0000	-0.0000	-0.00
0.300	1.0000	-0.0000	0.00
0.305	1.0001	+0.0001	0.00
0.310	1.0000	+0.0000	-0.00
0.315	1.0000	-0.0000	-0.00
0.320	0.9999	-0.0000	-0.00
0.325	1.0000	+0.0001	0.00
0.330	1.0001	+0.0000	0.00
0.335	0.9999	-0.0001	-0.00
0.340	0.9999	+0.0000	-0.00
0.345	1.0001	-0.0001	0.00
0.350	1.0000	+0.0001	0.00
0.355	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.360	1.0000	+0.0000	-0.00
0.365	0.9999	-0.0001	-0.00
0.370	1.0001	-0.0000	0.00
0.375	1.0000	+0.0001	-0.00
0.380	1.0000	+0.0000	0.00
0.385	0.9998	+0.0001	-0.00
0.390	1.0000	-0.0000	-0.00
0.395	1.0000	+0.0000	-0.00
0.400	1.0001	+0.0001	0.00
0.405	0.9999	+0.0001	-0.00
0.410	1.0000	-0.0001	-0.00
0.415	0.9999	+0.0001	-0.00
0.420	0.9999	-0.0000	-0.00
0.425	0.9999	-0.0001	-0.00
0.430	0.9999	-0.0000	-0.00
0.435	1.0001	+0.0000	0.00
0.440	1.0000	+0.0000	-0.00
0.445	0.9999	+0.0000	-0.00
0.450	1.0000	+0.0000	-0.00
0.455	1.0000	+0.0000	0.00
0.460	1.0001	+0.0000	0.00
0.465	0.9999	+0.0001	-0.00
0.470	0.9999	+0.0002	-0.00
0.475	0.9999	+0.0001	-0.00
0.480	1.0000	-0.0000	-0.00
0.485	1.0000	-0.0000	-0.00
0.490	1.0000	+0.0001	-0.00
0.495	0.9999	+0.0001	-0.00
0.500	1.0000	+0.0000	-0.00
0.505	1.0000	-0.0000	-0.00
0.510	1.0001	+0.0001	0.00
0.515	1.0000	+0.0001	0.00
0.520	1.0001	-0.0001	0.00
0.525	1.0001	+0.0000	0.00
0.530	0.9999	+0.0001	-0.00
0.535	0.9999	+0.0001	-0.00
0.540	0.9999	+0.0000	-0.00
0.545	1.0000	+0.0000	0.00
0.550	1.0000	-0.0000	0.00
0.555	1.0000	-0.0001	-0.00
0.560	1.0001	-0.0000	0.00
0.565	1.0001	+0.0000	0.00
0.570	1.0000	+0.0000	-0.00
0.575	1.0000	-0.0001	-0.00
0.580	0.9999	-0.0000	-0.00
0.585	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.590	1.0001	-0.0000	0.00
0.595	0.9999	+0.0000	-0.00
0.600	0.9999	-0.0000	-0.00
0.605	1.0000	-0.0000	-0.00
0.610	1.0001	+0.0000	0.00
0.615	0.9999	+0.0001	-0.00
0.620	1.0000	-0.0000	0.00
0.625	0.9999	-0.0000	-0.00
0.630	1.0000	-0.0001	-0.00
0.635	1.0000	+0.0001	-0.00
0.640	1.0000	-0.0000	-0.00
0.645	1.0000	+0.0000	0.00
0.650	1.0000	+0.0000	-0.00
0.655	1.0000	+0.0001	-0.00
0.660	0.9999	-0.0000	-0.00
0.665	0.9999	+0.0000	-0.00
0.670	0.9999	+0.0000	-0.00
0.675	1.0000	-0.0000	-0.00
0.680	1.0000	-0.0000	0.00
0.685	1.0000	+0.0001	0.00
0.690	1.0000	-0.0001	-0.00
0.695	1.0001	+0.0000	0.00
0.700	1.0001	+0.0001	0.00
0.705	1.0000	+0.0000	-0.00
0.710	1.0001	-0.0001	0.00
0.715	1.0001	-0.0000	0.00
0.720	1.0000	+0.0001	0.00
0.725	1.0000	+0.0000	-0.00
0.730	1.0000	+0.0000	0.00
0.735	1.0000	+0.0001	0.00
0.740	1.0000	+0.0000	-0.00
0.745	1.0000	-0.0000	-0.00
0.750	1.0001	+0.0001	0.00
0.755	1.0000	-0.0001	0.00
0.760	0.9999	+0.0000	-0.00
0.765	1.0000	+0.0000	0.00
0.770	1.0001	+0.0000	0.00
0.775	1.0000	-0.0001	0.00
0.780	1.0000	-0.0000	0.00
0.785	0.9999	-0.0000	-0.00
0.790	0.9999	+0.0000	-0.00
0.795	1.0000	-0.0000	-0.00
0.800	1.0001	-0.0001	0.00
0.805	1.0001	+0.0000	0.00
0.810	0.9999	+0.0001	-0.00
0.815	0.9999	+0.0000	-0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.820	1.0000	+0.0000	-0.00
0.825	1.0001	+0.0000	0.00
0.830	1.0000	+0.0000	0.00
0.835	0.9999	-0.0000	-0.00
0.840	0.9999	-0.0001	-0.00
0.845	1.0000	-0.0001	0.00
0.850	1.0001	+0.0000	0.00
0.855	1.0000	+0.0000	0.00
0.860	1.0000	+0.0001	0.00
0.865	1.0000	+0.0001	-0.00
0.870	0.9999	+0.0001	-0.00
0.875	1.0000	-0.0000	-0.00
0.880	1.0000	+0.0001	0.00
0.885	1.0000	+0.0001	0.00
0.890	1.0000	+0.0001	-0.00
0.895	1.0001	+0.0000	0.00
0.900	1.0000	-0.0000	-0.00
0.905	1.0000	+0.0000	-0.00
0.910	0.9999	+0.0001	-0.00
0.915	1.0000	+0.0001	0.00
0.920	1.0000	+0.0001	-0.00
0.925	0.9999	+0.0001	-0.00
0.930	1.0000	+0.0001	-0.00
0.935	1.0000	+0.0001	0.00
0.940	1.0000	+0.0001	-0.00
0.945	1.0000	-0.0000	-0.00
0.950	1.0001	+0.0000	0.00
0.955	1.0000	+0.0001	0.00
0.960	1.0000	+0.0000	-0.00
0.965	0.9999	-0.0000	-0.00
0.970	0.9999	+0.0001	-0.00
0.975	1.0000	+0.0001	0.00
0.980	1.0000	+0.0001	-0.00
0.985	0.9999	+0.0000	-0.00
0.990	1.0000	+0.0001	-0.00
0.995	1.0000	+0.0001	0.00
1.000	1.0001	-0.0000	0.00
1.005	1.0001	+0.0001	0.00
1.010	0.9999	-0.0000	-0.00
1.015	1.0000	+0.0000	0.00
1.020	1.0000	+0.0001	-0.00
1.025	0.9999	+0.0001	-0.00
1.030	1.0000	+0.0000	0.00
1.035	0.9999	-0.0001	-0.00
1.040	1.0000	+0.0001	-0.00
1.045	1.0000	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.050	0.9999	+0.0001	-0.00
1.055	1.0001	+0.0001	0.00
1.060	1.0000	+0.0002	-0.00
1.065	1.0001	+0.0000	0.00
1.070	1.0000	+0.0001	-0.00
1.075	1.0000	+0.0001	0.00
1.080	1.0000	+0.0000	0.00
1.085	1.0000	+0.0000	0.00
1.090	0.9999	+0.0001	-0.00
1.095	0.9999	+0.0001	-0.00
1.100	1.0000	+0.0001	0.00
1.105	0.9999	+0.0000	-0.00
1.110	1.0000	+0.0002	0.00
1.115	1.0000	+0.0002	-0.00
1.120	1.0000	+0.0000	-0.00
1.125	0.9999	-0.0000	-0.00
1.130	0.9999	+0.0002	-0.00
1.135	1.0001	+0.0000	0.00
1.140	1.0001	-0.0000	0.00
1.145	1.0001	+0.0001	0.00
1.150	1.0000	-0.0000	0.00
1.155	0.9999	+0.0001	-0.00
1.160	1.0001	-0.0000	0.00
1.165	1.0000	+0.0000	0.00
1.170	1.0000	+0.0000	0.00
1.175	1.0000	+0.0001	0.00
1.180	0.9999	-0.0000	-0.00
1.185	1.0000	+0.0001	0.00
1.190	1.0000	+0.0000	0.00
1.195	1.0000	+0.0001	-0.00
1.200	0.9999	+0.0001	-0.00
1.205	1.0001	+0.0000	0.00
1.210	0.9999	-0.0000	-0.00
1.215	1.0000	+0.0000	0.00
1.220	0.9999	+0.0000	-0.00
1.225	1.0000	+0.0000	-0.00
1.230	0.9999	+0.0000	-0.00
1.235	1.0000	+0.0001	-0.00
1.240	0.9999	-0.0001	-0.00
1.245	1.0000	+0.0000	-0.00
1.250	1.0000	+0.0001	0.00
1.255	1.0000	+0.0000	-0.00
1.260	0.9999	-0.0001	-0.00
1.265	1.0001	+0.0000	0.00
1.270	1.0000	+0.0001	0.00
1.275	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.280	0.9999	+0.0000	-0.00
1.285	0.9998	+0.0000	-0.00
1.290	1.0000	+0.0000	-0.00
1.295	1.0000	+0.0001	0.00
1.300	0.9999	-0.0000	-0.00
1.305	1.0000	-0.0000	0.00
1.310	0.9999	+0.0000	-0.00
1.315	1.0000	+0.0000	-0.00
1.320	1.0000	+0.0001	0.00
1.325	1.0000	-0.0000	0.00
1.330	1.0000	+0.0001	-0.00
1.335	0.9999	+0.0001	-0.00
1.340	0.9999	+0.0001	-0.00
1.345	1.0000	+0.0002	0.00
1.350	1.0001	+0.0000	0.00
1.355	1.0001	+0.0001	0.00
1.360	1.0000	-0.0000	0.00
1.365	1.0000	-0.0000	0.00
1.370	1.0001	+0.0001	0.00
1.375	0.9999	+0.0000	-0.00
1.380	1.0001	+0.0001	0.00
1.385	0.9999	+0.0001	-0.00
1.390	0.9999	-0.0000	-0.00
1.395	1.0000	-0.0000	-0.00
1.400	1.0000	-0.0001	-0.00
1.405	1.0000	+0.0002	0.00
1.410	0.9999	+0.0001	-0.00
1.415	1.0001	+0.0001	0.00
1.420	1.0001	-0.0001	0.00
1.425	1.0000	-0.0000	-0.00
1.430	1.0001	+0.0001	0.00
1.435	1.0000	+0.0002	0.00
1.440	1.0001	+0.0002	0.00
1.445	1.0000	+0.0001	0.00
1.450	0.9998	+0.0000	-0.00
1.455	1.0000	+0.0001	-0.00
1.460	0.9998	+0.0001	-0.00
1.465	0.9999	+0.0001	-0.00
1.470	1.0000	+0.0001	-0.00
1.475	1.0001	-0.0001	0.00
1.480	0.9999	+0.0001	-0.00
1.485	1.0000	-0.0000	-0.00
1.490	1.0000	-0.0000	-0.00
1.495	1.0000	+0.0001	0.00
1.500	0.9999	+0.0001	-0.00
1.505	1.0000	+0.0001	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.510	1.0002	+0.0001	0.00
1.515	0.9999	+0.0001	-0.00
1.520	1.0000	+0.0001	-0.00
1.525	1.0000	+0.0001	-0.00
1.530	1.0000	+0.0000	0.00
1.535	1.0000	+0.0001	-0.00
1.540	1.0001	-0.0000	0.00
1.545	1.0000	+0.0002	-0.00
1.550	1.0000	+0.0000	0.00
1.555	1.0001	-0.0000	0.00
1.560	0.9999	+0.0000	-0.00
1.565	1.0000	+0.0001	0.00
1.570	1.0000	+0.0001	0.00
1.575	1.0001	+0.0001	0.00
1.580	1.0000	+0.0001	-0.00
1.585	1.0000	-0.0000	0.00
1.590	1.0000	+0.0000	-0.00
1.595	0.9999	-0.0000	-0.00
1.600	1.0000	+0.0001	0.00
1.605	1.0000	+0.0000	0.00
1.610	1.0000	+0.0001	0.00
1.615	1.0000	+0.0001	0.00
1.620	0.9999	+0.0001	-0.00
1.625	1.0000	+0.0000	0.00
1.630	1.0000	-0.0000	0.00
1.635	0.9999	+0.0001	-0.00
1.640	0.9999	+0.0001	-0.00
1.645	1.0000	+0.0001	-0.00
1.650	1.0001	-0.0000	0.00
1.655	1.0001	+0.0000	0.00
1.660	1.0000	+0.0002	0.00
1.665	0.9998	+0.0001	-0.00
1.670	0.9997	+0.0001	-0.00
1.675	1.0001	+0.0001	0.00
1.680	1.0000	+0.0000	-0.00
1.685	1.0000	-0.0001	0.00
1.690	1.0001	+0.0000	0.00
1.695	0.9999	-0.0000	-0.00
1.700	1.0000	-0.0001	0.00
1.705	1.0001	+0.0001	0.00
1.710	0.9999	-0.0000	-0.00
1.715	1.0000	-0.0001	-0.00
1.720	1.0001	-0.0000	0.00
1.725	1.0001	+0.0001	0.00
1.730	0.9999	-0.0001	-0.00
1.735	1.0001	+0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.740	1.0001	+0.0001	0.00
1.745	1.0000	+0.0002	-0.00
1.750	1.0000	-0.0001	0.00
1.755	1.0000	-0.0001	-0.00
1.760	0.9999	-0.0000	-0.00
1.765	0.9999	-0.0000	-0.00
1.770	0.9999	+0.0002	-0.00
1.775	0.9998	+0.0001	-0.00
1.780	0.9999	-0.0000	-0.00
1.785	0.9999	+0.0001	-0.00
1.790	1.0000	+0.0000	0.00
1.795	0.9999	-0.0001	-0.00
1.800	1.0000	-0.0001	-0.00
1.805	1.0001	+0.0001	0.00
1.810	1.0000	-0.0000	-0.00
1.815	1.0000	-0.0000	-0.00
1.820	0.9999	+0.0000	-0.00
1.825	0.9999	-0.0001	-0.00
1.830	0.9999	+0.0000	-0.00
1.835	0.9999	-0.0000	-0.00
1.840	1.0000	+0.0000	0.00
1.845	1.0000	-0.0001	0.00
1.850	1.0000	+0.0001	0.00
1.855	0.9999	-0.0001	-0.00
1.860	1.0001	+0.0001	0.00
1.865	0.9999	+0.0000	-0.00
1.870	0.9999	+0.0001	-0.00
1.875	1.0000	+0.0000	-0.00
1.880	0.9999	-0.0000	-0.00
1.885	1.0001	+0.0000	0.00
1.890	1.0000	-0.0001	0.00
1.895	1.0000	-0.0000	0.00
1.900	0.9999	+0.0001	-0.00
1.905	1.0000	+0.0001	0.00
1.910	0.9999	+0.0000	-0.00
1.915	1.0000	+0.0001	-0.00
1.920	1.0000	+0.0000	-0.00
1.925	1.0000	-0.0000	-0.00
1.930	1.0000	+0.0000	0.00
1.935	0.9998	+0.0000	-0.00
1.940	1.0001	-0.0001	0.00
1.945	1.0000	+0.0001	0.00
1.950	1.0000	+0.0001	0.00
1.955	1.0000	+0.0001	0.00
1.960	1.0000	+0.0000	-0.00
1.965	1.0000	+0.0000	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	$\lvert S_{11}\rvert$ **(dB)**
1.970	1.0000	-0.0001	-0.00
1.975	1.0000	+0.0000	-0.00
1.980	1.0001	+0.0000	0.00
1.985	1.0001	-0.0001	0.00
1.990	0.9999	-0.0001	-0.00
1.995	1.0000	+0.0001	-0.00
2.000	1.0001	+0.0001	0.00
2.005	1.0000	+0.0000	-0.00
2.010	0.9999	+0.0002	-0.00
2.015	0.9999	+0.0000	-0.00
2.020	1.0001	+0.0001	0.00
2.025	1.0000	+0.0000	-0.00
2.030	0.9999	-0.0001	-0.00
2.035	1.0001	-0.0001	0.00
2.040	1.0000	-0.0000	0.00
2.045	1.0001	+0.0001	0.00
2.050	0.9998	+0.0001	-0.00
2.055	0.9999	+0.0001	-0.00
2.060	1.0000	+0.0001	0.00
2.065	0.9999	+0.0000	-0.00
2.070	1.0000	-0.0001	0.00
2.075	1.0000	+0.0000	0.00
2.080	1.0000	+0.0000	0.00
2.085	0.9999	-0.0001	-0.00
2.090	1.0000	+0.0000	-0.00
2.095	0.9999	-0.0000	-0.00
2.100	0.9998	+0.0000	-0.00
2.105	1.0000	-0.0000	0.00
2.110	0.9997	+0.0000	-0.00
2.115	1.0000	+0.0001	-0.00
2.120	1.0001	+0.0001	0.00
2.125	1.0001	+0.0001	0.00
2.130	1.0001	+0.0000	0.00
2.135	1.0000	+0.0001	-0.00
2.140	0.9999	-0.0001	-0.00
2.145	1.0000	-0.0000	-0.00
2.150	0.9999	-0.0002	-0.00
2.155	1.0001	-0.0001	0.00
2.160	1.0000	+0.0001	0.00
2.165	0.9998	-0.0000	-0.00
2.170	0.9999	+0.0000	-0.00
2.175	0.9999	+0.0001	-0.00
2.180	0.9999	+0.0000	-0.00
2.185	1.0000	+0.0000	-0.00
2.190	0.9999	+0.0001	-0.00
2.195	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.200	0.9999	+0.0001	-0.00
2.205	1.0001	-0.0000	0.00
2.210	1.0000	+0.0001	-0.00
2.215	1.0000	+0.0001	-0.00
2.220	0.9999	+0.0001	-0.00
2.225	1.0000	+0.0002	-0.00
2.230	1.0000	+0.0001	0.00
2.235	1.0000	-0.0001	-0.00
2.240	1.0000	-0.0000	-0.00
2.245	0.9999	+0.0001	-0.00
2.250	0.9999	-0.0000	-0.00
2.255	1.0000	+0.0001	0.00
2.260	0.9999	+0.0002	-0.00
2.265	0.9999	+0.0001	-0.00
2.270	1.0000	-0.0000	0.00
2.275	1.0001	+0.0001	0.00
2.280	1.0001	+0.0000	0.00
2.285	0.9998	+0.0001	-0.00
2.290	0.9999	+0.0000	-0.00
2.295	1.0000	-0.0001	0.00
2.300	1.0000	+0.0000	-0.00
2.305	1.0001	+0.0001	0.00
2.310	1.0000	-0.0001	-0.00
2.315	0.9999	-0.0001	-0.00
2.320	0.9999	+0.0000	-0.00
2.325	1.0003	-0.0001	0.00
2.330	0.9999	+0.0001	-0.00
2.335	1.0000	+0.0001	0.00
2.340	1.0000	-0.0000	-0.00
2.345	1.0001	+0.0002	0.00
2.350	1.0000	+0.0001	0.00
2.355	1.0001	-0.0001	0.00
2.360	0.9999	+0.0000	-0.00
2.365	1.0000	+0.0002	0.00
2.370	0.9999	+0.0001	-0.00
2.375	1.0000	+0.0000	-0.00
2.380	0.9999	+0.0000	-0.00
2.385	0.9998	+0.0002	-0.00
2.390	0.9998	+0.0001	-0.00
2.395	0.9997	+0.0002	-0.00
2.400	1.0000	+0.0001	-0.00
2.405	1.0000	+0.0001	0.00
2.410	0.9998	-0.0001	-0.00
2.415	1.0000	+0.0001	0.00
2.420	0.9999	+0.0001	-0.00
2.425	1.0000	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.430	1.0000	+0.0002	0.00
2.435	1.0000	-0.0000	-0.00
2.440	0.9999	+0.0001	-0.00
2.445	0.9999	-0.0001	-0.00
2.450	0.9999	-0.0001	-0.00
2.455	0.9999	-0.0001	-0.00
2.460	1.0000	-0.0001	0.00
2.465	0.9999	-0.0000	-0.00
2.470	1.0000	+0.0001	-0.00
2.475	1.0000	+0.0001	0.00
2.480	1.0000	+0.0001	-0.00
2.485	0.9999	-0.0000	-0.00
2.490	0.9999	+0.0000	-0.00
2.495	1.0000	+0.0000	0.00
2.500	0.9998	+0.0000	-0.00
2.505	1.0000	-0.0000	-0.00
2.510	0.9999	-0.0000	-0.00
2.515	1.0000	+0.0000	-0.00
2.520	0.9999	+0.0000	-0.00
2.525	1.0000	+0.0002	-0.00
2.530	1.0001	+0.0001	0.00
2.535	0.9998	+0.0000	-0.00
2.540	0.9999	+0.0001	-0.00
2.545	1.0000	-0.0000	-0.00
2.550	1.0000	+0.0001	-0.00
2.555	0.9999	-0.0000	-0.00
2.560	0.9999	+0.0001	-0.00
2.565	0.9999	+0.0000	-0.00
2.570	1.0000	+0.0000	-0.00
2.575	0.9999	-0.0001	-0.00
2.580	0.9999	+0.0000	-0.00
2.585	0.9998	+0.0002	-0.00
2.590	1.0000	+0.0000	0.00
2.595	1.0000	-0.0001	-0.00
2.600	1.0000	-0.0000	-0.00
2.605	1.0001	+0.0002	0.00
2.610	0.9998	-0.0001	-0.00
2.615	1.0001	-0.0002	0.00
2.620	1.0000	-0.0000	0.00
2.625	1.0000	+0.0000	-0.00
2.630	0.9998	-0.0001	-0.00
2.635	1.0000	+0.0001	-0.00
2.640	0.9999	+0.0001	-0.00
2.645	0.9999	-0.0000	-0.00
2.650	1.0000	+0.0002	0.00
2.655	0.9998	+0.0001	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.660	1.0000	+0.0001	0.00
2.665	1.0000	-0.0000	0.00
2.670	1.0000	+0.0002	-0.00
2.675	1.0001	-0.0001	0.00
2.680	1.0000	-0.0000	0.00
2.685	1.0001	+0.0001	0.00
2.690	1.0000	+0.0003	0.00
2.695	0.9999	+0.0001	-0.00
2.700	1.0000	+0.0000	-0.00
2.705	0.9998	+0.0000	-0.00
2.710	1.0000	+0.0001	-0.00
2.715	1.0000	-0.0002	0.00
2.720	0.9999	+0.0000	-0.00
2.725	1.0001	+0.0002	0.00
2.730	1.0000	+0.0003	0.00
2.735	0.9998	+0.0003	-0.00
2.740	1.0000	-0.0000	0.00
2.745	1.0000	+0.0000	-0.00
2.750	0.9998	+0.0002	-0.00
2.755	0.9999	+0.0002	-0.00
2.760	0.9999	+0.0002	-0.00
2.765	1.0000	+0.0001	0.00
2.770	0.9998	+0.0001	-0.00
2.775	1.0000	+0.0001	-0.00
2.780	1.0000	+0.0001	0.00
2.785	0.9999	+0.0001	-0.00
2.790	1.0000	+0.0001	-0.00
2.795	0.9998	+0.0000	-0.00
2.800	1.0000	+0.0001	0.00
2.805	1.0000	-0.0000	-0.00
2.810	1.0000	+0.0002	-0.00
2.815	1.0000	+0.0000	-0.00
2.820	0.9999	+0.0000	-0.00
2.825	1.0000	+0.0002	0.00
2.830	0.9998	+0.0002	-0.00
2.835	0.9999	+0.0001	-0.00
2.840	0.9999	+0.0003	-0.00
2.845	0.9999	+0.0000	-0.00
2.850	1.0001	+0.0001	0.00
2.855	0.9998	+0.0002	-0.00
2.860	1.0001	+0.0000	0.00
2.865	0.9999	+0.0003	-0.00
2.870	1.0000	+0.0001	-0.00
2.875	1.0000	+0.0001	-0.00
2.880	0.9999	+0.0002	-0.00
2.885	1.0000	-0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.890	1.0000	+0.0004	-0.00
2.895	0.9999	+0.0002	-0.00
2.900	1.0000	+0.0001	-0.00
2.905	1.0000	+0.0002	-0.00
2.910	0.9998	+0.0001	-0.00
2.915	1.0000	+0.0001	-0.00
2.920	0.9999	+0.0001	-0.00
2.925	1.0000	-0.0001	0.00
2.930	0.9999	+0.0002	-0.00
2.935	1.0001	+0.0001	0.00
2.940	0.9998	+0.0001	-0.00
2.945	0.9998	+0.0000	-0.00
2.950	1.0000	+0.0000	0.00
2.955	1.0000	-0.0001	-0.00
2.960	0.9999	+0.0003	-0.00
2.965	0.9999	+0.0000	-0.00
2.970	0.9999	+0.0001	-0.00
2.975	1.0001	+0.0001	0.00
2.980	0.9997	-0.0002	-0.00
2.985	0.9999	+0.0002	-0.00
2.990	0.9998	+0.0002	-0.00
2.995	1.0000	-0.0000	0.00
3.000	0.9999	+0.0002	-0.00
3.005	0.9998	+0.0003	-0.00
3.010	1.0003	-0.0002	0.00
3.015	1.0000	+0.0002	-0.00
3.020	0.9996	+0.0004	-0.00
3.025	1.0000	+0.0001	0.00
3.030	0.9999	+0.0002	-0.00
3.035	0.9999	-0.0000	-0.00
3.040	0.9999	+0.0002	-0.00
3.045	1.0003	+0.0000	0.00
3.050	1.0001	+0.0001	0.00
3.055	1.0000	+0.0000	0.00
3.060	0.9997	+0.0001	-0.00
3.065	1.0000	-0.0003	0.00
3.070	0.9999	+0.0003	-0.00
3.075	0.9998	+0.0001	-0.00
3.080	0.9998	+0.0001	-0.00
3.085	1.0000	-0.0000	0.00
3.090	1.0000	+0.0001	-0.00
3.095	0.9999	+0.0001	-0.00
3.100	0.9998	-0.0001	-0.00
3.105	0.9999	-0.0001	-0.00
3.110	0.9999	-0.0001	-0.00
3.115	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.120	0.9999	+0.0001	-0.00
3.125	1.0001	+0.0001	0.00
3.130	1.0000	+0.0001	0.00
3.135	1.0000	+0.0000	-0.00
3.140	0.9999	+0.0000	-0.00
3.145	1.0000	+0.0001	0.00
3.150	0.9999	+0.0001	-0.00
3.155	0.9999	-0.0000	-0.00
3.160	0.9999	-0.0000	-0.00
3.165	1.0001	+0.0003	0.00
3.170	0.9999	-0.0000	-0.00
3.175	0.9999	+0.0002	-0.00
3.180	1.0000	-0.0001	-0.00
3.185	1.0001	-0.0001	0.00
3.190	1.0000	+0.0002	0.00
3.195	0.9999	+0.0001	-0.00
3.200	0.9999	+0.0002	-0.00
3.205	1.0000	+0.0000	-0.00
3.210	1.0000	+0.0002	-0.00
3.215	1.0000	+0.0000	-0.00
3.220	0.9998	+0.0001	-0.00
3.225	0.9999	+0.0000	-0.00
3.230	1.0001	-0.0002	0.00
3.235	0.9998	-0.0000	-0.00
3.240	0.9997	+0.0001	-0.00
3.245	1.0000	+0.0000	-0.00
3.250	0.9997	+0.0003	-0.00
3.255	0.9999	+0.0002	-0.00
3.260	0.9999	-0.0001	-0.00
3.265	0.9998	-0.0001	-0.00
3.270	1.0000	+0.0000	0.00
3.275	1.0000	+0.0001	0.00
3.280	1.0000	-0.0001	-0.00
3.285	0.9999	+0.0001	-0.00
3.290	0.9999	+0.0001	-0.00
3.295	0.9999	+0.0001	-0.00
3.300	1.0000	-0.0003	-0.00
3.305	1.0001	+0.0001	0.00
3.310	1.0001	+0.0002	0.00
3.315	1.0000	+0.0000	-0.00
3.320	1.0000	+0.0001	-0.00
3.325	1.0003	-0.0000	0.00
3.330	0.9997	-0.0001	-0.00
3.335	1.0002	+0.0001	0.00
3.340	1.0000	-0.0000	0.00
3.345	1.0001	-0.0001	0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.350	0.9999	+0.0002	-0.00
3.355	1.0000	-0.0001	0.00
3.360	0.9997	+0.0003	-0.00
3.365	1.0000	+0.0003	0.00
3.370	1.0000	+0.0001	-0.00
3.375	0.9998	+0.0001	-0.00
3.380	0.9998	+0.0001	-0.00
3.385	0.9999	+0.0001	-0.00
3.390	1.0000	+0.0001	-0.00
3.395	0.9999	-0.0000	-0.00
3.400	0.9999	+0.0003	-0.00
3.405	0.9999	+0.0002	-0.00
3.410	0.9997	+0.0002	-0.00
3.415	0.9999	+0.0004	-0.00
3.420	0.9999	+0.0003	-0.00
3.425	1.0000	+0.0001	0.00
3.430	0.9999	+0.0001	-0.00
3.435	0.9998	+0.0002	-0.00
3.440	1.0000	+0.0002	-0.00
3.445	0.9999	+0.0003	-0.00
3.450	1.0000	+0.0003	-0.00
3.455	0.9998	+0.0003	-0.00
3.460	1.0000	+0.0003	-0.00
3.465	0.9997	+0.0002	-0.00
3.470	0.9998	+0.0003	-0.00
3.475	0.9999	+0.0000	-0.00
3.480	1.0001	+0.0002	0.00
3.485	1.0003	-0.0000	0.00
3.490	1.0000	+0.0002	-0.00
3.495	0.9999	-0.0000	-0.00
3.500	1.0001	+0.0001	0.00
3.505	0.9998	+0.0000	-0.00
3.510	1.0000	+0.0000	0.00
3.515	1.0001	+0.0002	0.00
3.520	1.0000	+0.0003	0.00
3.525	1.0000	+0.0005	0.00
3.530	1.0001	+0.0000	0.00
3.535	1.0003	+0.0001	0.00
3.540	1.0000	+0.0006	0.00
3.545	0.9999	+0.0000	-0.00
3.550	0.9996	+0.0004	-0.00
3.555	1.0002	-0.0002	0.00
3.560	1.0000	+0.0001	-0.00
3.565	1.0003	-0.0004	0.00
3.570	1.0000	+0.0003	-0.00
3.575	0.9995	+0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.580	0.9996	+0.0006	-0.00
3.585	1.0008	-0.0003	0.01
3.590	1.0000	-0.0000	-0.00
3.595	1.0003	-0.0002	0.00
3.600	0.9998	+0.0003	-0.00
3.605	1.0004	+0.0001	0.00
3.610	1.0003	+0.0003	0.00
3.615	1.0000	+0.0001	0.00
3.620	0.9999	+0.0000	-0.00
3.625	1.0007	-0.0009	0.01
3.630	1.0002	+0.0005	0.00
3.635	0.9999	+0.0001	-0.00
3.640	0.9998	+0.0000	-0.00
3.645	0.9996	+0.0001	-0.00
3.650	1.0001	+0.0005	0.00
3.655	1.0002	+0.0001	0.00
3.660	0.9999	+0.0003	-0.00
3.665	1.0002	+0.0001	0.00
3.670	0.9999	+0.0004	-0.00
3.675	1.0005	+0.0004	0.00
3.680	1.0000	+0.0002	0.00
3.685	0.9998	+0.0003	-0.00
3.690	1.0000	+0.0001	0.00
3.695	0.9999	+0.0000	-0.00
3.700	0.9999	+0.0001	-0.00
3.705	1.0000	-0.0000	0.00
3.710	1.0000	+0.0001	0.00
3.715	0.9999	+0.0000	-0.00
3.720	1.0001	-0.0001	0.00
3.725	0.9999	+0.0001	-0.00
3.730	1.0000	+0.0001	0.00
3.735	1.0001	+0.0002	0.00
3.740	1.0000	+0.0002	-0.00
3.745	0.9998	-0.0000	-0.00
3.750	1.0000	+0.0001	0.00
3.755	1.0001	+0.0000	0.00
3.760	1.0000	-0.0000	0.00
3.765	0.9999	+0.0003	-0.00
3.770	1.0000	+0.0002	0.00
3.775	1.0001	+0.0001	0.00
3.780	0.9996	+0.0011	-0.00
3.785	0.9999	+0.0005	-0.00
3.790	0.9998	+0.0000	-0.00
3.795	1.0000	-0.0001	0.00
3.800	0.9997	-0.0000	-0.00
3.805	0.9999	-0.0003	-0.00

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.810	1.0000	+0.0001	-0.00
3.815	1.0003	-0.0000	0.00
3.820	0.9999	+0.0001	-0.00
3.825	1.0003	+0.0002	0.00
3.830	1.0001	+0.0001	0.00
3.835	1.0000	+0.0001	0.00
3.840	1.0001	-0.0001	0.00
3.845	1.0000	+0.0001	-0.00
3.850	0.9998	-0.0001	-0.00
3.855	0.9999	+0.0003	-0.00
3.860	1.0002	+0.0002	0.00
3.865	0.9999	+0.0005	-0.00
3.870	1.0001	+0.0000	0.00
3.875	0.9999	+0.0001	-0.00
3.880	1.0000	-0.0000	0.00
3.885	0.9999	+0.0001	-0.00
3.890	0.9998	-0.0002	-0.00
3.895	1.0002	-0.0003	0.00
3.900	1.0001	+0.0002	0.00
3.905	0.9996	+0.0002	-0.00
3.910	0.9999	+0.0003	-0.00
3.915	1.0000	+0.0000	-0.00
3.920	0.9998	+0.0001	-0.00
3.925	1.0001	+0.0001	0.00
3.930	1.0001	-0.0000	0.00
3.935	1.0000	+0.0001	-0.00
3.940	1.0000	-0.0000	-0.00
3.945	0.9998	-0.0002	-0.00
3.950	1.0000	+0.0001	-0.00
3.955	0.9998	+0.0001	-0.00
3.960	1.0000	+0.0001	-0.00
3.965	0.9999	-0.0001	-0.00
3.970	0.9998	+0.0002	-0.00
3.975	1.0001	+0.0001	0.00
3.980	1.0000	+0.0001	0.00
3.985	1.0000	+0.0001	0.00
3.990	0.9998	+0.0000	-0.00
3.995	1.0001	+0.0001	0.00
4.000	0.9999	+0.0001	-0.00
4.005	1.0000	+0.0002	0.00
4.010	1.0000	+0.0002	0.00
4.015	0.9999	+0.0001	-0.00
4.020	0.9999	+0.0002	-0.00
4.025	1.0000	-0.0001	-0.00
4.030	1.0002	+0.0003	0.00
4.035	1.0000	+0.0002	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.040	1.0000	+0.0001	0.00
4.045	0.9999	+0.0002	-0.00
4.050	1.0002	-0.0001	0.00
4.055	1.0001	+0.0001	0.00
4.060	1.0001	+0.0002	0.00
4.065	0.9996	+0.0004	-0.00
4.070	0.9999	+0.0003	-0.00
4.075	1.0000	+0.0002	0.00
4.080	0.9999	+0.0001	-0.00
4.085	0.9998	+0.0003	-0.00
4.090	0.9999	+0.0000	-0.00
4.095	1.0000	+0.0003	-0.00
4.100	0.9998	+0.0001	-0.00
4.105	1.0005	+0.0003	0.00
4.110	0.9997	-0.0001	-0.00
4.115	1.0001	+0.0002	0.00
4.120	0.9998	+0.0003	-0.00
4.125	1.0001	-0.0002	0.00
4.130	0.9997	-0.0002	-0.00
4.135	1.0001	+0.0002	0.00
4.140	1.0002	+0.0003	0.00
4.145	1.0004	-0.0006	0.00
4.150	1.0002	+0.0004	0.00
4.155	0.9999	+0.0003	-0.00
4.160	0.9998	-0.0000	-0.00
4.165	1.0001	+0.0001	0.00
4.170	1.0000	+0.0002	-0.00
4.175	0.9998	+0.0002	-0.00
4.180	1.0000	+0.0000	-0.00
4.185	1.0001	+0.0000	0.00
4.190	1.0001	+0.0001	0.00
4.195	1.0001	+0.0001	0.00
4.200	1.0000	+0.0001	0.00
4.205	1.0001	-0.0003	0.00
4.210	1.0002	+0.0002	0.00
4.215	0.9999	+0.0003	-0.00
4.220	1.0000	+0.0001	0.00
4.225	1.0000	+0.0000	-0.00
4.230	0.9999	-0.0000	-0.00
4.235	1.0000	+0.0003	-0.00
4.240	0.9999	+0.0001	-0.00
4.245	1.0001	-0.0003	0.00
4.250	1.0001	+0.0001	0.00
4.255	1.0000	-0.0003	-0.00
4.260	1.0000	+0.0001	-0.00
4.265	1.0002	-0.0001	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.270	1.0000	+0.0000	-0.00
4.275	1.0000	+0.0001	0.00
4.280	1.0000	+0.0001	-0.00
4.285	0.9999	+0.0001	-0.00
4.290	1.0000	+0.0001	-0.00
4.295	1.0000	+0.0002	0.00
4.300	0.9998	+0.0000	-0.00
4.305	1.0000	+0.0001	0.00
4.310	1.0001	+0.0001	0.00
4.315	1.0000	+0.0001	-0.00
4.320	1.0001	+0.0001	0.00
4.325	1.0000	+0.0002	0.00
4.330	1.0001	+0.0000	0.00
4.335	1.0000	+0.0001	-0.00
4.340	1.0001	+0.0000	0.00
4.345	1.0000	-0.0001	0.00
4.350	1.0000	-0.0001	-0.00
4.355	0.9998	+0.0000	-0.00
4.360	1.0001	+0.0001	0.00
4.365	1.0000	+0.0002	0.00
4.370	0.9999	-0.0000	-0.00
4.375	1.0002	+0.0003	0.00
4.380	1.0001	+0.0000	0.00
4.385	1.0000	+0.0002	0.00
4.390	0.9999	-0.0001	-0.00
4.395	1.0001	+0.0001	0.00
4.400	0.9999	+0.0001	-0.00
4.405	1.0002	-0.0001	0.00
4.410	1.0000	-0.0001	-0.00
4.415	1.0001	-0.0000	0.00
4.420	0.9999	+0.0003	-0.00
4.425	0.9999	+0.0001	-0.00
4.430	1.0000	+0.0001	0.00
4.435	1.0003	+0.0001	0.00
4.440	0.9999	-0.0002	-0.00
4.445	1.0000	+0.0003	0.00
4.450	0.9997	+0.0001	-0.00
4.455	0.9994	+0.0003	-0.01
4.460	0.9998	-0.0000	-0.00
4.465	1.0002	+0.0003	0.00
4.470	1.0000	-0.0002	-0.00
4.475	1.0001	+0.0000	0.00
4.480	0.9999	-0.0000	-0.00
4.485	1.0000	+0.0004	-0.00
4.490	1.0001	+0.0001	0.00
4.495	1.0001	-0.0003	0.00

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.500	1.0000	-0.0001	0.00
4.505	1.0001	+0.0003	0.00
4.510	1.0003	+0.0004	0.00
4.515	0.9998	+0.0002	-0.00
4.520	0.9997	-0.0002	-0.00
4.525	1.0000	-0.0003	0.00
4.530	0.9999	-0.0000	-0.00
4.535	1.0000	-0.0002	0.00
4.540	1.0000	-0.0001	0.00
4.545	0.9999	+0.0000	-0.00
4.550	1.0001	+0.0000	0.00
4.555	1.0000	+0.0001	0.00
4.560	0.9999	+0.0000	-0.00
4.565	0.9999	+0.0001	-0.00
4.570	1.0000	-0.0001	0.00
4.575	1.0000	+0.0001	0.00
4.580	0.9999	+0.0000	-0.00
4.585	0.9999	+0.0001	-0.00
4.590	1.0001	+0.0001	0.00
4.595	1.0000	+0.0000	-0.00
4.600	1.0000	+0.0002	-0.00
4.605	1.0001	+0.0001	0.00
4.610	0.9999	+0.0002	-0.00
4.615	1.0000	+0.0001	-0.00
4.620	1.0000	+0.0001	0.00
4.625	0.9999	-0.0002	-0.00
4.630	1.0000	+0.0001	-0.00
4.635	1.0000	+0.0000	-0.00
4.640	1.0000	+0.0002	-0.00
4.645	0.9998	+0.0003	-0.00
4.650	1.0000	+0.0001	0.00
4.655	1.0001	+0.0003	0.00
4.660	0.9998	-0.0002	-0.00
4.665	0.9999	-0.0002	-0.00
4.670	0.9998	+0.0004	-0.00
4.675	1.0002	+0.0001	0.00
4.680	1.0000	+0.0003	-0.00
4.685	1.0001	+0.0001	0.00
4.690	0.9999	+0.0002	-0.00
4.695	1.0001	+0.0000	0.00
4.700	1.0000	+0.0002	-0.00
4.705	1.0006	-0.0003	0.01
4.710	0.9999	+0.0007	-0.00
4.715	1.0000	+0.0001	0.00
4.720	1.0003	-0.0000	0.00
4.725	0.9996	+0.0003	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.730	1.0000	-0.0000	-0.00
4.735	1.0001	+0.0000	0.00
4.740	1.0002	-0.0001	0.00
4.745	0.9999	+0.0004	-0.00
4.750	1.0004	+0.0001	0.00
4.755	1.0000	-0.0002	-0.00
4.760	1.0003	-0.0001	0.00
4.765	1.0004	-0.0002	0.00
4.770	0.9997	-0.0004	-0.00
4.775	1.0004	+0.0002	0.00
4.780	1.0001	-0.0001	0.00
4.785	0.9990	-0.0002	-0.01
4.790	1.0002	-0.0001	0.00
4.795	1.0001	+0.0001	0.00
4.800	1.0002	-0.0002	0.00
4.805	1.0001	-0.0003	0.00
4.810	1.0001	-0.0002	0.00
4.815	1.0007	-0.0002	0.01
4.820	0.9996	+0.0003	-0.00
4.825	1.0000	+0.0000	-0.00
4.830	0.9997	-0.0000	-0.00
4.835	1.0000	+0.0001	0.00
4.840	1.0001	-0.0001	0.00
4.845	1.0002	-0.0002	0.00
4.850	1.0000	+0.0002	-0.00
4.855	0.9998	+0.0001	-0.00
4.860	1.0000	-0.0001	0.00
4.865	1.0003	+0.0003	0.00
4.870	1.0000	+0.0002	0.00
4.875	1.0000	+0.0001	0.00
4.880	0.9999	-0.0001	-0.00
4.885	1.0000	+0.0001	0.00
4.890	0.9997	-0.0007	-0.00
4.895	0.9999	+0.0003	-0.00
4.900	0.9999	-0.0001	-0.00
4.905	0.9999	-0.0001	-0.00
4.910	1.0000	+0.0002	-0.00
4.915	1.0000	+0.0001	-0.00
4.920	1.0000	+0.0000	-0.00
4.925	1.0000	+0.0000	-0.00
4.930	1.0001	+0.0001	0.00
4.935	1.0000	+0.0001	0.00
4.940	1.0001	-0.0001	0.00
4.945	0.9999	+0.0001	-0.00
4.950	1.0000	-0.0000	0.00
4.955	1.0000	-0.0000	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.960	1.0001	-0.0001	0.00
4.965	0.9999	+0.0000	-0.00
4.970	1.0000	+0.0000	-0.00
4.975	1.0000	-0.0001	0.00
4.980	1.0000	+0.0001	0.00
4.985	0.9999	-0.0001	-0.00
4.990	1.0000	-0.0001	-0.00
4.995	1.0000	+0.0002	-0.00
5.000	1.0002	+0.0002	0.00
5.005	1.0000	+0.0001	-0.00
5.010	1.0000	+0.0001	0.00
5.015	0.9998	-0.0000	-0.00
5.020	1.0000	+0.0001	-0.00
5.025	0.9999	+0.0001	-0.00
5.030	1.0000	+0.0003	0.00
5.035	0.9998	+0.0001	-0.00
5.040	0.9999	+0.0001	-0.00
5.045	1.0000	-0.0001	-0.00
5.050	1.0001	+0.0001	0.00
5.055	0.9999	-0.0001	-0.00
5.060	1.0002	+0.0001	0.00
5.065	1.0002	+0.0003	0.00
5.070	0.9998	+0.0001	-0.00
5.075	0.9999	-0.0001	-0.00
5.080	1.0000	+0.0001	0.00
5.085	1.0000	+0.0003	0.00
5.090	1.0001	+0.0001	0.00
5.095	1.0000	-0.0003	-0.00
5.100	0.9999	-0.0002	-0.00
5.105	0.9995	+0.0000	-0.00
5.110	1.0002	-0.0000	0.00
5.115	0.9999	+0.0001	-0.00
5.120	1.0001	+0.0001	0.00
5.125	0.9997	+0.0000	-0.00
5.130	1.0001	-0.0000	0.00
5.135	1.0000	+0.0004	-0.00
5.140	1.0000	+0.0002	-0.00
5.145	1.0002	+0.0001	0.00
5.150	1.0000	+0.0001	0.00
5.155	1.0001	+0.0000	0.00
5.160	1.0001	-0.0000	0.00
5.165	1.0000	+0.0002	-0.00
5.170	1.0000	+0.0000	0.00
5.175	1.0001	+0.0001	0.00
5.180	1.0000	+0.0001	-0.00
5.185	1.0001	+0.0002	0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.190	1.0001	+0.0001	0.00
5.195	1.0000	+0.0001	-0.00
5.200	1.0000	+0.0001	-0.00
5.205	1.0000	+0.0000	-0.00
5.210	1.0000	+0.0000	0.00
5.215	1.0000	-0.0000	-0.00
5.220	1.0000	+0.0001	0.00
5.225	1.0000	+0.0001	-0.00
5.230	1.0001	+0.0001	0.00
5.235	1.0000	+0.0000	-0.00
5.240	1.0001	+0.0001	0.00
5.245	1.0000	+0.0001	0.00
5.250	1.0000	+0.0000	0.00
5.255	1.0000	+0.0002	-0.00
5.260	1.0001	-0.0002	0.00
5.265	0.9997	-0.0001	-0.00
5.270	1.0001	+0.0002	0.00
5.275	1.0001	+0.0002	0.00
5.280	1.0000	+0.0001	-0.00
5.285	1.0000	+0.0001	-0.00
5.290	1.0002	+0.0000	0.00
5.295	1.0004	+0.0000	0.00
5.300	1.0000	+0.0001	0.00
5.305	0.9999	-0.0003	-0.00
5.310	1.0001	+0.0003	0.00
5.315	1.0000	-0.0001	-0.00
5.320	1.0000	+0.0000	0.00
5.325	1.0000	+0.0001	-0.00
5.330	0.9999	+0.0000	-0.00
5.335	1.0000	+0.0002	-0.00
5.340	0.9998	+0.0002	-0.00
5.345	1.0001	+0.0002	0.00
5.350	1.0003	-0.0002	0.00
5.355	0.9998	+0.0003	-0.00
5.360	1.0004	+0.0001	0.00
5.365	0.9999	-0.0005	-0.00
5.370	1.0005	+0.0002	0.00
5.375	1.0000	+0.0002	-0.00
5.380	1.0002	-0.0002	0.00
5.385	1.0007	+0.0003	0.01
5.390	1.0000	+0.0004	-0.00
5.395	1.0000	-0.0002	0.00
5.400	1.0001	+0.0003	0.00
5.405	1.0004	-0.0001	0.00
5.410	1.0001	+0.0003	0.00
5.415	0.9999	-0.0005	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.420	1.0000	+0.0002	-0.00
5.425	0.9996	-0.0004	-0.00
5.430	1.0002	-0.0001	0.00
5.435	1.0000	-0.0001	0.00
5.440	0.9999	+0.0003	-0.00
5.445	0.9998	-0.0001	-0.00
5.450	1.0001	+0.0003	0.00
5.455	1.0002	-0.0003	0.00
5.460	1.0000	-0.0000	0.00
5.465	0.9995	-0.0002	-0.00
5.470	1.0001	+0.0002	0.00
5.475	1.0000	-0.0000	0.00
5.480	1.0001	+0.0001	0.00
5.485	1.0003	-0.0001	0.00
5.490	0.9998	+0.0003	-0.00
5.495	1.0002	-0.0003	0.00
5.500	1.0000	+0.0001	-0.00
5.505	1.0000	+0.0000	0.00
5.510	1.0000	-0.0001	-0.00
5.515	1.0000	-0.0001	0.00
5.520	1.0000	+0.0001	-0.00
5.525	1.0000	+0.0000	0.00
5.530	1.0000	+0.0000	0.00
5.535	1.0000	-0.0002	0.00
5.540	0.9999	+0.0001	-0.00
5.545	1.0000	+0.0001	0.00
5.550	0.9999	+0.0000	-0.00
5.555	1.0001	+0.0001	0.00
5.560	1.0000	+0.0000	-0.00
5.565	1.0001	-0.0000	0.00
5.570	1.0001	+0.0001	0.00
5.575	1.0001	+0.0000	0.00
5.580	1.0000	-0.0000	-0.00
5.585	1.0000	+0.0000	0.00
5.590	1.0000	+0.0000	0.00
5.595	1.0000	-0.0000	0.00
5.600	0.9999	+0.0000	-0.00
5.605	1.0001	-0.0001	0.00
5.610	1.0002	+0.0001	0.00
5.615	1.0001	+0.0002	0.00
5.620	1.0000	+0.0000	-0.00
5.625	0.9999	+0.0001	-0.00
5.630	1.0000	+0.0001	-0.00
5.635	1.0001	-0.0001	0.00
5.640	1.0001	-0.0000	0.00
5.645	1.0002	+0.0002	0.00

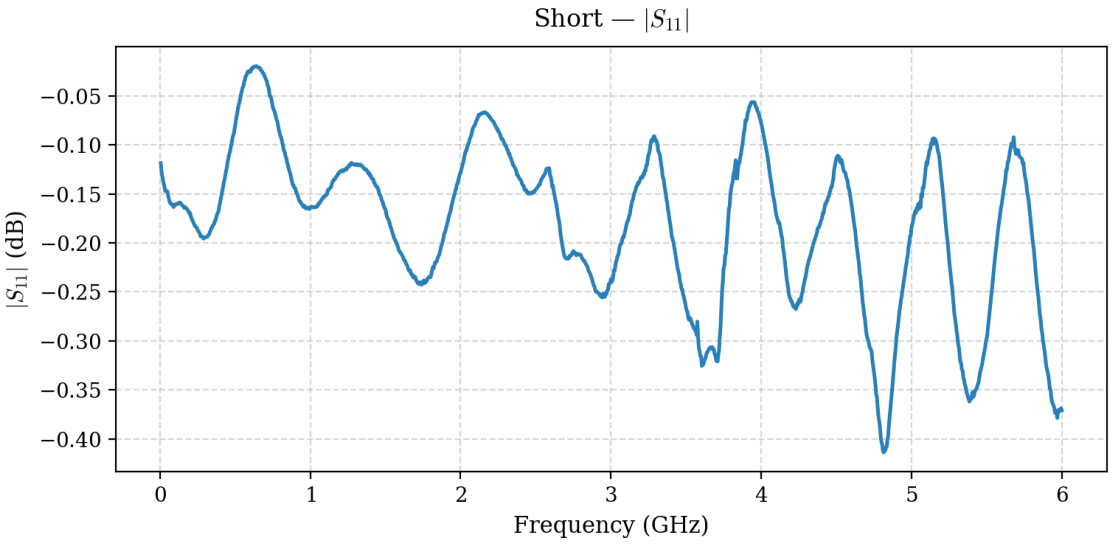
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.650	1.0001	-0.0001	0.00
5.655	0.9998	-0.0000	-0.00
5.660	0.9999	+0.0002	-0.00
5.665	1.0001	-0.0003	0.00
5.670	0.9997	+0.0004	-0.00
5.675	0.9999	+0.0001	-0.00
5.680	0.9999	+0.0003	-0.00
5.685	1.0000	-0.0003	-0.00
5.690	1.0001	-0.0003	0.00
5.695	0.9997	-0.0002	-0.00
5.700	1.0000	-0.0000	0.00
5.705	0.9995	+0.0003	-0.00
5.710	1.0000	-0.0001	0.00
5.715	1.0001	+0.0004	0.00
5.720	0.9999	-0.0001	-0.00
5.725	0.9997	-0.0001	-0.00
5.730	0.9998	-0.0001	-0.00
5.735	0.9999	+0.0001	-0.00
5.740	1.0000	-0.0000	-0.00
5.745	0.9999	-0.0000	-0.00
5.750	1.0002	-0.0000	0.00
5.755	0.9999	-0.0001	-0.00
5.760	1.0001	+0.0003	0.00
5.765	1.0003	-0.0001	0.00
5.770	1.0002	-0.0000	0.00
5.775	1.0001	-0.0000	0.00
5.780	1.0000	-0.0000	-0.00
5.785	1.0002	+0.0001	0.00
5.790	1.0000	+0.0001	0.00
5.795	0.9999	+0.0000	-0.00
5.800	1.0000	-0.0000	0.00
5.805	1.0002	+0.0002	0.00
5.810	1.0000	-0.0001	-0.00
5.815	0.9999	-0.0001	-0.00
5.820	1.0001	+0.0000	0.00
5.825	1.0001	-0.0001	0.00
5.830	1.0000	+0.0001	0.00
5.835	1.0000	+0.0001	-0.00
5.840	1.0000	+0.0001	-0.00
5.845	1.0000	-0.0001	-0.00
5.850	1.0000	+0.0000	0.00
5.855	1.0001	+0.0001	0.00
5.860	1.0000	+0.0001	-0.00
5.865	0.9999	-0.0000	-0.00
5.870	1.0000	+0.0002	-0.00
5.875	1.0000	+0.0000	-0.00

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.880	1.0001	+0.0001	0.00
5.885	1.0001	-0.0000	0.00
5.890	0.9999	+0.0002	-0.00
5.895	0.9999	+0.0000	-0.00
5.900	1.0003	+0.0002	0.00
5.905	0.9995	+0.0001	-0.00
5.910	1.0004	+0.0003	0.00
5.915	1.0000	+0.0001	-0.00
5.920	1.0002	-0.0001	0.00
5.925	1.0000	+0.0001	-0.00
5.930	1.0000	+0.0004	0.00
5.935	0.9999	+0.0004	-0.00
5.940	1.0000	+0.0002	-0.00
5.945	1.0005	+0.0001	0.00
5.950	0.9998	-0.0001	-0.00
5.955	0.9998	+0.0003	-0.00
5.960	0.9999	+0.0002	-0.00
5.965	1.0005	+0.0002	0.00
5.970	0.9997	+0.0002	-0.00
5.975	0.9997	-0.0000	-0.00
5.980	1.0002	-0.0005	0.00
5.985	0.9992	+0.0004	-0.01
5.990	0.9999	+0.0003	-0.00
5.995	0.9998	+0.0000	-0.00
6.000	1.0000	-0.0004	0.00

3.2 Short



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	-0.9864	+0.0007	-0.12
0.010	-0.9855	+0.0013	-0.13
0.015	-0.9849	+0.0016	-0.13
0.020	-0.9844	+0.0014	-0.14
0.025	-0.9841	+0.0014	-0.14
0.030	-0.9836	+0.0012	-0.14
0.035	-0.9832	+0.0011	-0.15
0.040	-0.9832	+0.0010	-0.15
0.045	-0.9832	+0.0007	-0.15
0.050	-0.9831	+0.0004	-0.15
0.055	-0.9826	+0.0002	-0.15
0.060	-0.9822	+0.0001	-0.16
0.065	-0.9819	-0.0001	-0.16
0.070	-0.9818	-0.0004	-0.16
0.075	-0.9817	-0.0009	-0.16
0.080	-0.9817	-0.0011	-0.16
0.085	-0.9816	-0.0014	-0.16
0.090	-0.9813	-0.0016	-0.16
0.095	-0.9815	-0.0017	-0.16
0.100	-0.9816	-0.0020	-0.16
0.105	-0.9816	-0.0020	-0.16
0.110	-0.9817	-0.0021	-0.16
0.115	-0.9817	-0.0022	-0.16
0.120	-0.9817	-0.0023	-0.16
0.125	-0.9818	-0.0023	-0.16
0.130	-0.9818	-0.0022	-0.16
0.135	-0.9819	-0.0021	-0.16
0.140	-0.9817	-0.0020	-0.16
0.145	-0.9816	-0.0021	-0.16
0.150	-0.9815	-0.0021	-0.16
0.155	-0.9814	-0.0021	-0.16
0.160	-0.9813	-0.0022	-0.16
0.165	-0.9814	-0.0022	-0.16
0.170	-0.9813	-0.0023	-0.16
0.175	-0.9812	-0.0023	-0.17
0.180	-0.9811	-0.0021	-0.17
0.185	-0.9810	-0.0022	-0.17
0.190	-0.9808	-0.0023	-0.17
0.195	-0.9806	-0.0025	-0.17
0.200	-0.9807	-0.0022	-0.17
0.205	-0.9803	-0.0024	-0.17
0.210	-0.9800	-0.0024	-0.18
0.215	-0.9798	-0.0027	-0.18
0.220	-0.9797	-0.0028	-0.18
0.225	-0.9795	-0.0030	-0.18
0.230	-0.9792	-0.0033	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.235	-0.9792	-0.0033	-0.18
0.240	-0.9790	-0.0034	-0.18
0.245	-0.9790	-0.0036	-0.18
0.250	-0.9787	-0.0038	-0.19
0.255	-0.9785	-0.0041	-0.19
0.260	-0.9784	-0.0044	-0.19
0.265	-0.9783	-0.0046	-0.19
0.270	-0.9780	-0.0050	-0.19
0.275	-0.9780	-0.0053	-0.19
0.280	-0.9780	-0.0057	-0.19
0.285	-0.9778	-0.0060	-0.19
0.290	-0.9777	-0.0061	-0.20
0.295	-0.9778	-0.0066	-0.19
0.300	-0.9778	-0.0071	-0.19
0.305	-0.9779	-0.0076	-0.19
0.310	-0.9779	-0.0082	-0.19
0.315	-0.9779	-0.0090	-0.19
0.320	-0.9780	-0.0096	-0.19
0.325	-0.9781	-0.0100	-0.19
0.330	-0.9783	-0.0105	-0.19
0.335	-0.9785	-0.0111	-0.19
0.340	-0.9788	-0.0116	-0.19
0.345	-0.9789	-0.0123	-0.18
0.350	-0.9791	-0.0129	-0.18
0.355	-0.9792	-0.0135	-0.18
0.360	-0.9793	-0.0141	-0.18
0.365	-0.9796	-0.0148	-0.18
0.370	-0.9797	-0.0154	-0.18
0.375	-0.9801	-0.0162	-0.17
0.380	-0.9804	-0.0167	-0.17
0.385	-0.9806	-0.0174	-0.17
0.390	-0.9807	-0.0180	-0.17
0.395	-0.9814	-0.0187	-0.16
0.400	-0.9817	-0.0196	-0.16
0.405	-0.9822	-0.0204	-0.15
0.410	-0.9825	-0.0213	-0.15
0.415	-0.9830	-0.0222	-0.15
0.420	-0.9834	-0.0229	-0.14
0.425	-0.9839	-0.0236	-0.14
0.430	-0.9842	-0.0246	-0.14
0.435	-0.9848	-0.0253	-0.13
0.440	-0.9852	-0.0261	-0.13
0.445	-0.9856	-0.0269	-0.12
0.450	-0.9862	-0.0276	-0.12
0.455	-0.9866	-0.0285	-0.11
0.460	-0.9870	-0.0291	-0.11

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	-0.9873	-0.0299	-0.11
0.470	-0.9877	-0.0306	-0.10
0.475	-0.9881	-0.0314	-0.10
0.480	-0.9886	-0.0321	-0.10
0.485	-0.9890	-0.0330	-0.09
0.490	-0.9892	-0.0337	-0.09
0.495	-0.9897	-0.0343	-0.09
0.500	-0.9902	-0.0350	-0.08
0.505	-0.9907	-0.0357	-0.08
0.510	-0.9912	-0.0364	-0.07
0.515	-0.9917	-0.0368	-0.07
0.520	-0.9922	-0.0373	-0.06
0.525	-0.9925	-0.0382	-0.06
0.530	-0.9928	-0.0387	-0.06
0.535	-0.9933	-0.0392	-0.05
0.540	-0.9937	-0.0396	-0.05
0.545	-0.9940	-0.0401	-0.05
0.550	-0.9944	-0.0405	-0.04
0.555	-0.9945	-0.0410	-0.04
0.560	-0.9950	-0.0414	-0.04
0.565	-0.9953	-0.0418	-0.03
0.570	-0.9956	-0.0423	-0.03
0.575	-0.9957	-0.0426	-0.03
0.580	-0.9960	-0.0429	-0.03
0.585	-0.9962	-0.0432	-0.03
0.590	-0.9961	-0.0435	-0.03
0.595	-0.9962	-0.0436	-0.02
0.600	-0.9961	-0.0438	-0.03
0.605	-0.9962	-0.0439	-0.02
0.610	-0.9965	-0.0442	-0.02
0.615	-0.9965	-0.0442	-0.02
0.620	-0.9966	-0.0441	-0.02
0.625	-0.9967	-0.0443	-0.02
0.630	-0.9967	-0.0441	-0.02
0.635	-0.9967	-0.0441	-0.02
0.640	-0.9967	-0.0441	-0.02
0.645	-0.9967	-0.0441	-0.02
0.650	-0.9966	-0.0440	-0.02
0.655	-0.9966	-0.0438	-0.02
0.660	-0.9966	-0.0436	-0.02
0.665	-0.9966	-0.0435	-0.02
0.670	-0.9963	-0.0432	-0.02
0.675	-0.9963	-0.0430	-0.02
0.680	-0.9961	-0.0428	-0.03
0.685	-0.9960	-0.0424	-0.03
0.690	-0.9959	-0.0424	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	-0.9957	-0.0421	-0.03
0.700	-0.9955	-0.0417	-0.03
0.705	-0.9953	-0.0413	-0.03
0.710	-0.9952	-0.0408	-0.03
0.715	-0.9947	-0.0404	-0.04
0.720	-0.9945	-0.0399	-0.04
0.725	-0.9942	-0.0394	-0.04
0.730	-0.9935	-0.0389	-0.05
0.735	-0.9934	-0.0385	-0.05
0.740	-0.9932	-0.0380	-0.05
0.745	-0.9928	-0.0375	-0.06
0.750	-0.9923	-0.0368	-0.06
0.755	-0.9920	-0.0362	-0.06
0.760	-0.9919	-0.0357	-0.07
0.765	-0.9916	-0.0349	-0.07
0.770	-0.9913	-0.0342	-0.07
0.775	-0.9909	-0.0335	-0.07
0.780	-0.9908	-0.0329	-0.08
0.785	-0.9903	-0.0321	-0.08
0.790	-0.9898	-0.0312	-0.08
0.795	-0.9895	-0.0307	-0.09
0.800	-0.9892	-0.0301	-0.09
0.805	-0.9890	-0.0292	-0.09
0.810	-0.9887	-0.0285	-0.10
0.815	-0.9883	-0.0279	-0.10
0.820	-0.9879	-0.0273	-0.10
0.825	-0.9875	-0.0264	-0.11
0.830	-0.9873	-0.0258	-0.11
0.835	-0.9869	-0.0250	-0.11
0.840	-0.9868	-0.0243	-0.11
0.845	-0.9862	-0.0236	-0.12
0.850	-0.9860	-0.0229	-0.12
0.855	-0.9856	-0.0223	-0.12
0.860	-0.9853	-0.0215	-0.13
0.865	-0.9850	-0.0208	-0.13
0.870	-0.9848	-0.0201	-0.13
0.875	-0.9843	-0.0195	-0.14
0.880	-0.9840	-0.0188	-0.14
0.885	-0.9837	-0.0184	-0.14
0.890	-0.9837	-0.0178	-0.14
0.895	-0.9834	-0.0173	-0.14
0.900	-0.9834	-0.0166	-0.14
0.905	-0.9831	-0.0162	-0.15
0.910	-0.9828	-0.0155	-0.15
0.915	-0.9826	-0.0152	-0.15
0.920	-0.9823	-0.0147	-0.15

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
0.925	-0.9821	-0.0141	-0.16
0.930	-0.9819	-0.0137	-0.16
0.935	-0.9819	-0.0133	-0.16
0.940	-0.9816	-0.0129	-0.16
0.945	-0.9816	-0.0125	-0.16
0.950	-0.9814	-0.0121	-0.16
0.955	-0.9813	-0.0120	-0.16
0.960	-0.9814	-0.0119	-0.16
0.965	-0.9813	-0.0117	-0.16
0.970	-0.9811	-0.0115	-0.16
0.975	-0.9812	-0.0112	-0.16
0.980	-0.9812	-0.0111	-0.16
0.985	-0.9811	-0.0108	-0.16
0.990	-0.9811	-0.0110	-0.17
0.995	-0.9811	-0.0109	-0.17
1.000	-0.9812	-0.0110	-0.16
1.005	-0.9813	-0.0108	-0.16
1.010	-0.9813	-0.0108	-0.16
1.015	-0.9812	-0.0111	-0.16
1.020	-0.9813	-0.0114	-0.16
1.025	-0.9813	-0.0114	-0.16
1.030	-0.9812	-0.0116	-0.16
1.035	-0.9813	-0.0118	-0.16
1.040	-0.9813	-0.0121	-0.16
1.045	-0.9813	-0.0124	-0.16
1.050	-0.9815	-0.0131	-0.16
1.055	-0.9817	-0.0134	-0.16
1.060	-0.9816	-0.0138	-0.16
1.065	-0.9818	-0.0143	-0.16
1.070	-0.9819	-0.0147	-0.16
1.075	-0.9821	-0.0150	-0.16
1.080	-0.9822	-0.0156	-0.15
1.085	-0.9822	-0.0163	-0.15
1.090	-0.9825	-0.0169	-0.15
1.095	-0.9824	-0.0173	-0.15
1.100	-0.9825	-0.0177	-0.15
1.105	-0.9826	-0.0185	-0.15
1.110	-0.9826	-0.0192	-0.15
1.115	-0.9830	-0.0197	-0.15
1.120	-0.9832	-0.0203	-0.15
1.125	-0.9831	-0.0210	-0.15
1.130	-0.9833	-0.0218	-0.14
1.135	-0.9833	-0.0226	-0.14
1.140	-0.9835	-0.0233	-0.14
1.145	-0.9838	-0.0241	-0.14
1.150	-0.9840	-0.0247	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	-0.9842	-0.0254	-0.14
1.160	-0.9841	-0.0262	-0.14
1.165	-0.9843	-0.0267	-0.13
1.170	-0.9844	-0.0275	-0.13
1.175	-0.9845	-0.0282	-0.13
1.180	-0.9845	-0.0290	-0.13
1.185	-0.9848	-0.0297	-0.13
1.190	-0.9848	-0.0305	-0.13
1.195	-0.9849	-0.0310	-0.13
1.200	-0.9849	-0.0315	-0.13
1.205	-0.9850	-0.0324	-0.13
1.210	-0.9850	-0.0330	-0.13
1.215	-0.9849	-0.0337	-0.13
1.220	-0.9849	-0.0342	-0.13
1.225	-0.9850	-0.0348	-0.13
1.230	-0.9850	-0.0354	-0.13
1.235	-0.9851	-0.0358	-0.12
1.240	-0.9850	-0.0362	-0.13
1.245	-0.9852	-0.0366	-0.12
1.250	-0.9852	-0.0371	-0.12
1.255	-0.9853	-0.0376	-0.12
1.260	-0.9855	-0.0379	-0.12
1.265	-0.9854	-0.0382	-0.12
1.270	-0.9857	-0.0386	-0.12
1.275	-0.9857	-0.0386	-0.12
1.280	-0.9856	-0.0388	-0.12
1.285	-0.9855	-0.0389	-0.12
1.290	-0.9855	-0.0390	-0.12
1.295	-0.9855	-0.0388	-0.12
1.300	-0.9855	-0.0386	-0.12
1.305	-0.9856	-0.0385	-0.12
1.310	-0.9855	-0.0387	-0.12
1.315	-0.9855	-0.0385	-0.12
1.320	-0.9854	-0.0382	-0.12
1.325	-0.9855	-0.0383	-0.12
1.330	-0.9855	-0.0380	-0.12
1.335	-0.9855	-0.0376	-0.12
1.340	-0.9854	-0.0372	-0.12
1.345	-0.9854	-0.0368	-0.12
1.350	-0.9853	-0.0364	-0.12
1.355	-0.9852	-0.0358	-0.12
1.360	-0.9851	-0.0351	-0.12
1.365	-0.9851	-0.0344	-0.12
1.370	-0.9851	-0.0338	-0.12
1.375	-0.9850	-0.0331	-0.13
1.380	-0.9849	-0.0324	-0.13

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.385	-0.9847	-0.0316	-0.13
1.390	-0.9849	-0.0307	-0.13
1.395	-0.9846	-0.0297	-0.13
1.400	-0.9846	-0.0286	-0.13
1.405	-0.9844	-0.0277	-0.13
1.410	-0.9843	-0.0267	-0.13
1.415	-0.9843	-0.0258	-0.13
1.420	-0.9842	-0.0246	-0.14
1.425	-0.9838	-0.0235	-0.14
1.430	-0.9838	-0.0221	-0.14
1.435	-0.9836	-0.0209	-0.14
1.440	-0.9835	-0.0198	-0.14
1.445	-0.9835	-0.0185	-0.14
1.450	-0.9833	-0.0173	-0.14
1.455	-0.9831	-0.0160	-0.15
1.460	-0.9827	-0.0146	-0.15
1.465	-0.9826	-0.0135	-0.15
1.470	-0.9824	-0.0121	-0.15
1.475	-0.9821	-0.0106	-0.16
1.480	-0.9820	-0.0092	-0.16
1.485	-0.9817	-0.0078	-0.16
1.490	-0.9814	-0.0068	-0.16
1.495	-0.9812	-0.0051	-0.16
1.500	-0.9811	-0.0037	-0.17
1.505	-0.9806	-0.0023	-0.17
1.510	-0.9803	-0.0010	-0.17
1.515	-0.9800	+0.0003	-0.18
1.520	-0.9799	+0.0015	-0.18
1.525	-0.9797	+0.0030	-0.18
1.530	-0.9793	+0.0043	-0.18
1.535	-0.9790	+0.0058	-0.18
1.540	-0.9787	+0.0071	-0.19
1.545	-0.9785	+0.0083	-0.19
1.550	-0.9782	+0.0095	-0.19
1.555	-0.9778	+0.0107	-0.19
1.560	-0.9776	+0.0118	-0.20
1.565	-0.9775	+0.0129	-0.20
1.570	-0.9773	+0.0139	-0.20
1.575	-0.9770	+0.0149	-0.20
1.580	-0.9768	+0.0157	-0.20
1.585	-0.9765	+0.0167	-0.21
1.590	-0.9762	+0.0175	-0.21
1.595	-0.9761	+0.0185	-0.21
1.600	-0.9759	+0.0191	-0.21
1.605	-0.9758	+0.0199	-0.21
1.610	-0.9756	+0.0207	-0.21

(continued on next page)

Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.615	-0.9755	+0.0215	-0.21
1.620	-0.9752	+0.0223	-0.22
1.625	-0.9750	+0.0230	-0.22
1.630	-0.9749	+0.0237	-0.22
1.635	-0.9747	+0.0242	-0.22
1.640	-0.9745	+0.0247	-0.22
1.645	-0.9743	+0.0252	-0.22
1.650	-0.9740	+0.0254	-0.23
1.655	-0.9737	+0.0257	-0.23
1.660	-0.9737	+0.0261	-0.23
1.665	-0.9737	+0.0263	-0.23
1.670	-0.9733	+0.0263	-0.23
1.675	-0.9732	+0.0263	-0.23
1.680	-0.9732	+0.0265	-0.23
1.685	-0.9729	+0.0265	-0.24
1.690	-0.9730	+0.0262	-0.23
1.695	-0.9730	+0.0259	-0.23
1.700	-0.9729	+0.0257	-0.24
1.705	-0.9726	+0.0254	-0.24
1.710	-0.9725	+0.0252	-0.24
1.715	-0.9723	+0.0249	-0.24
1.720	-0.9724	+0.0243	-0.24
1.725	-0.9722	+0.0237	-0.24
1.730	-0.9724	+0.0231	-0.24
1.735	-0.9725	+0.0226	-0.24
1.740	-0.9722	+0.0218	-0.24
1.745	-0.9725	+0.0209	-0.24
1.750	-0.9724	+0.0198	-0.24
1.755	-0.9728	+0.0190	-0.24
1.760	-0.9725	+0.0180	-0.24
1.765	-0.9725	+0.0169	-0.24
1.770	-0.9728	+0.0162	-0.24
1.775	-0.9727	+0.0151	-0.24
1.780	-0.9730	+0.0139	-0.24
1.785	-0.9731	+0.0131	-0.24
1.790	-0.9733	+0.0121	-0.23
1.795	-0.9733	+0.0108	-0.23
1.800	-0.9733	+0.0094	-0.24
1.805	-0.9736	+0.0082	-0.23
1.810	-0.9740	+0.0072	-0.23
1.815	-0.9744	+0.0061	-0.22
1.820	-0.9743	+0.0048	-0.23
1.825	-0.9746	+0.0038	-0.22
1.830	-0.9750	+0.0023	-0.22
1.835	-0.9750	+0.0010	-0.22
1.840	-0.9751	-0.0004	-0.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	-0.9754	-0.0015	-0.22
1.850	-0.9756	-0.0029	-0.21
1.855	-0.9758	-0.0041	-0.21
1.860	-0.9760	-0.0054	-0.21
1.865	-0.9763	-0.0066	-0.21
1.870	-0.9767	-0.0079	-0.20
1.875	-0.9770	-0.0091	-0.20
1.880	-0.9773	-0.0104	-0.20
1.885	-0.9777	-0.0115	-0.20
1.890	-0.9781	-0.0129	-0.19
1.895	-0.9783	-0.0140	-0.19
1.900	-0.9785	-0.0151	-0.19
1.905	-0.9789	-0.0162	-0.18
1.910	-0.9792	-0.0172	-0.18
1.915	-0.9795	-0.0182	-0.18
1.920	-0.9800	-0.0193	-0.17
1.925	-0.9803	-0.0203	-0.17
1.930	-0.9805	-0.0213	-0.17
1.935	-0.9808	-0.0221	-0.17
1.940	-0.9813	-0.0227	-0.16
1.945	-0.9816	-0.0238	-0.16
1.950	-0.9819	-0.0244	-0.16
1.955	-0.9823	-0.0251	-0.15
1.960	-0.9825	-0.0257	-0.15
1.965	-0.9829	-0.0263	-0.15
1.970	-0.9831	-0.0270	-0.14
1.975	-0.9835	-0.0273	-0.14
1.980	-0.9837	-0.0278	-0.14
1.985	-0.9841	-0.0283	-0.14
1.990	-0.9844	-0.0286	-0.13
1.995	-0.9848	-0.0289	-0.13
2.000	-0.9849	-0.0289	-0.13
2.005	-0.9853	-0.0289	-0.12
2.010	-0.9856	-0.0289	-0.12
2.015	-0.9860	-0.0290	-0.12
2.020	-0.9862	-0.0288	-0.12
2.025	-0.9868	-0.0285	-0.11
2.030	-0.9871	-0.0282	-0.11
2.035	-0.9875	-0.0281	-0.11
2.040	-0.9875	-0.0277	-0.11
2.045	-0.9880	-0.0273	-0.10
2.050	-0.9882	-0.0267	-0.10
2.055	-0.9884	-0.0263	-0.10
2.060	-0.9888	-0.0258	-0.09
2.065	-0.9891	-0.0250	-0.09
2.070	-0.9893	-0.0244	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	-0.9897	-0.0235	-0.09
2.080	-0.9901	-0.0228	-0.08
2.085	-0.9903	-0.0220	-0.08
2.090	-0.9905	-0.0213	-0.08
2.095	-0.9906	-0.0204	-0.08
2.100	-0.9909	-0.0195	-0.08
2.105	-0.9912	-0.0184	-0.08
2.110	-0.9914	-0.0174	-0.07
2.115	-0.9916	-0.0163	-0.07
2.120	-0.9919	-0.0151	-0.07
2.125	-0.9919	-0.0139	-0.07
2.130	-0.9920	-0.0130	-0.07
2.135	-0.9920	-0.0116	-0.07
2.140	-0.9922	-0.0104	-0.07
2.145	-0.9922	-0.0092	-0.07
2.150	-0.9922	-0.0079	-0.07
2.155	-0.9923	-0.0068	-0.07
2.160	-0.9923	-0.0056	-0.07
2.165	-0.9923	-0.0043	-0.07
2.170	-0.9923	-0.0031	-0.07
2.175	-0.9921	-0.0020	-0.07
2.180	-0.9921	-0.0006	-0.07
2.185	-0.9920	+0.0006	-0.07
2.190	-0.9918	+0.0018	-0.07
2.195	-0.9918	+0.0030	-0.07
2.200	-0.9916	+0.0043	-0.07
2.205	-0.9915	+0.0055	-0.07
2.210	-0.9913	+0.0066	-0.08
2.215	-0.9913	+0.0077	-0.08
2.220	-0.9912	+0.0088	-0.08
2.225	-0.9909	+0.0095	-0.08
2.230	-0.9908	+0.0106	-0.08
2.235	-0.9906	+0.0115	-0.08
2.240	-0.9904	+0.0124	-0.08
2.245	-0.9903	+0.0132	-0.08
2.250	-0.9900	+0.0139	-0.09
2.255	-0.9899	+0.0147	-0.09
2.260	-0.9898	+0.0155	-0.09
2.265	-0.9895	+0.0162	-0.09
2.270	-0.9892	+0.0167	-0.09
2.275	-0.9890	+0.0173	-0.09
2.280	-0.9890	+0.0177	-0.09
2.285	-0.9888	+0.0183	-0.10
2.290	-0.9886	+0.0185	-0.10
2.295	-0.9884	+0.0190	-0.10
2.300	-0.9883	+0.0194	-0.10

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.305	-0.9881	+0.0197	-0.10
2.310	-0.9878	+0.0198	-0.11
2.315	-0.9877	+0.0200	-0.11
2.320	-0.9873	+0.0199	-0.11
2.325	-0.9873	+0.0200	-0.11
2.330	-0.9869	+0.0200	-0.11
2.335	-0.9868	+0.0199	-0.11
2.340	-0.9865	+0.0200	-0.12
2.345	-0.9863	+0.0198	-0.12
2.350	-0.9860	+0.0195	-0.12
2.355	-0.9859	+0.0189	-0.12
2.360	-0.9857	+0.0183	-0.12
2.365	-0.9856	+0.0176	-0.12
2.370	-0.9854	+0.0170	-0.13
2.375	-0.9849	+0.0161	-0.13
2.380	-0.9849	+0.0155	-0.13
2.385	-0.9848	+0.0145	-0.13
2.390	-0.9846	+0.0137	-0.13
2.395	-0.9845	+0.0126	-0.14
2.400	-0.9844	+0.0118	-0.14
2.405	-0.9842	+0.0105	-0.14
2.410	-0.9839	+0.0094	-0.14
2.415	-0.9838	+0.0081	-0.14
2.420	-0.9838	+0.0069	-0.14
2.425	-0.9833	+0.0056	-0.15
2.430	-0.9834	+0.0043	-0.15
2.435	-0.9833	+0.0028	-0.15
2.440	-0.9831	+0.0012	-0.15
2.445	-0.9830	-0.0003	-0.15
2.450	-0.9829	-0.0018	-0.15
2.455	-0.9830	-0.0035	-0.15
2.460	-0.9829	-0.0052	-0.15
2.465	-0.9829	-0.0068	-0.15
2.470	-0.9829	-0.0084	-0.15
2.475	-0.9830	-0.0100	-0.15
2.480	-0.9830	-0.0115	-0.15
2.485	-0.9830	-0.0133	-0.15
2.490	-0.9830	-0.0151	-0.15
2.495	-0.9831	-0.0167	-0.15
2.500	-0.9832	-0.0183	-0.15
2.505	-0.9833	-0.0201	-0.14
2.510	-0.9834	-0.0218	-0.14
2.515	-0.9834	-0.0234	-0.14
2.520	-0.9835	-0.0251	-0.14
2.525	-0.9835	-0.0266	-0.14
2.530	-0.9838	-0.0282	-0.14

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	-0.9840	-0.0297	-0.14
2.540	-0.9841	-0.0311	-0.13
2.545	-0.9842	-0.0327	-0.13
2.550	-0.9843	-0.0341	-0.13
2.555	-0.9844	-0.0355	-0.13
2.560	-0.9845	-0.0369	-0.13
2.565	-0.9846	-0.0383	-0.13
2.570	-0.9849	-0.0394	-0.12
2.575	-0.9850	-0.0403	-0.12
2.580	-0.9849	-0.0411	-0.12
2.585	-0.9850	-0.0417	-0.12
2.590	-0.9849	-0.0423	-0.12
2.595	-0.9841	-0.0433	-0.13
2.600	-0.9839	-0.0439	-0.13
2.605	-0.9833	-0.0451	-0.14
2.610	-0.9826	-0.0460	-0.14
2.615	-0.9827	-0.0470	-0.14
2.620	-0.9819	-0.0482	-0.15
2.625	-0.9815	-0.0495	-0.15
2.630	-0.9809	-0.0506	-0.16
2.635	-0.9805	-0.0518	-0.16
2.640	-0.9799	-0.0528	-0.16
2.645	-0.9793	-0.0539	-0.17
2.650	-0.9786	-0.0547	-0.17
2.655	-0.9782	-0.0556	-0.18
2.660	-0.9775	-0.0567	-0.18
2.665	-0.9766	-0.0576	-0.19
2.670	-0.9759	-0.0586	-0.20
2.675	-0.9753	-0.0599	-0.20
2.680	-0.9747	-0.0610	-0.21
2.685	-0.9741	-0.0622	-0.21
2.690	-0.9737	-0.0634	-0.21
2.695	-0.9734	-0.0647	-0.22
2.700	-0.9734	-0.0656	-0.21
2.705	-0.9731	-0.0666	-0.22
2.710	-0.9731	-0.0676	-0.22
2.715	-0.9731	-0.0682	-0.22
2.720	-0.9730	-0.0690	-0.22
2.725	-0.9733	-0.0690	-0.21
2.730	-0.9733	-0.0693	-0.21
2.735	-0.9733	-0.0696	-0.21
2.740	-0.9734	-0.0697	-0.21
2.745	-0.9736	-0.0697	-0.21
2.750	-0.9738	-0.0698	-0.21
2.755	-0.9737	-0.0699	-0.21
2.760	-0.9735	-0.0698	-0.21

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	-0.9735	-0.0696	-0.21
2.770	-0.9733	-0.0695	-0.21
2.775	-0.9735	-0.0694	-0.21
2.780	-0.9735	-0.0692	-0.21
2.785	-0.9734	-0.0694	-0.21
2.790	-0.9734	-0.0692	-0.21
2.795	-0.9735	-0.0690	-0.21
2.800	-0.9734	-0.0689	-0.21
2.805	-0.9732	-0.0689	-0.21
2.810	-0.9730	-0.0684	-0.22
2.815	-0.9729	-0.0685	-0.22
2.820	-0.9728	-0.0683	-0.22
2.825	-0.9728	-0.0682	-0.22
2.830	-0.9725	-0.0678	-0.22
2.835	-0.9724	-0.0675	-0.22
2.840	-0.9723	-0.0672	-0.22
2.845	-0.9722	-0.0670	-0.22
2.850	-0.9718	-0.0667	-0.23
2.855	-0.9717	-0.0664	-0.23
2.860	-0.9715	-0.0663	-0.23
2.865	-0.9713	-0.0661	-0.23
2.870	-0.9711	-0.0658	-0.23
2.875	-0.9708	-0.0658	-0.24
2.880	-0.9705	-0.0657	-0.24
2.885	-0.9702	-0.0654	-0.24
2.890	-0.9701	-0.0653	-0.24
2.895	-0.9697	-0.0651	-0.25
2.900	-0.9695	-0.0650	-0.25
2.905	-0.9695	-0.0649	-0.25
2.910	-0.9694	-0.0649	-0.25
2.915	-0.9693	-0.0648	-0.25
2.920	-0.9692	-0.0648	-0.25
2.925	-0.9692	-0.0650	-0.25
2.930	-0.9690	-0.0653	-0.25
2.935	-0.9689	-0.0654	-0.25
2.940	-0.9688	-0.0657	-0.26
2.945	-0.9688	-0.0658	-0.26
2.950	-0.9692	-0.0661	-0.25
2.955	-0.9689	-0.0665	-0.25
2.960	-0.9687	-0.0671	-0.26
2.965	-0.9689	-0.0675	-0.25
2.970	-0.9690	-0.0678	-0.25
2.975	-0.9694	-0.0683	-0.25
2.980	-0.9694	-0.0688	-0.25
2.985	-0.9700	-0.0692	-0.24
2.990	-0.9700	-0.0699	-0.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.995	-0.9701	-0.0702	-0.24
3.000	-0.9705	-0.0711	-0.24
3.005	-0.9701	-0.0713	-0.24
3.010	-0.9702	-0.0721	-0.24
3.015	-0.9706	-0.0728	-0.23
3.020	-0.9713	-0.0733	-0.23
3.025	-0.9712	-0.0742	-0.23
3.030	-0.9718	-0.0752	-0.22
3.035	-0.9721	-0.0761	-0.22
3.040	-0.9723	-0.0774	-0.22
3.045	-0.9726	-0.0783	-0.21
3.050	-0.9726	-0.0794	-0.21
3.055	-0.9732	-0.0802	-0.21
3.060	-0.9735	-0.0817	-0.20
3.065	-0.9736	-0.0828	-0.20
3.070	-0.9736	-0.0837	-0.20
3.075	-0.9739	-0.0850	-0.20
3.080	-0.9741	-0.0861	-0.19
3.085	-0.9740	-0.0872	-0.19
3.090	-0.9743	-0.0879	-0.19
3.095	-0.9745	-0.0889	-0.19
3.100	-0.9748	-0.0900	-0.18
3.105	-0.9752	-0.0908	-0.18
3.110	-0.9754	-0.0917	-0.18
3.115	-0.9754	-0.0926	-0.18
3.120	-0.9758	-0.0934	-0.17
3.125	-0.9759	-0.0942	-0.17
3.130	-0.9764	-0.0950	-0.17
3.135	-0.9768	-0.0959	-0.16
3.140	-0.9771	-0.0970	-0.16
3.145	-0.9773	-0.0976	-0.16
3.150	-0.9776	-0.0983	-0.15
3.155	-0.9779	-0.0990	-0.15
3.160	-0.9780	-0.0996	-0.15
3.165	-0.9783	-0.1001	-0.15
3.170	-0.9786	-0.1007	-0.14
3.175	-0.9786	-0.1010	-0.14
3.180	-0.9789	-0.1017	-0.14
3.185	-0.9788	-0.1019	-0.14
3.190	-0.9790	-0.1024	-0.14
3.195	-0.9791	-0.1029	-0.14
3.200	-0.9793	-0.1036	-0.13
3.205	-0.9793	-0.1037	-0.13
3.210	-0.9792	-0.1040	-0.13
3.215	-0.9796	-0.1048	-0.13
3.220	-0.9795	-0.1050	-0.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	-0.9799	-0.1053	-0.13
3.230	-0.9798	-0.1063	-0.13
3.235	-0.9804	-0.1061	-0.12
3.240	-0.9805	-0.1063	-0.12
3.245	-0.9812	-0.1063	-0.11
3.250	-0.9816	-0.1067	-0.11
3.255	-0.9818	-0.1066	-0.11
3.260	-0.9823	-0.1064	-0.10
3.265	-0.9830	-0.1067	-0.10
3.270	-0.9834	-0.1062	-0.10
3.275	-0.9832	-0.1061	-0.10
3.280	-0.9836	-0.1056	-0.09
3.285	-0.9839	-0.1058	-0.09
3.290	-0.9839	-0.1052	-0.09
3.295	-0.9835	-0.1048	-0.10
3.300	-0.9836	-0.1044	-0.10
3.305	-0.9834	-0.1041	-0.10
3.310	-0.9831	-0.1035	-0.10
3.315	-0.9824	-0.1031	-0.11
3.320	-0.9824	-0.1018	-0.11
3.325	-0.9822	-0.1016	-0.11
3.330	-0.9817	-0.1005	-0.12
3.335	-0.9811	-0.0998	-0.12
3.340	-0.9807	-0.0985	-0.13
3.345	-0.9799	-0.0977	-0.13
3.350	-0.9799	-0.0964	-0.13
3.355	-0.9793	-0.0955	-0.14
3.360	-0.9789	-0.0943	-0.15
3.365	-0.9785	-0.0928	-0.15
3.370	-0.9779	-0.0915	-0.16
3.375	-0.9776	-0.0903	-0.16
3.380	-0.9770	-0.0893	-0.17
3.385	-0.9766	-0.0882	-0.17
3.390	-0.9760	-0.0866	-0.18
3.395	-0.9756	-0.0855	-0.18
3.400	-0.9752	-0.0841	-0.19
3.405	-0.9745	-0.0829	-0.19
3.410	-0.9743	-0.0817	-0.20
3.415	-0.9741	-0.0803	-0.20
3.420	-0.9737	-0.0793	-0.20
3.425	-0.9732	-0.0780	-0.21
3.430	-0.9727	-0.0768	-0.21
3.435	-0.9723	-0.0755	-0.22
3.440	-0.9718	-0.0744	-0.22
3.445	-0.9714	-0.0733	-0.23
3.450	-0.9712	-0.0722	-0.23

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	-0.9709	-0.0708	-0.23
3.460	-0.9703	-0.0697	-0.24
3.465	-0.9701	-0.0690	-0.24
3.470	-0.9696	-0.0675	-0.25
3.475	-0.9696	-0.0666	-0.25
3.480	-0.9691	-0.0657	-0.25
3.485	-0.9689	-0.0647	-0.26
3.490	-0.9685	-0.0638	-0.26
3.495	-0.9683	-0.0631	-0.26
3.500	-0.9678	-0.0619	-0.27
3.505	-0.9675	-0.0613	-0.27
3.510	-0.9674	-0.0603	-0.27
3.515	-0.9668	-0.0595	-0.28
3.520	-0.9667	-0.0587	-0.28
3.525	-0.9670	-0.0572	-0.28
3.530	-0.9670	-0.0555	-0.28
3.535	-0.9667	-0.0556	-0.28
3.540	-0.9664	-0.0554	-0.28
3.545	-0.9662	-0.0550	-0.28
3.550	-0.9660	-0.0546	-0.29
3.555	-0.9659	-0.0544	-0.29
3.560	-0.9657	-0.0541	-0.29
3.565	-0.9659	-0.0539	-0.29
3.570	-0.9653	-0.0535	-0.29
3.575	-0.9667	-0.0544	-0.28
3.580	-0.9650	-0.0537	-0.30
3.585	-0.9637	-0.0538	-0.31
3.590	-0.9634	-0.0542	-0.31
3.595	-0.9629	-0.0539	-0.32
3.600	-0.9628	-0.0536	-0.32
3.605	-0.9618	-0.0529	-0.33
3.610	-0.9619	-0.0523	-0.32
3.615	-0.9620	-0.0523	-0.32
3.620	-0.9623	-0.0524	-0.32
3.625	-0.9625	-0.0527	-0.32
3.630	-0.9628	-0.0526	-0.32
3.635	-0.9631	-0.0527	-0.31
3.640	-0.9634	-0.0526	-0.31
3.645	-0.9633	-0.0532	-0.31
3.650	-0.9636	-0.0533	-0.31
3.655	-0.9637	-0.0535	-0.31
3.660	-0.9637	-0.0535	-0.31
3.665	-0.9638	-0.0536	-0.31
3.670	-0.9637	-0.0540	-0.31
3.675	-0.9638	-0.0543	-0.31
3.680	-0.9635	-0.0546	-0.31

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	-0.9636	-0.0542	-0.31
3.690	-0.9631	-0.0539	-0.31
3.695	-0.9629	-0.0532	-0.31
3.700	-0.9625	-0.0526	-0.32
3.705	-0.9624	-0.0512	-0.32
3.710	-0.9624	-0.0501	-0.32
3.715	-0.9633	-0.0484	-0.31
3.720	-0.9639	-0.0465	-0.31
3.725	-0.9657	-0.0450	-0.29
3.730	-0.9668	-0.0434	-0.28
3.735	-0.9690	-0.0432	-0.27
3.740	-0.9709	-0.0423	-0.25
3.745	-0.9719	-0.0431	-0.24
3.750	-0.9734	-0.0430	-0.23
3.755	-0.9736	-0.0432	-0.22
3.760	-0.9748	-0.0433	-0.21
3.765	-0.9751	-0.0428	-0.21
3.770	-0.9763	-0.0423	-0.20
3.775	-0.9775	-0.0422	-0.19
3.780	-0.9788	-0.0424	-0.18
3.785	-0.9802	-0.0424	-0.17
3.790	-0.9806	-0.0422	-0.16
3.795	-0.9819	-0.0420	-0.15
3.800	-0.9826	-0.0420	-0.14
3.805	-0.9830	-0.0421	-0.14
3.810	-0.9834	-0.0415	-0.14
3.815	-0.9841	-0.0413	-0.13
3.820	-0.9846	-0.0407	-0.13
3.825	-0.9850	-0.0407	-0.12
3.830	-0.9859	-0.0400	-0.12
3.835	-0.9839	-0.0388	-0.13
3.840	-0.9839	-0.0374	-0.13
3.845	-0.9851	-0.0363	-0.12
3.850	-0.9858	-0.0356	-0.12
3.855	-0.9859	-0.0343	-0.12
3.860	-0.9865	-0.0334	-0.11
3.865	-0.9869	-0.0319	-0.11
3.870	-0.9876	-0.0307	-0.10
3.875	-0.9881	-0.0293	-0.10
3.880	-0.9888	-0.0279	-0.09
3.885	-0.9896	-0.0266	-0.09
3.890	-0.9901	-0.0252	-0.08
3.895	-0.9911	-0.0240	-0.07
3.900	-0.9910	-0.0224	-0.08
3.905	-0.9914	-0.0206	-0.07
3.910	-0.9921	-0.0191	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	-0.9923	-0.0175	-0.07
3.920	-0.9928	-0.0158	-0.06
3.925	-0.9929	-0.0140	-0.06
3.930	-0.9933	-0.0131	-0.06
3.935	-0.9934	-0.0115	-0.06
3.940	-0.9934	-0.0099	-0.06
3.945	-0.9935	-0.0083	-0.06
3.950	-0.9934	-0.0067	-0.06
3.955	-0.9935	-0.0050	-0.06
3.960	-0.9933	-0.0035	-0.06
3.965	-0.9929	-0.0020	-0.06
3.970	-0.9929	-0.0007	-0.06
3.975	-0.9925	+0.0011	-0.07
3.980	-0.9923	+0.0027	-0.07
3.985	-0.9921	+0.0042	-0.07
3.990	-0.9917	+0.0055	-0.07
3.995	-0.9914	+0.0071	-0.07
4.000	-0.9910	+0.0087	-0.08
4.005	-0.9906	+0.0099	-0.08
4.010	-0.9902	+0.0113	-0.08
4.015	-0.9896	+0.0125	-0.09
4.020	-0.9892	+0.0137	-0.09
4.025	-0.9886	+0.0146	-0.10
4.030	-0.9883	+0.0159	-0.10
4.035	-0.9878	+0.0169	-0.11
4.040	-0.9872	+0.0181	-0.11
4.045	-0.9867	+0.0191	-0.11
4.050	-0.9860	+0.0198	-0.12
4.055	-0.9853	+0.0206	-0.13
4.060	-0.9846	+0.0213	-0.13
4.065	-0.9841	+0.0216	-0.14
4.070	-0.9836	+0.0221	-0.14
4.075	-0.9829	+0.0227	-0.15
4.080	-0.9823	+0.0229	-0.15
4.085	-0.9819	+0.0230	-0.16
4.090	-0.9814	+0.0231	-0.16
4.095	-0.9807	+0.0232	-0.17
4.100	-0.9802	+0.0233	-0.17
4.105	-0.9799	+0.0234	-0.17
4.110	-0.9795	+0.0232	-0.18
4.115	-0.9792	+0.0232	-0.18
4.120	-0.9793	+0.0232	-0.18
4.125	-0.9789	+0.0232	-0.18
4.130	-0.9783	+0.0229	-0.19
4.135	-0.9778	+0.0232	-0.19
4.140	-0.9775	+0.0232	-0.20

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	-0.9764	+0.0230	-0.20
4.150	-0.9759	+0.0226	-0.21
4.155	-0.9752	+0.0223	-0.22
4.160	-0.9745	+0.0217	-0.22
4.165	-0.9737	+0.0203	-0.23
4.170	-0.9728	+0.0193	-0.24
4.175	-0.9724	+0.0176	-0.24
4.180	-0.9718	+0.0165	-0.25
4.185	-0.9712	+0.0151	-0.25
4.190	-0.9709	+0.0134	-0.26
4.195	-0.9707	+0.0117	-0.26
4.200	-0.9705	+0.0100	-0.26
4.205	-0.9705	+0.0082	-0.26
4.210	-0.9701	+0.0062	-0.26
4.215	-0.9699	+0.0041	-0.27
4.220	-0.9699	+0.0021	-0.27
4.225	-0.9700	+0.0000	-0.26
4.230	-0.9697	-0.0021	-0.27
4.235	-0.9699	-0.0045	-0.27
4.240	-0.9703	-0.0068	-0.26
4.245	-0.9701	-0.0093	-0.26
4.250	-0.9706	-0.0112	-0.26
4.255	-0.9707	-0.0137	-0.26
4.260	-0.9704	-0.0158	-0.26
4.265	-0.9706	-0.0181	-0.26
4.270	-0.9712	-0.0203	-0.25
4.275	-0.9713	-0.0228	-0.25
4.280	-0.9719	-0.0252	-0.24
4.285	-0.9722	-0.0276	-0.24
4.290	-0.9726	-0.0300	-0.24
4.295	-0.9728	-0.0322	-0.23
4.300	-0.9732	-0.0346	-0.23
4.305	-0.9737	-0.0370	-0.23
4.310	-0.9739	-0.0394	-0.22
4.315	-0.9742	-0.0416	-0.22
4.320	-0.9744	-0.0439	-0.22
4.325	-0.9745	-0.0463	-0.21
4.330	-0.9748	-0.0487	-0.21
4.335	-0.9750	-0.0509	-0.21
4.340	-0.9752	-0.0530	-0.21
4.345	-0.9754	-0.0552	-0.20
4.350	-0.9756	-0.0575	-0.20
4.355	-0.9757	-0.0594	-0.20
4.360	-0.9758	-0.0619	-0.19
4.365	-0.9760	-0.0640	-0.19
4.370	-0.9761	-0.0660	-0.19

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.375	-0.9765	-0.0681	-0.19
4.380	-0.9765	-0.0700	-0.18
4.385	-0.9764	-0.0720	-0.18
4.390	-0.9765	-0.0740	-0.18
4.395	-0.9767	-0.0757	-0.18
4.400	-0.9768	-0.0772	-0.18
4.405	-0.9769	-0.0789	-0.17
4.410	-0.9773	-0.0810	-0.17
4.415	-0.9771	-0.0826	-0.17
4.420	-0.9775	-0.0841	-0.17
4.425	-0.9779	-0.0854	-0.16
4.430	-0.9778	-0.0874	-0.16
4.435	-0.9780	-0.0887	-0.16
4.440	-0.9780	-0.0899	-0.16
4.445	-0.9782	-0.0918	-0.15
4.450	-0.9782	-0.0927	-0.15
4.455	-0.9785	-0.0945	-0.15
4.460	-0.9785	-0.0960	-0.15
4.465	-0.9788	-0.0978	-0.14
4.470	-0.9786	-0.0994	-0.14
4.475	-0.9790	-0.1015	-0.14
4.480	-0.9792	-0.1033	-0.13
4.485	-0.9800	-0.1049	-0.13
4.490	-0.9804	-0.1059	-0.12
4.495	-0.9807	-0.1074	-0.12
4.500	-0.9811	-0.1082	-0.11
4.505	-0.9811	-0.1090	-0.11
4.510	-0.9812	-0.1093	-0.11
4.515	-0.9809	-0.1096	-0.11
4.520	-0.9810	-0.1101	-0.11
4.525	-0.9804	-0.1102	-0.12
4.530	-0.9803	-0.1100	-0.12
4.535	-0.9807	-0.1099	-0.12
4.540	-0.9802	-0.1096	-0.12
4.545	-0.9798	-0.1089	-0.12
4.550	-0.9794	-0.1092	-0.13
4.555	-0.9794	-0.1089	-0.13
4.560	-0.9793	-0.1085	-0.13
4.565	-0.9789	-0.1081	-0.13
4.570	-0.9787	-0.1076	-0.13
4.575	-0.9785	-0.1072	-0.14
4.580	-0.9777	-0.1066	-0.14
4.585	-0.9777	-0.1060	-0.15
4.590	-0.9772	-0.1056	-0.15
4.595	-0.9767	-0.1051	-0.15
4.600	-0.9762	-0.1045	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	-0.9756	-0.1038	-0.17
4.610	-0.9751	-0.1035	-0.17
4.615	-0.9744	-0.1029	-0.18
4.620	-0.9738	-0.1024	-0.18
4.625	-0.9734	-0.1019	-0.19
4.630	-0.9725	-0.1013	-0.20
4.635	-0.9717	-0.1009	-0.20
4.640	-0.9711	-0.1005	-0.21
4.645	-0.9704	-0.0999	-0.22
4.650	-0.9699	-0.0993	-0.22
4.655	-0.9688	-0.0989	-0.23
4.660	-0.9683	-0.0987	-0.23
4.665	-0.9674	-0.0981	-0.24
4.670	-0.9666	-0.0982	-0.25
4.675	-0.9655	-0.0981	-0.26
4.680	-0.9649	-0.0979	-0.27
4.685	-0.9640	-0.0981	-0.27
4.690	-0.9636	-0.0982	-0.28
4.695	-0.9626	-0.0986	-0.29
4.700	-0.9619	-0.0986	-0.29
4.705	-0.9614	-0.0983	-0.30
4.710	-0.9612	-0.0988	-0.30
4.715	-0.9608	-0.0990	-0.30
4.720	-0.9605	-0.0984	-0.31
4.725	-0.9601	-0.0984	-0.31
4.730	-0.9601	-0.0978	-0.31
4.735	-0.9598	-0.0975	-0.31
4.740	-0.9590	-0.0968	-0.32
4.745	-0.9584	-0.0960	-0.33
4.750	-0.9578	-0.0954	-0.33
4.755	-0.9572	-0.0945	-0.34
4.760	-0.9565	-0.0943	-0.34
4.765	-0.9554	-0.0936	-0.35
4.770	-0.9551	-0.0935	-0.36
4.775	-0.9540	-0.0926	-0.37
4.780	-0.9530	-0.0933	-0.38
4.785	-0.9524	-0.0937	-0.38
4.790	-0.9518	-0.0935	-0.39
4.795	-0.9506	-0.0942	-0.40
4.800	-0.9499	-0.0951	-0.40
4.805	-0.9495	-0.0962	-0.41
4.810	-0.9487	-0.0977	-0.41
4.815	-0.9483	-0.0990	-0.41
4.820	-0.9483	-0.1004	-0.41
4.825	-0.9483	-0.1011	-0.41
4.830	-0.9484	-0.1033	-0.41

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	-0.9484	-0.1046	-0.41
4.840	-0.9490	-0.1057	-0.40
4.845	-0.9494	-0.1069	-0.40
4.850	-0.9505	-0.1081	-0.39
4.855	-0.9512	-0.1098	-0.38
4.860	-0.9520	-0.1108	-0.37
4.865	-0.9530	-0.1116	-0.36
4.870	-0.9537	-0.1125	-0.35
4.875	-0.9546	-0.1138	-0.34
4.880	-0.9555	-0.1146	-0.33
4.885	-0.9564	-0.1168	-0.32
4.890	-0.9574	-0.1162	-0.31
4.895	-0.9581	-0.1169	-0.31
4.900	-0.9593	-0.1178	-0.30
4.905	-0.9599	-0.1183	-0.29
4.910	-0.9605	-0.1188	-0.28
4.915	-0.9612	-0.1192	-0.28
4.920	-0.9620	-0.1197	-0.27
4.925	-0.9625	-0.1205	-0.26
4.930	-0.9631	-0.1210	-0.26
4.935	-0.9635	-0.1215	-0.25
4.940	-0.9641	-0.1220	-0.25
4.945	-0.9646	-0.1226	-0.24
4.950	-0.9652	-0.1233	-0.24
4.955	-0.9656	-0.1238	-0.23
4.960	-0.9663	-0.1240	-0.23
4.965	-0.9667	-0.1248	-0.22
4.970	-0.9673	-0.1253	-0.22
4.975	-0.9677	-0.1256	-0.21
4.980	-0.9686	-0.1261	-0.20
4.985	-0.9689	-0.1268	-0.20
4.990	-0.9698	-0.1266	-0.19
4.995	-0.9698	-0.1271	-0.19
5.000	-0.9707	-0.1270	-0.18
5.005	-0.9711	-0.1270	-0.18
5.010	-0.9714	-0.1270	-0.18
5.015	-0.9719	-0.1274	-0.17
5.020	-0.9723	-0.1267	-0.17
5.025	-0.9726	-0.1274	-0.17
5.030	-0.9725	-0.1266	-0.17
5.035	-0.9730	-0.1270	-0.16
5.040	-0.9730	-0.1257	-0.17
5.045	-0.9735	-0.1268	-0.16
5.050	-0.9730	-0.1260	-0.17
5.055	-0.9737	-0.1267	-0.16
5.060	-0.9732	-0.1262	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	-0.9747	-0.1258	-0.15
5.070	-0.9744	-0.1265	-0.15
5.075	-0.9756	-0.1257	-0.14
5.080	-0.9754	-0.1261	-0.14
5.085	-0.9762	-0.1252	-0.14
5.090	-0.9770	-0.1255	-0.13
5.095	-0.9780	-0.1245	-0.12
5.100	-0.9787	-0.1246	-0.12
5.105	-0.9785	-0.1234	-0.12
5.110	-0.9794	-0.1224	-0.11
5.115	-0.9799	-0.1210	-0.11
5.120	-0.9806	-0.1200	-0.11
5.125	-0.9810	-0.1182	-0.10
5.130	-0.9818	-0.1168	-0.10
5.135	-0.9819	-0.1149	-0.10
5.140	-0.9828	-0.1131	-0.09
5.145	-0.9824	-0.1106	-0.10
5.150	-0.9833	-0.1089	-0.09
5.155	-0.9833	-0.1067	-0.10
5.160	-0.9837	-0.1042	-0.09
5.165	-0.9835	-0.1020	-0.10
5.170	-0.9835	-0.0994	-0.10
5.175	-0.9831	-0.0970	-0.11
5.180	-0.9831	-0.0943	-0.11
5.185	-0.9828	-0.0918	-0.11
5.190	-0.9822	-0.0891	-0.12
5.195	-0.9815	-0.0867	-0.13
5.200	-0.9808	-0.0840	-0.14
5.205	-0.9804	-0.0815	-0.14
5.210	-0.9797	-0.0790	-0.15
5.215	-0.9790	-0.0765	-0.16
5.220	-0.9781	-0.0740	-0.17
5.225	-0.9773	-0.0716	-0.18
5.230	-0.9765	-0.0692	-0.18
5.235	-0.9758	-0.0670	-0.19
5.240	-0.9753	-0.0647	-0.20
5.245	-0.9745	-0.0625	-0.21
5.250	-0.9735	-0.0605	-0.22
5.255	-0.9727	-0.0583	-0.22
5.260	-0.9721	-0.0563	-0.23
5.265	-0.9713	-0.0542	-0.24
5.270	-0.9708	-0.0526	-0.24
5.275	-0.9699	-0.0508	-0.25
5.280	-0.9690	-0.0489	-0.26
5.285	-0.9680	-0.0473	-0.27
5.290	-0.9675	-0.0456	-0.28

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	-0.9669	-0.0443	-0.28
5.300	-0.9662	-0.0425	-0.29
5.305	-0.9658	-0.0410	-0.29
5.310	-0.9649	-0.0392	-0.30
5.315	-0.9646	-0.0379	-0.31
5.320	-0.9638	-0.0367	-0.31
5.325	-0.9631	-0.0350	-0.32
5.330	-0.9629	-0.0342	-0.32
5.335	-0.9623	-0.0326	-0.33
5.340	-0.9621	-0.0316	-0.33
5.345	-0.9614	-0.0308	-0.34
5.350	-0.9612	-0.0289	-0.34
5.355	-0.9606	-0.0282	-0.34
5.360	-0.9605	-0.0270	-0.35
5.365	-0.9600	-0.0262	-0.35
5.370	-0.9597	-0.0251	-0.35
5.375	-0.9594	-0.0242	-0.36
5.380	-0.9592	-0.0229	-0.36
5.385	-0.9589	-0.0221	-0.36
5.390	-0.9591	-0.0211	-0.36
5.395	-0.9592	-0.0206	-0.36
5.400	-0.9594	-0.0197	-0.36
5.405	-0.9601	-0.0189	-0.35
5.410	-0.9596	-0.0181	-0.36
5.415	-0.9596	-0.0178	-0.36
5.420	-0.9601	-0.0172	-0.35
5.425	-0.9603	-0.0164	-0.35
5.430	-0.9605	-0.0159	-0.35
5.435	-0.9606	-0.0157	-0.35
5.440	-0.9610	-0.0154	-0.34
5.445	-0.9611	-0.0151	-0.34
5.450	-0.9617	-0.0152	-0.34
5.455	-0.9622	-0.0153	-0.33
5.460	-0.9626	-0.0149	-0.33
5.465	-0.9629	-0.0152	-0.33
5.470	-0.9633	-0.0156	-0.32
5.475	-0.9639	-0.0155	-0.32
5.480	-0.9643	-0.0161	-0.31
5.485	-0.9648	-0.0161	-0.31
5.490	-0.9652	-0.0165	-0.31
5.495	-0.9658	-0.0167	-0.30
5.500	-0.9662	-0.0170	-0.30
5.505	-0.9667	-0.0174	-0.29
5.510	-0.9676	-0.0177	-0.29
5.515	-0.9683	-0.0180	-0.28
5.520	-0.9692	-0.0186	-0.27

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	-0.9699	-0.0194	-0.26
5.530	-0.9708	-0.0195	-0.26
5.535	-0.9717	-0.0202	-0.25
5.540	-0.9726	-0.0209	-0.24
5.545	-0.9735	-0.0217	-0.23
5.550	-0.9743	-0.0223	-0.22
5.555	-0.9752	-0.0231	-0.22
5.560	-0.9760	-0.0237	-0.21
5.565	-0.9767	-0.0244	-0.20
5.570	-0.9775	-0.0251	-0.19
5.575	-0.9783	-0.0259	-0.19
5.580	-0.9789	-0.0264	-0.18
5.585	-0.9796	-0.0270	-0.18
5.590	-0.9803	-0.0277	-0.17
5.595	-0.9808	-0.0282	-0.16
5.600	-0.9812	-0.0284	-0.16
5.605	-0.9818	-0.0289	-0.16
5.610	-0.9824	-0.0294	-0.15
5.615	-0.9831	-0.0297	-0.14
5.620	-0.9835	-0.0298	-0.14
5.625	-0.9842	-0.0301	-0.13
5.630	-0.9852	-0.0302	-0.13
5.635	-0.9852	-0.0303	-0.13
5.640	-0.9861	-0.0305	-0.12
5.645	-0.9862	-0.0309	-0.12
5.650	-0.9867	-0.0315	-0.11
5.655	-0.9873	-0.0316	-0.11
5.660	-0.9878	-0.0317	-0.10
5.665	-0.9882	-0.0323	-0.10
5.670	-0.9882	-0.0318	-0.10
5.675	-0.9882	-0.0321	-0.10
5.680	-0.9889	-0.0320	-0.09
5.685	-0.9878	-0.0321	-0.10
5.690	-0.9875	-0.0324	-0.10
5.695	-0.9870	-0.0322	-0.11
5.700	-0.9874	-0.0323	-0.11
5.705	-0.9872	-0.0322	-0.11
5.710	-0.9874	-0.0323	-0.11
5.715	-0.9868	-0.0324	-0.11
5.720	-0.9866	-0.0325	-0.11
5.725	-0.9866	-0.0326	-0.11
5.730	-0.9865	-0.0320	-0.11
5.735	-0.9864	-0.0327	-0.11
5.740	-0.9859	-0.0319	-0.12
5.745	-0.9859	-0.0321	-0.12
5.750	-0.9850	-0.0314	-0.13

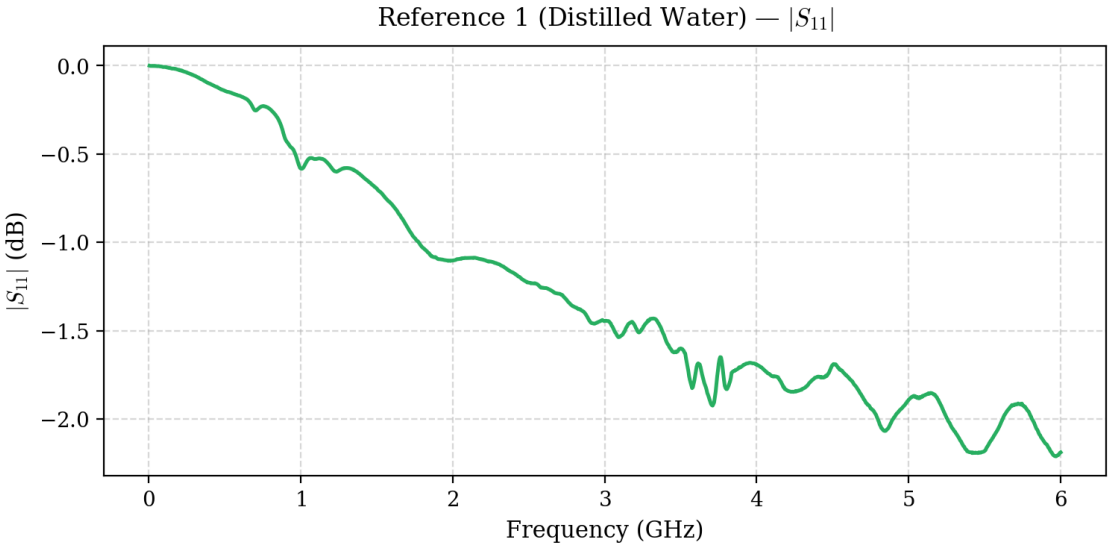
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	-0.9844	-0.0313	-0.13
5.760	-0.9840	-0.0311	-0.14
5.765	-0.9834	-0.0307	-0.14
5.770	-0.9825	-0.0305	-0.15
5.775	-0.9821	-0.0299	-0.15
5.780	-0.9816	-0.0295	-0.16
5.785	-0.9807	-0.0286	-0.17
5.790	-0.9799	-0.0287	-0.17
5.795	-0.9791	-0.0280	-0.18
5.800	-0.9782	-0.0275	-0.19
5.805	-0.9773	-0.0271	-0.20
5.810	-0.9767	-0.0265	-0.20
5.815	-0.9760	-0.0257	-0.21
5.820	-0.9749	-0.0255	-0.22
5.825	-0.9738	-0.0249	-0.23
5.830	-0.9729	-0.0242	-0.24
5.835	-0.9720	-0.0237	-0.24
5.840	-0.9713	-0.0233	-0.25
5.845	-0.9707	-0.0227	-0.26
5.850	-0.9696	-0.0221	-0.27
5.855	-0.9687	-0.0217	-0.27
5.860	-0.9681	-0.0212	-0.28
5.865	-0.9672	-0.0207	-0.29
5.870	-0.9667	-0.0202	-0.29
5.875	-0.9658	-0.0198	-0.30
5.880	-0.9653	-0.0195	-0.30
5.885	-0.9647	-0.0191	-0.31
5.890	-0.9641	-0.0188	-0.32
5.895	-0.9632	-0.0188	-0.32
5.900	-0.9628	-0.0186	-0.33
5.905	-0.9624	-0.0186	-0.33
5.910	-0.9620	-0.0185	-0.34
5.915	-0.9614	-0.0185	-0.34
5.920	-0.9605	-0.0185	-0.35
5.925	-0.9604	-0.0183	-0.35
5.930	-0.9595	-0.0188	-0.36
5.935	-0.9589	-0.0187	-0.36
5.940	-0.9588	-0.0188	-0.36
5.945	-0.9585	-0.0192	-0.37
5.950	-0.9583	-0.0195	-0.37
5.955	-0.9577	-0.0196	-0.37
5.960	-0.9579	-0.0197	-0.37
5.965	-0.9578	-0.0201	-0.37
5.970	-0.9571	-0.0209	-0.38
5.975	-0.9577	-0.0213	-0.37
5.980	-0.9580	-0.0225	-0.37

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.985	-0.9579	-0.0230	-0.37
5.990	-0.9579	-0.0243	-0.37
5.995	-0.9581	-0.0253	-0.37
6.000	-0.9578	-0.0261	-0.37

3.3 Reference 1 (Distilled Water)



Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
0.005	1.0000	-0.0051	0.00
0.010	0.9998	-0.0101	-0.00
0.015	0.9997	-0.0150	-0.00
0.020	0.9997	-0.0201	-0.00
0.025	0.9995	-0.0252	-0.00
0.030	0.9995	-0.0301	-0.00
0.035	0.9992	-0.0349	-0.00
0.040	0.9991	-0.0401	-0.00
0.045	0.9987	-0.0450	-0.00
0.050	0.9985	-0.0500	-0.00
0.055	0.9981	-0.0550	-0.00
0.060	0.9979	-0.0600	-0.00
0.065	0.9976	-0.0650	-0.00
0.070	0.9972	-0.0699	-0.00
0.075	0.9967	-0.0750	-0.00
0.080	0.9963	-0.0798	-0.00
0.085	0.9958	-0.0848	-0.01
0.090	0.9952	-0.0897	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.095	0.9946	-0.0947	-0.01
0.100	0.9942	-0.0996	-0.01
0.105	0.9937	-0.1046	-0.01
0.110	0.9931	-0.1094	-0.01
0.115	0.9925	-0.1144	-0.01
0.120	0.9917	-0.1193	-0.01
0.125	0.9910	-0.1242	-0.01
0.130	0.9902	-0.1292	-0.01
0.135	0.9895	-0.1341	-0.01
0.140	0.9888	-0.1389	-0.01
0.145	0.9879	-0.1440	-0.01
0.150	0.9871	-0.1487	-0.02
0.155	0.9862	-0.1536	-0.02
0.160	0.9855	-0.1583	-0.02
0.165	0.9847	-0.1633	-0.02
0.170	0.9837	-0.1681	-0.02
0.175	0.9829	-0.1730	-0.02
0.180	0.9818	-0.1779	-0.02
0.185	0.9808	-0.1827	-0.02
0.190	0.9797	-0.1876	-0.02
0.195	0.9786	-0.1924	-0.02
0.200	0.9776	-0.1973	-0.02
0.205	0.9763	-0.2020	-0.03
0.210	0.9753	-0.2070	-0.03
0.215	0.9740	-0.2117	-0.03
0.220	0.9730	-0.2166	-0.03
0.225	0.9715	-0.2213	-0.03
0.230	0.9703	-0.2261	-0.03
0.235	0.9689	-0.2308	-0.03
0.240	0.9677	-0.2356	-0.04
0.245	0.9663	-0.2405	-0.04
0.250	0.9650	-0.2453	-0.04
0.255	0.9634	-0.2499	-0.04
0.260	0.9621	-0.2546	-0.04
0.265	0.9606	-0.2594	-0.04
0.270	0.9591	-0.2641	-0.05
0.275	0.9576	-0.2688	-0.05
0.280	0.9561	-0.2735	-0.05
0.285	0.9544	-0.2781	-0.05
0.290	0.9529	-0.2828	-0.05
0.295	0.9513	-0.2874	-0.05
0.300	0.9496	-0.2920	-0.06
0.305	0.9480	-0.2968	-0.06
0.310	0.9463	-0.3013	-0.06
0.315	0.9445	-0.3058	-0.06
0.320	0.9427	-0.3104	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.325	0.9410	-0.3149	-0.07
0.330	0.9393	-0.3194	-0.07
0.335	0.9374	-0.3239	-0.07
0.340	0.9355	-0.3286	-0.07
0.345	0.9337	-0.3328	-0.08
0.350	0.9319	-0.3373	-0.08
0.355	0.9299	-0.3418	-0.08
0.360	0.9280	-0.3461	-0.08
0.365	0.9260	-0.3507	-0.09
0.370	0.9240	-0.3551	-0.09
0.375	0.9221	-0.3595	-0.09
0.380	0.9200	-0.3638	-0.09
0.385	0.9179	-0.3681	-0.10
0.390	0.9160	-0.3725	-0.10
0.395	0.9139	-0.3767	-0.10
0.400	0.9121	-0.3810	-0.10
0.405	0.9100	-0.3853	-0.10
0.410	0.9079	-0.3895	-0.11
0.415	0.9058	-0.3936	-0.11
0.420	0.9037	-0.3978	-0.11
0.425	0.9018	-0.4019	-0.11
0.430	0.8996	-0.4060	-0.11
0.435	0.8974	-0.4102	-0.12
0.440	0.8953	-0.4142	-0.12
0.445	0.8932	-0.4183	-0.12
0.450	0.8912	-0.4225	-0.12
0.455	0.8888	-0.4265	-0.12
0.460	0.8866	-0.4304	-0.13
0.465	0.8843	-0.4344	-0.13
0.470	0.8821	-0.4384	-0.13
0.475	0.8798	-0.4422	-0.13
0.480	0.8776	-0.4463	-0.14
0.485	0.8754	-0.4501	-0.14
0.490	0.8732	-0.4541	-0.14
0.495	0.8710	-0.4579	-0.14
0.500	0.8686	-0.4618	-0.14
0.505	0.8664	-0.4656	-0.14
0.510	0.8641	-0.4694	-0.15
0.515	0.8618	-0.4732	-0.15
0.520	0.8596	-0.4770	-0.15
0.525	0.8574	-0.4809	-0.15
0.530	0.8552	-0.4844	-0.15
0.535	0.8529	-0.4881	-0.15
0.540	0.8506	-0.4919	-0.15
0.545	0.8482	-0.4956	-0.15
0.550	0.8457	-0.4993	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.555	0.8434	-0.5030	-0.16
0.560	0.8409	-0.5067	-0.16
0.565	0.8386	-0.5103	-0.16
0.570	0.8361	-0.5139	-0.16
0.575	0.8337	-0.5176	-0.16
0.580	0.8313	-0.5212	-0.17
0.585	0.8288	-0.5246	-0.17
0.590	0.8264	-0.5282	-0.17
0.595	0.8241	-0.5317	-0.17
0.600	0.8216	-0.5352	-0.17
0.605	0.8190	-0.5387	-0.17
0.610	0.8164	-0.5422	-0.18
0.615	0.8138	-0.5457	-0.18
0.620	0.8113	-0.5492	-0.18
0.625	0.8086	-0.5526	-0.18
0.630	0.8059	-0.5561	-0.18
0.635	0.8033	-0.5595	-0.18
0.640	0.8006	-0.5628	-0.19
0.645	0.7980	-0.5662	-0.19
0.650	0.7952	-0.5694	-0.19
0.655	0.7922	-0.5726	-0.20
0.660	0.7893	-0.5758	-0.20
0.665	0.7864	-0.5789	-0.21
0.670	0.7832	-0.5819	-0.21
0.675	0.7802	-0.5847	-0.22
0.680	0.7773	-0.5872	-0.23
0.685	0.7742	-0.5895	-0.24
0.690	0.7716	-0.5917	-0.24
0.695	0.7690	-0.5937	-0.25
0.700	0.7669	-0.5961	-0.25
0.705	0.7648	-0.5986	-0.25
0.710	0.7630	-0.6012	-0.25
0.715	0.7612	-0.6043	-0.25
0.720	0.7593	-0.6073	-0.24
0.725	0.7571	-0.6107	-0.24
0.730	0.7549	-0.6142	-0.24
0.735	0.7525	-0.6175	-0.23
0.740	0.7500	-0.6210	-0.23
0.745	0.7474	-0.6244	-0.23
0.750	0.7447	-0.6278	-0.23
0.755	0.7420	-0.6310	-0.23
0.760	0.7390	-0.6341	-0.23
0.765	0.7361	-0.6376	-0.23
0.770	0.7331	-0.6409	-0.23
0.775	0.7298	-0.6440	-0.23
0.780	0.7268	-0.6473	-0.24

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.785	0.7235	-0.6506	-0.24
0.790	0.7203	-0.6536	-0.24
0.795	0.7170	-0.6567	-0.24
0.800	0.7138	-0.6596	-0.25
0.805	0.7104	-0.6626	-0.25
0.810	0.7070	-0.6656	-0.26
0.815	0.7036	-0.6685	-0.26
0.820	0.7000	-0.6715	-0.26
0.825	0.6966	-0.6743	-0.27
0.830	0.6929	-0.6772	-0.27
0.835	0.6894	-0.6800	-0.28
0.840	0.6856	-0.6827	-0.29
0.845	0.6821	-0.6854	-0.29
0.850	0.6782	-0.6881	-0.30
0.855	0.6742	-0.6907	-0.31
0.860	0.6702	-0.6932	-0.32
0.865	0.6662	-0.6956	-0.33
0.870	0.6621	-0.6978	-0.34
0.875	0.6579	-0.6999	-0.35
0.880	0.6537	-0.7017	-0.36
0.885	0.6496	-0.7031	-0.38
0.890	0.6459	-0.7045	-0.39
0.895	0.6423	-0.7056	-0.41
0.900	0.6390	-0.7069	-0.42
0.905	0.6362	-0.7084	-0.43
0.910	0.6332	-0.7101	-0.43
0.915	0.6302	-0.7121	-0.44
0.920	0.6270	-0.7140	-0.44
0.925	0.6238	-0.7157	-0.45
0.930	0.6208	-0.7176	-0.46
0.935	0.6177	-0.7195	-0.46
0.940	0.6146	-0.7217	-0.46
0.945	0.6113	-0.7239	-0.47
0.950	0.6077	-0.7263	-0.47
0.955	0.6039	-0.7283	-0.48
0.960	0.6001	-0.7300	-0.49
0.965	0.5964	-0.7315	-0.50
0.970	0.5926	-0.7330	-0.51
0.975	0.5889	-0.7340	-0.53
0.980	0.5855	-0.7348	-0.54
0.985	0.5823	-0.7353	-0.56
0.990	0.5798	-0.7359	-0.57
0.995	0.5771	-0.7364	-0.58
1.000	0.5753	-0.7372	-0.58
1.005	0.5737	-0.7384	-0.58
1.010	0.5719	-0.7398	-0.58

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.015	0.5702	-0.7416	-0.58
1.020	0.5686	-0.7437	-0.57
1.025	0.5668	-0.7461	-0.57
1.030	0.5648	-0.7488	-0.56
1.035	0.5625	-0.7515	-0.55
1.040	0.5602	-0.7544	-0.54
1.045	0.5575	-0.7572	-0.53
1.050	0.5547	-0.7599	-0.53
1.055	0.5518	-0.7626	-0.53
1.060	0.5488	-0.7651	-0.52
1.065	0.5454	-0.7675	-0.52
1.070	0.5422	-0.7698	-0.52
1.075	0.5390	-0.7720	-0.52
1.080	0.5358	-0.7740	-0.53
1.085	0.5327	-0.7759	-0.53
1.090	0.5297	-0.7777	-0.53
1.095	0.5268	-0.7798	-0.53
1.100	0.5237	-0.7817	-0.53
1.105	0.5209	-0.7838	-0.53
1.110	0.5179	-0.7858	-0.53
1.115	0.5149	-0.7879	-0.53
1.120	0.5119	-0.7901	-0.52
1.125	0.5087	-0.7921	-0.53
1.130	0.5054	-0.7941	-0.53
1.135	0.5023	-0.7958	-0.53
1.140	0.4992	-0.7978	-0.53
1.145	0.4959	-0.7997	-0.53
1.150	0.4925	-0.8015	-0.53
1.155	0.4893	-0.8032	-0.53
1.160	0.4861	-0.8047	-0.54
1.165	0.4828	-0.8062	-0.54
1.170	0.4794	-0.8076	-0.55
1.175	0.4764	-0.8089	-0.55
1.180	0.4732	-0.8102	-0.55
1.185	0.4702	-0.8114	-0.56
1.190	0.4673	-0.8123	-0.56
1.195	0.4645	-0.8132	-0.57
1.200	0.4618	-0.8141	-0.57
1.205	0.4590	-0.8149	-0.58
1.210	0.4563	-0.8157	-0.59
1.215	0.4539	-0.8165	-0.59
1.220	0.4515	-0.8172	-0.60
1.225	0.4492	-0.8183	-0.60
1.230	0.4471	-0.8194	-0.60
1.235	0.4448	-0.8204	-0.60
1.240	0.4428	-0.8217	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.245	0.4407	-0.8230	-0.60
1.250	0.4385	-0.8246	-0.59
1.255	0.4362	-0.8261	-0.59
1.260	0.4339	-0.8275	-0.59
1.265	0.4316	-0.8290	-0.59
1.270	0.4292	-0.8306	-0.58
1.275	0.4265	-0.8321	-0.58
1.280	0.4241	-0.8337	-0.58
1.285	0.4212	-0.8350	-0.58
1.290	0.4184	-0.8367	-0.58
1.295	0.4156	-0.8381	-0.58
1.300	0.4129	-0.8394	-0.58
1.305	0.4102	-0.8408	-0.58
1.310	0.4076	-0.8420	-0.58
1.315	0.4048	-0.8433	-0.58
1.320	0.4020	-0.8445	-0.58
1.325	0.3993	-0.8458	-0.58
1.330	0.3966	-0.8468	-0.58
1.335	0.3937	-0.8480	-0.58
1.340	0.3909	-0.8491	-0.59
1.345	0.3880	-0.8500	-0.59
1.350	0.3853	-0.8512	-0.59
1.355	0.3823	-0.8521	-0.59
1.360	0.3794	-0.8531	-0.60
1.365	0.3764	-0.8541	-0.60
1.370	0.3736	-0.8551	-0.60
1.375	0.3707	-0.8559	-0.60
1.380	0.3677	-0.8567	-0.61
1.385	0.3648	-0.8577	-0.61
1.390	0.3618	-0.8586	-0.61
1.395	0.3587	-0.8594	-0.62
1.400	0.3557	-0.8603	-0.62
1.405	0.3525	-0.8611	-0.63
1.410	0.3494	-0.8619	-0.63
1.415	0.3463	-0.8627	-0.63
1.420	0.3431	-0.8635	-0.64
1.425	0.3399	-0.8644	-0.64
1.430	0.3368	-0.8652	-0.64
1.435	0.3335	-0.8661	-0.65
1.440	0.3304	-0.8669	-0.65
1.445	0.3273	-0.8677	-0.65
1.450	0.3240	-0.8684	-0.66
1.455	0.3208	-0.8694	-0.66
1.460	0.3175	-0.8701	-0.67
1.465	0.3143	-0.8707	-0.67
1.470	0.3111	-0.8716	-0.67

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
1.475	0.3077	-0.8723	-0.68
1.480	0.3044	-0.8731	-0.68
1.485	0.3008	-0.8738	-0.69
1.490	0.2975	-0.8747	-0.69
1.495	0.2940	-0.8754	-0.69
1.500	0.2908	-0.8761	-0.70
1.505	0.2872	-0.8766	-0.70
1.510	0.2836	-0.8773	-0.71
1.515	0.2801	-0.8780	-0.71
1.520	0.2767	-0.8787	-0.71
1.525	0.2731	-0.8795	-0.72
1.530	0.2695	-0.8801	-0.72
1.535	0.2658	-0.8808	-0.72
1.540	0.2621	-0.8813	-0.73
1.545	0.2584	-0.8817	-0.74
1.550	0.2549	-0.8823	-0.74
1.555	0.2509	-0.8827	-0.75
1.560	0.2472	-0.8832	-0.75
1.565	0.2434	-0.8838	-0.76
1.570	0.2396	-0.8844	-0.76
1.575	0.2358	-0.8848	-0.77
1.580	0.2323	-0.8853	-0.77
1.585	0.2284	-0.8861	-0.77
1.590	0.2247	-0.8865	-0.78
1.595	0.2209	-0.8870	-0.78
1.600	0.2173	-0.8874	-0.78
1.605	0.2136	-0.8877	-0.79
1.610	0.2099	-0.8879	-0.80
1.615	0.2061	-0.8883	-0.80
1.620	0.2023	-0.8887	-0.81
1.625	0.1985	-0.8890	-0.81
1.630	0.1947	-0.8892	-0.82
1.635	0.1909	-0.8894	-0.82
1.640	0.1869	-0.8893	-0.83
1.645	0.1832	-0.8893	-0.84
1.650	0.1793	-0.8895	-0.84
1.655	0.1756	-0.8898	-0.85
1.660	0.1720	-0.8896	-0.86
1.665	0.1684	-0.8896	-0.86
1.670	0.1648	-0.8898	-0.87
1.675	0.1612	-0.8896	-0.88
1.680	0.1577	-0.8896	-0.88
1.685	0.1541	-0.8893	-0.89
1.690	0.1507	-0.8891	-0.90
1.695	0.1476	-0.8890	-0.90
1.700	0.1441	-0.8887	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.705	0.1408	-0.8886	-0.92
1.710	0.1374	-0.8883	-0.93
1.715	0.1340	-0.8880	-0.93
1.720	0.1310	-0.8878	-0.94
1.725	0.1276	-0.8876	-0.95
1.730	0.1247	-0.8875	-0.95
1.735	0.1215	-0.8870	-0.96
1.740	0.1185	-0.8868	-0.97
1.745	0.1157	-0.8867	-0.97
1.750	0.1128	-0.8866	-0.98
1.755	0.1101	-0.8865	-0.98
1.760	0.1074	-0.8861	-0.99
1.765	0.1048	-0.8860	-0.99
1.770	0.1023	-0.8860	-0.99
1.775	0.0999	-0.8858	-1.00
1.780	0.0976	-0.8855	-1.00
1.785	0.0950	-0.8850	-1.01
1.790	0.0923	-0.8846	-1.02
1.795	0.0899	-0.8843	-1.02
1.800	0.0875	-0.8841	-1.03
1.805	0.0852	-0.8839	-1.03
1.810	0.0829	-0.8838	-1.03
1.815	0.0808	-0.8836	-1.04
1.820	0.0785	-0.8834	-1.04
1.825	0.0766	-0.8831	-1.05
1.830	0.0747	-0.8830	-1.05
1.835	0.0725	-0.8826	-1.06
1.840	0.0706	-0.8824	-1.06
1.845	0.0688	-0.8822	-1.06
1.850	0.0667	-0.8815	-1.07
1.855	0.0648	-0.8815	-1.07
1.860	0.0628	-0.8809	-1.08
1.865	0.0608	-0.8811	-1.08
1.870	0.0590	-0.8809	-1.08
1.875	0.0569	-0.8809	-1.08
1.880	0.0551	-0.8809	-1.08
1.885	0.0535	-0.8812	-1.08
1.890	0.0522	-0.8810	-1.09
1.895	0.0504	-0.8807	-1.09
1.900	0.0486	-0.8804	-1.09
1.905	0.0470	-0.8803	-1.09
1.910	0.0452	-0.8804	-1.09
1.915	0.0434	-0.8804	-1.10
1.920	0.0419	-0.8805	-1.10
1.925	0.0403	-0.8804	-1.10
1.930	0.0389	-0.8805	-1.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.935	0.0374	-0.8804	-1.10
1.940	0.0359	-0.8804	-1.10
1.945	0.0345	-0.8804	-1.10
1.950	0.0330	-0.8803	-1.10
1.955	0.0316	-0.8803	-1.10
1.960	0.0302	-0.8802	-1.10
1.965	0.0288	-0.8803	-1.10
1.970	0.0274	-0.8801	-1.10
1.975	0.0262	-0.8803	-1.10
1.980	0.0248	-0.8804	-1.10
1.985	0.0235	-0.8802	-1.10
1.990	0.0220	-0.8804	-1.10
1.995	0.0206	-0.8805	-1.10
2.000	0.0190	-0.8805	-1.10
2.005	0.0175	-0.8807	-1.10
2.010	0.0159	-0.8808	-1.10
2.015	0.0145	-0.8809	-1.10
2.020	0.0130	-0.8810	-1.10
2.025	0.0112	-0.8813	-1.10
2.030	0.0095	-0.8815	-1.09
2.035	0.0079	-0.8815	-1.10
2.040	0.0061	-0.8816	-1.09
2.045	0.0044	-0.8817	-1.09
2.050	0.0026	-0.8817	-1.09
2.055	0.0007	-0.8820	-1.09
2.060	-0.0012	-0.8819	-1.09
2.065	-0.0031	-0.8821	-1.09
2.070	-0.0052	-0.8821	-1.09
2.075	-0.0073	-0.8821	-1.09
2.080	-0.0095	-0.8822	-1.09
2.085	-0.0117	-0.8821	-1.09
2.090	-0.0140	-0.8821	-1.09
2.095	-0.0163	-0.8821	-1.09
2.100	-0.0188	-0.8820	-1.09
2.105	-0.0212	-0.8820	-1.09
2.110	-0.0237	-0.8820	-1.09
2.115	-0.0261	-0.8819	-1.09
2.120	-0.0286	-0.8818	-1.09
2.125	-0.0312	-0.8818	-1.09
2.130	-0.0341	-0.8816	-1.09
2.135	-0.0366	-0.8814	-1.09
2.140	-0.0393	-0.8815	-1.09
2.145	-0.0419	-0.8814	-1.09
2.150	-0.0445	-0.8812	-1.09
2.155	-0.0473	-0.8809	-1.09
2.160	-0.0501	-0.8806	-1.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.165	-0.0529	-0.8804	-1.09
2.170	-0.0558	-0.8800	-1.09
2.175	-0.0585	-0.8798	-1.09
2.180	-0.0613	-0.8795	-1.09
2.185	-0.0641	-0.8791	-1.10
2.190	-0.0671	-0.8790	-1.10
2.195	-0.0700	-0.8785	-1.10
2.200	-0.0729	-0.8783	-1.10
2.205	-0.0759	-0.8778	-1.10
2.210	-0.0788	-0.8775	-1.10
2.215	-0.0817	-0.8770	-1.10
2.220	-0.0846	-0.8766	-1.10
2.225	-0.0877	-0.8761	-1.11
2.230	-0.0906	-0.8756	-1.11
2.235	-0.0936	-0.8752	-1.11
2.240	-0.0967	-0.8750	-1.11
2.245	-0.0995	-0.8745	-1.11
2.250	-0.1025	-0.8742	-1.11
2.255	-0.1056	-0.8737	-1.11
2.260	-0.1083	-0.8733	-1.11
2.265	-0.1114	-0.8728	-1.11
2.270	-0.1141	-0.8723	-1.11
2.275	-0.1170	-0.8716	-1.12
2.280	-0.1199	-0.8711	-1.12
2.285	-0.1228	-0.8707	-1.12
2.290	-0.1255	-0.8700	-1.12
2.295	-0.1283	-0.8695	-1.12
2.300	-0.1312	-0.8689	-1.12
2.305	-0.1339	-0.8684	-1.12
2.310	-0.1369	-0.8677	-1.13
2.315	-0.1394	-0.8669	-1.13
2.320	-0.1423	-0.8662	-1.13
2.325	-0.1449	-0.8656	-1.13
2.330	-0.1474	-0.8649	-1.14
2.335	-0.1503	-0.8643	-1.14
2.340	-0.1526	-0.8635	-1.14
2.345	-0.1553	-0.8627	-1.14
2.350	-0.1577	-0.8620	-1.15
2.355	-0.1601	-0.8612	-1.15
2.360	-0.1625	-0.8603	-1.15
2.365	-0.1646	-0.8598	-1.16
2.370	-0.1667	-0.8589	-1.16
2.375	-0.1690	-0.8584	-1.16
2.380	-0.1711	-0.8576	-1.16
2.385	-0.1732	-0.8570	-1.17
2.390	-0.1754	-0.8564	-1.17

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.395	-0.1773	-0.8557	-1.17
2.400	-0.1792	-0.8551	-1.17
2.405	-0.1813	-0.8544	-1.18
2.410	-0.1831	-0.8536	-1.18
2.415	-0.1849	-0.8531	-1.18
2.420	-0.1866	-0.8523	-1.18
2.425	-0.1882	-0.8517	-1.19
2.430	-0.1899	-0.8508	-1.19
2.435	-0.1914	-0.8503	-1.19
2.440	-0.1929	-0.8498	-1.20
2.445	-0.1945	-0.8490	-1.20
2.450	-0.1958	-0.8483	-1.20
2.455	-0.1969	-0.8476	-1.21
2.460	-0.1982	-0.8471	-1.21
2.465	-0.1995	-0.8464	-1.21
2.470	-0.2007	-0.8459	-1.22
2.475	-0.2018	-0.8453	-1.22
2.480	-0.2033	-0.8448	-1.22
2.485	-0.2044	-0.8443	-1.22
2.490	-0.2055	-0.8436	-1.23
2.495	-0.2066	-0.8434	-1.23
2.500	-0.2080	-0.8429	-1.23
2.505	-0.2090	-0.8426	-1.23
2.510	-0.2101	-0.8422	-1.23
2.515	-0.2111	-0.8417	-1.23
2.520	-0.2123	-0.8415	-1.23
2.525	-0.2133	-0.8411	-1.23
2.530	-0.2144	-0.8408	-1.23
2.535	-0.2155	-0.8405	-1.23
2.540	-0.2166	-0.8404	-1.23
2.545	-0.2174	-0.8400	-1.23
2.550	-0.2183	-0.8396	-1.23
2.555	-0.2192	-0.8392	-1.24
2.560	-0.2200	-0.8386	-1.24
2.565	-0.2204	-0.8382	-1.24
2.570	-0.2209	-0.8375	-1.25
2.575	-0.2213	-0.8371	-1.25
2.580	-0.2216	-0.8365	-1.26
2.585	-0.2221	-0.8364	-1.26
2.590	-0.2225	-0.8361	-1.26
2.595	-0.2229	-0.8361	-1.26
2.600	-0.2234	-0.8358	-1.26
2.605	-0.2241	-0.8356	-1.26
2.610	-0.2246	-0.8354	-1.26
2.615	-0.2253	-0.8353	-1.26
2.620	-0.2258	-0.8350	-1.26

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
2.625	-0.2263	-0.8346	-1.26
2.630	-0.2268	-0.8342	-1.26
2.635	-0.2273	-0.8339	-1.27
2.640	-0.2280	-0.8335	-1.27
2.645	-0.2285	-0.8331	-1.27
2.650	-0.2291	-0.8328	-1.27
2.655	-0.2297	-0.8323	-1.28
2.660	-0.2303	-0.8317	-1.28
2.665	-0.2309	-0.8314	-1.28
2.670	-0.2317	-0.8307	-1.29
2.675	-0.2321	-0.8305	-1.29
2.680	-0.2327	-0.8302	-1.29
2.685	-0.2330	-0.8300	-1.29
2.690	-0.2336	-0.8298	-1.29
2.695	-0.2341	-0.8296	-1.29
2.700	-0.2344	-0.8294	-1.29
2.705	-0.2349	-0.8292	-1.29
2.710	-0.2350	-0.8290	-1.29
2.715	-0.2357	-0.8284	-1.30
2.720	-0.2362	-0.8282	-1.30
2.725	-0.2369	-0.8276	-1.30
2.730	-0.2374	-0.8268	-1.31
2.735	-0.2383	-0.8261	-1.31
2.740	-0.2391	-0.8253	-1.32
2.745	-0.2402	-0.8247	-1.32
2.750	-0.2409	-0.8239	-1.33
2.755	-0.2419	-0.8230	-1.33
2.760	-0.2432	-0.8221	-1.34
2.765	-0.2444	-0.8213	-1.34
2.770	-0.2456	-0.8205	-1.35
2.775	-0.2467	-0.8196	-1.35
2.780	-0.2481	-0.8189	-1.35
2.785	-0.2493	-0.8181	-1.36
2.790	-0.2508	-0.8173	-1.36
2.795	-0.2522	-0.8168	-1.36
2.800	-0.2533	-0.8161	-1.37
2.805	-0.2550	-0.8155	-1.37
2.810	-0.2563	-0.8148	-1.37
2.815	-0.2576	-0.8143	-1.37
2.820	-0.2590	-0.8136	-1.37
2.825	-0.2603	-0.8127	-1.38
2.830	-0.2617	-0.8123	-1.38
2.835	-0.2630	-0.8116	-1.38
2.840	-0.2644	-0.8112	-1.38
2.845	-0.2657	-0.8103	-1.38
2.850	-0.2670	-0.8096	-1.39

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.855	-0.2682	-0.8089	-1.39
2.860	-0.2694	-0.8082	-1.39
2.865	-0.2705	-0.8074	-1.40
2.870	-0.2715	-0.8066	-1.40
2.875	-0.2724	-0.8057	-1.41
2.880	-0.2733	-0.8046	-1.41
2.885	-0.2745	-0.8035	-1.42
2.890	-0.2758	-0.8022	-1.43
2.895	-0.2773	-0.8009	-1.44
2.900	-0.2786	-0.7996	-1.44
2.905	-0.2803	-0.7985	-1.45
2.910	-0.2817	-0.7975	-1.46
2.915	-0.2836	-0.7966	-1.46
2.920	-0.2853	-0.7958	-1.46
2.925	-0.2870	-0.7952	-1.46
2.930	-0.2887	-0.7944	-1.46
2.935	-0.2900	-0.7940	-1.46
2.940	-0.2914	-0.7937	-1.46
2.945	-0.2929	-0.7934	-1.46
2.950	-0.2940	-0.7931	-1.45
2.955	-0.2953	-0.7929	-1.45
2.960	-0.2960	-0.7928	-1.45
2.965	-0.2971	-0.7926	-1.45
2.970	-0.2978	-0.7925	-1.45
2.975	-0.2985	-0.7925	-1.44
2.980	-0.2992	-0.7925	-1.44
2.985	-0.2998	-0.7925	-1.44
2.990	-0.3000	-0.7921	-1.44
2.995	-0.3003	-0.7916	-1.45
3.000	-0.3008	-0.7915	-1.45
3.005	-0.3044	-0.7903	-1.44
3.010	-0.3045	-0.7902	-1.44
3.015	-0.3047	-0.7900	-1.44
3.020	-0.3049	-0.7899	-1.45
3.025	-0.3046	-0.7900	-1.45
3.030	-0.3048	-0.7895	-1.45
3.035	-0.3048	-0.7890	-1.45
3.040	-0.3047	-0.7885	-1.46
3.045	-0.3047	-0.7874	-1.47
3.050	-0.3048	-0.7868	-1.48
3.055	-0.3049	-0.7859	-1.48
3.060	-0.3048	-0.7849	-1.49
3.065	-0.3054	-0.7837	-1.50
3.070	-0.3057	-0.7827	-1.51
3.075	-0.3065	-0.7819	-1.52
3.080	-0.3070	-0.7808	-1.53

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.085	-0.3079	-0.7797	-1.53
3.090	-0.3092	-0.7787	-1.54
3.095	-0.3105	-0.7783	-1.54
3.100	-0.3117	-0.7779	-1.53
3.105	-0.3131	-0.7779	-1.53
3.110	-0.3143	-0.7777	-1.53
3.115	-0.3156	-0.7774	-1.52
3.120	-0.3168	-0.7774	-1.52
3.125	-0.3180	-0.7778	-1.51
3.130	-0.3188	-0.7780	-1.51
3.135	-0.3196	-0.7788	-1.50
3.140	-0.3202	-0.7797	-1.48
3.145	-0.3210	-0.7800	-1.48
3.150	-0.3213	-0.7808	-1.47
3.155	-0.3217	-0.7814	-1.46
3.160	-0.3218	-0.7817	-1.46
3.165	-0.3219	-0.7820	-1.46
3.170	-0.3220	-0.7822	-1.45
3.175	-0.3220	-0.7825	-1.45
3.180	-0.3214	-0.7829	-1.45
3.185	-0.3216	-0.7824	-1.45
3.190	-0.3213	-0.7819	-1.46
3.195	-0.3210	-0.7813	-1.47
3.200	-0.3215	-0.7803	-1.47
3.205	-0.3214	-0.7792	-1.48
3.210	-0.3218	-0.7784	-1.49
3.215	-0.3221	-0.7772	-1.50
3.220	-0.3232	-0.7760	-1.51
3.225	-0.3243	-0.7753	-1.51
3.230	-0.3258	-0.7749	-1.51
3.235	-0.3269	-0.7750	-1.50
3.240	-0.3281	-0.7750	-1.50
3.245	-0.3291	-0.7754	-1.49
3.250	-0.3305	-0.7752	-1.49
3.255	-0.3315	-0.7758	-1.48
3.260	-0.3325	-0.7760	-1.47
3.265	-0.3338	-0.7761	-1.46
3.270	-0.3347	-0.7762	-1.46
3.275	-0.3355	-0.7765	-1.45
3.280	-0.3364	-0.7764	-1.45
3.285	-0.3374	-0.7770	-1.44
3.290	-0.3383	-0.7764	-1.44
3.295	-0.3394	-0.7765	-1.44
3.300	-0.3401	-0.7767	-1.43
3.305	-0.3404	-0.7766	-1.43
3.310	-0.3414	-0.7763	-1.43

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.315	-0.3421	-0.7760	-1.43
3.320	-0.3425	-0.7759	-1.43
3.325	-0.3434	-0.7754	-1.43
3.330	-0.3437	-0.7752	-1.43
3.335	-0.3442	-0.7747	-1.43
3.340	-0.3448	-0.7740	-1.44
3.345	-0.3457	-0.7729	-1.45
3.350	-0.3462	-0.7719	-1.45
3.355	-0.3467	-0.7710	-1.46
3.360	-0.3474	-0.7698	-1.47
3.365	-0.3478	-0.7688	-1.48
3.370	-0.3490	-0.7674	-1.48
3.375	-0.3494	-0.7660	-1.49
3.380	-0.3506	-0.7644	-1.50
3.385	-0.3515	-0.7628	-1.52
3.390	-0.3526	-0.7614	-1.52
3.395	-0.3537	-0.7595	-1.54
3.400	-0.3549	-0.7577	-1.55
3.405	-0.3560	-0.7560	-1.56
3.410	-0.3576	-0.7542	-1.57
3.415	-0.3591	-0.7527	-1.58
3.420	-0.3607	-0.7515	-1.58
3.425	-0.3621	-0.7497	-1.59
3.430	-0.3637	-0.7481	-1.60
3.435	-0.3654	-0.7468	-1.60
3.440	-0.3671	-0.7452	-1.61
3.445	-0.3693	-0.7434	-1.62
3.450	-0.3713	-0.7422	-1.62
3.455	-0.3729	-0.7410	-1.62
3.460	-0.3748	-0.7402	-1.62
3.465	-0.3768	-0.7396	-1.62
3.470	-0.3784	-0.7383	-1.62
3.475	-0.3801	-0.7377	-1.62
3.480	-0.3815	-0.7372	-1.62
3.485	-0.3836	-0.7370	-1.61
3.490	-0.3849	-0.7370	-1.60
3.495	-0.3859	-0.7365	-1.60
3.500	-0.3866	-0.7365	-1.60
3.505	-0.3870	-0.7360	-1.60
3.510	-0.3877	-0.7355	-1.60
3.515	-0.3879	-0.7348	-1.61
3.520	-0.3877	-0.7343	-1.61
3.525	-0.3879	-0.7332	-1.62
3.530	-0.3875	-0.7328	-1.63
3.535	-0.3867	-0.7303	-1.66
3.540	-0.3866	-0.7279	-1.68

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
3.545	-0.3866	-0.7253	-1.70
3.550	-0.3873	-0.7224	-1.73
3.555	-0.3877	-0.7190	-1.76
3.560	-0.3894	-0.7159	-1.78
3.565	-0.3909	-0.7134	-1.79
3.570	-0.3936	-0.7102	-1.81
3.575	-0.3960	-0.7071	-1.83
3.580	-0.4000	-0.7054	-1.82
3.585	-0.4046	-0.7048	-1.80
3.590	-0.4077	-0.7051	-1.78
3.595	-0.4106	-0.7063	-1.76
3.600	-0.4126	-0.7081	-1.73
3.605	-0.4139	-0.7100	-1.70
3.610	-0.4137	-0.7118	-1.69
3.615	-0.4130	-0.7127	-1.68
3.620	-0.4117	-0.7129	-1.69
3.625	-0.4102	-0.7136	-1.69
3.630	-0.4091	-0.7128	-1.70
3.635	-0.4074	-0.7115	-1.73
3.640	-0.4067	-0.7098	-1.74
3.645	-0.4064	-0.7082	-1.76
3.650	-0.4063	-0.7063	-1.78
3.655	-0.4066	-0.7043	-1.80
3.660	-0.4070	-0.7029	-1.81
3.665	-0.4075	-0.7013	-1.82
3.670	-0.4080	-0.6994	-1.83
3.675	-0.4083	-0.6971	-1.85
3.680	-0.4090	-0.6956	-1.86
3.685	-0.4095	-0.6939	-1.88
3.690	-0.4106	-0.6917	-1.89
3.695	-0.4123	-0.6893	-1.90
3.700	-0.4140	-0.6870	-1.92
3.705	-0.4166	-0.6848	-1.92
3.710	-0.4195	-0.6827	-1.92
3.715	-0.4231	-0.6814	-1.92
3.720	-0.4272	-0.6800	-1.91
3.725	-0.4314	-0.6805	-1.88
3.730	-0.4369	-0.6803	-1.85
3.735	-0.4404	-0.6828	-1.80
3.740	-0.4438	-0.6855	-1.76
3.745	-0.4451	-0.6885	-1.73
3.750	-0.4457	-0.6924	-1.69
3.755	-0.4450	-0.6956	-1.66
3.760	-0.4432	-0.6982	-1.65
3.765	-0.4403	-0.6999	-1.65
3.770	-0.4369	-0.6998	-1.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.775	-0.4339	-0.6980	-1.70
3.780	-0.4311	-0.6954	-1.74
3.785	-0.4307	-0.6915	-1.78
3.790	-0.4317	-0.6874	-1.81
3.795	-0.4343	-0.6844	-1.82
3.800	-0.4370	-0.6820	-1.83
3.805	-0.4394	-0.6804	-1.83
3.810	-0.4425	-0.6794	-1.82
3.815	-0.4452	-0.6787	-1.81
3.820	-0.4479	-0.6784	-1.80
3.825	-0.4503	-0.6782	-1.79
3.830	-0.4526	-0.6788	-1.77
3.835	-0.4534	-0.6816	-1.74
3.840	-0.4549	-0.6811	-1.73
3.845	-0.4559	-0.6808	-1.73
3.850	-0.4572	-0.6803	-1.73
3.855	-0.4583	-0.6795	-1.73
3.860	-0.4599	-0.6791	-1.72
3.865	-0.4613	-0.6782	-1.72
3.870	-0.4627	-0.6775	-1.72
3.875	-0.4642	-0.6768	-1.72
3.880	-0.4657	-0.6761	-1.71
3.885	-0.4677	-0.6755	-1.71
3.890	-0.4691	-0.6742	-1.71
3.895	-0.4708	-0.6734	-1.71
3.900	-0.4726	-0.6725	-1.70
3.905	-0.4743	-0.6716	-1.70
3.910	-0.4761	-0.6708	-1.70
3.915	-0.4782	-0.6698	-1.69
3.920	-0.4798	-0.6687	-1.69
3.925	-0.4818	-0.6675	-1.69
3.930	-0.4834	-0.6667	-1.69
3.935	-0.4855	-0.6652	-1.69
3.940	-0.4869	-0.6644	-1.68
3.945	-0.4889	-0.6630	-1.68
3.950	-0.4906	-0.6620	-1.68
3.955	-0.4925	-0.6606	-1.68
3.960	-0.4942	-0.6593	-1.68
3.965	-0.4959	-0.6579	-1.68
3.970	-0.4975	-0.6566	-1.68
3.975	-0.4991	-0.6552	-1.69
3.980	-0.5012	-0.6536	-1.68
3.985	-0.5028	-0.6523	-1.69
3.990	-0.5046	-0.6508	-1.69
3.995	-0.5061	-0.6491	-1.69
4.000	-0.5079	-0.6477	-1.69

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.005	-0.5095	-0.6459	-1.70
4.010	-0.5114	-0.6442	-1.70
4.015	-0.5127	-0.6430	-1.70
4.020	-0.5145	-0.6412	-1.70
4.025	-0.5158	-0.6397	-1.71
4.030	-0.5174	-0.6383	-1.71
4.035	-0.5191	-0.6364	-1.71
4.040	-0.5205	-0.6347	-1.71
4.045	-0.5219	-0.6331	-1.72
4.050	-0.5233	-0.6314	-1.72
4.055	-0.5247	-0.6298	-1.73
4.060	-0.5259	-0.6282	-1.73
4.065	-0.5273	-0.6268	-1.73
4.070	-0.5285	-0.6251	-1.74
4.075	-0.5296	-0.6238	-1.74
4.080	-0.5309	-0.6225	-1.74
4.085	-0.5322	-0.6210	-1.75
4.090	-0.5331	-0.6198	-1.75
4.095	-0.5340	-0.6185	-1.75
4.100	-0.5353	-0.6170	-1.76
4.105	-0.5364	-0.6159	-1.76
4.110	-0.5375	-0.6148	-1.76
4.115	-0.5389	-0.6138	-1.76
4.120	-0.5399	-0.6128	-1.76
4.125	-0.5416	-0.6113	-1.76
4.130	-0.5426	-0.6103	-1.76
4.135	-0.5438	-0.6089	-1.76
4.140	-0.5450	-0.6077	-1.76
4.145	-0.5458	-0.6060	-1.77
4.150	-0.5468	-0.6043	-1.78
4.155	-0.5475	-0.6029	-1.78
4.160	-0.5480	-0.6016	-1.79
4.165	-0.5480	-0.6004	-1.80
4.170	-0.5481	-0.5994	-1.81
4.175	-0.5476	-0.5984	-1.82
4.180	-0.5479	-0.5979	-1.82
4.185	-0.5474	-0.5971	-1.83
4.190	-0.5476	-0.5965	-1.83
4.195	-0.5473	-0.5966	-1.83
4.200	-0.5471	-0.5963	-1.84
4.205	-0.5469	-0.5962	-1.84
4.210	-0.5468	-0.5961	-1.84
4.215	-0.5468	-0.5960	-1.84
4.220	-0.5464	-0.5962	-1.84
4.225	-0.5463	-0.5960	-1.85
4.230	-0.5460	-0.5964	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.235	-0.5459	-0.5966	-1.84
4.240	-0.5456	-0.5967	-1.85
4.245	-0.5452	-0.5971	-1.85
4.250	-0.5450	-0.5976	-1.84
4.255	-0.5447	-0.5980	-1.84
4.260	-0.5442	-0.5982	-1.84
4.265	-0.5438	-0.5988	-1.84
4.270	-0.5437	-0.5992	-1.84
4.275	-0.5431	-0.5996	-1.84
4.280	-0.5431	-0.6000	-1.84
4.285	-0.5425	-0.6007	-1.84
4.290	-0.5423	-0.6010	-1.84
4.295	-0.5418	-0.6015	-1.83
4.300	-0.5418	-0.6022	-1.83
4.305	-0.5416	-0.6026	-1.83
4.310	-0.5413	-0.6032	-1.83
4.315	-0.5410	-0.6037	-1.82
4.320	-0.5406	-0.6043	-1.82
4.325	-0.5405	-0.6051	-1.82
4.330	-0.5401	-0.6056	-1.81
4.335	-0.5400	-0.6064	-1.81
4.340	-0.5399	-0.6072	-1.80
4.345	-0.5396	-0.6082	-1.80
4.350	-0.5395	-0.6089	-1.79
4.355	-0.5390	-0.6098	-1.79
4.360	-0.5386	-0.6107	-1.78
4.365	-0.5381	-0.6117	-1.78
4.370	-0.5378	-0.6124	-1.78
4.375	-0.5370	-0.6135	-1.77
4.380	-0.5367	-0.6143	-1.77
4.385	-0.5362	-0.6152	-1.77
4.390	-0.5353	-0.6161	-1.76
4.395	-0.5347	-0.6168	-1.76
4.400	-0.5343	-0.6175	-1.76
4.405	-0.5337	-0.6180	-1.76
4.410	-0.5330	-0.6186	-1.76
4.415	-0.5322	-0.6190	-1.76
4.420	-0.5321	-0.6192	-1.76
4.425	-0.5315	-0.6193	-1.77
4.430	-0.5311	-0.6200	-1.76
4.435	-0.5308	-0.6199	-1.77
4.440	-0.5307	-0.6200	-1.77
4.445	-0.5304	-0.6206	-1.76
4.450	-0.5304	-0.6205	-1.76
4.455	-0.5306	-0.6214	-1.75
4.460	-0.5297	-0.6218	-1.76

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.465	-0.5295	-0.6227	-1.75
4.470	-0.5292	-0.6239	-1.74
4.475	-0.5289	-0.6248	-1.74
4.480	-0.5285	-0.6263	-1.73
4.485	-0.5285	-0.6275	-1.72
4.490	-0.5279	-0.6293	-1.71
4.495	-0.5279	-0.6306	-1.70
4.500	-0.5274	-0.6315	-1.69
4.505	-0.5270	-0.6326	-1.69
4.510	-0.5266	-0.6327	-1.69
4.515	-0.5259	-0.6331	-1.69
4.520	-0.5255	-0.6337	-1.69
4.525	-0.5246	-0.6335	-1.70
4.530	-0.5248	-0.6328	-1.70
4.535	-0.5244	-0.6322	-1.71
4.540	-0.5246	-0.6318	-1.71
4.545	-0.5245	-0.6307	-1.72
4.550	-0.5249	-0.6304	-1.72
4.555	-0.5251	-0.6297	-1.72
4.560	-0.5255	-0.6293	-1.73
4.565	-0.5255	-0.6286	-1.73
4.570	-0.5258	-0.6277	-1.74
4.575	-0.5263	-0.6268	-1.74
4.580	-0.5264	-0.6261	-1.75
4.585	-0.5269	-0.6250	-1.75
4.590	-0.5273	-0.6245	-1.75
4.595	-0.5275	-0.6237	-1.76
4.600	-0.5282	-0.6227	-1.76
4.605	-0.5283	-0.6218	-1.77
4.610	-0.5289	-0.6209	-1.77
4.615	-0.5295	-0.6196	-1.78
4.620	-0.5299	-0.6189	-1.78
4.625	-0.5305	-0.6178	-1.78
4.630	-0.5307	-0.6172	-1.79
4.635	-0.5309	-0.6161	-1.80
4.640	-0.5309	-0.6151	-1.80
4.645	-0.5312	-0.6143	-1.81
4.650	-0.5314	-0.6136	-1.81
4.655	-0.5318	-0.6126	-1.82
4.660	-0.5317	-0.6118	-1.82
4.665	-0.5317	-0.6109	-1.83
4.670	-0.5317	-0.6102	-1.84
4.675	-0.5315	-0.6095	-1.84
4.680	-0.5313	-0.6086	-1.85
4.685	-0.5309	-0.6083	-1.86
4.690	-0.5307	-0.6076	-1.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.695	-0.5305	-0.6068	-1.87
4.700	-0.5301	-0.6063	-1.88
4.705	-0.5299	-0.6054	-1.89
4.710	-0.5300	-0.6048	-1.89
4.715	-0.5299	-0.6039	-1.90
4.720	-0.5299	-0.6028	-1.91
4.725	-0.5303	-0.6021	-1.91
4.730	-0.5310	-0.6007	-1.92
4.735	-0.5316	-0.6000	-1.92
4.740	-0.5325	-0.5987	-1.92
4.745	-0.5329	-0.5975	-1.93
4.750	-0.5337	-0.5957	-1.94
4.755	-0.5348	-0.5945	-1.94
4.760	-0.5354	-0.5933	-1.95
4.765	-0.5361	-0.5924	-1.95
4.770	-0.5363	-0.5916	-1.95
4.775	-0.5361	-0.5908	-1.96
4.780	-0.5363	-0.5898	-1.97
4.785	-0.5361	-0.5891	-1.98
4.790	-0.5351	-0.5887	-1.99
4.795	-0.5345	-0.5880	-2.00
4.800	-0.5337	-0.5874	-2.01
4.805	-0.5329	-0.5868	-2.02
4.810	-0.5320	-0.5858	-2.03
4.815	-0.5312	-0.5858	-2.04
4.820	-0.5301	-0.5853	-2.05
4.825	-0.5298	-0.5852	-2.05
4.830	-0.5291	-0.5850	-2.06
4.835	-0.5287	-0.5849	-2.07
4.840	-0.5284	-0.5849	-2.07
4.845	-0.5285	-0.5848	-2.07
4.850	-0.5283	-0.5851	-2.07
4.855	-0.5284	-0.5856	-2.06
4.860	-0.5286	-0.5860	-2.06
4.865	-0.5292	-0.5859	-2.05
4.870	-0.5293	-0.5867	-2.05
4.875	-0.5298	-0.5874	-2.04
4.880	-0.5302	-0.5880	-2.03
4.885	-0.5308	-0.5885	-2.02
4.890	-0.5315	-0.5892	-2.01
4.895	-0.5319	-0.5897	-2.00
4.900	-0.5325	-0.5899	-2.00
4.905	-0.5329	-0.5904	-1.99
4.910	-0.5335	-0.5903	-1.99
4.915	-0.5341	-0.5908	-1.98
4.920	-0.5348	-0.5910	-1.97

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.925	-0.5352	-0.5911	-1.97
4.930	-0.5356	-0.5913	-1.96
4.935	-0.5361	-0.5916	-1.96
4.940	-0.5366	-0.5917	-1.95
4.945	-0.5370	-0.5920	-1.95
4.950	-0.5371	-0.5922	-1.94
4.955	-0.5377	-0.5923	-1.94
4.960	-0.5381	-0.5925	-1.93
4.965	-0.5384	-0.5928	-1.93
4.970	-0.5387	-0.5931	-1.92
4.975	-0.5392	-0.5933	-1.92
4.980	-0.5397	-0.5936	-1.91
4.985	-0.5399	-0.5939	-1.91
4.990	-0.5405	-0.5944	-1.90
4.995	-0.5408	-0.5946	-1.90
5.000	-0.5414	-0.5949	-1.89
5.005	-0.5418	-0.5951	-1.89
5.010	-0.5425	-0.5951	-1.88
5.015	-0.5426	-0.5953	-1.88
5.020	-0.5431	-0.5954	-1.87
5.025	-0.5432	-0.5956	-1.87
5.030	-0.5437	-0.5950	-1.87
5.035	-0.5438	-0.5955	-1.87
5.040	-0.5439	-0.5948	-1.87
5.045	-0.5438	-0.5952	-1.87
5.050	-0.5436	-0.5945	-1.88
5.055	-0.5436	-0.5949	-1.88
5.060	-0.5433	-0.5943	-1.88
5.065	-0.5438	-0.5945	-1.88
5.070	-0.5434	-0.5941	-1.88
5.075	-0.5444	-0.5940	-1.88
5.080	-0.5440	-0.5941	-1.88
5.085	-0.5452	-0.5936	-1.87
5.090	-0.5452	-0.5935	-1.87
5.095	-0.5464	-0.5933	-1.87
5.100	-0.5468	-0.5929	-1.87
5.105	-0.5477	-0.5922	-1.87
5.110	-0.5484	-0.5921	-1.86
5.115	-0.5493	-0.5913	-1.86
5.120	-0.5500	-0.5909	-1.86
5.125	-0.5512	-0.5900	-1.86
5.130	-0.5520	-0.5898	-1.85
5.135	-0.5527	-0.5888	-1.86
5.140	-0.5538	-0.5879	-1.86
5.145	-0.5552	-0.5870	-1.85
5.150	-0.5559	-0.5862	-1.85

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.155	-0.5573	-0.5846	-1.85
5.160	-0.5585	-0.5832	-1.86
5.165	-0.5595	-0.5817	-1.86
5.170	-0.5609	-0.5804	-1.86
5.175	-0.5623	-0.5785	-1.87
5.180	-0.5632	-0.5766	-1.87
5.185	-0.5643	-0.5752	-1.88
5.190	-0.5655	-0.5727	-1.89
5.195	-0.5665	-0.5710	-1.89
5.200	-0.5676	-0.5685	-1.90
5.205	-0.5687	-0.5665	-1.91
5.210	-0.5695	-0.5645	-1.92
5.215	-0.5706	-0.5623	-1.93
5.220	-0.5715	-0.5600	-1.94
5.225	-0.5728	-0.5574	-1.95
5.230	-0.5736	-0.5553	-1.96
5.235	-0.5745	-0.5531	-1.97
5.240	-0.5755	-0.5509	-1.97
5.245	-0.5764	-0.5489	-1.98
5.250	-0.5772	-0.5466	-1.99
5.255	-0.5784	-0.5443	-2.00
5.260	-0.5792	-0.5421	-2.01
5.265	-0.5803	-0.5401	-2.02
5.270	-0.5814	-0.5381	-2.02
5.275	-0.5822	-0.5360	-2.03
5.280	-0.5829	-0.5341	-2.04
5.285	-0.5837	-0.5320	-2.05
5.290	-0.5848	-0.5303	-2.05
5.295	-0.5856	-0.5283	-2.06
5.300	-0.5863	-0.5267	-2.07
5.305	-0.5874	-0.5250	-2.07
5.310	-0.5878	-0.5233	-2.08
5.315	-0.5886	-0.5217	-2.09
5.320	-0.5894	-0.5198	-2.09
5.325	-0.5900	-0.5182	-2.10
5.330	-0.5905	-0.5167	-2.11
5.335	-0.5910	-0.5150	-2.11
5.340	-0.5915	-0.5133	-2.12
5.345	-0.5925	-0.5116	-2.13
5.350	-0.5926	-0.5101	-2.14
5.355	-0.5933	-0.5083	-2.14
5.360	-0.5937	-0.5067	-2.15
5.365	-0.5946	-0.5051	-2.16
5.370	-0.5950	-0.5034	-2.16
5.375	-0.5959	-0.5014	-2.17
5.380	-0.5967	-0.5001	-2.17

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.385	-0.5972	-0.4983	-2.18
5.390	-0.5981	-0.4971	-2.18
5.395	-0.5988	-0.4959	-2.19
5.400	-0.5999	-0.4944	-2.19
5.405	-0.6007	-0.4932	-2.19
5.410	-0.6017	-0.4920	-2.19
5.415	-0.6026	-0.4909	-2.19
5.420	-0.6033	-0.4895	-2.19
5.425	-0.6046	-0.4884	-2.19
5.430	-0.6051	-0.4874	-2.19
5.435	-0.6057	-0.4866	-2.19
5.440	-0.6066	-0.4858	-2.19
5.445	-0.6072	-0.4848	-2.19
5.450	-0.6079	-0.4839	-2.19
5.455	-0.6084	-0.4831	-2.19
5.460	-0.6093	-0.4825	-2.19
5.465	-0.6098	-0.4816	-2.19
5.470	-0.6104	-0.4811	-2.19
5.475	-0.6109	-0.4803	-2.19
5.480	-0.6117	-0.4797	-2.19
5.485	-0.6125	-0.4788	-2.19
5.490	-0.6131	-0.4782	-2.19
5.495	-0.6139	-0.4775	-2.18
5.500	-0.6147	-0.4769	-2.18
5.505	-0.6154	-0.4764	-2.18
5.510	-0.6168	-0.4762	-2.17
5.515	-0.6177	-0.4761	-2.16
5.520	-0.6186	-0.4760	-2.15
5.525	-0.6193	-0.4760	-2.15
5.530	-0.6203	-0.4760	-2.14
5.535	-0.6210	-0.4761	-2.13
5.540	-0.6217	-0.4762	-2.12
5.545	-0.6221	-0.4763	-2.12
5.550	-0.6229	-0.4765	-2.11
5.555	-0.6236	-0.4766	-2.10
5.560	-0.6243	-0.4768	-2.10
5.565	-0.6250	-0.4768	-2.09
5.570	-0.6259	-0.4768	-2.08
5.575	-0.6266	-0.4770	-2.08
5.580	-0.6273	-0.4771	-2.07
5.585	-0.6281	-0.4770	-2.06
5.590	-0.6290	-0.4771	-2.05
5.595	-0.6299	-0.4771	-2.04
5.600	-0.6309	-0.4772	-2.04
5.605	-0.6319	-0.4774	-2.03
5.610	-0.6330	-0.4775	-2.02

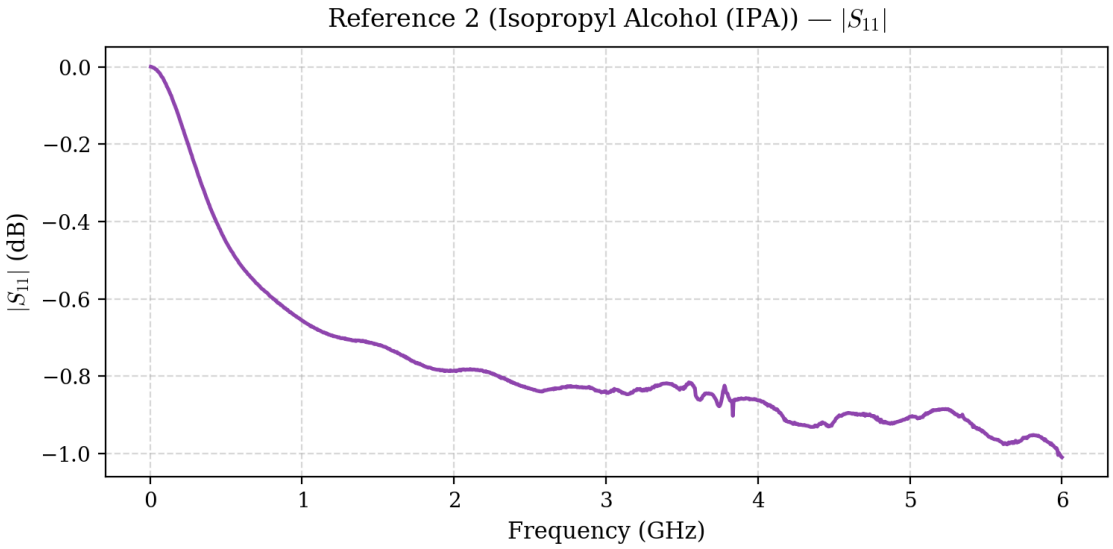
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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.615	-0.6337	-0.4779	-2.01
5.620	-0.6348	-0.4780	-2.00
5.625	-0.6356	-0.4785	-1.99
5.630	-0.6364	-0.4787	-1.98
5.635	-0.6370	-0.4789	-1.97
5.640	-0.6376	-0.4791	-1.96
5.645	-0.6382	-0.4794	-1.96
5.650	-0.6389	-0.4794	-1.95
5.655	-0.6393	-0.4795	-1.95
5.660	-0.6399	-0.4795	-1.94
5.665	-0.6404	-0.4797	-1.94
5.670	-0.6410	-0.4798	-1.93
5.675	-0.6415	-0.4797	-1.93
5.680	-0.6424	-0.4796	-1.92
5.685	-0.6421	-0.4794	-1.92
5.690	-0.6425	-0.4791	-1.92
5.695	-0.6430	-0.4791	-1.92
5.700	-0.6430	-0.4789	-1.92
5.705	-0.6436	-0.4786	-1.92
5.710	-0.6435	-0.4784	-1.92
5.715	-0.6439	-0.4784	-1.92
5.720	-0.6442	-0.4784	-1.91
5.725	-0.6441	-0.4781	-1.92
5.730	-0.6445	-0.4775	-1.92
5.735	-0.6446	-0.4774	-1.91
5.740	-0.6447	-0.4767	-1.92
5.745	-0.6453	-0.4768	-1.91
5.750	-0.6450	-0.4758	-1.92
5.755	-0.6450	-0.4754	-1.92
5.760	-0.6448	-0.4748	-1.93
5.765	-0.6454	-0.4741	-1.93
5.770	-0.6450	-0.4735	-1.94
5.775	-0.6453	-0.4721	-1.94
5.780	-0.6448	-0.4717	-1.95
5.785	-0.6452	-0.4705	-1.95
5.790	-0.6452	-0.4698	-1.96
5.795	-0.6450	-0.4685	-1.97
5.800	-0.6448	-0.4671	-1.98
5.805	-0.6448	-0.4661	-1.99
5.810	-0.6443	-0.4649	-2.00
5.815	-0.6444	-0.4637	-2.00
5.820	-0.6445	-0.4623	-2.01
5.825	-0.6443	-0.4613	-2.02
5.830	-0.6442	-0.4598	-2.03
5.835	-0.6441	-0.4584	-2.04
5.840	-0.6438	-0.4569	-2.05

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.845	-0.6441	-0.4555	-2.06
5.850	-0.6441	-0.4544	-2.07
5.855	-0.6438	-0.4530	-2.08
5.860	-0.6440	-0.4514	-2.09
5.865	-0.6441	-0.4504	-2.09
5.870	-0.6442	-0.4487	-2.10
5.875	-0.6443	-0.4477	-2.11
5.880	-0.6446	-0.4463	-2.11
5.885	-0.6447	-0.4450	-2.12
5.890	-0.6447	-0.4439	-2.13
5.895	-0.6447	-0.4429	-2.13
5.900	-0.6449	-0.4419	-2.14
5.905	-0.6451	-0.4410	-2.14
5.910	-0.6452	-0.4404	-2.15
5.915	-0.6449	-0.4393	-2.15
5.920	-0.6449	-0.4381	-2.16
5.925	-0.6450	-0.4373	-2.17
5.930	-0.6447	-0.4361	-2.18
5.935	-0.6446	-0.4351	-2.18
5.940	-0.6448	-0.4341	-2.19
5.945	-0.6444	-0.4334	-2.20
5.950	-0.6445	-0.4322	-2.20
5.955	-0.6448	-0.4312	-2.21
5.960	-0.6451	-0.4303	-2.21
5.965	-0.6454	-0.4296	-2.21
5.970	-0.6460	-0.4286	-2.21
5.975	-0.6468	-0.4279	-2.21
5.980	-0.6469	-0.4282	-2.20
5.985	-0.6475	-0.4276	-2.20
5.990	-0.6480	-0.4277	-2.20
5.995	-0.6486	-0.4279	-2.19
6.000	-0.6488	-0.4281	-2.19

3.4 Reference 2 (Isopropyl Alcohol (IPA))



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0001	-0.0016	0.00
0.010	1.0000	-0.0028	0.00
0.015	0.9999	-0.0040	-0.00
0.020	0.9998	-0.0053	-0.00
0.025	0.9997	-0.0064	-0.00
0.030	0.9996	-0.0077	-0.00
0.035	0.9993	-0.0089	-0.01
0.040	0.9992	-0.0102	-0.01
0.045	0.9991	-0.0112	-0.01
0.050	0.9986	-0.0125	-0.01
0.055	0.9985	-0.0137	-0.01
0.060	0.9982	-0.0148	-0.01
0.065	0.9980	-0.0161	-0.02
0.070	0.9975	-0.0172	-0.02
0.075	0.9970	-0.0183	-0.02
0.080	0.9968	-0.0194	-0.03
0.085	0.9965	-0.0206	-0.03
0.090	0.9959	-0.0216	-0.03
0.095	0.9955	-0.0225	-0.04
0.100	0.9950	-0.0237	-0.04
0.105	0.9946	-0.0248	-0.04
0.110	0.9939	-0.0257	-0.05
0.115	0.9935	-0.0268	-0.05
0.120	0.9931	-0.0278	-0.06
0.125	0.9925	-0.0287	-0.06
0.130	0.9919	-0.0297	-0.07

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.135	0.9914	-0.0305	-0.07
0.140	0.9910	-0.0315	-0.07
0.145	0.9902	-0.0325	-0.08
0.150	0.9896	-0.0334	-0.09
0.155	0.9890	-0.0341	-0.09
0.160	0.9883	-0.0351	-0.10
0.165	0.9878	-0.0358	-0.10
0.170	0.9871	-0.0366	-0.11
0.175	0.9866	-0.0374	-0.11
0.180	0.9858	-0.0382	-0.12
0.185	0.9852	-0.0388	-0.12
0.190	0.9844	-0.0397	-0.13
0.195	0.9837	-0.0405	-0.14
0.200	0.9831	-0.0412	-0.14
0.205	0.9824	-0.0418	-0.15
0.210	0.9817	-0.0424	-0.15
0.215	0.9809	-0.0433	-0.16
0.220	0.9804	-0.0438	-0.16
0.225	0.9795	-0.0444	-0.17
0.230	0.9788	-0.0449	-0.18
0.235	0.9782	-0.0455	-0.18
0.240	0.9775	-0.0461	-0.19
0.245	0.9768	-0.0468	-0.19
0.250	0.9762	-0.0473	-0.20
0.255	0.9754	-0.0477	-0.21
0.260	0.9746	-0.0482	-0.21
0.265	0.9740	-0.0487	-0.22
0.270	0.9733	-0.0493	-0.22
0.275	0.9728	-0.0496	-0.23
0.280	0.9718	-0.0502	-0.24
0.285	0.9712	-0.0505	-0.24
0.290	0.9705	-0.0509	-0.25
0.295	0.9699	-0.0514	-0.25
0.300	0.9692	-0.0517	-0.26
0.305	0.9686	-0.0521	-0.26
0.310	0.9679	-0.0525	-0.27
0.315	0.9672	-0.0527	-0.28
0.320	0.9665	-0.0530	-0.28
0.325	0.9660	-0.0535	-0.29
0.330	0.9652	-0.0538	-0.29
0.335	0.9645	-0.0541	-0.30
0.340	0.9639	-0.0543	-0.31
0.345	0.9634	-0.0545	-0.31
0.350	0.9627	-0.0548	-0.32
0.355	0.9622	-0.0552	-0.32
0.360	0.9615	-0.0554	-0.33

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.365	0.9609	-0.0556	-0.33
0.370	0.9603	-0.0558	-0.34
0.375	0.9597	-0.0560	-0.34
0.380	0.9592	-0.0563	-0.35
0.385	0.9585	-0.0564	-0.35
0.390	0.9581	-0.0566	-0.36
0.395	0.9573	-0.0569	-0.36
0.400	0.9568	-0.0570	-0.37
0.405	0.9564	-0.0572	-0.37
0.410	0.9557	-0.0572	-0.38
0.415	0.9553	-0.0574	-0.38
0.420	0.9548	-0.0576	-0.39
0.425	0.9543	-0.0577	-0.39
0.430	0.9538	-0.0579	-0.39
0.435	0.9532	-0.0580	-0.40
0.440	0.9528	-0.0581	-0.40
0.445	0.9524	-0.0583	-0.41
0.450	0.9519	-0.0584	-0.41
0.455	0.9513	-0.0584	-0.42
0.460	0.9510	-0.0585	-0.42
0.465	0.9504	-0.0586	-0.42
0.470	0.9500	-0.0587	-0.43
0.475	0.9495	-0.0589	-0.43
0.480	0.9491	-0.0589	-0.44
0.485	0.9487	-0.0590	-0.44
0.490	0.9482	-0.0590	-0.44
0.495	0.9478	-0.0592	-0.45
0.500	0.9474	-0.0592	-0.45
0.505	0.9470	-0.0592	-0.46
0.510	0.9466	-0.0592	-0.46
0.515	0.9462	-0.0594	-0.46
0.520	0.9458	-0.0594	-0.47
0.525	0.9455	-0.0594	-0.47
0.530	0.9451	-0.0595	-0.47
0.535	0.9448	-0.0596	-0.48
0.540	0.9445	-0.0596	-0.48
0.545	0.9442	-0.0597	-0.48
0.550	0.9437	-0.0597	-0.49
0.555	0.9434	-0.0598	-0.49
0.560	0.9430	-0.0599	-0.49
0.565	0.9427	-0.0599	-0.49
0.570	0.9424	-0.0600	-0.50
0.575	0.9422	-0.0599	-0.50
0.580	0.9419	-0.0600	-0.50
0.585	0.9414	-0.0599	-0.51
0.590	0.9412	-0.0600	-0.51

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.595	0.9409	-0.0601	-0.51
0.600	0.9406	-0.0601	-0.51
0.605	0.9403	-0.0602	-0.52
0.610	0.9400	-0.0602	-0.52
0.615	0.9398	-0.0602	-0.52
0.620	0.9395	-0.0602	-0.52
0.625	0.9392	-0.0603	-0.53
0.630	0.9390	-0.0602	-0.53
0.635	0.9388	-0.0603	-0.53
0.640	0.9385	-0.0604	-0.53
0.645	0.9382	-0.0604	-0.54
0.650	0.9380	-0.0605	-0.54
0.655	0.9378	-0.0604	-0.54
0.660	0.9375	-0.0604	-0.54
0.665	0.9373	-0.0604	-0.54
0.670	0.9370	-0.0605	-0.55
0.675	0.9368	-0.0605	-0.55
0.680	0.9366	-0.0607	-0.55
0.685	0.9363	-0.0606	-0.55
0.690	0.9361	-0.0606	-0.56
0.695	0.9358	-0.0606	-0.56
0.700	0.9356	-0.0606	-0.56
0.705	0.9355	-0.0607	-0.56
0.710	0.9352	-0.0608	-0.56
0.715	0.9350	-0.0608	-0.57
0.720	0.9349	-0.0608	-0.57
0.725	0.9345	-0.0608	-0.57
0.730	0.9343	-0.0607	-0.57
0.735	0.9341	-0.0609	-0.57
0.740	0.9340	-0.0608	-0.58
0.745	0.9338	-0.0609	-0.58
0.750	0.9336	-0.0610	-0.58
0.755	0.9334	-0.0610	-0.58
0.760	0.9332	-0.0610	-0.58
0.765	0.9331	-0.0610	-0.58
0.770	0.9329	-0.0611	-0.58
0.775	0.9327	-0.0610	-0.59
0.780	0.9325	-0.0612	-0.59
0.785	0.9322	-0.0612	-0.59
0.790	0.9320	-0.0611	-0.59
0.795	0.9318	-0.0613	-0.59
0.800	0.9317	-0.0613	-0.60
0.805	0.9315	-0.0614	-0.60
0.810	0.9312	-0.0614	-0.60
0.815	0.9312	-0.0614	-0.60
0.820	0.9310	-0.0615	-0.60

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.825	0.9307	-0.0615	-0.60
0.830	0.9307	-0.0615	-0.60
0.835	0.9303	-0.0617	-0.61
0.840	0.9301	-0.0617	-0.61
0.845	0.9301	-0.0616	-0.61
0.850	0.9299	-0.0617	-0.61
0.855	0.9298	-0.0619	-0.61
0.860	0.9295	-0.0618	-0.62
0.865	0.9294	-0.0621	-0.62
0.870	0.9293	-0.0621	-0.62
0.875	0.9291	-0.0620	-0.62
0.880	0.9287	-0.0620	-0.62
0.885	0.9286	-0.0621	-0.62
0.890	0.9286	-0.0622	-0.62
0.895	0.9283	-0.0623	-0.63
0.900	0.9282	-0.0621	-0.63
0.905	0.9281	-0.0623	-0.63
0.910	0.9278	-0.0624	-0.63
0.915	0.9277	-0.0625	-0.63
0.920	0.9274	-0.0624	-0.63
0.925	0.9275	-0.0625	-0.63
0.930	0.9273	-0.0626	-0.64
0.935	0.9270	-0.0627	-0.64
0.940	0.9269	-0.0627	-0.64
0.945	0.9267	-0.0628	-0.64
0.950	0.9266	-0.0627	-0.64
0.955	0.9265	-0.0628	-0.64
0.960	0.9262	-0.0629	-0.65
0.965	0.9262	-0.0630	-0.65
0.970	0.9260	-0.0629	-0.65
0.975	0.9258	-0.0630	-0.65
0.980	0.9258	-0.0630	-0.65
0.985	0.9256	-0.0631	-0.65
0.990	0.9254	-0.0632	-0.65
0.995	0.9253	-0.0633	-0.65
1.000	0.9251	-0.0633	-0.66
1.005	0.9250	-0.0634	-0.66
1.010	0.9249	-0.0636	-0.66
1.015	0.9246	-0.0636	-0.66
1.020	0.9245	-0.0636	-0.66
1.025	0.9244	-0.0636	-0.66
1.030	0.9243	-0.0637	-0.66
1.035	0.9241	-0.0638	-0.66
1.040	0.9240	-0.0638	-0.67
1.045	0.9239	-0.0639	-0.67
1.050	0.9237	-0.0639	-0.67

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.055	0.9235	-0.0640	-0.67
1.060	0.9235	-0.0640	-0.67
1.065	0.9233	-0.0640	-0.67
1.070	0.9233	-0.0642	-0.67
1.075	0.9232	-0.0643	-0.67
1.080	0.9230	-0.0643	-0.67
1.085	0.9229	-0.0643	-0.68
1.090	0.9228	-0.0644	-0.68
1.095	0.9226	-0.0645	-0.68
1.100	0.9226	-0.0646	-0.68
1.105	0.9223	-0.0647	-0.68
1.110	0.9223	-0.0647	-0.68
1.115	0.9222	-0.0647	-0.68
1.120	0.9222	-0.0648	-0.68
1.125	0.9220	-0.0648	-0.68
1.130	0.9219	-0.0649	-0.68
1.135	0.9218	-0.0651	-0.69
1.140	0.9216	-0.0650	-0.69
1.145	0.9217	-0.0651	-0.69
1.150	0.9216	-0.0650	-0.69
1.155	0.9215	-0.0652	-0.69
1.160	0.9213	-0.0653	-0.69
1.165	0.9214	-0.0654	-0.69
1.170	0.9212	-0.0654	-0.69
1.175	0.9210	-0.0655	-0.69
1.180	0.9210	-0.0657	-0.69
1.185	0.9210	-0.0655	-0.69
1.190	0.9208	-0.0656	-0.69
1.195	0.9208	-0.0657	-0.69
1.200	0.9208	-0.0657	-0.69
1.205	0.9206	-0.0658	-0.70
1.210	0.9206	-0.0658	-0.70
1.215	0.9205	-0.0659	-0.70
1.220	0.9205	-0.0658	-0.70
1.225	0.9204	-0.0659	-0.70
1.230	0.9204	-0.0660	-0.70
1.235	0.9202	-0.0661	-0.70
1.240	0.9202	-0.0661	-0.70
1.245	0.9203	-0.0661	-0.70
1.250	0.9201	-0.0663	-0.70
1.255	0.9200	-0.0663	-0.70
1.260	0.9200	-0.0663	-0.70
1.265	0.9200	-0.0664	-0.70
1.270	0.9199	-0.0664	-0.70
1.275	0.9199	-0.0665	-0.70
1.280	0.9198	-0.0665	-0.70

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.285	0.9199	-0.0666	-0.70
1.290	0.9197	-0.0666	-0.70
1.295	0.9197	-0.0667	-0.70
1.300	0.9196	-0.0669	-0.71
1.305	0.9197	-0.0669	-0.70
1.310	0.9196	-0.0669	-0.71
1.315	0.9196	-0.0669	-0.71
1.320	0.9194	-0.0670	-0.71
1.325	0.9195	-0.0671	-0.71
1.330	0.9193	-0.0671	-0.71
1.335	0.9193	-0.0672	-0.71
1.340	0.9193	-0.0671	-0.71
1.345	0.9192	-0.0672	-0.71
1.350	0.9193	-0.0674	-0.71
1.355	0.9191	-0.0673	-0.71
1.360	0.9193	-0.0674	-0.71
1.365	0.9193	-0.0675	-0.71
1.370	0.9193	-0.0675	-0.71
1.375	0.9193	-0.0675	-0.71
1.380	0.9192	-0.0676	-0.71
1.385	0.9191	-0.0677	-0.71
1.390	0.9190	-0.0679	-0.71
1.395	0.9192	-0.0680	-0.71
1.400	0.9192	-0.0680	-0.71
1.405	0.9190	-0.0680	-0.71
1.410	0.9190	-0.0681	-0.71
1.415	0.9189	-0.0681	-0.71
1.420	0.9190	-0.0683	-0.71
1.425	0.9187	-0.0682	-0.71
1.430	0.9187	-0.0684	-0.71
1.435	0.9187	-0.0683	-0.71
1.440	0.9187	-0.0685	-0.71
1.445	0.9186	-0.0687	-0.71
1.450	0.9185	-0.0687	-0.71
1.455	0.9186	-0.0689	-0.71
1.460	0.9185	-0.0691	-0.71
1.465	0.9184	-0.0691	-0.71
1.470	0.9183	-0.0691	-0.72
1.475	0.9182	-0.0692	-0.72
1.480	0.9182	-0.0692	-0.72
1.485	0.9181	-0.0695	-0.72
1.490	0.9181	-0.0695	-0.72
1.495	0.9180	-0.0696	-0.72
1.500	0.9179	-0.0698	-0.72
1.505	0.9179	-0.0699	-0.72
1.510	0.9178	-0.0699	-0.72

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.515	0.9178	-0.0701	-0.72
1.520	0.9177	-0.0703	-0.72
1.525	0.9176	-0.0704	-0.72
1.530	0.9175	-0.0704	-0.72
1.535	0.9174	-0.0705	-0.72
1.540	0.9173	-0.0707	-0.72
1.545	0.9172	-0.0709	-0.72
1.550	0.9172	-0.0709	-0.72
1.555	0.9171	-0.0710	-0.73
1.560	0.9169	-0.0712	-0.73
1.565	0.9168	-0.0714	-0.73
1.570	0.9166	-0.0714	-0.73
1.575	0.9167	-0.0716	-0.73
1.580	0.9165	-0.0715	-0.73
1.585	0.9163	-0.0718	-0.73
1.590	0.9162	-0.0719	-0.73
1.595	0.9161	-0.0721	-0.73
1.600	0.9161	-0.0721	-0.73
1.605	0.9160	-0.0723	-0.74
1.610	0.9158	-0.0723	-0.74
1.615	0.9157	-0.0725	-0.74
1.620	0.9157	-0.0727	-0.74
1.625	0.9157	-0.0729	-0.74
1.630	0.9153	-0.0730	-0.74
1.635	0.9153	-0.0731	-0.74
1.640	0.9152	-0.0731	-0.74
1.645	0.9150	-0.0734	-0.74
1.650	0.9148	-0.0736	-0.75
1.655	0.9148	-0.0736	-0.75
1.660	0.9147	-0.0738	-0.75
1.665	0.9147	-0.0740	-0.75
1.670	0.9143	-0.0741	-0.75
1.675	0.9143	-0.0743	-0.75
1.680	0.9143	-0.0743	-0.75
1.685	0.9139	-0.0744	-0.75
1.690	0.9141	-0.0746	-0.75
1.695	0.9138	-0.0746	-0.75
1.700	0.9137	-0.0748	-0.75
1.705	0.9137	-0.0749	-0.76
1.710	0.9134	-0.0750	-0.76
1.715	0.9132	-0.0750	-0.76
1.720	0.9132	-0.0754	-0.76
1.725	0.9130	-0.0753	-0.76
1.730	0.9130	-0.0754	-0.76
1.735	0.9129	-0.0756	-0.76
1.740	0.9128	-0.0757	-0.76

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.745	0.9125	-0.0760	-0.77
1.750	0.9124	-0.0759	-0.77
1.755	0.9125	-0.0761	-0.77
1.760	0.9123	-0.0762	-0.77
1.765	0.9123	-0.0763	-0.77
1.770	0.9121	-0.0764	-0.77
1.775	0.9122	-0.0766	-0.77
1.780	0.9120	-0.0766	-0.77
1.785	0.9119	-0.0768	-0.77
1.790	0.9118	-0.0768	-0.77
1.795	0.9117	-0.0769	-0.77
1.800	0.9115	-0.0770	-0.77
1.805	0.9115	-0.0772	-0.77
1.810	0.9113	-0.0772	-0.78
1.815	0.9113	-0.0774	-0.78
1.820	0.9112	-0.0774	-0.78
1.825	0.9111	-0.0776	-0.78
1.830	0.9110	-0.0777	-0.78
1.835	0.9110	-0.0779	-0.78
1.840	0.9110	-0.0780	-0.78
1.845	0.9109	-0.0781	-0.78
1.850	0.9108	-0.0780	-0.78
1.855	0.9107	-0.0782	-0.78
1.860	0.9106	-0.0780	-0.78
1.865	0.9106	-0.0781	-0.78
1.870	0.9106	-0.0782	-0.78
1.875	0.9105	-0.0784	-0.78
1.880	0.9104	-0.0786	-0.78
1.885	0.9104	-0.0787	-0.78
1.890	0.9104	-0.0788	-0.78
1.895	0.9103	-0.0789	-0.78
1.900	0.9103	-0.0790	-0.78
1.905	0.9103	-0.0790	-0.78
1.910	0.9101	-0.0791	-0.79
1.915	0.9102	-0.0793	-0.78
1.920	0.9101	-0.0793	-0.79
1.925	0.9102	-0.0795	-0.78
1.930	0.9100	-0.0794	-0.79
1.935	0.9100	-0.0796	-0.79
1.940	0.9099	-0.0797	-0.79
1.945	0.9100	-0.0799	-0.79
1.950	0.9100	-0.0799	-0.79
1.955	0.9100	-0.0800	-0.79
1.960	0.9099	-0.0800	-0.79
1.965	0.9098	-0.0802	-0.79
1.970	0.9099	-0.0803	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.975	0.9100	-0.0804	-0.79
1.980	0.9099	-0.0804	-0.79
1.985	0.9099	-0.0804	-0.79
1.990	0.9098	-0.0806	-0.79
1.995	0.9099	-0.0806	-0.79
2.000	0.9099	-0.0808	-0.79
2.005	0.9099	-0.0807	-0.79
2.010	0.9099	-0.0809	-0.79
2.015	0.9097	-0.0810	-0.79
2.020	0.9099	-0.0811	-0.79
2.025	0.9100	-0.0812	-0.79
2.030	0.9099	-0.0812	-0.79
2.035	0.9101	-0.0814	-0.78
2.040	0.9101	-0.0814	-0.78
2.045	0.9099	-0.0816	-0.79
2.050	0.9101	-0.0816	-0.78
2.055	0.9102	-0.0816	-0.78
2.060	0.9101	-0.0819	-0.78
2.065	0.9102	-0.0818	-0.78
2.070	0.9100	-0.0821	-0.78
2.075	0.9100	-0.0821	-0.78
2.080	0.9101	-0.0822	-0.78
2.085	0.9101	-0.0824	-0.78
2.090	0.9101	-0.0824	-0.78
2.095	0.9100	-0.0825	-0.78
2.100	0.9100	-0.0828	-0.78
2.105	0.9102	-0.0829	-0.78
2.110	0.9100	-0.0828	-0.78
2.115	0.9101	-0.0830	-0.78
2.120	0.9099	-0.0831	-0.78
2.125	0.9100	-0.0833	-0.78
2.130	0.9100	-0.0835	-0.78
2.135	0.9100	-0.0835	-0.78
2.140	0.9099	-0.0837	-0.78
2.145	0.9098	-0.0839	-0.78
2.150	0.9098	-0.0840	-0.78
2.155	0.9099	-0.0842	-0.78
2.160	0.9097	-0.0843	-0.78
2.165	0.9097	-0.0844	-0.78
2.170	0.9096	-0.0846	-0.79
2.175	0.9097	-0.0847	-0.78
2.180	0.9096	-0.0849	-0.79
2.185	0.9096	-0.0850	-0.79
2.190	0.9094	-0.0852	-0.79
2.195	0.9093	-0.0855	-0.79
2.200	0.9092	-0.0856	-0.79

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.205	0.9092	-0.0857	-0.79
2.210	0.9093	-0.0859	-0.79
2.215	0.9091	-0.0860	-0.79
2.220	0.9090	-0.0862	-0.79
2.225	0.9090	-0.0864	-0.79
2.230	0.9089	-0.0866	-0.79
2.235	0.9089	-0.0869	-0.79
2.240	0.9087	-0.0869	-0.79
2.245	0.9086	-0.0872	-0.79
2.250	0.9084	-0.0874	-0.79
2.255	0.9085	-0.0874	-0.79
2.260	0.9084	-0.0877	-0.79
2.265	0.9083	-0.0879	-0.79
2.270	0.9081	-0.0880	-0.80
2.275	0.9080	-0.0882	-0.80
2.280	0.9079	-0.0884	-0.80
2.285	0.9078	-0.0886	-0.80
2.290	0.9077	-0.0888	-0.80
2.295	0.9077	-0.0888	-0.80
2.300	0.9073	-0.0889	-0.80
2.305	0.9075	-0.0894	-0.80
2.310	0.9071	-0.0893	-0.81
2.315	0.9072	-0.0895	-0.80
2.320	0.9070	-0.0898	-0.81
2.325	0.9069	-0.0901	-0.81
2.330	0.9070	-0.0903	-0.80
2.335	0.9068	-0.0904	-0.81
2.340	0.9066	-0.0904	-0.81
2.345	0.9066	-0.0908	-0.81
2.350	0.9065	-0.0910	-0.81
2.355	0.9062	-0.0912	-0.81
2.360	0.9062	-0.0913	-0.81
2.365	0.9062	-0.0914	-0.81
2.370	0.9059	-0.0916	-0.81
2.375	0.9057	-0.0916	-0.82
2.380	0.9058	-0.0920	-0.81
2.385	0.9055	-0.0921	-0.82
2.390	0.9056	-0.0923	-0.82
2.395	0.9054	-0.0926	-0.82
2.400	0.9053	-0.0929	-0.82
2.405	0.9051	-0.0929	-0.82
2.410	0.9050	-0.0931	-0.82
2.415	0.9050	-0.0933	-0.82
2.420	0.9050	-0.0935	-0.82
2.425	0.9047	-0.0936	-0.82
2.430	0.9047	-0.0936	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.435	0.9046	-0.0938	-0.82
2.440	0.9044	-0.0940	-0.83
2.445	0.9044	-0.0942	-0.83
2.450	0.9044	-0.0943	-0.83
2.455	0.9042	-0.0945	-0.83
2.460	0.9042	-0.0946	-0.83
2.465	0.9042	-0.0947	-0.83
2.470	0.9040	-0.0948	-0.83
2.475	0.9040	-0.0949	-0.83
2.480	0.9039	-0.0951	-0.83
2.485	0.9039	-0.0952	-0.83
2.490	0.9037	-0.0953	-0.83
2.495	0.9037	-0.0955	-0.83
2.500	0.9036	-0.0954	-0.83
2.505	0.9035	-0.0957	-0.83
2.510	0.9034	-0.0958	-0.83
2.515	0.9034	-0.0959	-0.83
2.520	0.9033	-0.0961	-0.83
2.525	0.9031	-0.0962	-0.84
2.530	0.9032	-0.0964	-0.84
2.535	0.9031	-0.0964	-0.84
2.540	0.9030	-0.0965	-0.84
2.545	0.9029	-0.0967	-0.84
2.550	0.9028	-0.0968	-0.84
2.555	0.9028	-0.0968	-0.84
2.560	0.9028	-0.0970	-0.84
2.565	0.9027	-0.0970	-0.84
2.570	0.9027	-0.0970	-0.84
2.575	0.9027	-0.0971	-0.84
2.580	0.9026	-0.0971	-0.84
2.585	0.9028	-0.0972	-0.84
2.590	0.9028	-0.0972	-0.84
2.595	0.9029	-0.0975	-0.84
2.600	0.9029	-0.0976	-0.84
2.605	0.9029	-0.0976	-0.84
2.610	0.9029	-0.0977	-0.84
2.615	0.9030	-0.0979	-0.84
2.620	0.9031	-0.0981	-0.83
2.625	0.9030	-0.0982	-0.83
2.630	0.9031	-0.0984	-0.83
2.635	0.9032	-0.0981	-0.83
2.640	0.9031	-0.0985	-0.83
2.645	0.9031	-0.0985	-0.83
2.650	0.9032	-0.0987	-0.83
2.655	0.9030	-0.0987	-0.83
2.660	0.9032	-0.0990	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.665	0.9033	-0.0989	-0.83
2.670	0.9031	-0.0988	-0.83
2.675	0.9032	-0.0993	-0.83
2.680	0.9032	-0.0993	-0.83
2.685	0.9032	-0.0995	-0.83
2.690	0.9032	-0.0995	-0.83
2.695	0.9032	-0.0997	-0.83
2.700	0.9033	-0.0999	-0.83
2.705	0.9032	-0.0999	-0.83
2.710	0.9035	-0.1001	-0.83
2.715	0.9035	-0.1002	-0.83
2.720	0.9035	-0.1003	-0.83
2.725	0.9034	-0.1003	-0.83
2.730	0.9034	-0.1004	-0.83
2.735	0.9035	-0.1005	-0.83
2.740	0.9034	-0.1005	-0.83
2.745	0.9037	-0.1008	-0.83
2.750	0.9036	-0.1007	-0.83
2.755	0.9036	-0.1008	-0.83
2.760	0.9037	-0.1008	-0.83
2.765	0.9035	-0.1010	-0.83
2.770	0.9037	-0.1010	-0.83
2.775	0.9034	-0.1011	-0.83
2.780	0.9035	-0.1011	-0.83
2.785	0.9035	-0.1013	-0.83
2.790	0.9035	-0.1012	-0.83
2.795	0.9035	-0.1015	-0.83
2.800	0.9035	-0.1015	-0.83
2.805	0.9035	-0.1017	-0.83
2.810	0.9033	-0.1018	-0.83
2.815	0.9034	-0.1020	-0.83
2.820	0.9033	-0.1020	-0.83
2.825	0.9032	-0.1020	-0.83
2.830	0.9032	-0.1021	-0.83
2.835	0.9033	-0.1022	-0.83
2.840	0.9032	-0.1024	-0.83
2.845	0.9032	-0.1027	-0.83
2.850	0.9031	-0.1026	-0.83
2.855	0.9031	-0.1027	-0.83
2.860	0.9031	-0.1030	-0.83
2.865	0.9030	-0.1031	-0.83
2.870	0.9030	-0.1033	-0.83
2.875	0.9030	-0.1033	-0.83
2.880	0.9030	-0.1034	-0.83
2.885	0.9031	-0.1036	-0.83
2.890	0.9032	-0.1036	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.895	0.9028	-0.1036	-0.83
2.900	0.9028	-0.1037	-0.83
2.905	0.9029	-0.1038	-0.83
2.910	0.9027	-0.1039	-0.83
2.915	0.9026	-0.1041	-0.83
2.920	0.9025	-0.1041	-0.83
2.925	0.9023	-0.1042	-0.84
2.930	0.9021	-0.1042	-0.84
2.935	0.9022	-0.1045	-0.84
2.940	0.9020	-0.1045	-0.84
2.945	0.9019	-0.1049	-0.84
2.950	0.9019	-0.1051	-0.84
2.955	0.9020	-0.1049	-0.84
2.960	0.9018	-0.1057	-0.84
2.965	0.9016	-0.1053	-0.84
2.970	0.9014	-0.1056	-0.84
2.975	0.9016	-0.1058	-0.84
2.980	0.9015	-0.1060	-0.84
2.985	0.9015	-0.1060	-0.84
2.990	0.9017	-0.1063	-0.84
2.995	0.9016	-0.1063	-0.84
3.000	0.9016	-0.1066	-0.84
3.005	0.9012	-0.1064	-0.84
3.010	0.9012	-0.1069	-0.84
3.015	0.9011	-0.1068	-0.84
3.020	0.9013	-0.1071	-0.84
3.025	0.9014	-0.1073	-0.84
3.030	0.9014	-0.1071	-0.84
3.035	0.9018	-0.1073	-0.84
3.040	0.9017	-0.1073	-0.84
3.045	0.9019	-0.1075	-0.84
3.050	0.9015	-0.1076	-0.84
3.055	0.9022	-0.1075	-0.83
3.060	0.9019	-0.1074	-0.84
3.065	0.9020	-0.1072	-0.84
3.070	0.9020	-0.1072	-0.84
3.075	0.9019	-0.1071	-0.84
3.080	0.9019	-0.1071	-0.84
3.085	0.9016	-0.1072	-0.84
3.090	0.9017	-0.1070	-0.84
3.095	0.9016	-0.1070	-0.84
3.100	0.9014	-0.1070	-0.84
3.105	0.9011	-0.1072	-0.84
3.110	0.9012	-0.1073	-0.84
3.115	0.9010	-0.1073	-0.84
3.120	0.9010	-0.1074	-0.84

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.125	0.9009	-0.1076	-0.84
3.130	0.9008	-0.1077	-0.85
3.135	0.9006	-0.1081	-0.85
3.140	0.9007	-0.1081	-0.85
3.145	0.9006	-0.1084	-0.85
3.150	0.9006	-0.1085	-0.85
3.155	0.9007	-0.1088	-0.85
3.160	0.9007	-0.1088	-0.85
3.165	0.9010	-0.1090	-0.84
3.170	0.9009	-0.1090	-0.84
3.175	0.9009	-0.1092	-0.84
3.180	0.9012	-0.1095	-0.84
3.185	0.9014	-0.1098	-0.84
3.190	0.9015	-0.1093	-0.84
3.195	0.9015	-0.1096	-0.84
3.200	0.9015	-0.1097	-0.84
3.205	0.9019	-0.1097	-0.83
3.210	0.9018	-0.1095	-0.83
3.215	0.9018	-0.1094	-0.83
3.220	0.9019	-0.1090	-0.83
3.225	0.9019	-0.1095	-0.83
3.230	0.9015	-0.1094	-0.84
3.235	0.9016	-0.1096	-0.84
3.240	0.9014	-0.1098	-0.84
3.245	0.9015	-0.1097	-0.84
3.250	0.9017	-0.1103	-0.83
3.255	0.9014	-0.1103	-0.84
3.260	0.9016	-0.1104	-0.83
3.265	0.9018	-0.1106	-0.83
3.270	0.9014	-0.1106	-0.84
3.275	0.9016	-0.1108	-0.83
3.280	0.9018	-0.1109	-0.83
3.285	0.9017	-0.1116	-0.83
3.290	0.9015	-0.1111	-0.83
3.295	0.9016	-0.1115	-0.83
3.300	0.9019	-0.1115	-0.83
3.305	0.9021	-0.1116	-0.83
3.310	0.9021	-0.1117	-0.83
3.315	0.9020	-0.1120	-0.83
3.320	0.9023	-0.1120	-0.83
3.325	0.9026	-0.1121	-0.82
3.330	0.9023	-0.1125	-0.83
3.335	0.9023	-0.1126	-0.83
3.340	0.9025	-0.1127	-0.82
3.345	0.9024	-0.1128	-0.82
3.350	0.9024	-0.1130	-0.82

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.355	0.9026	-0.1129	-0.82
3.360	0.9026	-0.1131	-0.82
3.365	0.9028	-0.1131	-0.82
3.370	0.9028	-0.1135	-0.82
3.375	0.9029	-0.1131	-0.82
3.380	0.9029	-0.1132	-0.82
3.385	0.9029	-0.1135	-0.82
3.390	0.9029	-0.1135	-0.82
3.395	0.9029	-0.1136	-0.82
3.400	0.9029	-0.1138	-0.82
3.405	0.9028	-0.1137	-0.82
3.410	0.9029	-0.1137	-0.82
3.415	0.9028	-0.1139	-0.82
3.420	0.9027	-0.1138	-0.82
3.425	0.9027	-0.1138	-0.82
3.430	0.9027	-0.1139	-0.82
3.435	0.9027	-0.1140	-0.82
3.440	0.9024	-0.1139	-0.82
3.445	0.9023	-0.1143	-0.82
3.450	0.9020	-0.1147	-0.83
3.455	0.9021	-0.1145	-0.83
3.460	0.9020	-0.1148	-0.83
3.465	0.9016	-0.1149	-0.83
3.470	0.9016	-0.1151	-0.83
3.475	0.9014	-0.1152	-0.83
3.480	0.9012	-0.1153	-0.83
3.485	0.9009	-0.1154	-0.84
3.490	0.9014	-0.1162	-0.83
3.495	0.9013	-0.1166	-0.83
3.500	0.9011	-0.1166	-0.83
3.505	0.9014	-0.1165	-0.83
3.510	0.9010	-0.1167	-0.83
3.515	0.9015	-0.1171	-0.83
3.520	0.9012	-0.1172	-0.83
3.525	0.9016	-0.1181	-0.83
3.530	0.9021	-0.1180	-0.82
3.535	0.9020	-0.1180	-0.82
3.540	0.9022	-0.1179	-0.82
3.545	0.9027	-0.1177	-0.82
3.550	0.9023	-0.1175	-0.82
3.555	0.9026	-0.1175	-0.82
3.560	0.9026	-0.1169	-0.82
3.565	0.9025	-0.1167	-0.82
3.570	0.9020	-0.1167	-0.82
3.575	0.9020	-0.1163	-0.82
3.580	0.9016	-0.1155	-0.83

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.585	0.9014	-0.1153	-0.83
3.590	0.8994	-0.1153	-0.85
3.595	0.8989	-0.1157	-0.85
3.600	0.8986	-0.1162	-0.86
3.605	0.8982	-0.1166	-0.86
3.610	0.8981	-0.1171	-0.86
3.615	0.8979	-0.1183	-0.86
3.620	0.8977	-0.1189	-0.86
3.625	0.8980	-0.1191	-0.86
3.630	0.8979	-0.1198	-0.86
3.635	0.8985	-0.1198	-0.85
3.640	0.8989	-0.1196	-0.85
3.645	0.8989	-0.1197	-0.85
3.650	0.8993	-0.1197	-0.85
3.655	0.8992	-0.1199	-0.85
3.660	0.8994	-0.1197	-0.84
3.665	0.8994	-0.1197	-0.84
3.670	0.8993	-0.1196	-0.85
3.675	0.8995	-0.1195	-0.84
3.680	0.8993	-0.1194	-0.85
3.685	0.8993	-0.1193	-0.85
3.690	0.8993	-0.1194	-0.85
3.695	0.8993	-0.1194	-0.85
3.700	0.8992	-0.1190	-0.85
3.705	0.8990	-0.1189	-0.85
3.710	0.8987	-0.1190	-0.85
3.715	0.8984	-0.1187	-0.86
3.720	0.8980	-0.1185	-0.86
3.725	0.8977	-0.1188	-0.86
3.730	0.8970	-0.1189	-0.87
3.735	0.8964	-0.1194	-0.87
3.740	0.8960	-0.1198	-0.88
3.745	0.8958	-0.1207	-0.88
3.750	0.8958	-0.1217	-0.88
3.755	0.8963	-0.1227	-0.87
3.760	0.8970	-0.1235	-0.86
3.765	0.8980	-0.1239	-0.85
3.770	0.8995	-0.1235	-0.84
3.775	0.9001	-0.1229	-0.83
3.780	0.9011	-0.1229	-0.82
3.785	0.9007	-0.1210	-0.83
3.790	0.8996	-0.1202	-0.84
3.795	0.8994	-0.1202	-0.84
3.800	0.8993	-0.1202	-0.85
3.805	0.8984	-0.1200	-0.85
3.810	0.8980	-0.1203	-0.86

(continued on next page)

Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.815	0.8973	-0.1207	-0.86
3.820	0.8974	-0.1206	-0.86
3.825	0.8970	-0.1210	-0.87
3.830	0.8970	-0.1214	-0.87
3.835	0.8927	-0.1242	-0.90
3.840	0.8971	-0.1227	-0.86
3.845	0.8973	-0.1228	-0.86
3.850	0.8970	-0.1230	-0.86
3.855	0.8973	-0.1231	-0.86
3.860	0.8972	-0.1233	-0.86
3.865	0.8971	-0.1236	-0.86
3.870	0.8974	-0.1236	-0.86
3.875	0.8972	-0.1239	-0.86
3.880	0.8973	-0.1238	-0.86
3.885	0.8973	-0.1242	-0.86
3.890	0.8974	-0.1244	-0.86
3.895	0.8973	-0.1244	-0.86
3.900	0.8973	-0.1245	-0.86
3.905	0.8972	-0.1249	-0.86
3.910	0.8972	-0.1247	-0.86
3.915	0.8972	-0.1252	-0.86
3.920	0.8973	-0.1254	-0.86
3.925	0.8972	-0.1256	-0.86
3.930	0.8972	-0.1258	-0.86
3.935	0.8971	-0.1258	-0.86
3.940	0.8971	-0.1263	-0.86
3.945	0.8968	-0.1265	-0.86
3.950	0.8968	-0.1267	-0.86
3.955	0.8969	-0.1269	-0.86
3.960	0.8968	-0.1271	-0.86
3.965	0.8967	-0.1273	-0.86
3.970	0.8967	-0.1275	-0.86
3.975	0.8968	-0.1279	-0.86
3.980	0.8966	-0.1280	-0.86
3.985	0.8966	-0.1282	-0.86
3.990	0.8964	-0.1284	-0.86
3.995	0.8963	-0.1288	-0.86
4.000	0.8962	-0.1290	-0.86
4.005	0.8962	-0.1293	-0.86
4.010	0.8961	-0.1297	-0.86
4.015	0.8961	-0.1298	-0.86
4.020	0.8958	-0.1300	-0.87
4.025	0.8956	-0.1301	-0.87
4.030	0.8957	-0.1306	-0.87
4.035	0.8956	-0.1309	-0.87
4.040	0.8954	-0.1310	-0.87

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.045	0.8950	-0.1311	-0.87
4.050	0.8950	-0.1317	-0.87
4.055	0.8947	-0.1319	-0.87
4.060	0.8947	-0.1323	-0.87
4.065	0.8942	-0.1324	-0.88
4.070	0.8944	-0.1326	-0.87
4.075	0.8942	-0.1330	-0.88
4.080	0.8943	-0.1334	-0.88
4.085	0.8939	-0.1334	-0.88
4.090	0.8937	-0.1338	-0.88
4.095	0.8938	-0.1342	-0.88
4.100	0.8936	-0.1343	-0.88
4.105	0.8931	-0.1346	-0.88
4.110	0.8930	-0.1350	-0.88
4.115	0.8930	-0.1353	-0.88
4.120	0.8926	-0.1355	-0.89
4.125	0.8928	-0.1358	-0.89
4.130	0.8925	-0.1362	-0.89
4.135	0.8921	-0.1366	-0.89
4.140	0.8917	-0.1365	-0.89
4.145	0.8917	-0.1369	-0.89
4.150	0.8914	-0.1375	-0.90
4.155	0.8910	-0.1378	-0.90
4.160	0.8907	-0.1378	-0.90
4.165	0.8904	-0.1383	-0.91
4.170	0.8900	-0.1384	-0.91
4.175	0.8894	-0.1382	-0.91
4.180	0.8897	-0.1385	-0.91
4.185	0.8891	-0.1387	-0.92
4.190	0.8890	-0.1390	-0.92
4.195	0.8889	-0.1392	-0.92
4.200	0.8888	-0.1390	-0.92
4.205	0.8888	-0.1395	-0.92
4.210	0.8887	-0.1399	-0.92
4.215	0.8886	-0.1394	-0.92
4.220	0.8885	-0.1399	-0.92
4.225	0.8884	-0.1395	-0.92
4.230	0.8881	-0.1401	-0.92
4.235	0.8881	-0.1402	-0.92
4.240	0.8884	-0.1405	-0.92
4.245	0.8882	-0.1406	-0.92
4.250	0.8880	-0.1407	-0.92
4.255	0.8881	-0.1409	-0.92
4.260	0.8878	-0.1411	-0.93
4.265	0.8879	-0.1409	-0.92
4.270	0.8877	-0.1413	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.275	0.8877	-0.1414	-0.93
4.280	0.8877	-0.1415	-0.93
4.285	0.8876	-0.1417	-0.93
4.290	0.8876	-0.1418	-0.93
4.295	0.8875	-0.1418	-0.93
4.300	0.8875	-0.1420	-0.93
4.305	0.8875	-0.1421	-0.93
4.310	0.8874	-0.1422	-0.93
4.315	0.8873	-0.1423	-0.93
4.320	0.8873	-0.1423	-0.93
4.325	0.8872	-0.1424	-0.93
4.330	0.8872	-0.1425	-0.93
4.335	0.8871	-0.1426	-0.93
4.340	0.8870	-0.1427	-0.93
4.345	0.8869	-0.1430	-0.93
4.350	0.8868	-0.1431	-0.93
4.355	0.8868	-0.1432	-0.93
4.360	0.8868	-0.1434	-0.93
4.365	0.8868	-0.1437	-0.93
4.370	0.8868	-0.1436	-0.93
4.375	0.8871	-0.1440	-0.93
4.380	0.8871	-0.1441	-0.93
4.385	0.8869	-0.1445	-0.93
4.390	0.8867	-0.1444	-0.93
4.395	0.8870	-0.1447	-0.93
4.400	0.8872	-0.1448	-0.93
4.405	0.8874	-0.1453	-0.92
4.410	0.8874	-0.1451	-0.92
4.415	0.8873	-0.1450	-0.92
4.420	0.8877	-0.1450	-0.92
4.425	0.8878	-0.1449	-0.92
4.430	0.8875	-0.1451	-0.92
4.435	0.8874	-0.1454	-0.92
4.440	0.8874	-0.1454	-0.92
4.445	0.8873	-0.1455	-0.92
4.450	0.8868	-0.1454	-0.93
4.455	0.8870	-0.1457	-0.93
4.460	0.8866	-0.1456	-0.93
4.465	0.8866	-0.1453	-0.93
4.470	0.8868	-0.1454	-0.93
4.475	0.8868	-0.1450	-0.93
4.480	0.8870	-0.1447	-0.93
4.485	0.8868	-0.1451	-0.93
4.490	0.8874	-0.1452	-0.92
4.495	0.8876	-0.1454	-0.92
4.500	0.8878	-0.1456	-0.92

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.505	0.8880	-0.1456	-0.92
4.510	0.8884	-0.1461	-0.91
4.515	0.8887	-0.1461	-0.91
4.520	0.8888	-0.1463	-0.91
4.525	0.8891	-0.1461	-0.90
4.530	0.8891	-0.1464	-0.91
4.535	0.8895	-0.1465	-0.90
4.540	0.8893	-0.1462	-0.90
4.545	0.8893	-0.1464	-0.90
4.550	0.8893	-0.1463	-0.90
4.555	0.8895	-0.1465	-0.90
4.560	0.8896	-0.1465	-0.90
4.565	0.8898	-0.1465	-0.90
4.570	0.8897	-0.1467	-0.90
4.575	0.8898	-0.1466	-0.90
4.580	0.8899	-0.1466	-0.90
4.585	0.8899	-0.1468	-0.90
4.590	0.8899	-0.1468	-0.90
4.595	0.8900	-0.1470	-0.90
4.600	0.8900	-0.1469	-0.90
4.605	0.8899	-0.1472	-0.90
4.610	0.8899	-0.1471	-0.90
4.615	0.8898	-0.1474	-0.90
4.620	0.8897	-0.1473	-0.90
4.625	0.8898	-0.1477	-0.90
4.630	0.8897	-0.1476	-0.90
4.635	0.8895	-0.1478	-0.90
4.640	0.8896	-0.1480	-0.90
4.645	0.8894	-0.1478	-0.90
4.650	0.8896	-0.1481	-0.90
4.655	0.8893	-0.1481	-0.90
4.660	0.8894	-0.1482	-0.90
4.665	0.8894	-0.1484	-0.90
4.670	0.8894	-0.1487	-0.90
4.675	0.8892	-0.1488	-0.90
4.680	0.8892	-0.1486	-0.90
4.685	0.8892	-0.1487	-0.90
4.690	0.8895	-0.1487	-0.90
4.695	0.8891	-0.1488	-0.90
4.700	0.8894	-0.1490	-0.90
4.705	0.8888	-0.1496	-0.90
4.710	0.8892	-0.1495	-0.90
4.715	0.8892	-0.1494	-0.90
4.720	0.8890	-0.1496	-0.90
4.725	0.8890	-0.1493	-0.90
4.730	0.8890	-0.1493	-0.90

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.735	0.8893	-0.1492	-0.90
4.740	0.8891	-0.1493	-0.90
4.745	0.8887	-0.1495	-0.90
4.750	0.8888	-0.1494	-0.90
4.755	0.8888	-0.1497	-0.90
4.760	0.8884	-0.1500	-0.91
4.765	0.8887	-0.1497	-0.90
4.770	0.8882	-0.1501	-0.91
4.775	0.8878	-0.1507	-0.91
4.780	0.8876	-0.1506	-0.91
4.785	0.8882	-0.1509	-0.91
4.790	0.8874	-0.1508	-0.91
4.795	0.8874	-0.1511	-0.91
4.800	0.8874	-0.1512	-0.91
4.805	0.8868	-0.1510	-0.92
4.810	0.8875	-0.1513	-0.91
4.815	0.8870	-0.1509	-0.92
4.820	0.8867	-0.1511	-0.92
4.825	0.8870	-0.1508	-0.92
4.830	0.8866	-0.1512	-0.92
4.835	0.8870	-0.1510	-0.92
4.840	0.8869	-0.1508	-0.92
4.845	0.8867	-0.1507	-0.92
4.850	0.8867	-0.1508	-0.92
4.855	0.8867	-0.1503	-0.92
4.860	0.8866	-0.1505	-0.92
4.865	0.8868	-0.1499	-0.92
4.870	0.8865	-0.1503	-0.92
4.875	0.8867	-0.1503	-0.92
4.880	0.8868	-0.1504	-0.92
4.885	0.8865	-0.1504	-0.92
4.890	0.8867	-0.1505	-0.92
4.895	0.8866	-0.1504	-0.92
4.900	0.8868	-0.1505	-0.92
4.905	0.8868	-0.1505	-0.92
4.910	0.8868	-0.1507	-0.92
4.915	0.8869	-0.1508	-0.92
4.920	0.8870	-0.1509	-0.92
4.925	0.8870	-0.1510	-0.92
4.930	0.8870	-0.1512	-0.92
4.935	0.8870	-0.1513	-0.92
4.940	0.8872	-0.1514	-0.92
4.945	0.8871	-0.1516	-0.92
4.950	0.8872	-0.1517	-0.91
4.955	0.8874	-0.1518	-0.91
4.960	0.8873	-0.1520	-0.91

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.965	0.8873	-0.1520	-0.91
4.970	0.8875	-0.1522	-0.91
4.975	0.8876	-0.1522	-0.91
4.980	0.8876	-0.1526	-0.91
4.985	0.8874	-0.1528	-0.91
4.990	0.8877	-0.1529	-0.91
4.995	0.8878	-0.1533	-0.91
5.000	0.8879	-0.1534	-0.90
5.005	0.8879	-0.1537	-0.90
5.010	0.8878	-0.1540	-0.91
5.015	0.8879	-0.1542	-0.90
5.020	0.8877	-0.1543	-0.91
5.025	0.8877	-0.1549	-0.90
5.030	0.8876	-0.1551	-0.91
5.035	0.8876	-0.1549	-0.90
5.040	0.8875	-0.1552	-0.91
5.045	0.8873	-0.1551	-0.91
5.050	0.8871	-0.1551	-0.91
5.055	0.8872	-0.1547	-0.91
5.060	0.8874	-0.1548	-0.91
5.065	0.8873	-0.1544	-0.91
5.070	0.8875	-0.1545	-0.91
5.075	0.8876	-0.1545	-0.91
5.080	0.8876	-0.1545	-0.91
5.085	0.8876	-0.1543	-0.91
5.090	0.8881	-0.1543	-0.90
5.095	0.8882	-0.1543	-0.90
5.100	0.8883	-0.1542	-0.90
5.105	0.8887	-0.1542	-0.90
5.110	0.8886	-0.1544	-0.90
5.115	0.8884	-0.1545	-0.90
5.120	0.8888	-0.1545	-0.89
5.125	0.8887	-0.1546	-0.90
5.130	0.8889	-0.1546	-0.89
5.135	0.8889	-0.1546	-0.89
5.140	0.8892	-0.1547	-0.89
5.145	0.8891	-0.1550	-0.89
5.150	0.8894	-0.1550	-0.89
5.155	0.8893	-0.1552	-0.89
5.160	0.8893	-0.1553	-0.89
5.165	0.8894	-0.1554	-0.89
5.170	0.8893	-0.1555	-0.89
5.175	0.8893	-0.1557	-0.89
5.180	0.8893	-0.1556	-0.89
5.185	0.8894	-0.1558	-0.89
5.190	0.8895	-0.1558	-0.89

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.195	0.8893	-0.1559	-0.89
5.200	0.8894	-0.1559	-0.89
5.205	0.8894	-0.1560	-0.89
5.210	0.8894	-0.1562	-0.89
5.215	0.8895	-0.1563	-0.89
5.220	0.8894	-0.1564	-0.89
5.225	0.8894	-0.1566	-0.89
5.230	0.8893	-0.1566	-0.89
5.235	0.8894	-0.1569	-0.88
5.240	0.8892	-0.1571	-0.89
5.245	0.8893	-0.1570	-0.89
5.250	0.8893	-0.1573	-0.89
5.255	0.8892	-0.1575	-0.89
5.260	0.8889	-0.1574	-0.89
5.265	0.8890	-0.1574	-0.89
5.270	0.8888	-0.1577	-0.89
5.275	0.8885	-0.1579	-0.89
5.280	0.8885	-0.1580	-0.89
5.285	0.8882	-0.1580	-0.89
5.290	0.8881	-0.1585	-0.89
5.295	0.8878	-0.1583	-0.90
5.300	0.8877	-0.1587	-0.90
5.305	0.8875	-0.1588	-0.90
5.310	0.8874	-0.1593	-0.90
5.315	0.8873	-0.1593	-0.90
5.320	0.8868	-0.1596	-0.90
5.325	0.8870	-0.1595	-0.90
5.330	0.8868	-0.1598	-0.90
5.335	0.8868	-0.1602	-0.90
5.340	0.8868	-0.1603	-0.90
5.345	0.8872	-0.1606	-0.90
5.350	0.8866	-0.1604	-0.91
5.355	0.8863	-0.1606	-0.91
5.360	0.8859	-0.1610	-0.91
5.365	0.8858	-0.1607	-0.91
5.370	0.8857	-0.1612	-0.91
5.375	0.8855	-0.1609	-0.91
5.380	0.8855	-0.1612	-0.91
5.385	0.8853	-0.1615	-0.92
5.390	0.8851	-0.1614	-0.92
5.395	0.8847	-0.1616	-0.92
5.400	0.8846	-0.1617	-0.92
5.405	0.8844	-0.1623	-0.92
5.410	0.8842	-0.1622	-0.93
5.415	0.8834	-0.1628	-0.93
5.420	0.8838	-0.1628	-0.93

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.425	0.8834	-0.1633	-0.93
5.430	0.8833	-0.1630	-0.93
5.435	0.8830	-0.1633	-0.93
5.440	0.8831	-0.1636	-0.93
5.445	0.8827	-0.1635	-0.94
5.450	0.8827	-0.1640	-0.94
5.455	0.8828	-0.1641	-0.94
5.460	0.8824	-0.1642	-0.94
5.465	0.8823	-0.1641	-0.94
5.470	0.8822	-0.1647	-0.94
5.475	0.8818	-0.1646	-0.94
5.480	0.8819	-0.1648	-0.94
5.485	0.8816	-0.1649	-0.94
5.490	0.8815	-0.1652	-0.95
5.495	0.8812	-0.1652	-0.95
5.500	0.8811	-0.1653	-0.95
5.505	0.8809	-0.1654	-0.95
5.510	0.8806	-0.1656	-0.95
5.515	0.8804	-0.1657	-0.96
5.520	0.8801	-0.1660	-0.96
5.525	0.8801	-0.1659	-0.96
5.530	0.8800	-0.1662	-0.96
5.535	0.8798	-0.1663	-0.96
5.540	0.8798	-0.1665	-0.96
5.545	0.8795	-0.1668	-0.96
5.550	0.8794	-0.1668	-0.96
5.555	0.8793	-0.1669	-0.96
5.560	0.8792	-0.1671	-0.96
5.565	0.8792	-0.1671	-0.96
5.570	0.8792	-0.1672	-0.96
5.575	0.8791	-0.1673	-0.97
5.580	0.8791	-0.1674	-0.96
5.585	0.8789	-0.1676	-0.97
5.590	0.8787	-0.1676	-0.97
5.595	0.8785	-0.1679	-0.97
5.600	0.8784	-0.1679	-0.97
5.605	0.8782	-0.1681	-0.97
5.610	0.8780	-0.1685	-0.97
5.615	0.8776	-0.1687	-0.98
5.620	0.8776	-0.1691	-0.98
5.625	0.8778	-0.1693	-0.97
5.630	0.8775	-0.1697	-0.98
5.635	0.8775	-0.1700	-0.98
5.640	0.8773	-0.1702	-0.98
5.645	0.8774	-0.1706	-0.98
5.650	0.8777	-0.1709	-0.97

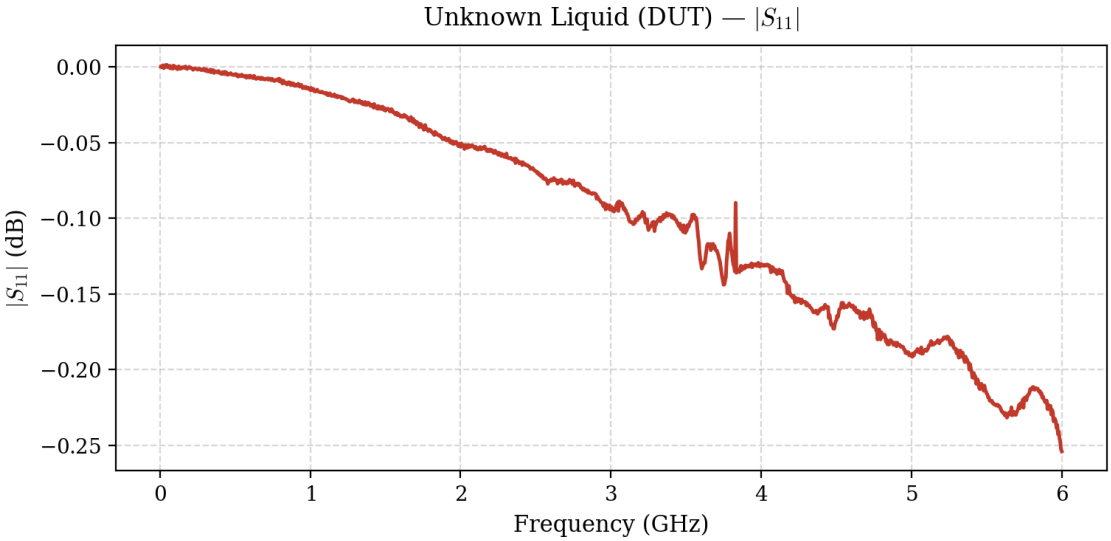
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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.655	0.8773	-0.1712	-0.97
5.660	0.8776	-0.1713	-0.97
5.665	0.8776	-0.1712	-0.97
5.670	0.8775	-0.1721	-0.97
5.675	0.8773	-0.1720	-0.97
5.680	0.8775	-0.1724	-0.97
5.685	0.8772	-0.1726	-0.97
5.690	0.8773	-0.1732	-0.97
5.695	0.8777	-0.1734	-0.97
5.700	0.8772	-0.1737	-0.97
5.705	0.8770	-0.1741	-0.97
5.710	0.8773	-0.1740	-0.97
5.715	0.8771	-0.1742	-0.97
5.720	0.8773	-0.1747	-0.97
5.725	0.8772	-0.1752	-0.97
5.730	0.8772	-0.1754	-0.97
5.735	0.8774	-0.1754	-0.97
5.740	0.8777	-0.1759	-0.96
5.745	0.8780	-0.1759	-0.96
5.750	0.8779	-0.1765	-0.96
5.755	0.8778	-0.1764	-0.96
5.760	0.8777	-0.1768	-0.96
5.765	0.8779	-0.1768	-0.96
5.770	0.8776	-0.1770	-0.96
5.775	0.8779	-0.1771	-0.96
5.780	0.8779	-0.1776	-0.96
5.785	0.8781	-0.1779	-0.95
5.790	0.8781	-0.1781	-0.95
5.795	0.8781	-0.1781	-0.95
5.800	0.8781	-0.1786	-0.95
5.805	0.8780	-0.1785	-0.95
5.810	0.8781	-0.1788	-0.95
5.815	0.8781	-0.1789	-0.95
5.820	0.8780	-0.1790	-0.95
5.825	0.8779	-0.1793	-0.95
5.830	0.8779	-0.1795	-0.95
5.835	0.8778	-0.1795	-0.95
5.840	0.8777	-0.1797	-0.95
5.845	0.8777	-0.1801	-0.95
5.850	0.8775	-0.1800	-0.96
5.855	0.8774	-0.1802	-0.96
5.860	0.8772	-0.1805	-0.96
5.865	0.8772	-0.1805	-0.96
5.870	0.8768	-0.1807	-0.96
5.875	0.8767	-0.1809	-0.96
5.880	0.8766	-0.1811	-0.96

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Frequency (GHz)	Re(S_{11})	Im(S_{11})	$\lvert S_{11}\rvert$ (dB)
5.885	0.8762	-0.1812	-0.97
5.890	0.8764	-0.1817	-0.96
5.895	0.8760	-0.1818	-0.97
5.900	0.8758	-0.1818	-0.97
5.905	0.8757	-0.1820	-0.97
5.910	0.8754	-0.1823	-0.97
5.915	0.8753	-0.1825	-0.97
5.920	0.8751	-0.1828	-0.97
5.925	0.8747	-0.1829	-0.98
5.930	0.8749	-0.1833	-0.97
5.935	0.8744	-0.1833	-0.98
5.940	0.8744	-0.1835	-0.98
5.945	0.8743	-0.1831	-0.98
5.950	0.8739	-0.1838	-0.98
5.955	0.8737	-0.1835	-0.99
5.960	0.8734	-0.1838	-0.99
5.965	0.8735	-0.1840	-0.99
5.970	0.8729	-0.1842	-0.99
5.975	0.8716	-0.1840	-1.00
5.980	0.8722	-0.1839	-1.00
5.985	0.8715	-0.1848	-1.00
5.990	0.8713	-0.1851	-1.01
5.995	0.8709	-0.1854	-1.01
6.000	0.8707	-0.1854	-1.01

3.5 Unknown Liquid (DUT)



Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.005	1.0000	-0.0000	0.00
0.010	1.0000	-0.0002	-0.00
0.015	1.0001	-0.0006	0.00
0.020	1.0001	-0.0007	0.00
0.025	0.9999	-0.0008	-0.00
0.030	1.0000	-0.0009	-0.00
0.035	1.0001	-0.0012	0.00
0.040	1.0000	-0.0012	0.00
0.045	1.0002	-0.0015	0.00
0.050	1.0001	-0.0017	0.00
0.055	1.0000	-0.0017	-0.00
0.060	1.0000	-0.0018	0.00
0.065	1.0001	-0.0021	0.00
0.070	0.9999	-0.0022	-0.00
0.075	1.0000	-0.0024	-0.00
0.080	1.0000	-0.0026	0.00
0.085	0.9999	-0.0027	-0.00
0.090	1.0001	-0.0029	0.00
0.095	1.0001	-0.0032	0.00
0.100	0.9999	-0.0032	-0.00
0.105	1.0000	-0.0035	0.00
0.110	0.9999	-0.0035	-0.00
0.115	0.9998	-0.0037	-0.00
0.120	1.0000	-0.0038	-0.00
0.125	1.0000	-0.0040	0.00
0.130	0.9999	-0.0042	-0.00
0.135	0.9998	-0.0044	-0.00
0.140	0.9999	-0.0045	-0.00
0.145	1.0000	-0.0046	-0.00
0.150	1.0000	-0.0048	-0.00
0.155	0.9999	-0.0049	-0.00
0.160	0.9999	-0.0052	-0.00
0.165	1.0000	-0.0052	0.00
0.170	1.0000	-0.0053	0.00
0.175	0.9999	-0.0055	-0.00
0.180	0.9998	-0.0057	-0.00
0.185	0.9999	-0.0060	-0.00
0.190	0.9999	-0.0061	-0.00
0.195	1.0000	-0.0063	0.00
0.200	0.9999	-0.0064	-0.00
0.205	0.9999	-0.0064	-0.00
0.210	0.9999	-0.0067	-0.00
0.215	0.9999	-0.0069	-0.00
0.220	0.9998	-0.0070	-0.00
0.225	0.9998	-0.0073	-0.00
0.230	0.9998	-0.0074	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.235	0.9999	-0.0076	-0.00
0.240	0.9998	-0.0077	-0.00
0.245	0.9998	-0.0079	-0.00
0.250	0.9998	-0.0080	-0.00
0.255	0.9998	-0.0081	-0.00
0.260	0.9998	-0.0083	-0.00
0.265	0.9998	-0.0085	-0.00
0.270	0.9998	-0.0087	-0.00
0.275	0.9997	-0.0088	-0.00
0.280	0.9997	-0.0090	-0.00
0.285	0.9998	-0.0091	-0.00
0.290	0.9998	-0.0094	-0.00
0.295	0.9997	-0.0094	-0.00
0.300	0.9998	-0.0095	-0.00
0.305	0.9996	-0.0098	-0.00
0.310	0.9998	-0.0099	-0.00
0.315	0.9997	-0.0101	-0.00
0.320	0.9996	-0.0102	-0.00
0.325	0.9996	-0.0106	-0.00
0.330	0.9996	-0.0106	-0.00
0.335	0.9997	-0.0107	-0.00
0.340	0.9995	-0.0109	-0.00
0.345	0.9997	-0.0111	-0.00
0.350	0.9996	-0.0112	-0.00
0.355	0.9997	-0.0114	-0.00
0.360	0.9995	-0.0116	-0.00
0.365	0.9996	-0.0117	-0.00
0.370	0.9996	-0.0117	-0.00
0.375	0.9996	-0.0120	-0.00
0.380	0.9995	-0.0122	-0.00
0.385	0.9995	-0.0123	-0.00
0.390	0.9996	-0.0125	-0.00
0.395	0.9995	-0.0127	-0.00
0.400	0.9995	-0.0128	-0.00
0.405	0.9994	-0.0129	-0.00
0.410	0.9994	-0.0130	-0.00
0.415	0.9995	-0.0132	-0.00
0.420	0.9994	-0.0133	-0.00
0.425	0.9994	-0.0136	-0.00
0.430	0.9995	-0.0136	-0.00
0.435	0.9994	-0.0139	-0.00
0.440	0.9995	-0.0140	-0.00
0.445	0.9994	-0.0142	-0.00
0.450	0.9993	-0.0143	-0.01
0.455	0.9994	-0.0145	-0.00
0.460	0.9993	-0.0146	-0.00

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.465	0.9994	-0.0149	-0.00
0.470	0.9994	-0.0149	-0.00
0.475	0.9993	-0.0151	-0.00
0.480	0.9993	-0.0153	-0.01
0.485	0.9993	-0.0154	-0.01
0.490	0.9993	-0.0156	-0.01
0.495	0.9993	-0.0156	-0.01
0.500	0.9993	-0.0158	-0.01
0.505	0.9993	-0.0159	-0.00
0.510	0.9992	-0.0161	-0.01
0.515	0.9993	-0.0164	-0.01
0.520	0.9991	-0.0165	-0.01
0.525	0.9992	-0.0166	-0.01
0.530	0.9992	-0.0167	-0.01
0.535	0.9992	-0.0168	-0.01
0.540	0.9992	-0.0172	-0.01
0.545	0.9993	-0.0172	-0.01
0.550	0.9992	-0.0174	-0.01
0.555	0.9991	-0.0175	-0.01
0.560	0.9991	-0.0177	-0.01
0.565	0.9991	-0.0179	-0.01
0.570	0.9992	-0.0180	-0.01
0.575	0.9990	-0.0182	-0.01
0.580	0.9991	-0.0182	-0.01
0.585	0.9991	-0.0185	-0.01
0.590	0.9991	-0.0186	-0.01
0.595	0.9990	-0.0188	-0.01
0.600	0.9992	-0.0189	-0.01
0.605	0.9991	-0.0191	-0.01
0.610	0.9990	-0.0192	-0.01
0.615	0.9990	-0.0194	-0.01
0.620	0.9990	-0.0194	-0.01
0.625	0.9991	-0.0196	-0.01
0.630	0.9990	-0.0198	-0.01
0.635	0.9990	-0.0199	-0.01
0.640	0.9989	-0.0200	-0.01
0.645	0.9990	-0.0203	-0.01
0.650	0.9989	-0.0204	-0.01
0.655	0.9990	-0.0206	-0.01
0.660	0.9989	-0.0207	-0.01
0.665	0.9989	-0.0208	-0.01
0.670	0.9989	-0.0209	-0.01
0.675	0.9990	-0.0210	-0.01
0.680	0.9989	-0.0212	-0.01
0.685	0.9988	-0.0214	-0.01
0.690	0.9989	-0.0215	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.695	0.9988	-0.0216	-0.01
0.700	0.9988	-0.0218	-0.01
0.705	0.9988	-0.0219	-0.01
0.710	0.9989	-0.0220	-0.01
0.715	0.9989	-0.0222	-0.01
0.720	0.9988	-0.0224	-0.01
0.725	0.9988	-0.0225	-0.01
0.730	0.9989	-0.0227	-0.01
0.735	0.9988	-0.0227	-0.01
0.740	0.9987	-0.0230	-0.01
0.745	0.9988	-0.0231	-0.01
0.750	0.9988	-0.0233	-0.01
0.755	0.9987	-0.0235	-0.01
0.760	0.9986	-0.0236	-0.01
0.765	0.9987	-0.0237	-0.01
0.770	0.9987	-0.0238	-0.01
0.775	0.9987	-0.0239	-0.01
0.780	0.9987	-0.0241	-0.01
0.785	0.9987	-0.0243	-0.01
0.790	0.9988	-0.0244	-0.01
0.795	0.9986	-0.0247	-0.01
0.800	0.9987	-0.0247	-0.01
0.805	0.9986	-0.0249	-0.01
0.810	0.9985	-0.0251	-0.01
0.815	0.9984	-0.0252	-0.01
0.820	0.9986	-0.0252	-0.01
0.825	0.9985	-0.0254	-0.01
0.830	0.9984	-0.0256	-0.01
0.835	0.9985	-0.0257	-0.01
0.840	0.9984	-0.0260	-0.01
0.845	0.9983	-0.0261	-0.01
0.850	0.9984	-0.0262	-0.01
0.855	0.9985	-0.0264	-0.01
0.860	0.9985	-0.0265	-0.01
0.865	0.9984	-0.0267	-0.01
0.870	0.9983	-0.0268	-0.01
0.875	0.9984	-0.0269	-0.01
0.880	0.9983	-0.0272	-0.01
0.885	0.9984	-0.0273	-0.01
0.890	0.9982	-0.0274	-0.01
0.895	0.9983	-0.0276	-0.01
0.900	0.9982	-0.0278	-0.01
0.905	0.9982	-0.0280	-0.01
0.910	0.9982	-0.0280	-0.01
0.915	0.9982	-0.0281	-0.01
0.920	0.9983	-0.0284	-0.01

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
0.925	0.9982	-0.0286	-0.01
0.930	0.9981	-0.0286	-0.01
0.935	0.9981	-0.0289	-0.01
0.940	0.9982	-0.0291	-0.01
0.945	0.9981	-0.0291	-0.01
0.950	0.9980	-0.0294	-0.01
0.955	0.9980	-0.0294	-0.01
0.960	0.9980	-0.0296	-0.01
0.965	0.9979	-0.0298	-0.01
0.970	0.9980	-0.0299	-0.01
0.975	0.9979	-0.0301	-0.01
0.980	0.9979	-0.0304	-0.01
0.985	0.9980	-0.0305	-0.01
0.990	0.9979	-0.0305	-0.01
0.995	0.9978	-0.0307	-0.01
1.000	0.9978	-0.0309	-0.02
1.005	0.9979	-0.0310	-0.01
1.010	0.9978	-0.0312	-0.02
1.015	0.9978	-0.0314	-0.02
1.020	0.9978	-0.0315	-0.01
1.025	0.9977	-0.0318	-0.02
1.030	0.9976	-0.0318	-0.02
1.035	0.9976	-0.0320	-0.02
1.040	0.9977	-0.0321	-0.02
1.045	0.9978	-0.0324	-0.01
1.050	0.9976	-0.0325	-0.02
1.055	0.9976	-0.0325	-0.02
1.060	0.9976	-0.0327	-0.02
1.065	0.9975	-0.0328	-0.02
1.070	0.9975	-0.0330	-0.02
1.075	0.9976	-0.0332	-0.02
1.080	0.9975	-0.0334	-0.02
1.085	0.9975	-0.0336	-0.02
1.090	0.9975	-0.0338	-0.02
1.095	0.9974	-0.0338	-0.02
1.100	0.9974	-0.0340	-0.02
1.105	0.9974	-0.0342	-0.02
1.110	0.9974	-0.0343	-0.02
1.115	0.9973	-0.0345	-0.02
1.120	0.9972	-0.0346	-0.02
1.125	0.9973	-0.0348	-0.02
1.130	0.9974	-0.0350	-0.02
1.135	0.9973	-0.0350	-0.02
1.140	0.9972	-0.0352	-0.02
1.145	0.9972	-0.0353	-0.02
1.150	0.9972	-0.0354	-0.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.155	0.9972	-0.0356	-0.02
1.160	0.9973	-0.0358	-0.02
1.165	0.9971	-0.0358	-0.02
1.170	0.9971	-0.0360	-0.02
1.175	0.9971	-0.0363	-0.02
1.180	0.9970	-0.0364	-0.02
1.185	0.9971	-0.0366	-0.02
1.190	0.9971	-0.0368	-0.02
1.195	0.9971	-0.0368	-0.02
1.200	0.9970	-0.0369	-0.02
1.205	0.9970	-0.0372	-0.02
1.210	0.9969	-0.0372	-0.02
1.215	0.9970	-0.0373	-0.02
1.220	0.9969	-0.0375	-0.02
1.225	0.9969	-0.0376	-0.02
1.230	0.9969	-0.0377	-0.02
1.235	0.9969	-0.0379	-0.02
1.240	0.9968	-0.0380	-0.02
1.245	0.9968	-0.0382	-0.02
1.250	0.9967	-0.0383	-0.02
1.255	0.9967	-0.0383	-0.02
1.260	0.9966	-0.0385	-0.02
1.265	0.9966	-0.0385	-0.02
1.270	0.9966	-0.0388	-0.02
1.275	0.9967	-0.0389	-0.02
1.280	0.9968	-0.0390	-0.02
1.285	0.9968	-0.0391	-0.02
1.290	0.9966	-0.0392	-0.02
1.295	0.9965	-0.0393	-0.02
1.300	0.9967	-0.0395	-0.02
1.305	0.9966	-0.0397	-0.02
1.310	0.9965	-0.0399	-0.02
1.315	0.9965	-0.0399	-0.02
1.320	0.9965	-0.0402	-0.02
1.325	0.9966	-0.0402	-0.02
1.330	0.9965	-0.0405	-0.02
1.335	0.9965	-0.0405	-0.02
1.340	0.9964	-0.0406	-0.02
1.345	0.9965	-0.0408	-0.02
1.350	0.9963	-0.0409	-0.02
1.355	0.9964	-0.0409	-0.02
1.360	0.9963	-0.0411	-0.02
1.365	0.9965	-0.0412	-0.02
1.370	0.9963	-0.0413	-0.02
1.375	0.9962	-0.0415	-0.03
1.380	0.9963	-0.0416	-0.02

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.385	0.9964	-0.0418	-0.02
1.390	0.9963	-0.0418	-0.02
1.395	0.9963	-0.0419	-0.02
1.400	0.9961	-0.0421	-0.03
1.405	0.9962	-0.0421	-0.03
1.410	0.9962	-0.0423	-0.02
1.415	0.9962	-0.0425	-0.03
1.420	0.9960	-0.0425	-0.03
1.425	0.9963	-0.0428	-0.02
1.430	0.9961	-0.0428	-0.03
1.435	0.9960	-0.0431	-0.03
1.440	0.9959	-0.0432	-0.03
1.445	0.9961	-0.0433	-0.03
1.450	0.9959	-0.0434	-0.03
1.455	0.9961	-0.0436	-0.03
1.460	0.9960	-0.0437	-0.03
1.465	0.9959	-0.0440	-0.03
1.470	0.9958	-0.0440	-0.03
1.475	0.9960	-0.0442	-0.03
1.480	0.9960	-0.0441	-0.03
1.485	0.9958	-0.0444	-0.03
1.490	0.9957	-0.0445	-0.03
1.495	0.9959	-0.0447	-0.03
1.500	0.9957	-0.0449	-0.03
1.505	0.9957	-0.0451	-0.03
1.510	0.9957	-0.0452	-0.03
1.515	0.9958	-0.0453	-0.03
1.520	0.9957	-0.0455	-0.03
1.525	0.9957	-0.0456	-0.03
1.530	0.9956	-0.0458	-0.03
1.535	0.9956	-0.0460	-0.03
1.540	0.9957	-0.0460	-0.03
1.545	0.9956	-0.0462	-0.03
1.550	0.9955	-0.0464	-0.03
1.555	0.9955	-0.0465	-0.03
1.560	0.9954	-0.0466	-0.03
1.565	0.9953	-0.0468	-0.03
1.570	0.9954	-0.0471	-0.03
1.575	0.9954	-0.0473	-0.03
1.580	0.9952	-0.0473	-0.03
1.585	0.9953	-0.0476	-0.03
1.590	0.9952	-0.0477	-0.03
1.595	0.9952	-0.0479	-0.03
1.600	0.9951	-0.0482	-0.03
1.605	0.9951	-0.0483	-0.03
1.610	0.9951	-0.0483	-0.03

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.615	0.9950	-0.0486	-0.03
1.620	0.9951	-0.0488	-0.03
1.625	0.9952	-0.0489	-0.03
1.630	0.9950	-0.0492	-0.03
1.635	0.9951	-0.0493	-0.03
1.640	0.9950	-0.0495	-0.03
1.645	0.9949	-0.0496	-0.03
1.650	0.9948	-0.0498	-0.03
1.655	0.9948	-0.0499	-0.03
1.660	0.9946	-0.0501	-0.04
1.665	0.9948	-0.0505	-0.03
1.670	0.9949	-0.0507	-0.03
1.675	0.9947	-0.0507	-0.03
1.680	0.9945	-0.0508	-0.04
1.685	0.9947	-0.0510	-0.03
1.690	0.9945	-0.0512	-0.04
1.695	0.9946	-0.0515	-0.04
1.700	0.9945	-0.0516	-0.04
1.705	0.9943	-0.0517	-0.04
1.710	0.9945	-0.0520	-0.04
1.715	0.9944	-0.0520	-0.04
1.720	0.9943	-0.0522	-0.04
1.725	0.9940	-0.0526	-0.04
1.730	0.9942	-0.0526	-0.04
1.735	0.9943	-0.0530	-0.04
1.740	0.9941	-0.0530	-0.04
1.745	0.9940	-0.0531	-0.04
1.750	0.9941	-0.0534	-0.04
1.755	0.9938	-0.0534	-0.04
1.760	0.9940	-0.0537	-0.04
1.765	0.9941	-0.0537	-0.04
1.770	0.9938	-0.0540	-0.04
1.775	0.9938	-0.0542	-0.04
1.780	0.9937	-0.0544	-0.04
1.785	0.9936	-0.0546	-0.04
1.790	0.9938	-0.0547	-0.04
1.795	0.9937	-0.0548	-0.04
1.800	0.9937	-0.0551	-0.04
1.805	0.9936	-0.0553	-0.04
1.810	0.9935	-0.0553	-0.04
1.815	0.9934	-0.0554	-0.04
1.820	0.9933	-0.0556	-0.04
1.825	0.9933	-0.0557	-0.05
1.830	0.9935	-0.0560	-0.04
1.835	0.9934	-0.0562	-0.04
1.840	0.9934	-0.0562	-0.04

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
1.845	0.9933	-0.0565	-0.04
1.850	0.9932	-0.0565	-0.05
1.855	0.9932	-0.0566	-0.04
1.860	0.9931	-0.0567	-0.05
1.865	0.9932	-0.0569	-0.05
1.870	0.9931	-0.0570	-0.05
1.875	0.9930	-0.0571	-0.05
1.880	0.9929	-0.0573	-0.05
1.885	0.9929	-0.0574	-0.05
1.890	0.9928	-0.0577	-0.05
1.895	0.9929	-0.0578	-0.05
1.900	0.9929	-0.0579	-0.05
1.905	0.9927	-0.0580	-0.05
1.910	0.9927	-0.0581	-0.05
1.915	0.9928	-0.0583	-0.05
1.920	0.9928	-0.0584	-0.05
1.925	0.9928	-0.0585	-0.05
1.930	0.9925	-0.0586	-0.05
1.935	0.9926	-0.0589	-0.05
1.940	0.9926	-0.0588	-0.05
1.945	0.9924	-0.0590	-0.05
1.950	0.9926	-0.0592	-0.05
1.955	0.9924	-0.0593	-0.05
1.960	0.9925	-0.0595	-0.05
1.965	0.9924	-0.0596	-0.05
1.970	0.9925	-0.0597	-0.05
1.975	0.9924	-0.0598	-0.05
1.980	0.9923	-0.0598	-0.05
1.985	0.9922	-0.0600	-0.05
1.990	0.9924	-0.0601	-0.05
1.995	0.9923	-0.0603	-0.05
2.000	0.9921	-0.0603	-0.05
2.005	0.9924	-0.0605	-0.05
2.010	0.9922	-0.0606	-0.05
2.015	0.9922	-0.0606	-0.05
2.020	0.9922	-0.0606	-0.05
2.025	0.9919	-0.0607	-0.05
2.030	0.9922	-0.0610	-0.05
2.035	0.9921	-0.0612	-0.05
2.040	0.9922	-0.0613	-0.05
2.045	0.9920	-0.0613	-0.05
2.050	0.9920	-0.0615	-0.05
2.055	0.9922	-0.0616	-0.05
2.060	0.9920	-0.0618	-0.05
2.065	0.9920	-0.0617	-0.05
2.070	0.9921	-0.0619	-0.05

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.075	0.9921	-0.0620	-0.05
2.080	0.9921	-0.0622	-0.05
2.085	0.9920	-0.0621	-0.05
2.090	0.9919	-0.0624	-0.05
2.095	0.9919	-0.0626	-0.05
2.100	0.9920	-0.0626	-0.05
2.105	0.9918	-0.0626	-0.05
2.110	0.9919	-0.0628	-0.05
2.115	0.9917	-0.0629	-0.05
2.120	0.9918	-0.0631	-0.05
2.125	0.9919	-0.0632	-0.05
2.130	0.9919	-0.0632	-0.05
2.135	0.9917	-0.0634	-0.05
2.140	0.9917	-0.0636	-0.05
2.145	0.9918	-0.0637	-0.05
2.150	0.9918	-0.0640	-0.05
2.155	0.9917	-0.0642	-0.05
2.160	0.9919	-0.0643	-0.05
2.165	0.9917	-0.0644	-0.05
2.170	0.9917	-0.0644	-0.05
2.175	0.9915	-0.0646	-0.06
2.180	0.9916	-0.0648	-0.05
2.185	0.9915	-0.0650	-0.06
2.190	0.9915	-0.0652	-0.06
2.195	0.9916	-0.0653	-0.05
2.200	0.9915	-0.0654	-0.06
2.205	0.9915	-0.0657	-0.05
2.210	0.9914	-0.0657	-0.06
2.215	0.9915	-0.0659	-0.06
2.220	0.9913	-0.0662	-0.06
2.225	0.9913	-0.0663	-0.06
2.230	0.9914	-0.0665	-0.06
2.235	0.9912	-0.0665	-0.06
2.240	0.9913	-0.0667	-0.06
2.245	0.9912	-0.0669	-0.06
2.250	0.9910	-0.0673	-0.06
2.255	0.9912	-0.0673	-0.06
2.260	0.9911	-0.0677	-0.06
2.265	0.9911	-0.0676	-0.06
2.270	0.9909	-0.0680	-0.06
2.275	0.9910	-0.0682	-0.06
2.280	0.9909	-0.0683	-0.06
2.285	0.9910	-0.0687	-0.06
2.290	0.9908	-0.0687	-0.06
2.295	0.9909	-0.0689	-0.06
2.300	0.9908	-0.0689	-0.06

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.305	0.9910	-0.0693	-0.06
2.310	0.9907	-0.0693	-0.06
2.315	0.9907	-0.0697	-0.06
2.320	0.9906	-0.0699	-0.06
2.325	0.9906	-0.0700	-0.06
2.330	0.9906	-0.0702	-0.06
2.335	0.9906	-0.0705	-0.06
2.340	0.9905	-0.0707	-0.06
2.345	0.9904	-0.0709	-0.06
2.350	0.9904	-0.0712	-0.06
2.355	0.9904	-0.0713	-0.06
2.360	0.9905	-0.0715	-0.06
2.365	0.9901	-0.0718	-0.06
2.370	0.9904	-0.0720	-0.06
2.375	0.9902	-0.0722	-0.06
2.380	0.9902	-0.0723	-0.06
2.385	0.9902	-0.0723	-0.06
2.390	0.9900	-0.0727	-0.06
2.395	0.9901	-0.0727	-0.06
2.400	0.9901	-0.0729	-0.06
2.405	0.9900	-0.0732	-0.06
2.410	0.9898	-0.0735	-0.06
2.415	0.9900	-0.0737	-0.06
2.420	0.9899	-0.0738	-0.06
2.425	0.9899	-0.0740	-0.06
2.430	0.9898	-0.0741	-0.06
2.435	0.9898	-0.0742	-0.06
2.440	0.9898	-0.0744	-0.07
2.445	0.9897	-0.0746	-0.06
2.450	0.9897	-0.0747	-0.07
2.455	0.9896	-0.0749	-0.07
2.460	0.9896	-0.0751	-0.07
2.465	0.9894	-0.0753	-0.07
2.470	0.9894	-0.0755	-0.07
2.475	0.9894	-0.0756	-0.07
2.480	0.9894	-0.0757	-0.07
2.485	0.9893	-0.0758	-0.07
2.490	0.9892	-0.0761	-0.07
2.495	0.9892	-0.0763	-0.07
2.500	0.9891	-0.0764	-0.07
2.505	0.9891	-0.0765	-0.07
2.510	0.9890	-0.0767	-0.07
2.515	0.9890	-0.0768	-0.07
2.520	0.9889	-0.0770	-0.07
2.525	0.9888	-0.0771	-0.07
2.530	0.9888	-0.0773	-0.07

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.535	0.9887	-0.0775	-0.07
2.540	0.9887	-0.0776	-0.07
2.545	0.9887	-0.0777	-0.07
2.550	0.9885	-0.0780	-0.07
2.555	0.9885	-0.0781	-0.07
2.560	0.9884	-0.0781	-0.07
2.565	0.9885	-0.0782	-0.07
2.570	0.9884	-0.0784	-0.07
2.575	0.9883	-0.0784	-0.08
2.580	0.9880	-0.0785	-0.08
2.585	0.9882	-0.0786	-0.08
2.590	0.9882	-0.0786	-0.08
2.595	0.9883	-0.0787	-0.07
2.600	0.9882	-0.0789	-0.08
2.605	0.9883	-0.0791	-0.07
2.610	0.9882	-0.0792	-0.08
2.615	0.9882	-0.0794	-0.07
2.620	0.9884	-0.0795	-0.07
2.625	0.9882	-0.0796	-0.07
2.630	0.9883	-0.0794	-0.07
2.635	0.9882	-0.0797	-0.07
2.640	0.9882	-0.0797	-0.08
2.645	0.9879	-0.0800	-0.08
2.650	0.9881	-0.0801	-0.08
2.655	0.9882	-0.0801	-0.07
2.660	0.9880	-0.0801	-0.08
2.665	0.9881	-0.0802	-0.08
2.670	0.9882	-0.0805	-0.07
2.675	0.9881	-0.0807	-0.07
2.680	0.9879	-0.0806	-0.08
2.685	0.9879	-0.0809	-0.08
2.690	0.9879	-0.0811	-0.08
2.695	0.9880	-0.0811	-0.08
2.700	0.9879	-0.0813	-0.08
2.705	0.9878	-0.0812	-0.08
2.710	0.9880	-0.0813	-0.08
2.715	0.9878	-0.0815	-0.08
2.720	0.9879	-0.0817	-0.08
2.725	0.9881	-0.0817	-0.07
2.730	0.9880	-0.0818	-0.08
2.735	0.9879	-0.0819	-0.08
2.740	0.9878	-0.0819	-0.08
2.745	0.9877	-0.0820	-0.08
2.750	0.9880	-0.0820	-0.07
2.755	0.9878	-0.0822	-0.08
2.760	0.9876	-0.0822	-0.08

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.765	0.9877	-0.0824	-0.08
2.770	0.9877	-0.0824	-0.08
2.775	0.9877	-0.0826	-0.08
2.780	0.9877	-0.0826	-0.08
2.785	0.9876	-0.0828	-0.08
2.790	0.9877	-0.0828	-0.08
2.795	0.9875	-0.0829	-0.08
2.800	0.9874	-0.0830	-0.08
2.805	0.9874	-0.0831	-0.08
2.810	0.9872	-0.0832	-0.08
2.815	0.9872	-0.0834	-0.08
2.820	0.9873	-0.0836	-0.08
2.825	0.9872	-0.0836	-0.08
2.830	0.9871	-0.0837	-0.08
2.835	0.9871	-0.0839	-0.08
2.840	0.9871	-0.0840	-0.08
2.845	0.9869	-0.0840	-0.08
2.850	0.9870	-0.0842	-0.08
2.855	0.9869	-0.0845	-0.08
2.860	0.9868	-0.0846	-0.08
2.865	0.9868	-0.0850	-0.08
2.870	0.9867	-0.0852	-0.08
2.875	0.9867	-0.0850	-0.08
2.880	0.9868	-0.0852	-0.08
2.885	0.9867	-0.0853	-0.08
2.890	0.9867	-0.0854	-0.08
2.895	0.9867	-0.0855	-0.08
2.900	0.9867	-0.0855	-0.08
2.905	0.9863	-0.0856	-0.09
2.910	0.9865	-0.0858	-0.09
2.915	0.9864	-0.0857	-0.09
2.920	0.9862	-0.0859	-0.09
2.925	0.9861	-0.0862	-0.09
2.930	0.9861	-0.0863	-0.09
2.935	0.9860	-0.0865	-0.09
2.940	0.9859	-0.0866	-0.09
2.945	0.9858	-0.0868	-0.09
2.950	0.9856	-0.0868	-0.09
2.955	0.9858	-0.0870	-0.09
2.960	0.9857	-0.0874	-0.09
2.965	0.9853	-0.0876	-0.09
2.970	0.9857	-0.0878	-0.09
2.975	0.9855	-0.0879	-0.09
2.980	0.9854	-0.0879	-0.09
2.985	0.9853	-0.0881	-0.09
2.990	0.9856	-0.0885	-0.09

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
2.995	0.9852	-0.0886	-0.09
3.000	0.9853	-0.0888	-0.09
3.005	0.9854	-0.0888	-0.09
3.010	0.9851	-0.0891	-0.09
3.015	0.9852	-0.0893	-0.09
3.020	0.9850	-0.0891	-0.10
3.025	0.9851	-0.0891	-0.10
3.030	0.9852	-0.0896	-0.09
3.035	0.9854	-0.0897	-0.09
3.040	0.9855	-0.0900	-0.09
3.045	0.9857	-0.0899	-0.09
3.050	0.9850	-0.0901	-0.10
3.055	0.9857	-0.0898	-0.09
3.060	0.9856	-0.0902	-0.09
3.065	0.9854	-0.0903	-0.09
3.070	0.9856	-0.0900	-0.09
3.075	0.9856	-0.0900	-0.09
3.080	0.9854	-0.0899	-0.09
3.085	0.9851	-0.0900	-0.09
3.090	0.9852	-0.0900	-0.09
3.095	0.9850	-0.0898	-0.10
3.100	0.9850	-0.0902	-0.09
3.105	0.9848	-0.0901	-0.10
3.110	0.9847	-0.0903	-0.10
3.115	0.9845	-0.0904	-0.10
3.120	0.9843	-0.0903	-0.10
3.125	0.9841	-0.0905	-0.10
3.130	0.9841	-0.0908	-0.10
3.135	0.9841	-0.0908	-0.10
3.140	0.9840	-0.0911	-0.10
3.145	0.9839	-0.0913	-0.10
3.150	0.9838	-0.0915	-0.10
3.155	0.9840	-0.0916	-0.10
3.160	0.9839	-0.0919	-0.10
3.165	0.9842	-0.0920	-0.10
3.170	0.9841	-0.0920	-0.10
3.175	0.9841	-0.0923	-0.10
3.180	0.9842	-0.0923	-0.10
3.185	0.9844	-0.0925	-0.10
3.190	0.9842	-0.0925	-0.10
3.195	0.9844	-0.0924	-0.10
3.200	0.9845	-0.0925	-0.10
3.205	0.9844	-0.0923	-0.10
3.210	0.9847	-0.0922	-0.10
3.215	0.9845	-0.0922	-0.10
3.220	0.9846	-0.0923	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.225	0.9842	-0.0923	-0.10
3.230	0.9839	-0.0925	-0.10
3.235	0.9839	-0.0924	-0.10
3.240	0.9841	-0.0924	-0.10
3.245	0.9836	-0.0926	-0.11
3.250	0.9833	-0.0927	-0.11
3.255	0.9834	-0.0929	-0.11
3.260	0.9836	-0.0928	-0.11
3.265	0.9835	-0.0935	-0.11
3.270	0.9836	-0.0932	-0.11
3.275	0.9838	-0.0933	-0.10
3.280	0.9836	-0.0935	-0.10
3.285	0.9838	-0.0936	-0.10
3.290	0.9831	-0.0939	-0.11
3.295	0.9835	-0.0938	-0.10
3.300	0.9836	-0.0939	-0.10
3.305	0.9838	-0.0940	-0.10
3.310	0.9839	-0.0942	-0.10
3.315	0.9839	-0.0946	-0.10
3.320	0.9838	-0.0944	-0.10
3.325	0.9840	-0.0950	-0.10
3.330	0.9838	-0.0946	-0.10
3.335	0.9840	-0.0948	-0.10
3.340	0.9841	-0.0949	-0.10
3.345	0.9839	-0.0946	-0.10
3.350	0.9842	-0.0950	-0.10
3.355	0.9840	-0.0954	-0.10
3.360	0.9842	-0.0953	-0.10
3.365	0.9841	-0.0952	-0.10
3.370	0.9844	-0.0952	-0.10
3.375	0.9842	-0.0955	-0.10
3.380	0.9841	-0.0954	-0.10
3.385	0.9842	-0.0955	-0.10
3.390	0.9842	-0.0955	-0.10
3.395	0.9842	-0.0956	-0.10
3.400	0.9841	-0.0957	-0.10
3.405	0.9840	-0.0956	-0.10
3.410	0.9841	-0.0957	-0.10
3.415	0.9839	-0.0957	-0.10
3.420	0.9840	-0.0956	-0.10
3.425	0.9838	-0.0960	-0.10
3.430	0.9838	-0.0960	-0.10
3.435	0.9837	-0.0961	-0.10
3.440	0.9835	-0.0963	-0.10
3.445	0.9835	-0.0960	-0.10
3.450	0.9833	-0.0962	-0.10

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.455	0.9832	-0.0966	-0.11
3.460	0.9834	-0.0966	-0.10
3.465	0.9829	-0.0970	-0.11
3.470	0.9829	-0.0970	-0.11
3.475	0.9830	-0.0972	-0.11
3.480	0.9827	-0.0972	-0.11
3.485	0.9832	-0.0974	-0.10
3.490	0.9826	-0.0981	-0.11
3.495	0.9826	-0.0978	-0.11
3.500	0.9827	-0.0983	-0.11
3.505	0.9827	-0.0983	-0.11
3.510	0.9831	-0.0991	-0.10
3.515	0.9829	-0.0988	-0.11
3.520	0.9830	-0.0991	-0.10
3.525	0.9832	-0.0993	-0.10
3.530	0.9833	-0.0994	-0.10
3.535	0.9833	-0.0994	-0.10
3.540	0.9838	-0.0997	-0.10
3.545	0.9836	-0.0996	-0.10
3.550	0.9838	-0.0998	-0.10
3.555	0.9838	-0.0992	-0.10
3.560	0.9835	-0.0988	-0.10
3.565	0.9837	-0.0987	-0.10
3.570	0.9835	-0.0985	-0.10
3.575	0.9829	-0.0980	-0.11
3.580	0.9823	-0.0984	-0.11
3.585	0.9819	-0.0978	-0.12
3.590	0.9810	-0.0984	-0.12
3.595	0.9804	-0.0993	-0.13
3.600	0.9800	-0.1000	-0.13
3.605	0.9796	-0.1003	-0.13
3.610	0.9798	-0.1009	-0.13
3.615	0.9799	-0.1013	-0.13
3.620	0.9800	-0.1017	-0.13
3.625	0.9798	-0.1026	-0.13
3.630	0.9803	-0.1027	-0.13
3.635	0.9807	-0.1031	-0.12
3.640	0.9811	-0.1030	-0.12
3.645	0.9813	-0.1028	-0.12
3.650	0.9812	-0.1033	-0.12
3.655	0.9812	-0.1030	-0.12
3.660	0.9812	-0.1027	-0.12
3.665	0.9808	-0.1029	-0.12
3.670	0.9812	-0.1026	-0.12
3.675	0.9812	-0.1028	-0.12
3.680	0.9813	-0.1027	-0.12

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.685	0.9809	-0.1027	-0.12
3.690	0.9811	-0.1027	-0.12
3.695	0.9808	-0.1027	-0.12
3.700	0.9808	-0.1028	-0.12
3.705	0.9808	-0.1026	-0.12
3.710	0.9806	-0.1027	-0.12
3.715	0.9804	-0.1026	-0.12
3.720	0.9802	-0.1025	-0.13
3.725	0.9799	-0.1028	-0.13
3.730	0.9796	-0.1024	-0.13
3.735	0.9791	-0.1026	-0.14
3.740	0.9787	-0.1030	-0.14
3.745	0.9785	-0.1034	-0.14
3.750	0.9781	-0.1037	-0.14
3.755	0.9780	-0.1048	-0.14
3.760	0.9782	-0.1054	-0.14
3.765	0.9784	-0.1062	-0.14
3.770	0.9796	-0.1067	-0.13
3.775	0.9800	-0.1079	-0.12
3.780	0.9811	-0.1066	-0.11
3.785	0.9813	-0.1056	-0.11
3.790	0.9818	-0.1054	-0.11
3.795	0.9812	-0.1049	-0.12
3.800	0.9807	-0.1044	-0.12
3.805	0.9804	-0.1044	-0.12
3.810	0.9799	-0.1044	-0.13
3.815	0.9795	-0.1044	-0.13
3.820	0.9794	-0.1045	-0.13
3.825	0.9790	-0.1046	-0.14
3.830	0.9843	-0.1036	-0.09
3.835	0.9787	-0.1061	-0.14
3.840	0.9789	-0.1063	-0.13
3.845	0.9789	-0.1066	-0.13
3.850	0.9789	-0.1063	-0.13
3.855	0.9787	-0.1065	-0.14
3.860	0.9789	-0.1067	-0.13
3.865	0.9791	-0.1072	-0.13
3.870	0.9788	-0.1068	-0.13
3.875	0.9789	-0.1072	-0.13
3.880	0.9789	-0.1074	-0.13
3.885	0.9791	-0.1074	-0.13
3.890	0.9789	-0.1078	-0.13
3.895	0.9791	-0.1077	-0.13
3.900	0.9790	-0.1077	-0.13
3.905	0.9790	-0.1078	-0.13
3.910	0.9790	-0.1081	-0.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
3.915	0.9789	-0.1083	-0.13
3.920	0.9789	-0.1085	-0.13
3.925	0.9790	-0.1084	-0.13
3.930	0.9790	-0.1085	-0.13
3.935	0.9791	-0.1089	-0.13
3.940	0.9788	-0.1090	-0.13
3.945	0.9789	-0.1092	-0.13
3.950	0.9790	-0.1091	-0.13
3.955	0.9789	-0.1096	-0.13
3.960	0.9790	-0.1096	-0.13
3.965	0.9789	-0.1100	-0.13
3.970	0.9788	-0.1102	-0.13
3.975	0.9789	-0.1103	-0.13
3.980	0.9790	-0.1105	-0.13
3.985	0.9789	-0.1108	-0.13
3.990	0.9787	-0.1109	-0.13
3.995	0.9787	-0.1111	-0.13
4.000	0.9788	-0.1113	-0.13
4.005	0.9787	-0.1116	-0.13
4.010	0.9786	-0.1118	-0.13
4.015	0.9786	-0.1120	-0.13
4.020	0.9786	-0.1123	-0.13
4.025	0.9786	-0.1124	-0.13
4.030	0.9785	-0.1128	-0.13
4.035	0.9786	-0.1129	-0.13
4.040	0.9785	-0.1132	-0.13
4.045	0.9785	-0.1135	-0.13
4.050	0.9784	-0.1136	-0.13
4.055	0.9784	-0.1141	-0.13
4.060	0.9782	-0.1142	-0.13
4.065	0.9778	-0.1146	-0.14
4.070	0.9782	-0.1146	-0.13
4.075	0.9782	-0.1147	-0.13
4.080	0.9780	-0.1151	-0.13
4.085	0.9778	-0.1151	-0.14
4.090	0.9779	-0.1156	-0.13
4.095	0.9777	-0.1158	-0.14
4.100	0.9778	-0.1160	-0.13
4.105	0.9775	-0.1163	-0.14
4.110	0.9776	-0.1168	-0.14
4.115	0.9776	-0.1167	-0.14
4.120	0.9777	-0.1172	-0.13
4.125	0.9774	-0.1170	-0.14
4.130	0.9773	-0.1179	-0.14
4.135	0.9775	-0.1179	-0.14
4.140	0.9775	-0.1183	-0.13

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.145	0.9770	-0.1189	-0.14
4.150	0.9768	-0.1191	-0.14
4.155	0.9765	-0.1195	-0.14
4.160	0.9765	-0.1197	-0.14
4.165	0.9764	-0.1199	-0.14
4.170	0.9762	-0.1203	-0.14
4.175	0.9755	-0.1206	-0.15
4.180	0.9761	-0.1205	-0.14
4.185	0.9760	-0.1208	-0.14
4.190	0.9754	-0.1208	-0.15
4.195	0.9754	-0.1211	-0.15
4.200	0.9752	-0.1214	-0.15
4.205	0.9752	-0.1217	-0.15
4.210	0.9750	-0.1217	-0.15
4.215	0.9751	-0.1220	-0.15
4.220	0.9750	-0.1224	-0.15
4.225	0.9747	-0.1218	-0.16
4.230	0.9749	-0.1226	-0.15
4.235	0.9746	-0.1227	-0.16
4.240	0.9746	-0.1232	-0.16
4.245	0.9748	-0.1232	-0.15
4.250	0.9746	-0.1232	-0.15
4.255	0.9747	-0.1234	-0.15
4.260	0.9746	-0.1235	-0.15
4.265	0.9744	-0.1237	-0.16
4.270	0.9746	-0.1239	-0.15
4.275	0.9744	-0.1240	-0.16
4.280	0.9744	-0.1242	-0.16
4.285	0.9744	-0.1243	-0.16
4.290	0.9744	-0.1245	-0.16
4.295	0.9744	-0.1244	-0.16
4.300	0.9741	-0.1246	-0.16
4.305	0.9742	-0.1248	-0.16
4.310	0.9740	-0.1248	-0.16
4.315	0.9740	-0.1250	-0.16
4.320	0.9740	-0.1251	-0.16
4.325	0.9739	-0.1251	-0.16
4.330	0.9738	-0.1253	-0.16
4.335	0.9737	-0.1253	-0.16
4.340	0.9736	-0.1255	-0.16
4.345	0.9735	-0.1256	-0.16
4.350	0.9734	-0.1257	-0.16
4.355	0.9734	-0.1259	-0.16
4.360	0.9734	-0.1260	-0.16
4.365	0.9735	-0.1262	-0.16
4.370	0.9733	-0.1263	-0.16

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
4.375	0.9732	-0.1263	-0.16
4.380	0.9733	-0.1266	-0.16
4.385	0.9734	-0.1270	-0.16
4.390	0.9733	-0.1270	-0.16
4.395	0.9735	-0.1270	-0.16
4.400	0.9735	-0.1271	-0.16
4.405	0.9735	-0.1273	-0.16
4.410	0.9735	-0.1273	-0.16
4.415	0.9734	-0.1275	-0.16
4.420	0.9737	-0.1272	-0.16
4.425	0.9738	-0.1274	-0.16
4.430	0.9736	-0.1277	-0.16
4.435	0.9735	-0.1277	-0.16
4.440	0.9736	-0.1278	-0.16
4.445	0.9736	-0.1276	-0.16
4.450	0.9727	-0.1277	-0.17
4.455	0.9730	-0.1275	-0.16
4.460	0.9726	-0.1278	-0.17
4.465	0.9723	-0.1274	-0.17
4.470	0.9723	-0.1277	-0.17
4.475	0.9722	-0.1274	-0.17
4.480	0.9719	-0.1276	-0.17
4.485	0.9719	-0.1275	-0.17
4.490	0.9722	-0.1276	-0.17
4.495	0.9725	-0.1274	-0.17
4.500	0.9726	-0.1275	-0.17
4.505	0.9729	-0.1280	-0.16
4.510	0.9728	-0.1277	-0.17
4.515	0.9730	-0.1280	-0.16
4.520	0.9734	-0.1279	-0.16
4.525	0.9734	-0.1278	-0.16
4.530	0.9734	-0.1281	-0.16
4.535	0.9738	-0.1278	-0.16
4.540	0.9736	-0.1279	-0.16
4.545	0.9738	-0.1281	-0.16
4.550	0.9734	-0.1282	-0.16
4.555	0.9735	-0.1280	-0.16
4.560	0.9736	-0.1281	-0.16
4.565	0.9735	-0.1282	-0.16
4.570	0.9735	-0.1283	-0.16
4.575	0.9736	-0.1282	-0.16
4.580	0.9736	-0.1280	-0.16
4.585	0.9738	-0.1283	-0.16
4.590	0.9737	-0.1283	-0.16
4.595	0.9736	-0.1283	-0.16
4.600	0.9735	-0.1283	-0.16

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.605	0.9735	-0.1284	-0.16
4.610	0.9736	-0.1285	-0.16
4.615	0.9734	-0.1286	-0.16
4.620	0.9732	-0.1286	-0.16
4.625	0.9734	-0.1286	-0.16
4.630	0.9731	-0.1289	-0.16
4.635	0.9730	-0.1288	-0.16
4.640	0.9731	-0.1289	-0.16
4.645	0.9729	-0.1293	-0.16
4.650	0.9730	-0.1294	-0.16
4.655	0.9730	-0.1292	-0.16
4.660	0.9725	-0.1295	-0.17
4.665	0.9727	-0.1300	-0.16
4.670	0.9726	-0.1301	-0.16
4.675	0.9728	-0.1299	-0.16
4.680	0.9729	-0.1300	-0.16
4.685	0.9723	-0.1301	-0.17
4.690	0.9727	-0.1302	-0.16
4.695	0.9725	-0.1302	-0.17
4.700	0.9727	-0.1305	-0.16
4.705	0.9723	-0.1312	-0.17
4.710	0.9727	-0.1306	-0.16
4.715	0.9727	-0.1308	-0.16
4.720	0.9730	-0.1306	-0.16
4.725	0.9728	-0.1304	-0.16
4.730	0.9723	-0.1309	-0.17
4.735	0.9726	-0.1306	-0.16
4.740	0.9724	-0.1309	-0.17
4.745	0.9718	-0.1304	-0.17
4.750	0.9716	-0.1308	-0.17
4.755	0.9719	-0.1313	-0.17
4.760	0.9717	-0.1315	-0.17
4.765	0.9712	-0.1319	-0.17
4.770	0.9713	-0.1318	-0.17
4.775	0.9705	-0.1319	-0.18
4.780	0.9711	-0.1324	-0.17
4.785	0.9712	-0.1323	-0.17
4.790	0.9705	-0.1328	-0.18
4.795	0.9704	-0.1329	-0.18
4.800	0.9709	-0.1332	-0.18
4.805	0.9706	-0.1331	-0.18
4.810	0.9706	-0.1329	-0.18
4.815	0.9706	-0.1335	-0.18
4.820	0.9708	-0.1328	-0.18
4.825	0.9706	-0.1327	-0.18
4.830	0.9702	-0.1333	-0.18

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
4.835	0.9703	-0.1334	-0.18
4.840	0.9700	-0.1331	-0.18
4.845	0.9702	-0.1331	-0.18
4.850	0.9700	-0.1329	-0.18
4.855	0.9703	-0.1328	-0.18
4.860	0.9704	-0.1329	-0.18
4.865	0.9703	-0.1324	-0.18
4.870	0.9702	-0.1330	-0.18
4.875	0.9701	-0.1330	-0.18
4.880	0.9700	-0.1330	-0.18
4.885	0.9701	-0.1324	-0.18
4.890	0.9701	-0.1333	-0.18
4.895	0.9701	-0.1332	-0.18
4.900	0.9698	-0.1335	-0.18
4.905	0.9701	-0.1336	-0.18
4.910	0.9699	-0.1336	-0.18
4.915	0.9699	-0.1337	-0.18
4.920	0.9699	-0.1338	-0.18
4.925	0.9698	-0.1341	-0.18
4.930	0.9697	-0.1340	-0.19
4.935	0.9697	-0.1342	-0.18
4.940	0.9697	-0.1343	-0.18
4.945	0.9696	-0.1343	-0.19
4.950	0.9694	-0.1345	-0.19
4.955	0.9694	-0.1346	-0.19
4.960	0.9692	-0.1345	-0.19
4.965	0.9692	-0.1345	-0.19
4.970	0.9691	-0.1348	-0.19
4.975	0.9692	-0.1349	-0.19
4.980	0.9692	-0.1350	-0.19
4.985	0.9689	-0.1352	-0.19
4.990	0.9690	-0.1352	-0.19
4.995	0.9690	-0.1351	-0.19
5.000	0.9688	-0.1355	-0.19
5.005	0.9688	-0.1354	-0.19
5.010	0.9690	-0.1353	-0.19
5.015	0.9689	-0.1353	-0.19
5.020	0.9690	-0.1354	-0.19
5.025	0.9691	-0.1354	-0.19
5.030	0.9691	-0.1356	-0.19
5.035	0.9692	-0.1357	-0.19
5.040	0.9693	-0.1358	-0.19
5.045	0.9690	-0.1360	-0.19
5.050	0.9691	-0.1359	-0.19
5.055	0.9690	-0.1363	-0.19
5.060	0.9689	-0.1361	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.065	0.9693	-0.1363	-0.19
5.070	0.9689	-0.1359	-0.19
5.075	0.9690	-0.1356	-0.19
5.080	0.9691	-0.1355	-0.19
5.085	0.9693	-0.1353	-0.19
5.090	0.9693	-0.1354	-0.19
5.095	0.9693	-0.1352	-0.19
5.100	0.9692	-0.1353	-0.19
5.105	0.9693	-0.1353	-0.19
5.110	0.9694	-0.1355	-0.19
5.115	0.9696	-0.1355	-0.18
5.120	0.9696	-0.1356	-0.18
5.125	0.9697	-0.1357	-0.18
5.130	0.9699	-0.1358	-0.18
5.135	0.9696	-0.1359	-0.18
5.140	0.9698	-0.1359	-0.18
5.145	0.9696	-0.1358	-0.18
5.150	0.9696	-0.1360	-0.18
5.155	0.9697	-0.1360	-0.18
5.160	0.9697	-0.1361	-0.18
5.165	0.9698	-0.1361	-0.18
5.170	0.9698	-0.1361	-0.18
5.175	0.9699	-0.1361	-0.18
5.180	0.9698	-0.1362	-0.18
5.185	0.9697	-0.1362	-0.18
5.190	0.9698	-0.1362	-0.18
5.195	0.9698	-0.1364	-0.18
5.200	0.9699	-0.1364	-0.18
5.205	0.9699	-0.1366	-0.18
5.210	0.9699	-0.1366	-0.18
5.215	0.9700	-0.1368	-0.18
5.220	0.9700	-0.1370	-0.18
5.225	0.9699	-0.1371	-0.18
5.230	0.9700	-0.1371	-0.18
5.235	0.9699	-0.1372	-0.18
5.240	0.9700	-0.1374	-0.18
5.245	0.9698	-0.1377	-0.18
5.250	0.9698	-0.1377	-0.18
5.255	0.9695	-0.1376	-0.18
5.260	0.9693	-0.1378	-0.18
5.265	0.9696	-0.1378	-0.18
5.270	0.9695	-0.1381	-0.18
5.275	0.9694	-0.1385	-0.18
5.280	0.9690	-0.1386	-0.19
5.285	0.9689	-0.1388	-0.19
5.290	0.9688	-0.1390	-0.19

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.295	0.9688	-0.1392	-0.19
5.300	0.9685	-0.1393	-0.19
5.305	0.9687	-0.1390	-0.19
5.310	0.9684	-0.1397	-0.19
5.315	0.9681	-0.1397	-0.19
5.320	0.9683	-0.1406	-0.19
5.325	0.9685	-0.1407	-0.19
5.330	0.9680	-0.1405	-0.19
5.335	0.9681	-0.1409	-0.19
5.340	0.9683	-0.1409	-0.19
5.345	0.9679	-0.1414	-0.19
5.350	0.9677	-0.1415	-0.19
5.355	0.9675	-0.1417	-0.20
5.360	0.9678	-0.1421	-0.19
5.365	0.9675	-0.1418	-0.19
5.370	0.9676	-0.1422	-0.19
5.375	0.9674	-0.1421	-0.20
5.380	0.9674	-0.1421	-0.20
5.385	0.9668	-0.1421	-0.20
5.390	0.9671	-0.1429	-0.20
5.395	0.9668	-0.1428	-0.20
5.400	0.9665	-0.1432	-0.20
5.405	0.9661	-0.1433	-0.21
5.410	0.9660	-0.1437	-0.21
5.415	0.9663	-0.1441	-0.20
5.420	0.9659	-0.1443	-0.21
5.425	0.9661	-0.1440	-0.20
5.430	0.9654	-0.1445	-0.21
5.435	0.9653	-0.1450	-0.21
5.440	0.9657	-0.1448	-0.21
5.445	0.9654	-0.1453	-0.21
5.450	0.9649	-0.1458	-0.21
5.455	0.9648	-0.1458	-0.21
5.460	0.9650	-0.1459	-0.21
5.465	0.9652	-0.1462	-0.21
5.470	0.9649	-0.1462	-0.21
5.475	0.9650	-0.1465	-0.21
5.480	0.9646	-0.1466	-0.21
5.485	0.9649	-0.1469	-0.21
5.490	0.9646	-0.1468	-0.21
5.495	0.9645	-0.1469	-0.21
5.500	0.9643	-0.1469	-0.22
5.505	0.9642	-0.1472	-0.22
5.510	0.9640	-0.1473	-0.22
5.515	0.9639	-0.1475	-0.22
5.520	0.9637	-0.1477	-0.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.525	0.9636	-0.1479	-0.22
5.530	0.9636	-0.1480	-0.22
5.535	0.9636	-0.1483	-0.22
5.540	0.9634	-0.1484	-0.22
5.545	0.9633	-0.1487	-0.22
5.550	0.9632	-0.1488	-0.22
5.555	0.9632	-0.1489	-0.22
5.560	0.9631	-0.1490	-0.22
5.565	0.9631	-0.1491	-0.22
5.570	0.9631	-0.1492	-0.22
5.575	0.9631	-0.1493	-0.22
5.580	0.9630	-0.1494	-0.22
5.585	0.9630	-0.1496	-0.22
5.590	0.9629	-0.1496	-0.23
5.595	0.9627	-0.1496	-0.23
5.600	0.9625	-0.1499	-0.23
5.605	0.9624	-0.1501	-0.23
5.610	0.9624	-0.1502	-0.23
5.615	0.9623	-0.1505	-0.23
5.620	0.9621	-0.1508	-0.23
5.625	0.9622	-0.1511	-0.23
5.630	0.9620	-0.1513	-0.23
5.635	0.9618	-0.1515	-0.23
5.640	0.9619	-0.1516	-0.23
5.645	0.9619	-0.1521	-0.23
5.650	0.9621	-0.1520	-0.23
5.655	0.9620	-0.1522	-0.23
5.660	0.9619	-0.1523	-0.23
5.665	0.9624	-0.1527	-0.23
5.670	0.9618	-0.1528	-0.23
5.675	0.9620	-0.1530	-0.23
5.680	0.9620	-0.1530	-0.23
5.685	0.9617	-0.1534	-0.23
5.690	0.9618	-0.1536	-0.23
5.695	0.9617	-0.1536	-0.23
5.700	0.9618	-0.1541	-0.23
5.705	0.9619	-0.1542	-0.23
5.710	0.9620	-0.1544	-0.23
5.715	0.9622	-0.1548	-0.22
5.720	0.9620	-0.1546	-0.23
5.725	0.9623	-0.1548	-0.22
5.730	0.9622	-0.1553	-0.22
5.735	0.9622	-0.1556	-0.22
5.740	0.9625	-0.1557	-0.22
5.745	0.9619	-0.1556	-0.23
5.750	0.9625	-0.1561	-0.22

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Frequency (GHz)	**Re(S_{11})**	**Im(S_{11})**	**$\lvert S_{11}\rvert$ (dB)**
5.755	0.9625	-0.1563	-0.22
5.760	0.9626	-0.1561	-0.22
5.765	0.9623	-0.1565	-0.22
5.770	0.9626	-0.1566	-0.22
5.775	0.9627	-0.1565	-0.22
5.780	0.9627	-0.1569	-0.22
5.785	0.9628	-0.1571	-0.22
5.790	0.9630	-0.1572	-0.21
5.795	0.9629	-0.1574	-0.21
5.800	0.9629	-0.1576	-0.21
5.805	0.9631	-0.1577	-0.21
5.810	0.9631	-0.1577	-0.21
5.815	0.9629	-0.1579	-0.21
5.820	0.9628	-0.1580	-0.21
5.825	0.9630	-0.1582	-0.21
5.830	0.9629	-0.1583	-0.21
5.835	0.9629	-0.1584	-0.21
5.840	0.9628	-0.1587	-0.21
5.845	0.9626	-0.1588	-0.21
5.850	0.9626	-0.1588	-0.21
5.855	0.9626	-0.1590	-0.21
5.860	0.9623	-0.1591	-0.22
5.865	0.9624	-0.1593	-0.22
5.870	0.9620	-0.1595	-0.22
5.875	0.9621	-0.1597	-0.22
5.880	0.9619	-0.1598	-0.22
5.885	0.9617	-0.1601	-0.22
5.890	0.9615	-0.1602	-0.22
5.895	0.9615	-0.1609	-0.22
5.900	0.9616	-0.1608	-0.22
5.905	0.9609	-0.1612	-0.23
5.910	0.9612	-0.1613	-0.22
5.915	0.9608	-0.1613	-0.23
5.920	0.9608	-0.1618	-0.23
5.925	0.9610	-0.1620	-0.22
5.930	0.9603	-0.1619	-0.23
5.935	0.9607	-0.1621	-0.23
5.940	0.9605	-0.1623	-0.23
5.945	0.9597	-0.1624	-0.23
5.950	0.9600	-0.1624	-0.23
5.955	0.9599	-0.1625	-0.23
5.960	0.9597	-0.1629	-0.23
5.965	0.9592	-0.1629	-0.24
5.970	0.9593	-0.1632	-0.24
5.975	0.9586	-0.1632	-0.24
5.980	0.9587	-0.1637	-0.24

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Frequency (GHz)	**Re**(S_{11})	**Im**(S_{11})	$\lvert S_{11}\rvert$ **(dB)**
5.985	0.9581	-0.1646	-0.25
5.990	0.9579	-0.1641	-0.25
5.995	0.9573	-0.1645	-0.25
6.000	0.9571	-0.1646	-0.25